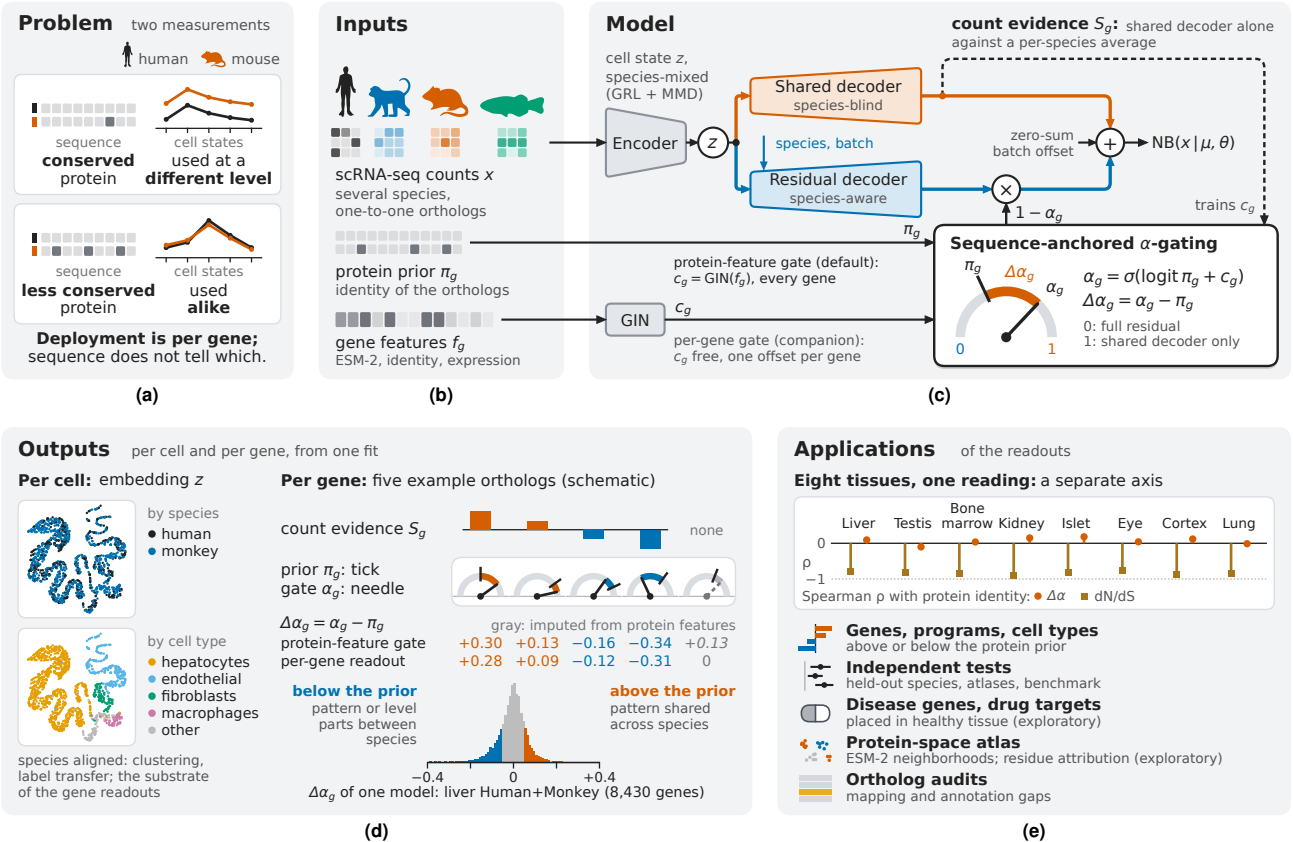
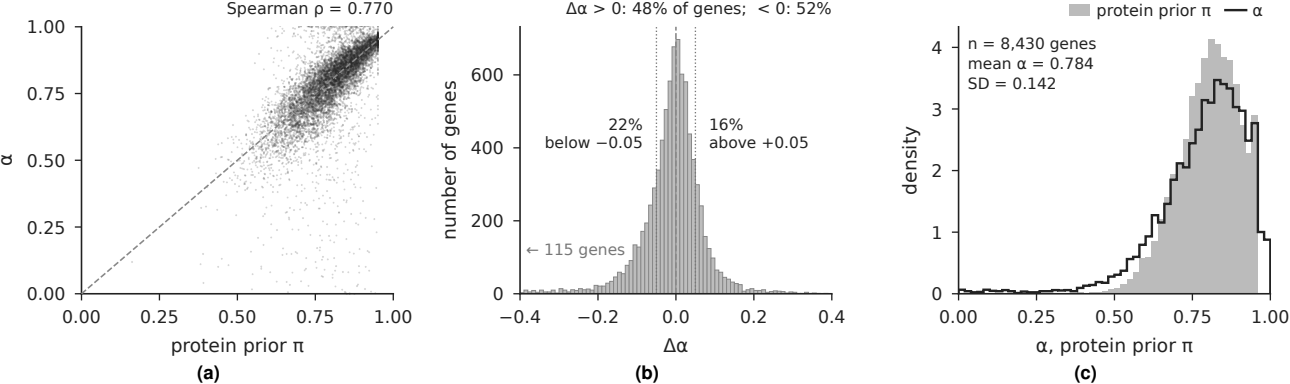
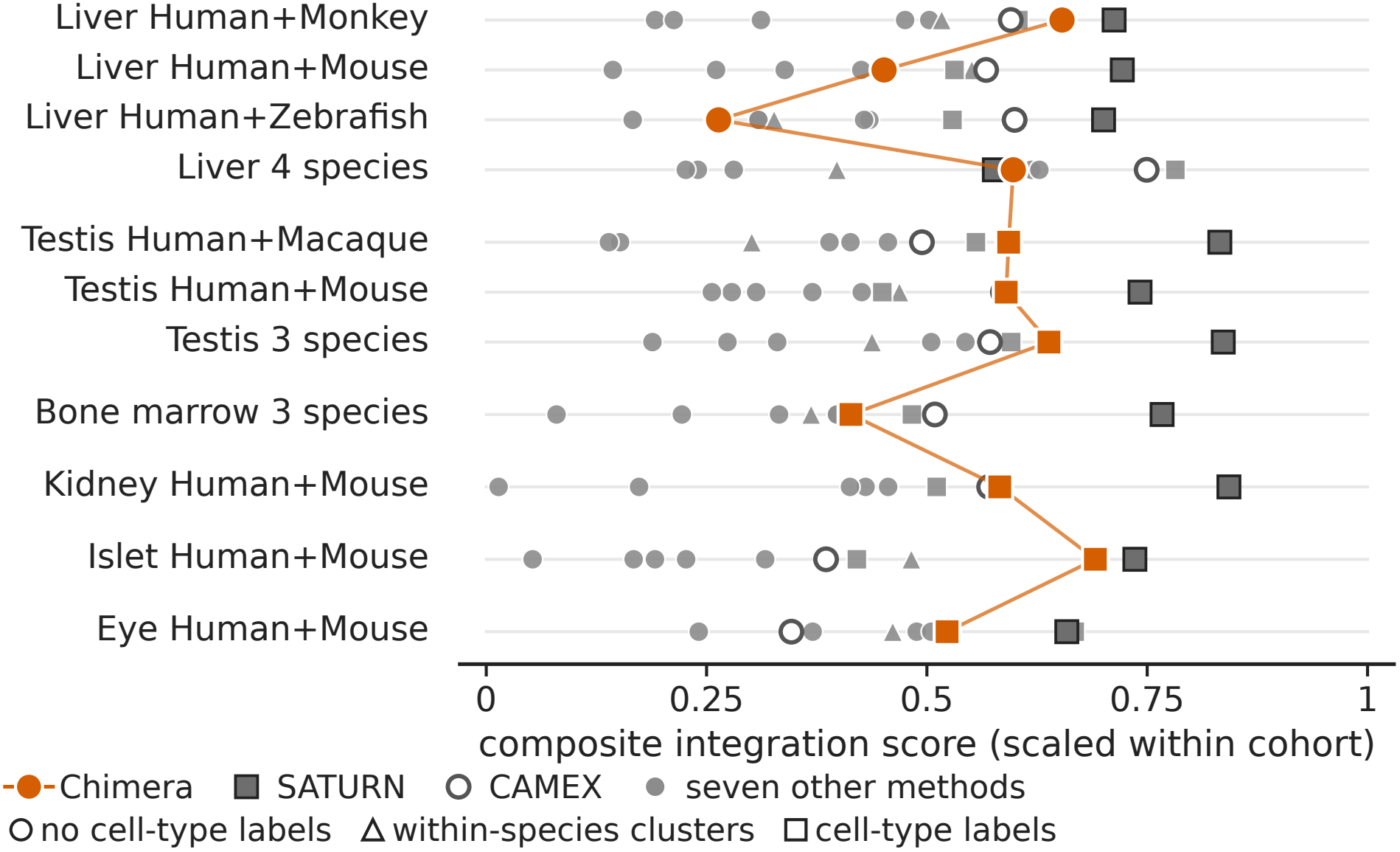
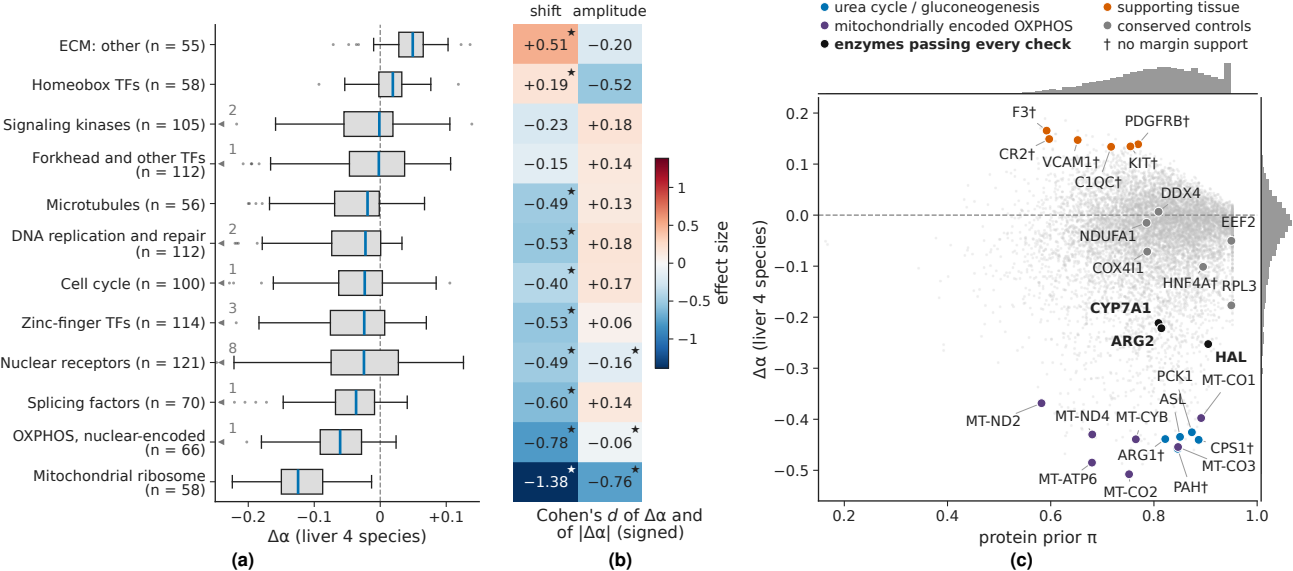
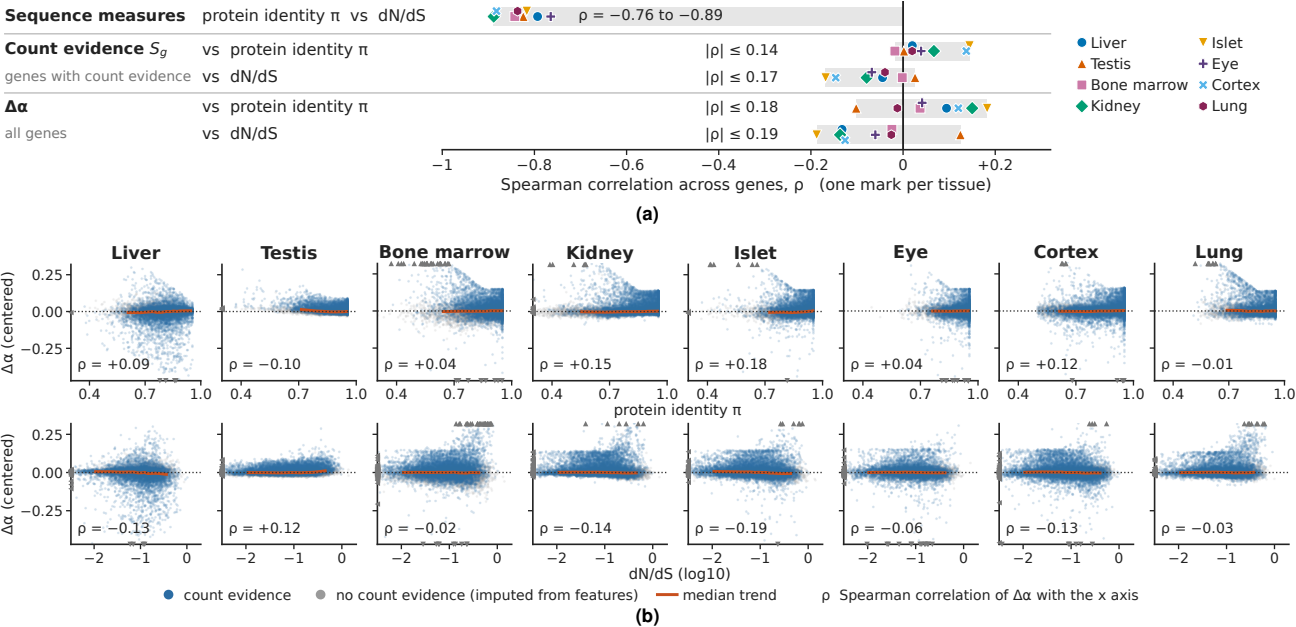
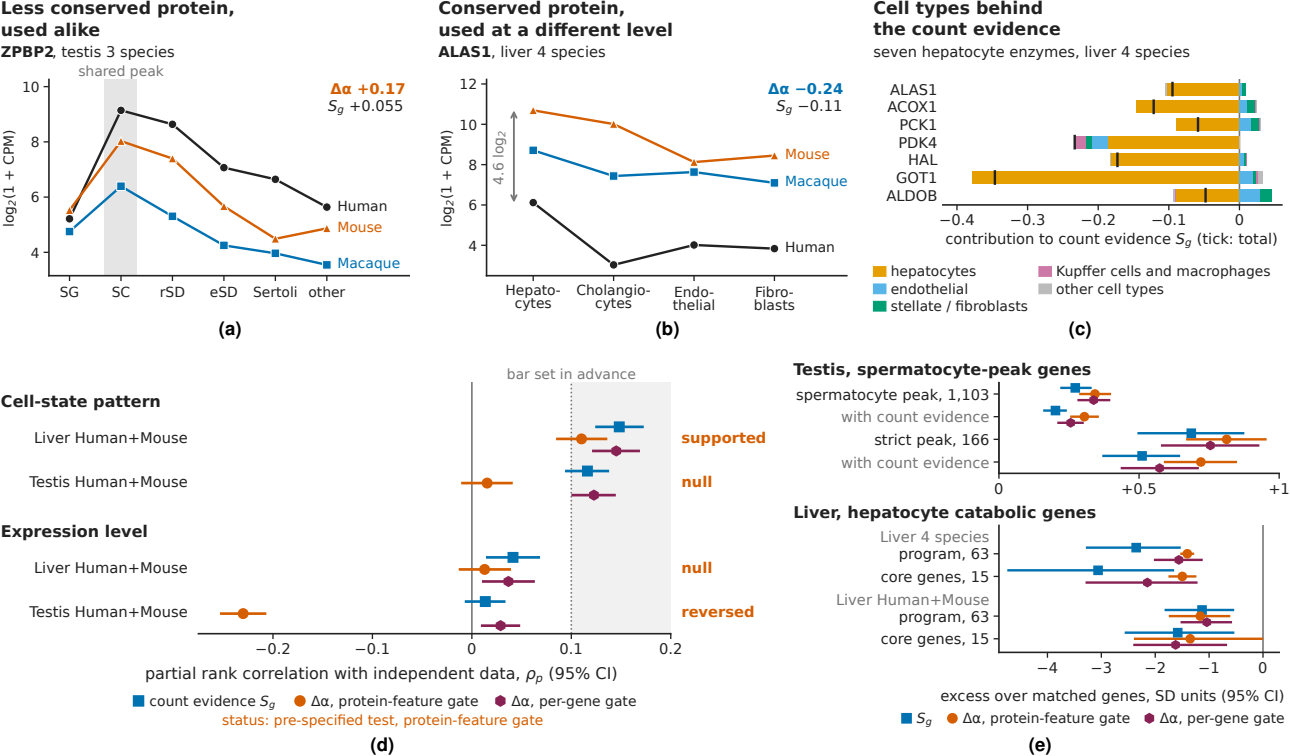

Problem
two measurements
human
mouse
sequence
conserved
protein
cell states
used at a
different level
sequence
less conserved
protein
cell states
used
alike
Deployment is per gene;
sequence does not tell which.
(a)
Inputs
scRNA-seq counts x
several species,
one-to-one orthologs
protein prior π_g
identity of the orthologs
gene features f_g
ESM-2, identity, expression
(b)
Model
cell state z,
species-mixed
(GRL + MMD)
Encoder
z
Shared decoder
species-blind
species, batch
Residual decoder
species-aware
count evidence S_g: shared decoder alone
against a per-species average
zero-sum
batch offset
NB(x | μ, θ)
1 − α_g
trains c_g
π_g
protein-feature gate (default):
c_g = GIN(f_g), every gene
GIN
c_g
per-gene gate (companion):
c_g free, one offset per gene
Sequence-anchored α-gating
π_g
Δα_g
α_g
α_g = σ(logit π_g + c_g)
Δα_g = α_g − π_g
0: full residual
1: shared decoder only
0
1
(c)
Outputs
per cell and per gene, from one fit
Per cell: embedding z
by species
human
monkey
by cell type
hepatocytes
endothelial
fibroblasts
macrophages
other
species aligned: clustering,
label transfer; the substrate
of the gene readouts
Per gene: five example orthologs (schematic)
count evidence S_g
none
prior π_g: tick
gate α_g: needle
Δα_g = α_g − π_g
protein-feature gate
per-gene readout
gray: imputed from protein features
+0.30 +0.13 −0.16 −0.34 +0.13
+0.28 +0.09 −0.12 −0.31 0
below the prior
pattern or level
parts between
species
above the prior
pattern shared
across species
−0.4
0
+0.4
Δα_g of one model: liver Human+Monkey (8,430 genes)
(d)
Applications
of the readouts
Eight tissues, one reading: a separate axis
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
0
ρ
−1
Spearman ρ with protein identity: Δα dN/dS
Genes, programs, cell types
above or below the protein prior
Independent tests
held-out species, atlases, benchmark
Disease genes, drug targets
placed in healthy tissue (exploratory)
Protein-space atlas
ESM-2 neighborhoods; residue attribution (exploratory)
Ortholog audits
mapping and annotation gaps
(e)

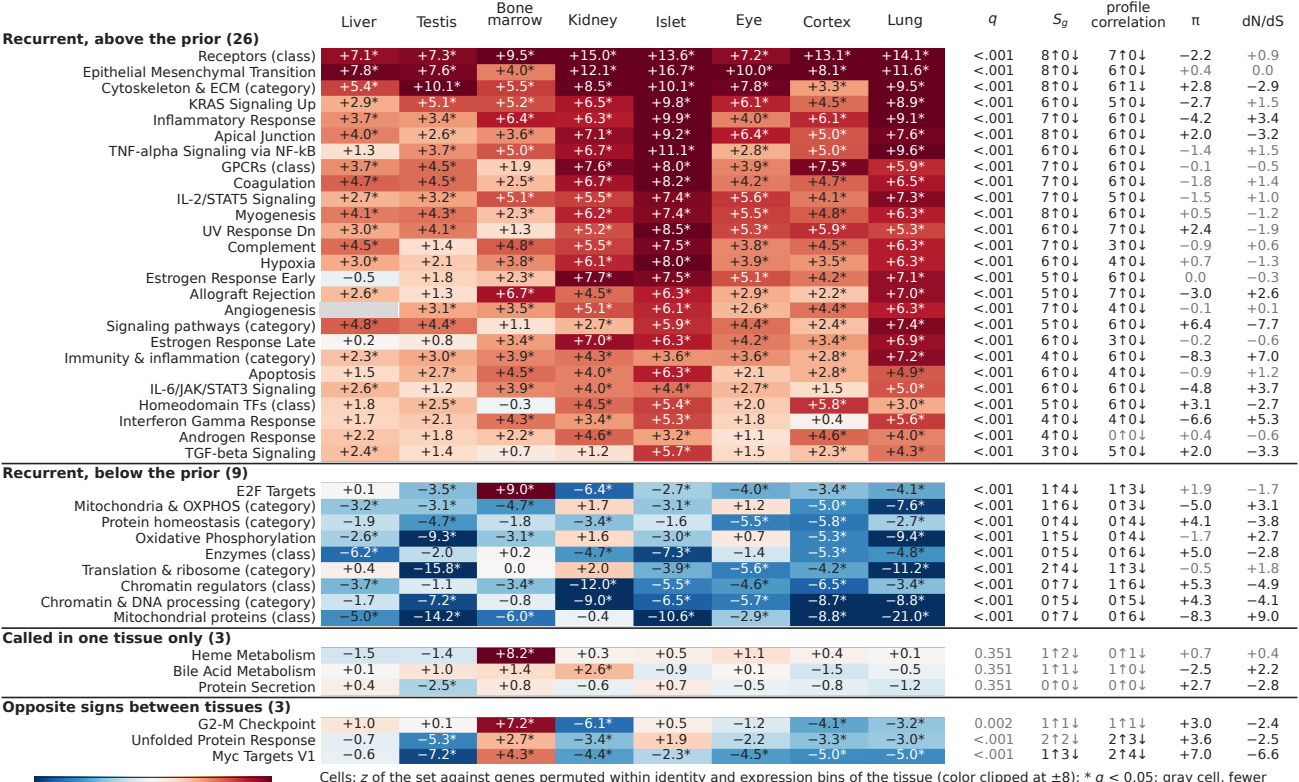

Spearman ρ = 0.770
1.00
0.75
0.50
0.25
0.00
α
0.00
0.25
0.50
0.75
1.00
protein prior π
(a)
Δα > 0: 48% of genes; < 0: 52%
number of genes
600
400
200
0
22%
below −0.05
16%
above +0.05
← 115 genes
−0.4
−0.2
0.0
0.2
0.4
Δα
(b)
protein prior π
α
n = 8,430 genes
mean α = 0.784
SD = 0.142
density
4
3
2
1
0
0.00
0.25
0.50
0.75
1.00
α, protein prior π
(c)


Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
Bone marrow 3 species
Kidney Human+Mouse
Islet Human+Mouse
Eye Human+Mouse
0
0.25
0.5
0.75
1
composite integration score (scaled within cohort)
Chimera
SATURN
CAMEX
seven other methods
no cell-type labels
within-species clusters
cell-type labels

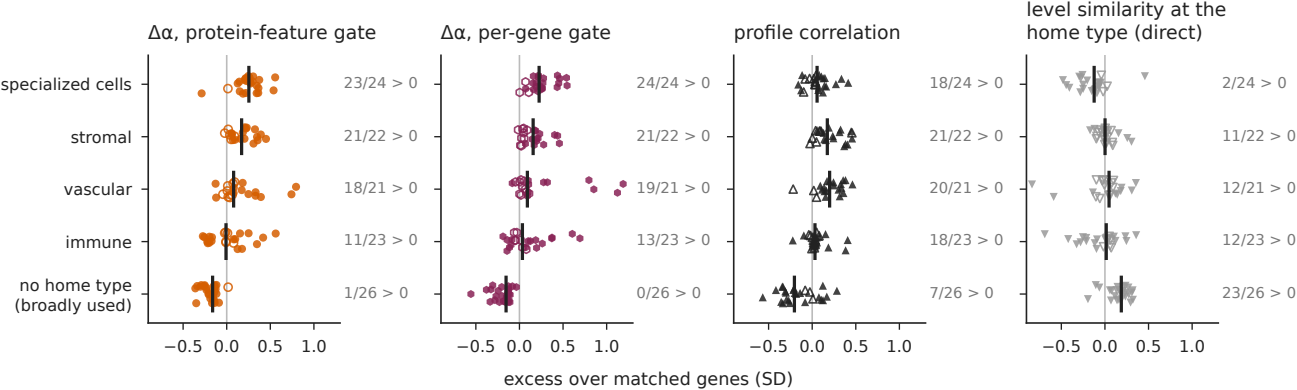

ECM: other (n = 55)
Homeobox TFs (n = 58)
Signaling kinases (n = 105)
Forkhead and other TFs (n = 112)
Microtubules (n = 56)
DNA replication and repair (n = 112)
Cell cycle (n = 100)
Zinc-finger TFs (n = 114)
Nuclear receptors (n = 121)
Splicing factors (n = 70)
OXPHOS, nuclear-encoded (n = 66)
Mitochondrial ribosome (n = 58)
2
1
2
1
3
8
1
1
−0.2
−0.1
0
+0.1
Δα (liver 4 species)
(a)
shift
amplitude
+0.51★
−0.20
+0.19★
−0.52
−0.23
+0.18
−0.15
+0.14
−0.49★
+0.13
−0.53★
+0.18
−0.40★
+0.17
−0.53★
+0.06
−0.49★
−0.16★
−0.60★
+0.14
−0.78★
−0.06★
−1.38★
−0.76★
1
0.5
0
−0.5
−1
effect size
Cohen's d of Δα and of |Δα| (signed)
(b)
urea cycle / gluconeogenesis
mitochondrially encoded OXPHOS
enzymes passing every check
supporting tissue
conserved controls
† no margin support
F3†
PDGFRB†
CR2†
VCAM1†
KIT†
C1QC†
DDX4
EEF2
NDUFA1
COX4I1
HNF4A†
RPL3
CYP7A1
ARG2
HAL
PCK1
MT-CO1
ASL
MT-ND2
MT-ND4
MT-CYB
CPS1†
ARG1†
MT-CO3
MT-ATP6
MT-CO2
PAH†
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
Δα (liver 4 species)
0.2
0.4
0.6
0.8
1.0
protein prior π
(c)

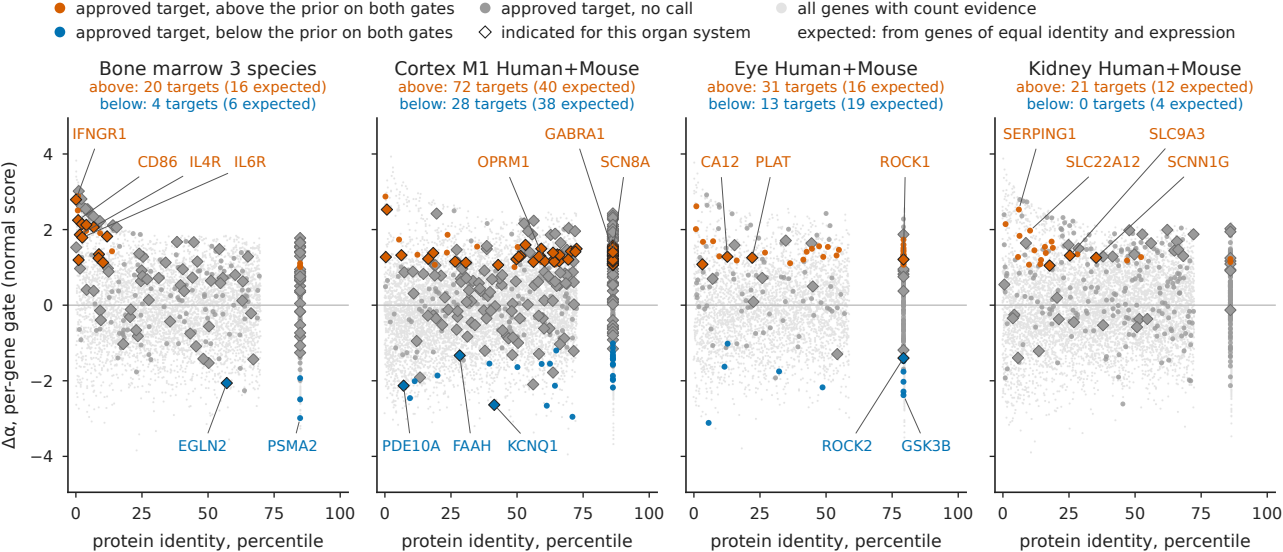

Sequence measures
protein identity π vs dN/dS
ρ = −0.76 to −0.89
Count evidence S_g
vs protein identity π
|ρ| ≤ 0.14
genes with count evidence
vs dN/dS
|ρ| ≤ 0.17
Δα
vs protein identity π
|ρ| ≤ 0.18
all genes
vs dN/dS
|ρ| ≤ 0.19
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
−1
−0.8
−0.6
−0.4
−0.2
0
+0.2
Spearman correlation across genes, ρ (one mark per tissue)
(a)
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
Δα (centered)
0.25
0.00
−0.25
ρ = +0.09
ρ = −0.10
ρ = +0.04
ρ = +0.15
ρ = +0.18
ρ = +0.04
ρ = +0.12
ρ = −0.01
0.4
0.7
1.0
protein identity π
ρ = −0.13
ρ = +0.12
ρ = −0.02
ρ = −0.14
ρ = −0.19
ρ = −0.06
ρ = −0.13
ρ = −0.03
−2
−1
0
dN/dS (log10)
count evidence
no count evidence (imputed from features)
median trend
ρ Spearman correlation of Δα with the x axis
(b)

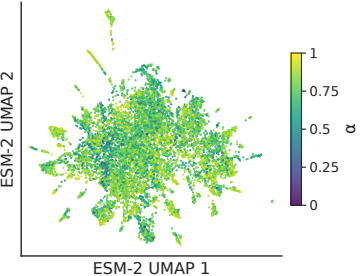

Less conserved protein, used alike
ZPBP2, testis 3 species
shared peak
Δα +0.17
S_g +0.055
log2(1 + CPM)
Human
Mouse
Macaque
SG
SC
rSD
eSD
Sertoli
other
(a)
Conserved protein, used at a different level
ALAS1, liver 4 species
Δα −0.24
S_g −0.11
4.6 log2
Mouse
Macaque
Human
Hepato-cytes
Cholangio-cytes
Endo-thelial
Fibro-blasts
(b)
Cell types behind the count evidence
seven hepatocyte enzymes, liver 4 species
ALAS1
ACOX1
PCK1
PDK4
HAL
GOT1
ALDOB
contribution to count evidence S_g (tick: total)
hepatocytes
endothelial
stellate / fibroblasts
Kupffer cells and macrophages
other cell types
(c)
bar set in advance
Cell-state pattern
Liver Human+Mouse
Testis Human+Mouse
supported
null
Expression level
Liver Human+Mouse
Testis Human+Mouse
null
reversed
partial rank correlation with independent data, ρ_p (95% CI)
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
status: pre-specified test, protein-feature gate
(d)
Testis, spermatocyte-peak genes
spermatocyte peak, 1,103
with count evidence
strict peak, 166
with count evidence
Liver, hepatocyte catabolic genes
Liver 4 species
program, 63
core genes, 15
Liver Human+Mouse
program, 63
core genes, 15
excess over matched genes, SD units (95% CI)
S_g
Δα, protein-feature gate
Δα, per-gene gate
(e)



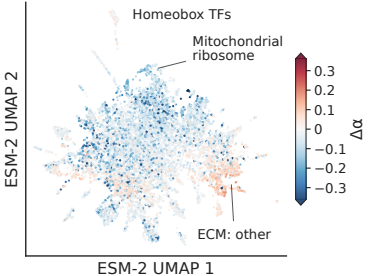
Cells: z of the set against genes permuted within identity and expression bins of the tissue (color clipped at ±8); * $q < 0.05$; gray cell, fewer than 15 genes. Right: q, recurrence across tissues; S_g and profile correlation, tissues with the set above (↑) and below (↓); π and dN/dS, mean z over tissues. Gray type: not recurrent under every control (q), on that readout (S_g, profile correlation), or |mean z| below 1.96 (π, dN/dS).


Δα, protein-feature gate
Δα, per-gene gate
profile correlation
level similarity at the home type (direct)
specialized cells
stromal
vascular
immune
no home type (broadly used)
23/24 > 0
21/22 > 0
18/21 > 0
11/23 > 0
1/26 > 0
24/24 > 0
21/22 > 0
19/21 > 0
13/23 > 0
0/26 > 0
18/24 > 0
21/22 > 0
20/21 > 0
18/23 > 0
7/26 > 0
2/24 > 0
11/22 > 0
12/21 > 0
12/23 > 0
23/26 > 0
−0.5 0.0 0.5 1.0
excess over matched genes (SD)

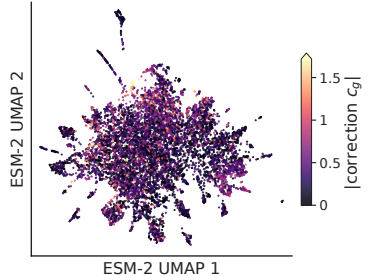

approved target, above the prior on both gates
approved target, below the prior on both gates
approved target, no call
indicated for this organ system
all genes with count evidence
expected: from genes of equal identity and expression
Bone marrow 3 species
above: 20 targets (16 expected)
below: 4 targets (6 expected)
Cortex M1 Human+Mouse
above: 72 targets (40 expected)
below: 28 targets (38 expected)
Eye Human+Mouse
above: 31 targets (16 expected)
below: 13 targets (19 expected)
Kidney Human+Mouse
above: 21 targets (12 expected)
below: 0 targets (4 expected)
IFNGR1
CD86
IL4R
IL6R
EGLN2
PSMA2
GABRA1
OPRM1
SCN8A
PDE10A
FAAH
KCNQ1
CA12
PLAT
ROCK1
ROCK2
GSK3B
SERPING1
SLC9A3
SLC22A12
SCNN1G
Δα, per-gene gate (normal score)
protein identity, percentile



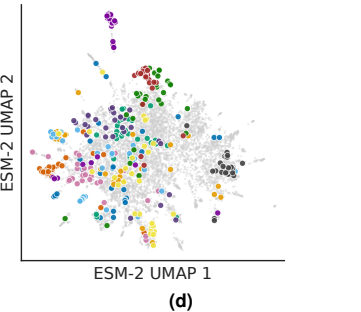
(a)



(b)



(c)



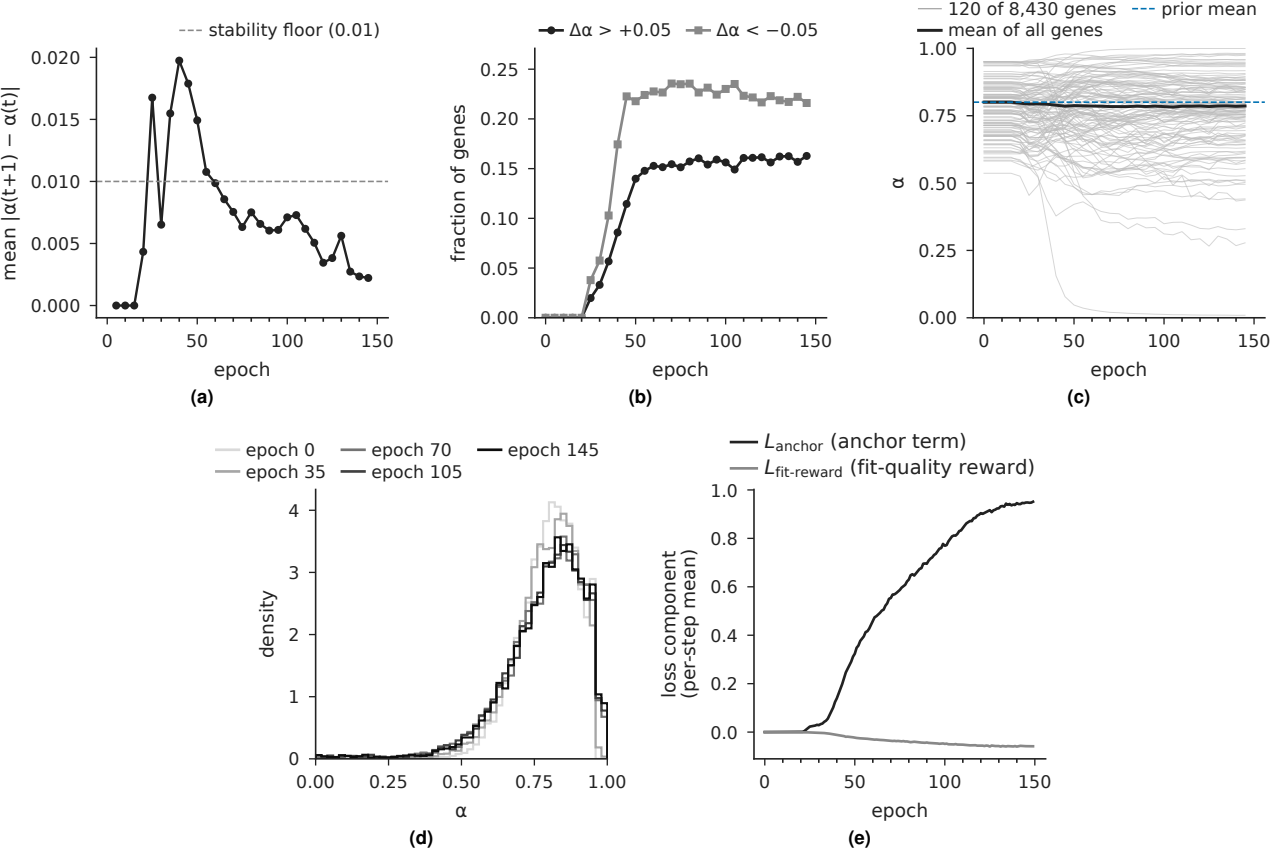
(d)

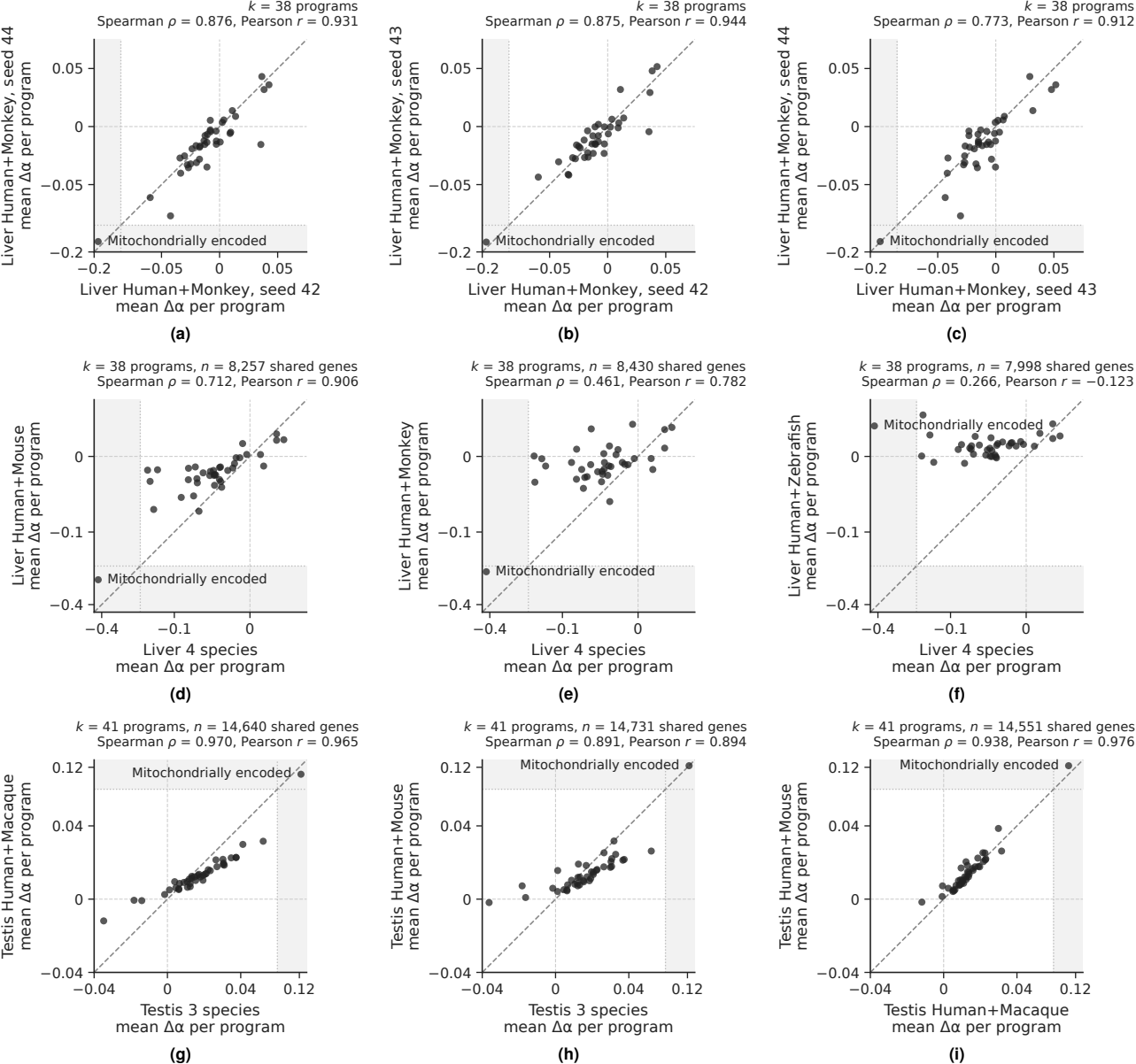
- Nuclear receptors (n = 121)
- Zinc-finger TFs (n = 114)
- DNA replication and repair (n = 112)
- Forkhead and other TFs (n = 112)
- Signaling kinases (n = 105)
- Cell cycle (n = 100)
- Splicing factors (n = 70)
- OXPHOS, nuclear-encoded (n = 66)
- Homeobox TFs (n = 58)
- Mitochondrial ribosome (n = 58)
- Microtubules (n = 56)
- ECM: other (n = 55)

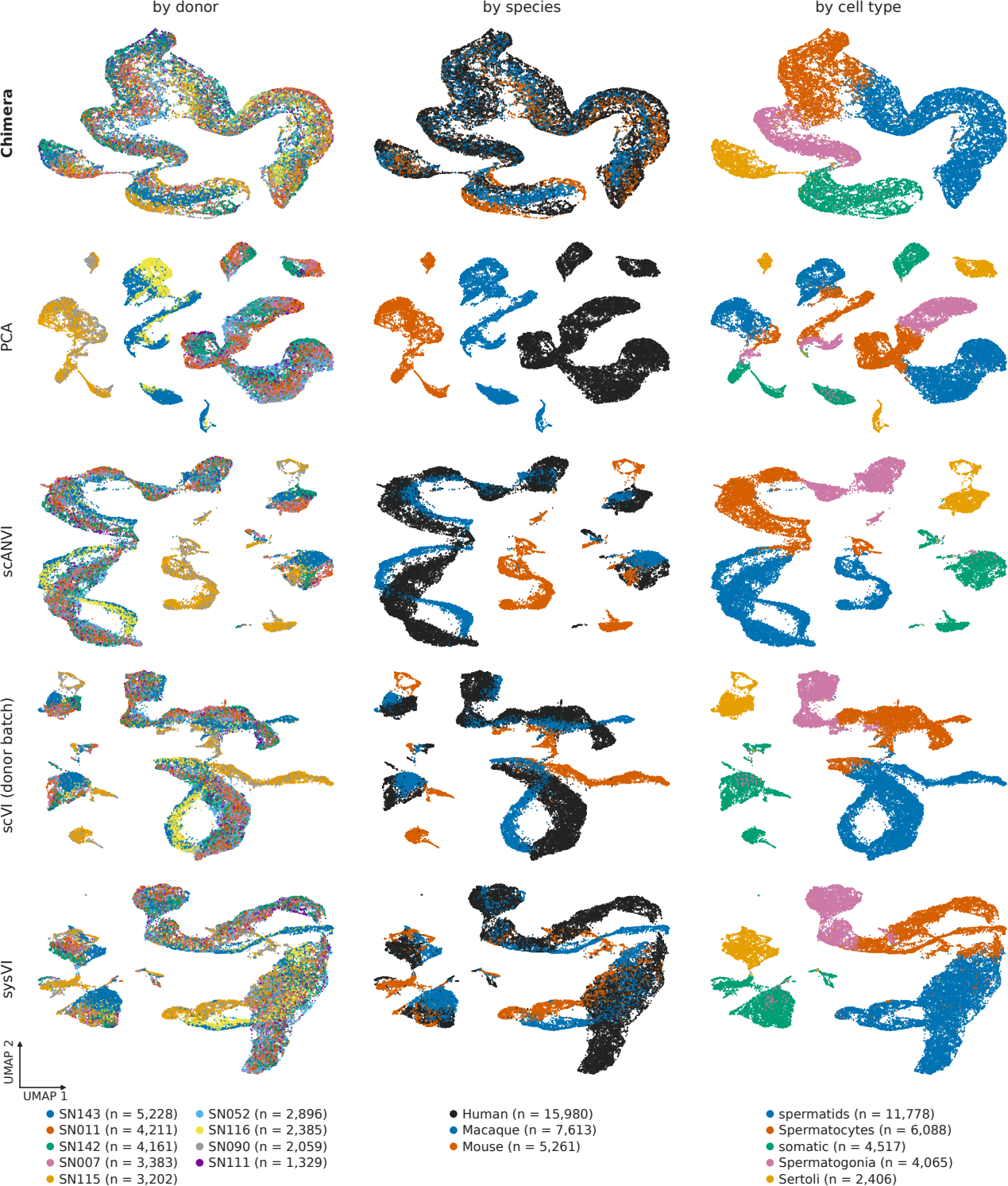
	1	2	3	4	5	6	7	8	9	10	11	12
Human+Monkey	−.010	−.009	−.008	−.007	−.007	−.006	−.004	−.001	+.001	+.004	+.006	+.010
Human+Mouse	−.033	−.031	−.031	−.029	−.028	−.027	−.018	−.017	−.016	−.009	+.004	+.008
Human+Zebrafish	−.003	+.001	+.002	+.003	+.004	+.005	+.005	+.005	+.010	+.017	+.018	+.019
4 species	−.062	−.045	−.045	−.029	−.023	−.015	−.014	−.012	−.006	+.006	+.012	+.019

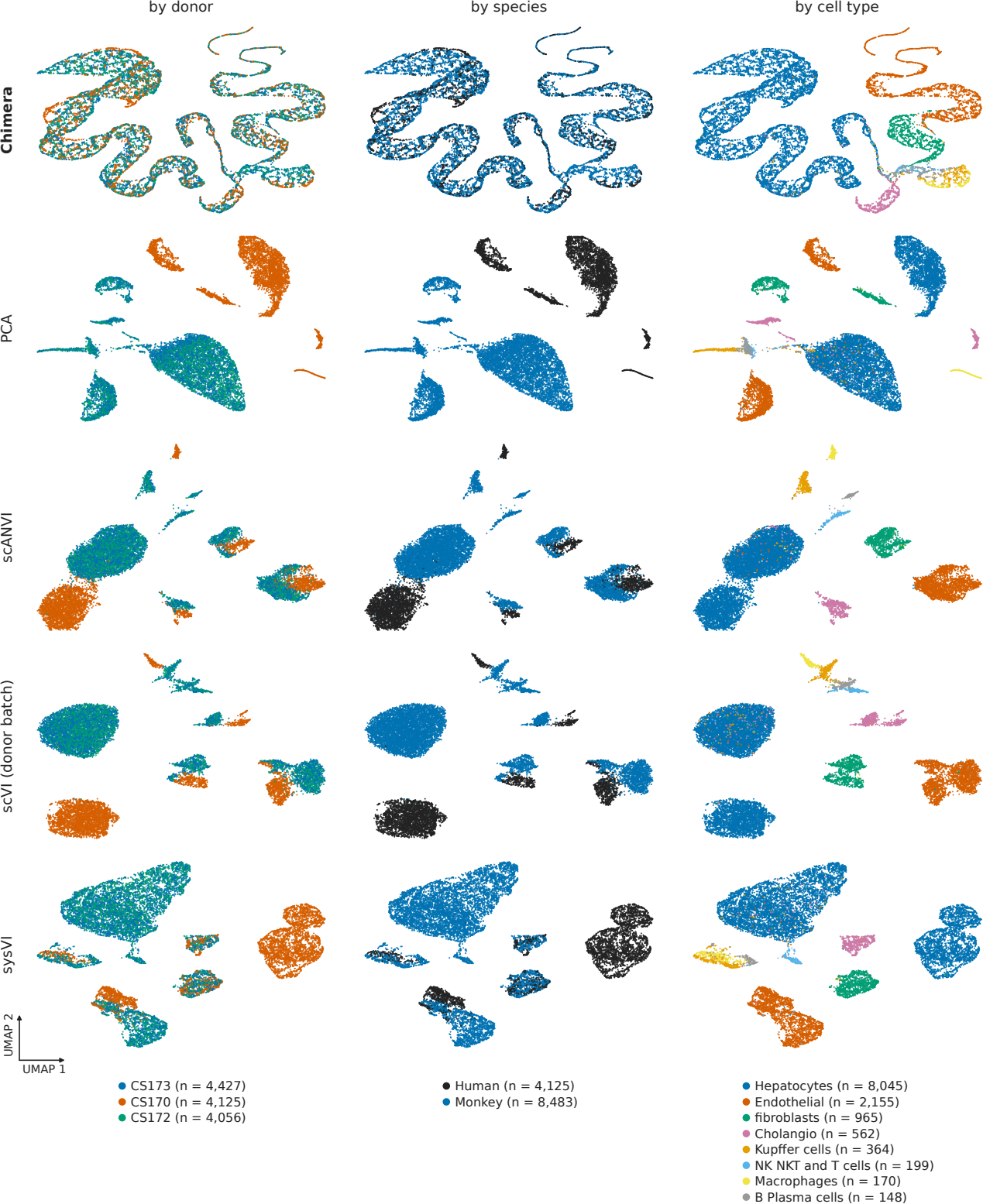
median Δα of 12 protein clusters, sorted within the cohort
(*k*-means on the ESM-2 components, one fit per cohort)

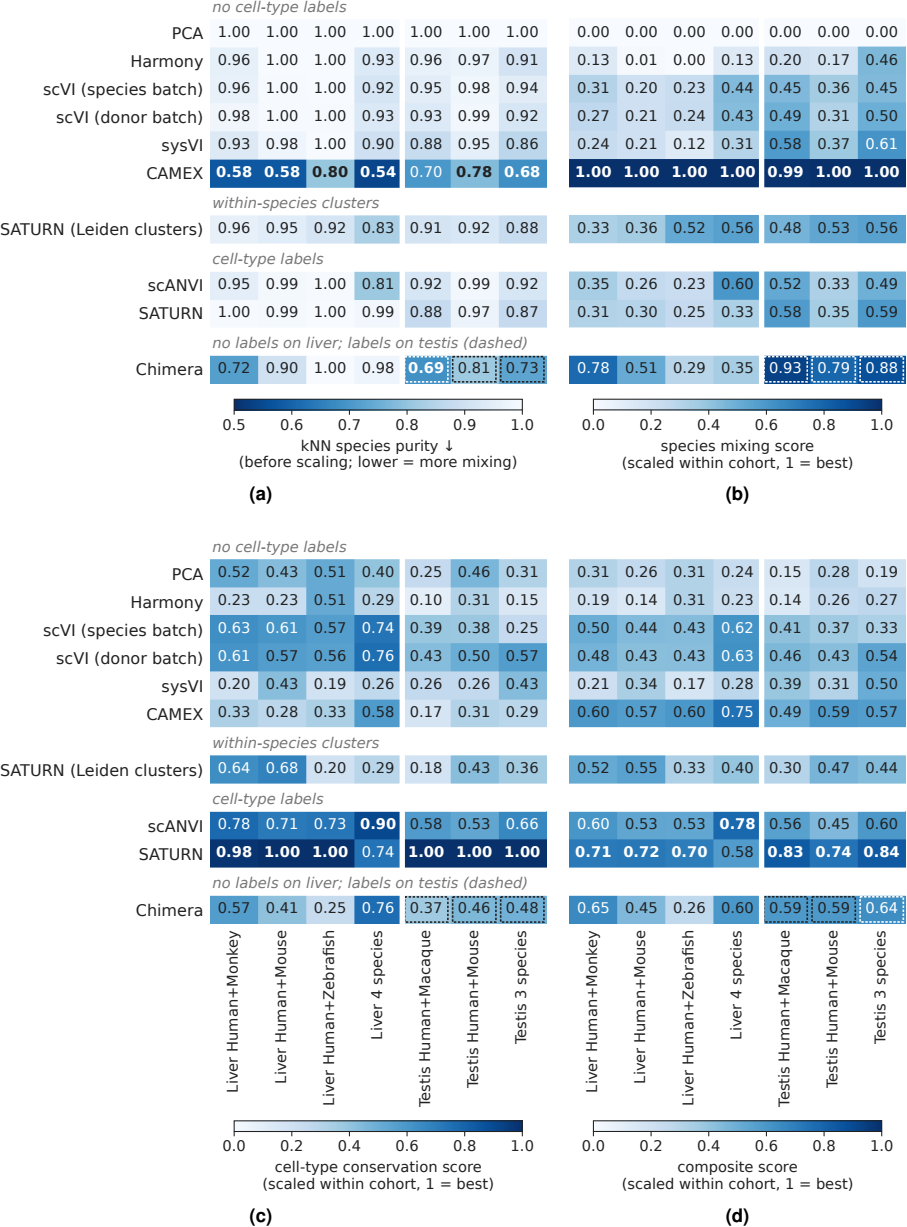
(e)

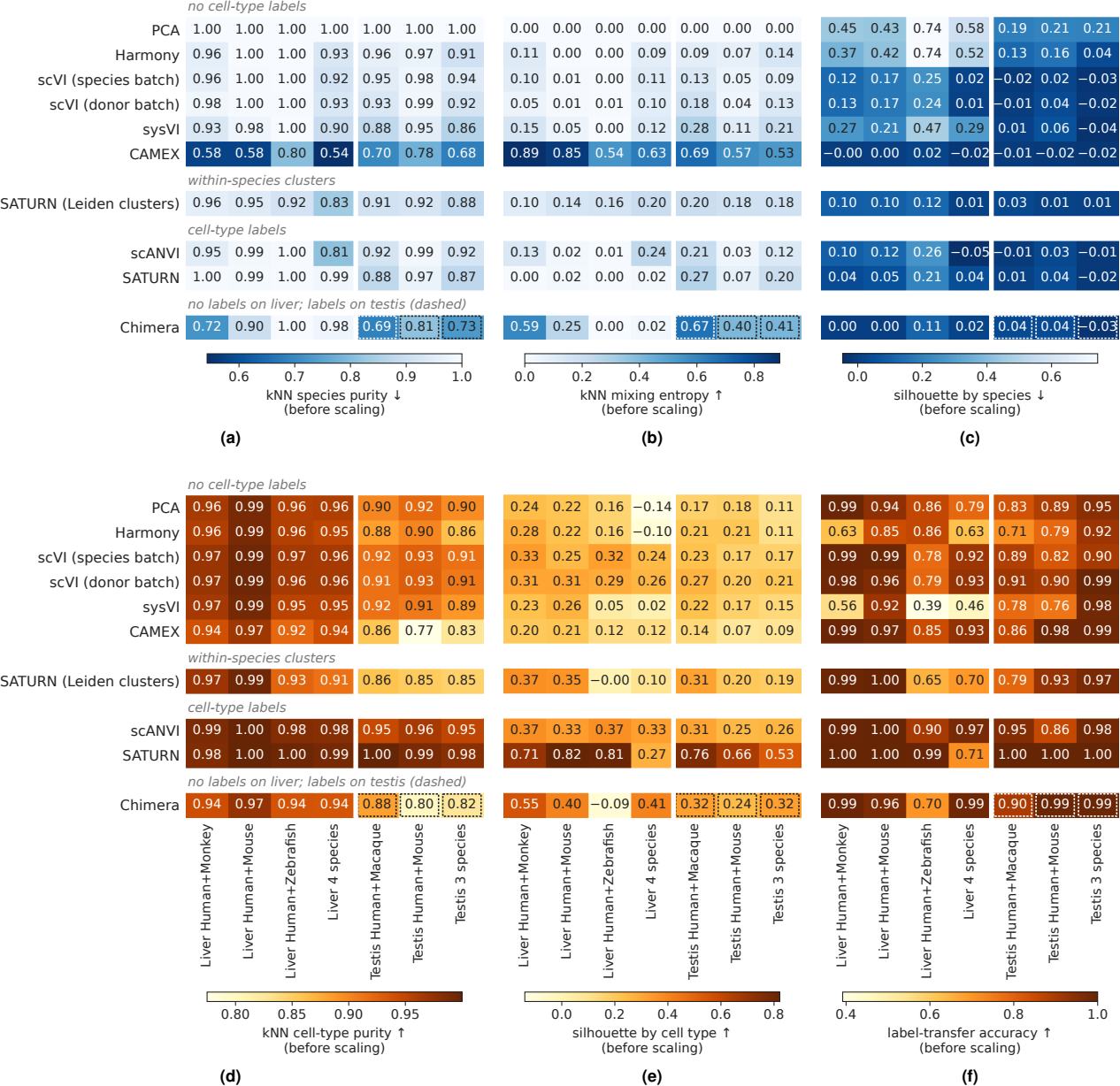

stability floor (0.01)
mean |α(t+1) − α(t)|
epoch
(a)
Δα > +0.05
Δα < −0.05
fraction of genes
epoch
(b)
120 of 8,430 genes
prior mean
mean of all genes
α
epoch
(c)
epoch 0
epoch 35
epoch 70
epoch 105
epoch 145
density
α
(d)
L_{anchor} (anchor term)
$L_{\text{fit-reward}}$ (fit-quality reward)
loss component (per-step mean)
epoch
(e)

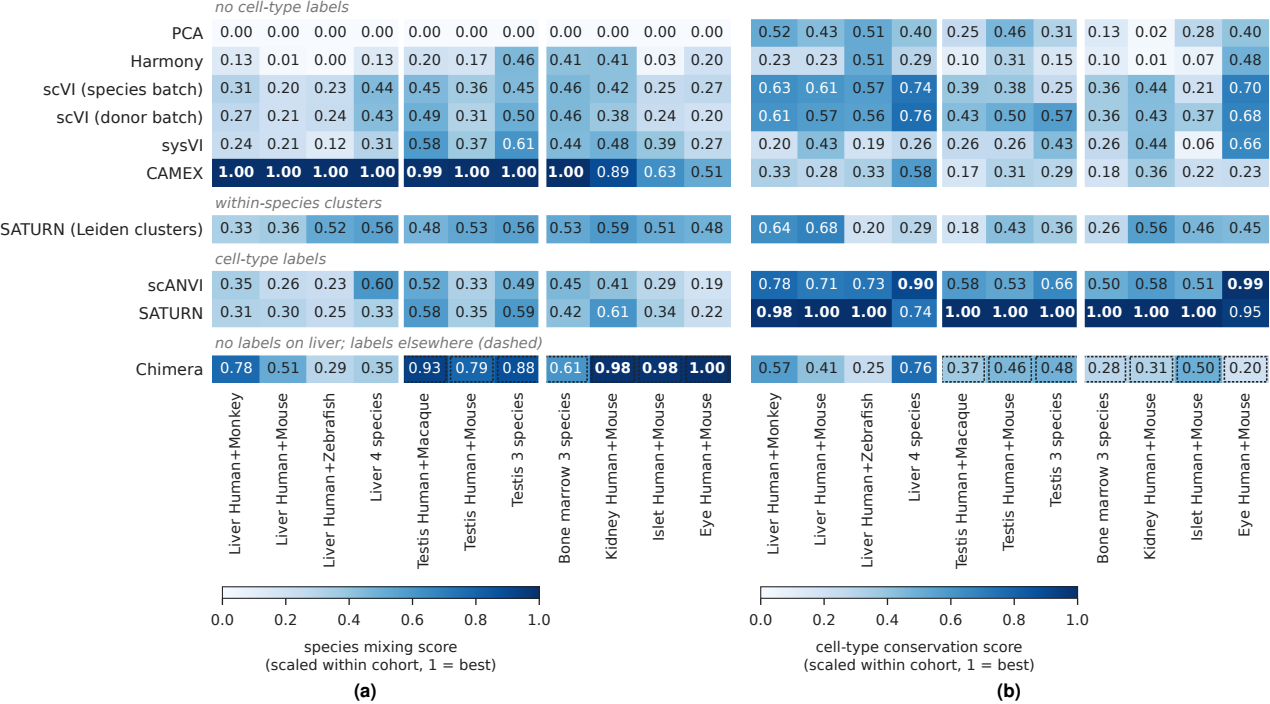

k = 38 programs
Spearman ρ = 0.876, Pearson r = 0.931
Liver Human+Monkey, seed 44 mean Δα per program
Liver Human+Monkey, seed 42 mean Δα per program
Mitochondrially encoded
(a)
k = 38 programs
Spearman ρ = 0.875, Pearson r = 0.944
Liver Human+Monkey, seed 43 mean Δα per program
Liver Human+Monkey, seed 42 mean Δα per program
Mitochondrially encoded
(b)
k = 38 programs
Spearman ρ = 0.773, Pearson r = 0.912
Liver Human+Monkey, seed 44 mean Δα per program
Liver Human+Monkey, seed 43 mean Δα per program
Mitochondrially encoded
(c)
k = 38 programs, n = 8,257 shared genes
Spearman ρ = 0.712, Pearson r = 0.906
Liver Human+Mouse mean Δα per program
Liver 4 species mean Δα per program
Mitochondrially encoded
(d)
k = 38 programs, n = 8,430 shared genes
Spearman ρ = 0.461, Pearson r = 0.782
Liver Human+Monkey mean Δα per program
Liver 4 species mean Δα per program
Mitochondrially encoded
(e)
k = 38 programs, n = 7,998 shared genes
Spearman ρ = 0.266, Pearson r = −0.123
Liver Human+Zebrafish mean Δα per program
Liver 4 species mean Δα per program
Mitochondrially encoded
(f)
k = 41 programs, n = 14,640 shared genes
Spearman ρ = 0.970, Pearson r = 0.965
Testis Human+Macaque mean Δα per program
Testis 3 species mean Δα per program
Mitochondrially encoded
(g)
k = 41 programs, n = 14,731 shared genes
Spearman ρ = 0.891, Pearson r = 0.894
Testis Human+Mouse mean Δα per program
Testis 3 species mean Δα per program
Mitochondrially encoded
(h)
k = 41 programs, n = 14,551 shared genes
Spearman ρ = 0.938, Pearson r = 0.976
Testis Human+Mouse mean Δα per program
Testis Human+Macaque mean Δα per program
Mitochondrially encoded
(i)

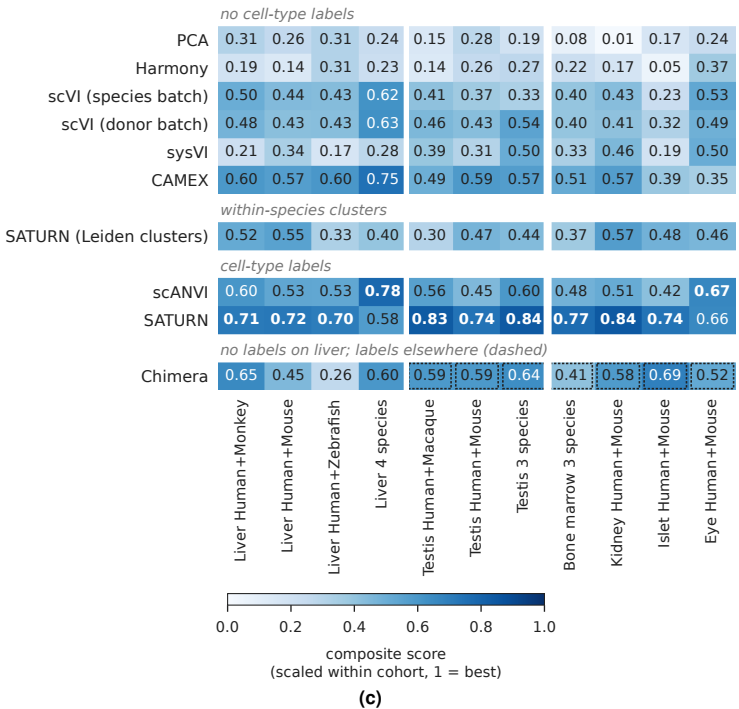

by donor
by species
by cell type
Chimera
PCA
scANVI
scVI (donor batch)
sysVI
UMAP 2
UMAP 1
SN143 (n = 5,228)
SN011 (n = 4,211)
SN142 (n = 4,161)
SN007 (n = 3,383)
SN115 (n = 3,202)
SN052 (n = 2,896)
SN116 (n = 2,385)
SN090 (n = 2,059)
SN111 (n = 1,329)
Human (n = 15,980)
Macaque (n = 7,613)
Mouse (n = 5,261)
spermatids (n = 11,778)
Spermatocytes (n = 6,088)
somatic (n = 4,517)
Spermatogonia (n = 4,065)
Sertoli (n = 2,406)

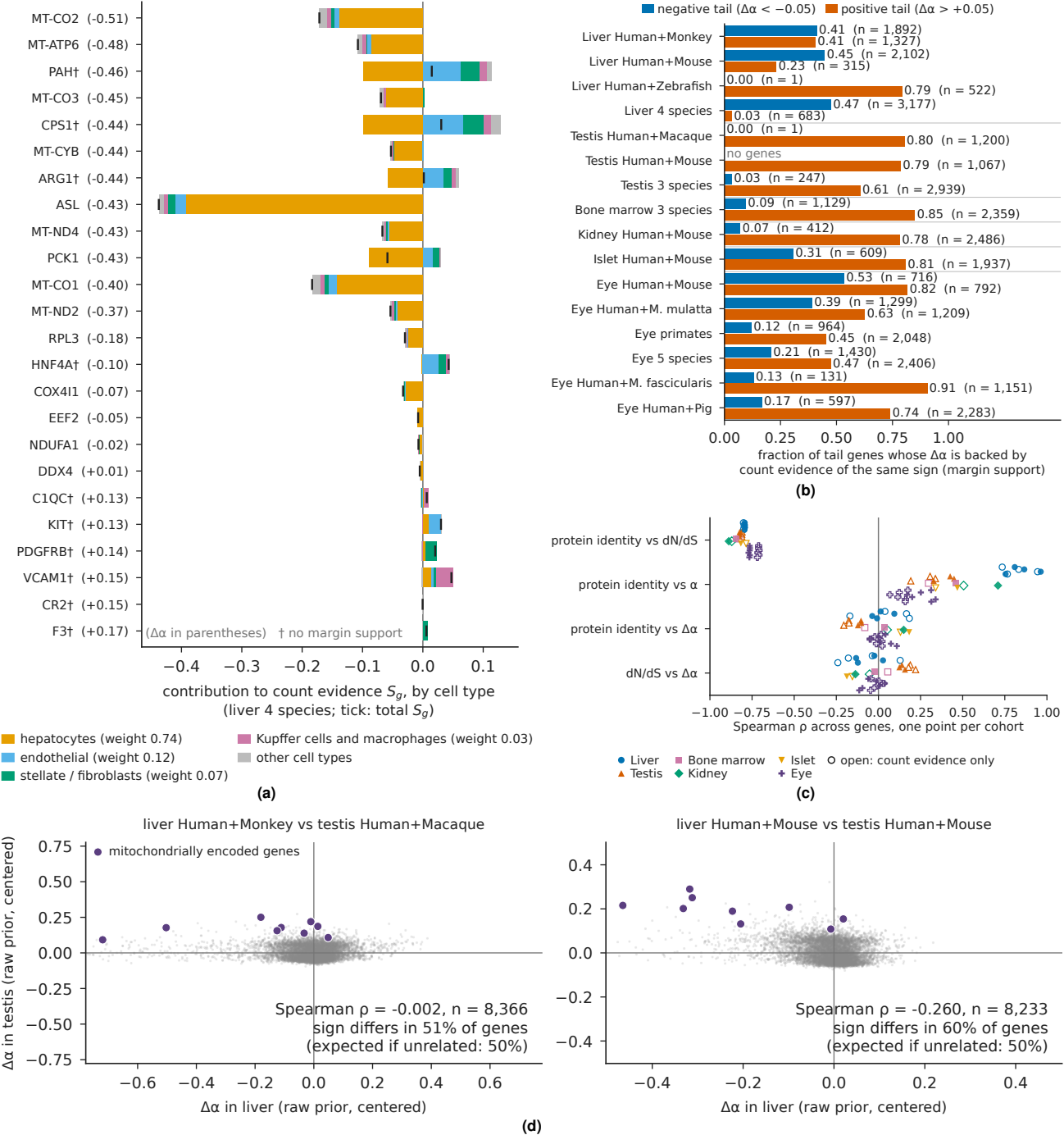

by donor
by species
by cell type
Chimera
PCA
scANVI
scVI (donor batch)
sysVI
UMAP 2
UMAP 1
CS173 (n = 4,427)
CS170 (n = 4,125)
CS172 (n = 4,056)
Human (n = 4,125)
Monkey (n = 8,483)
Hepatocytes (n = 8,045)
Endothelial (n = 2,155)
fibroblasts (n = 965)
Cholangio (n = 562)
Kupffer cells (n = 364)
NK NKT and T cells (n = 199)
Macrophages (n = 170)
B Plasma cells (n = 148)

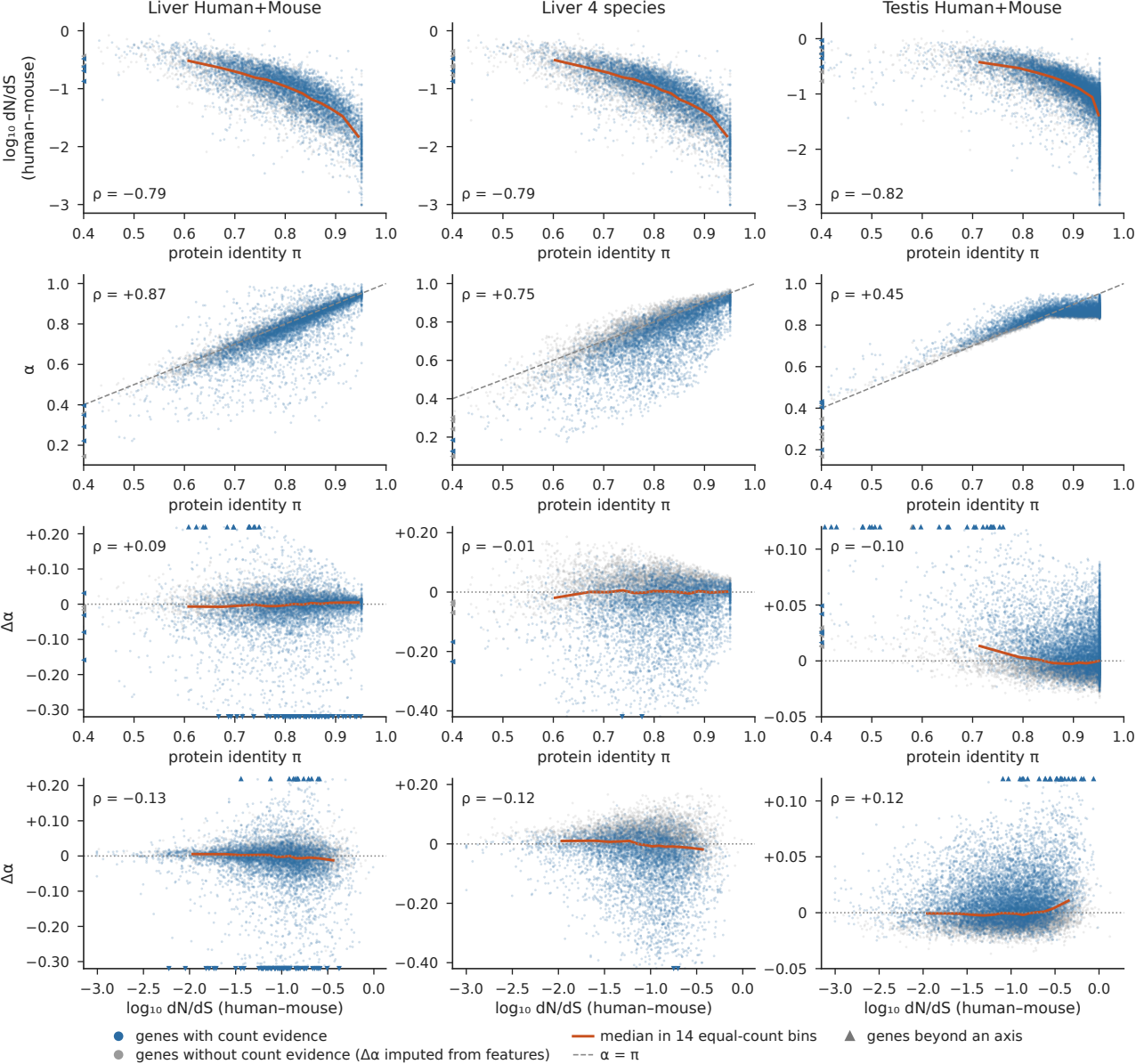

no cell-type labels
PCA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Harmony 0.96 1.00 1.00 0.93 0.96 0.97 0.91 0.13 0.01 0.00 0.13 0.20 0.17 0.46
scVI (species batch) 0.96 1.00 1.00 0.92 0.95 0.98 0.94 0.31 0.20 0.23 0.44 0.45 0.36 0.45
scVI (donor batch) 0.98 1.00 1.00 0.93 0.93 0.99 0.92 0.27 0.21 0.24 0.43 0.49 0.31 0.50
sysVI 0.93 0.98 1.00 0.90 0.88 0.95 0.86 0.24 0.21 0.12 0.31 0.58 0.37 0.61
CAMEX 0.58 0.58 0.80 0.54 0.70 0.78 0.68 1.00 1.00 1.00 1.00 0.99 1.00 1.00
within-species clusters
SATURN (Leiden clusters) 0.96 0.95 0.92 0.83 0.91 0.92 0.88 0.33 0.36 0.52 0.56 0.48 0.53 0.56
cell-type labels
scANVI 0.95 0.99 1.00 0.81 0.92 0.99 0.92 0.35 0.26 0.23 0.60 0.52 0.33 0.49
SATURN 1.00 0.99 1.00 0.99 0.88 0.97 0.87 0.31 0.30 0.25 0.33 0.58 0.35 0.59
no labels on liver; labels on testis (dashed)
Chimera 0.72 0.90 1.00 0.98 0.69 0.81 0.73 0.78 0.51 0.29 0.35 0.93 0.79 0.88
0.5 0.6 0.7 0.8 0.9 1.0
kNN species purity ↓
(before scaling; lower = more mixing)
(a)
0.0 0.2 0.4 0.6 0.8 1.0
species mixing score
(scaled within cohort, 1 = best)
(b)
no cell-type labels
PCA 0.52 0.43 0.51 0.40 0.25 0.46 0.31 0.31 0.26 0.31 0.24 0.15 0.28 0.19
Harmony 0.23 0.23 0.51 0.29 0.10 0.31 0.15 0.19 0.14 0.31 0.23 0.14 0.26 0.27
scVI (species batch) 0.63 0.61 0.57 0.74 0.39 0.38 0.25 0.50 0.44 0.43 0.62 0.41 0.37 0.33
scVI (donor batch) 0.61 0.57 0.56 0.76 0.43 0.50 0.57 0.48 0.43 0.43 0.63 0.46 0.43 0.54
sysVI 0.20 0.43 0.19 0.26 0.26 0.26 0.43 0.21 0.34 0.17 0.28 0.39 0.31 0.50
CAMEX 0.33 0.28 0.33 0.58 0.17 0.31 0.29 0.60 0.57 0.60 0.75 0.49 0.59 0.57
within-species clusters
SATURN (Leiden clusters) 0.64 0.68 0.20 0.29 0.18 0.43 0.36 0.52 0.55 0.33 0.40 0.30 0.47 0.44
cell-type labels
scANVI 0.78 0.71 0.73 0.90 0.58 0.53 0.66 0.60 0.53 0.53 0.78 0.56 0.45 0.60
SATURN 0.98 1.00 1.00 0.74 1.00 1.00 1.00 0.71 0.72 0.70 0.58 0.83 0.74 0.84
no labels on liver; labels on testis (dashed)
Chimera 0.57 0.41 0.25 0.76 0.37 0.46 0.48 0.65 0.45 0.26 0.60 0.59 0.59 0.64
Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
0.0 0.2 0.4 0.6 0.8 1.0
cell-type conservation score
(scaled within cohort, 1 = best)
(c)
0.0 0.2 0.4 0.6 0.8 1.0
composite score
(scaled within cohort, 1 = best)
(d)

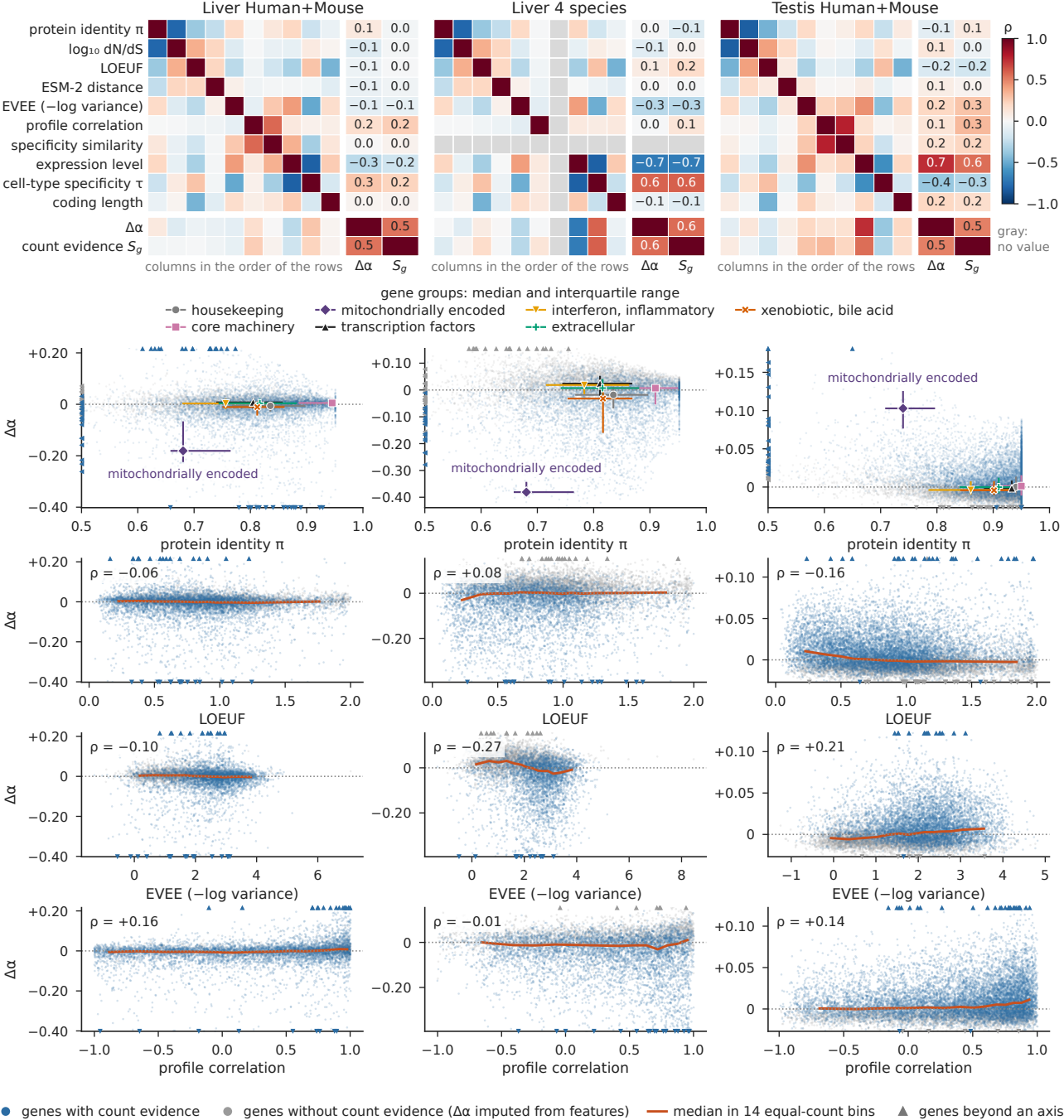

no cell-type labels
PCA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.45 0.43 0.74 0.58 0.19 0.21 0.21
Harmony 0.96 1.00 1.00 0.93 0.96 0.97 0.91 0.11 0.00 0.00 0.09 0.09 0.07 0.14 0.37 0.42 0.74 0.52 0.13 0.16 0.04
scVI (species batch) 0.96 1.00 1.00 0.92 0.95 0.98 0.94 0.10 0.01 0.00 0.11 0.13 0.05 0.09 0.12 0.17 0.25 0.02 −0.02 0.02 −0.03
scVI (donor batch) 0.98 1.00 1.00 0.93 0.93 0.99 0.92 0.05 0.01 0.01 0.10 0.18 0.04 0.13 0.13 0.17 0.24 0.01 −0.01 0.04 −0.02
sysVI 0.93 0.98 1.00 0.90 0.88 0.95 0.86 0.15 0.05 0.00 0.12 0.28 0.11 0.21 0.27 0.21 0.47 0.29 0.01 0.06 −0.04
CAMEX 0.58 0.58 0.80 0.54 0.70 0.78 0.68 0.89 0.85 0.54 0.63 0.69 0.57 0.53 −0.00 0.00 0.02 −0.02 −0.01 −0.02 −0.02
within-species clusters
SATURN (Leiden clusters) 0.96 0.95 0.92 0.83 0.91 0.92 0.88 0.10 0.14 0.16 0.20 0.20 0.18 0.18 0.10 0.10 0.12 0.01 0.03 0.01 0.01
cell-type labels
scANVI 0.95 0.99 1.00 0.81 0.92 0.99 0.92 0.13 0.02 0.01 0.24 0.21 0.03 0.12 0.10 0.12 0.26 −0.05 −0.01 0.03 −0.01
SATURN 1.00 0.99 1.00 0.99 0.88 0.97 0.87 0.00 0.02 0.00 0.02 0.27 0.07 0.20 0.04 0.05 0.21 0.04 0.01 0.04 −0.02
no labels on liver; labels on testis (dashed)
Chimera 0.72 0.90 1.00 0.98 0.69 0.81 0.73 0.59 0.25 0.00 0.02 0.67 0.40 0.41 0.00 0.00 0.11 0.02 0.04 0.04 −0.03
0.6 0.7 0.8 0.9 1.0
kNN species purity ↓ (before scaling)
(a)
0.0 0.2 0.4 0.6 0.8
kNN mixing entropy ↑ (before scaling)
(b)
0.0 0.2 0.4 0.6
silhouette by species ↓ (before scaling)
(c)
no cell-type labels
PCA 0.96 0.99 0.96 0.96 0.90 0.92 0.90 0.24 0.22 0.16 −0.14 0.17 0.18 0.11 0.99 0.94 0.86 0.79 0.83 0.89 0.95
Harmony 0.96 0.99 0.96 0.95 0.88 0.90 0.86 0.28 0.22 0.16 −0.10 0.21 0.21 0.11 0.63 0.85 0.86 0.63 0.71 0.79 0.92
scVI (species batch) 0.97 0.99 0.97 0.96 0.92 0.93 0.91 0.33 0.25 0.32 0.24 0.23 0.17 0.17 0.99 0.99 0.78 0.92 0.89 0.82 0.90
scVI (donor batch) 0.97 0.99 0.96 0.96 0.91 0.93 0.91 0.31 0.31 0.29 0.26 0.27 0.20 0.21 0.98 0.96 0.79 0.93 0.91 0.90 0.99
sysVI 0.97 0.99 0.95 0.95 0.92 0.91 0.89 0.23 0.26 0.05 0.02 0.22 0.17 0.15 0.56 0.92 0.39 0.46 0.78 0.76 0.98
CAMEX 0.94 0.97 0.92 0.94 0.86 0.77 0.83 0.20 0.21 0.12 0.12 0.14 0.07 0.09 0.99 0.97 0.85 0.93 0.86 0.98 0.99
within-species clusters
SATURN (Leiden clusters) 0.97 0.99 0.93 0.91 0.86 0.85 0.85 0.37 0.35 −0.00 0.10 0.31 0.20 0.19 0.99 1.00 0.65 0.70 0.79 0.93 0.97
cell-type labels
scANVI 0.99 1.00 0.98 0.98 0.95 0.96 0.95 0.37 0.33 0.37 0.33 0.31 0.25 0.26 0.99 1.00 0.90 0.97 0.95 0.86 0.98
SATURN 0.98 1.00 1.00 0.99 1.00 0.99 0.98 0.71 0.82 0.81 0.27 0.76 0.66 0.53 1.00 1.00 0.99 0.71 1.00 1.00 1.00
no labels on liver; labels on testis (dashed)
Chimera 0.94 0.97 0.94 0.94 0.88 0.80 0.82 0.55 0.40 −0.09 0.41 0.32 0.24 0.32 0.99 0.96 0.70 0.99 0.90 0.99 0.99
Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
0.80 0.85 0.90 0.95
kNN cell-type purity ↑ (before scaling)
(d)
0.0 0.2 0.4 0.6 0.8
silhouette by cell type ↑ (before scaling)
(e)
0.4 0.6 0.8 1.0
label-transfer accuracy ↑ (before scaling)
(f)

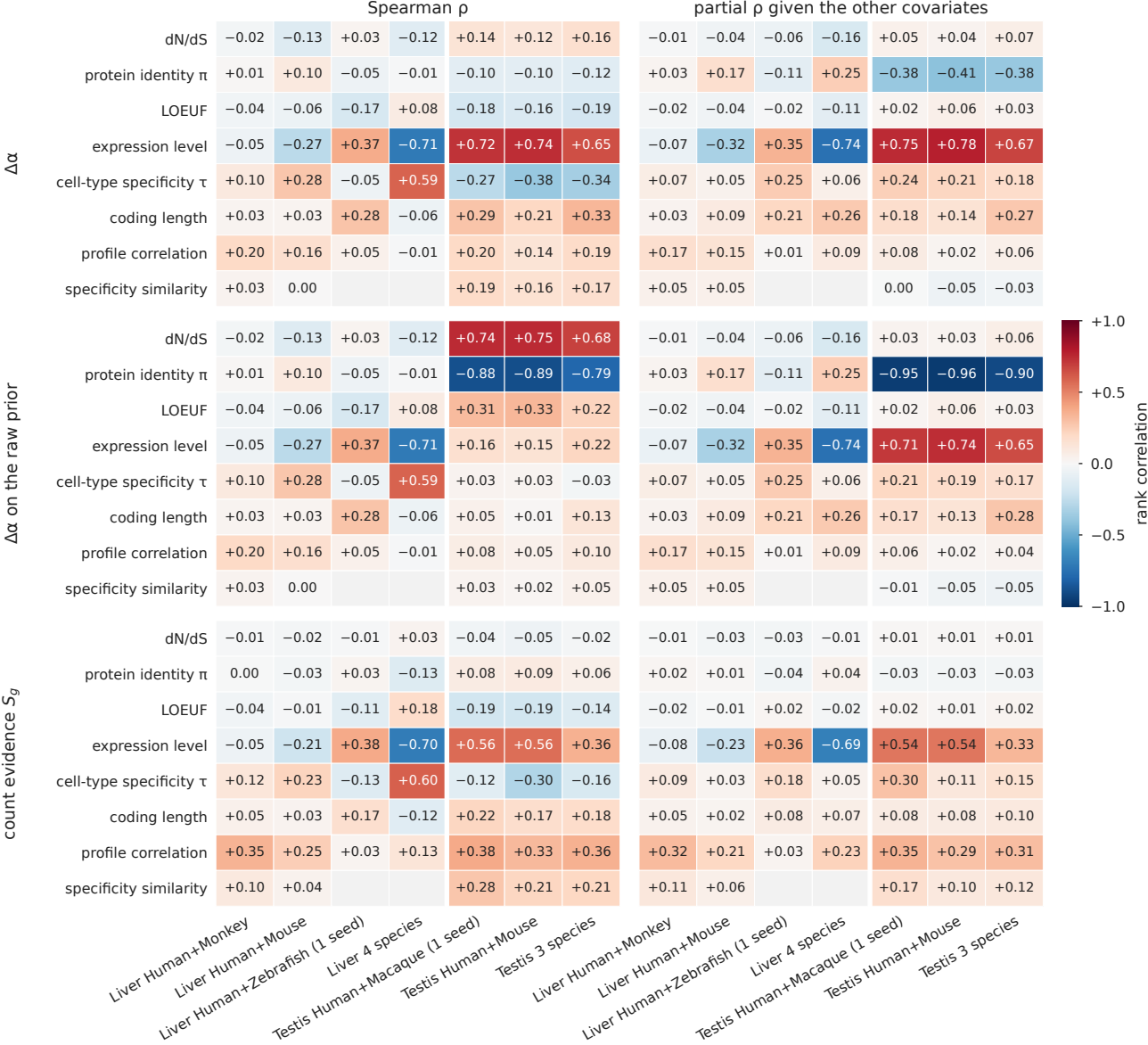

no cell-type labels
PCA 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Harmony 0.13 0.01 0.00 0.13 0.20 0.17 0.46 0.41 0.41 0.03 0.20
scVI (species batch) 0.31 0.20 0.23 0.44 0.45 0.36 0.45 0.46 0.42 0.25 0.27
scVI (donor batch) 0.27 0.21 0.24 0.43 0.49 0.31 0.50 0.46 0.38 0.24 0.20
sysVI 0.24 0.21 0.12 0.31 0.58 0.37 0.61 0.44 0.48 0.39 0.27
CAMEX 1.00 1.00 1.00 1.00 0.99 1.00 1.00 1.00 0.89 0.63 0.51
within-species clusters
SATURN (Leiden clusters) 0.33 0.36 0.52 0.56 0.48 0.53 0.56 0.53 0.59 0.51 0.48
cell-type labels
scANVI 0.35 0.26 0.23 0.60 0.52 0.33 0.49 0.45 0.41 0.29 0.19
SATURN 0.31 0.30 0.25 0.33 0.58 0.35 0.59 0.42 0.61 0.34 0.22
no labels on liver; labels elsewhere (dashed)
Chimera 0.78 0.51 0.29 0.35 0.93 0.79 0.88 0.61 0.98 0.98 1.00
Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
Bone marrow 3 species
Kidney Human+Mouse
Islet Human+Mouse
Eye Human+Mouse
0.0 0.2 0.4 0.6 0.8 1.0
species mixing score
(scaled within cohort, 1 = best)
(a)
0.52 0.43 0.51 0.40 0.25 0.46 0.31 0.13 0.02 0.28 0.40
0.23 0.23 0.51 0.29 0.10 0.31 0.15 0.10 0.01 0.07 0.48
0.63 0.61 0.57 0.74 0.39 0.38 0.25 0.36 0.44 0.21 0.70
0.61 0.57 0.56 0.76 0.43 0.50 0.57 0.36 0.43 0.37 0.68
0.20 0.43 0.19 0.26 0.26 0.26 0.43 0.26 0.44 0.06 0.66
0.33 0.28 0.33 0.58 0.17 0.31 0.29 0.18 0.36 0.22 0.23
0.64 0.68 0.20 0.29 0.18 0.43 0.36 0.26 0.56 0.46 0.45
0.78 0.71 0.73 0.90 0.58 0.53 0.66 0.50 0.58 0.51 0.99
0.98 1.00 1.00 0.74 1.00 1.00 1.00 1.00 1.00 1.00 0.95
0.57 0.41 0.25 0.76 0.37 0.46 0.48 0.28 0.31 0.50 0.20
Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
Bone marrow 3 species
Kidney Human+Mouse
Islet Human+Mouse
Eye Human+Mouse
0.0 0.2 0.4 0.6 0.8 1.0
cell-type conservation score
(scaled within cohort, 1 = best)
(b)


no cell-type labels
PCA 0.31 0.26 0.31 0.24 0.15 0.28 0.19 0.08 0.01 0.17 0.24
Harmony 0.19 0.14 0.31 0.23 0.14 0.26 0.27 0.22 0.17 0.05 0.37
scVI (species batch) 0.50 0.44 0.43 0.62 0.41 0.37 0.33 0.40 0.43 0.23 0.53
scVI (donor batch) 0.48 0.43 0.43 0.63 0.46 0.43 0.54 0.40 0.41 0.32 0.49
sysVI 0.21 0.34 0.17 0.28 0.39 0.31 0.50 0.33 0.46 0.19 0.50
CAMEX 0.60 0.57 0.60 0.75 0.49 0.59 0.57 0.51 0.57 0.39 0.35
within-species clusters
SATURN (Leiden clusters) 0.52 0.55 0.33 0.40 0.30 0.47 0.44 0.37 0.57 0.48 0.46
cell-type labels
scANVI 0.60 0.53 0.53 0.78 0.56 0.45 0.60 0.48 0.51 0.42 0.67
SATURN 0.71 0.72 0.70 0.58 0.83 0.74 0.84 0.77 0.84 0.74 0.66
no labels on liver; labels elsewhere (dashed)
Chimera 0.65 0.45 0.26 0.60 0.59 0.59 0.64 0.41 0.58 0.69 0.52
Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
Bone marrow 3 species
Kidney Human+Mouse
Islet Human+Mouse
Eye Human+Mouse
0.0 0.2 0.4 0.6 0.8 1.0
composite score
(scaled within cohort, 1 = best)
(c)


MT-CO2 (-0.51)
MT-ATP6 (-0.48)
PAH† (-0.46)
MT-CO3 (-0.45)
CPS1† (-0.44)
MT-CYB (-0.44)
ARG1† (-0.44)
ASL (-0.43)
MT-ND4 (-0.43)
PCK1 (-0.43)
MT-CO1 (-0.40)
MT-ND2 (-0.37)
RPL3 (-0.18)
HNF4A† (-0.10)
COX4I1 (-0.07)
EEF2 (-0.05)
NDUFA1 (-0.02)
DDX4 (+0.01)
C1QC† (+0.13)
KIT† (+0.13)
PDGFRB† (+0.14)
VCAM1† (+0.15)
CR2† (+0.15)
F3† (+0.17)
(Δα in parentheses) † no margin support
−0.4 −0.3 −0.2 −0.1 0.0 0.1
contribution to count evidence S_g, by cell type
(liver 4 species; tick: total S_g)
hepatocytes (weight 0.74)
endothelial (weight 0.12)
stellate / fibroblasts (weight 0.07)
Kupffer cells and macrophages (weight 0.03)
other cell types
(a)
negative tail (Δα < −0.05)
positive tail (Δα > +0.05)
Liver Human+Monkey
0.41 (n = 1,892)
0.41 (n = 1,327)
Liver Human+Mouse
0.45 (n = 2,102)
0.23 (n = 315)
Liver Human+Zebrafish
0.00 (n = 1)
0.79 (n = 522)
Liver 4 species
0.47 (n = 3,177)
0.03 (n = 683)
Testis Human+Macaque
0.00 (n = 1)
0.80 (n = 1,200)
Testis Human+Mouse
no genes
0.79 (n = 1,067)
Testis 3 species
0.03 (n = 247)
0.61 (n = 2,939)
Bone marrow 3 species
0.09 (n = 1,129)
0.85 (n = 2,359)
Kidney Human+Mouse
0.07 (n = 412)
0.78 (n = 2,486)
Islet Human+Mouse
0.31 (n = 609)
0.81 (n = 1,937)
Eye Human+Mouse
0.53 (n = 716)
0.82 (n = 792)
Eye Human+M. mulatta
0.39 (n = 1,299)
0.63 (n = 1,209)
Eye primates
0.12 (n = 964)
0.45 (n = 2,048)
Eye 5 species
0.21 (n = 1,430)
0.47 (n = 2,406)
Eye Human+M. fascicularis
0.13 (n = 131)
0.91 (n = 1,151)
Eye Human+Pig
0.17 (n = 597)
0.74 (n = 2,283)
0.00 0.25 0.50 0.75 1.00
fraction of tail genes whose Δα is backed by count evidence of the same sign (margin support)
(b)
protein identity vs dN/dS
protein identity vs α
protein identity vs Δα
dN/dS vs Δα
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
Spearman ρ across genes, one point per cohort
Liver
Bone marrow
Islet
open: count evidence only
Testis
Kidney
Eye
(c)
liver Human+Monkey vs testis Human+Macaque
mitochondrially encoded genes
Spearman ρ = -0.002, n = 8,366
sign differs in 51% of genes
(expected if unrelated: 50%)
Δα in testis (raw prior, centered)
Δα in liver (raw prior, centered)
liver Human+Mouse vs testis Human+Mouse
Spearman ρ = -0.260, n = 8,233
sign differs in 60% of genes
(expected if unrelated: 50%)
Δα in liver (raw prior, centered)
(d)

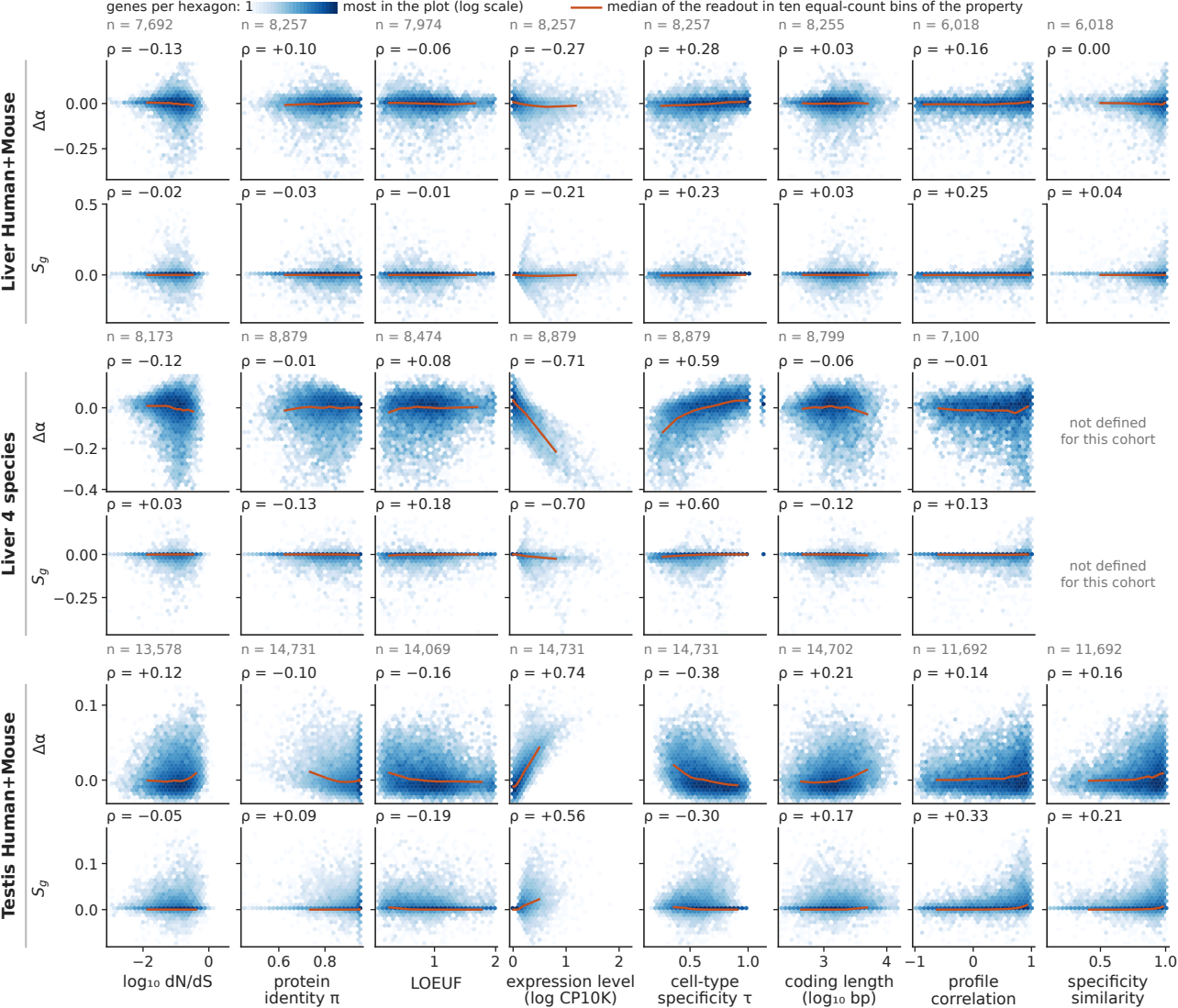

Liver Human+Mouse
Liver 4 species
Testis Human+Mouse
log₁₀ dN/dS (human–mouse)
α
Δα
protein identity π
log₁₀ dN/dS (human–mouse)
ρ = −0.79
ρ = −0.79
ρ = −0.82
ρ = +0.87
ρ = +0.75
ρ = +0.45
ρ = +0.09
ρ = −0.01
ρ = −0.10
ρ = −0.13
ρ = −0.12
ρ = +0.12
genes with count evidence
genes without count evidence (Δα imputed from features)
median in 14 equal-count bins
α = π
genes beyond an axis

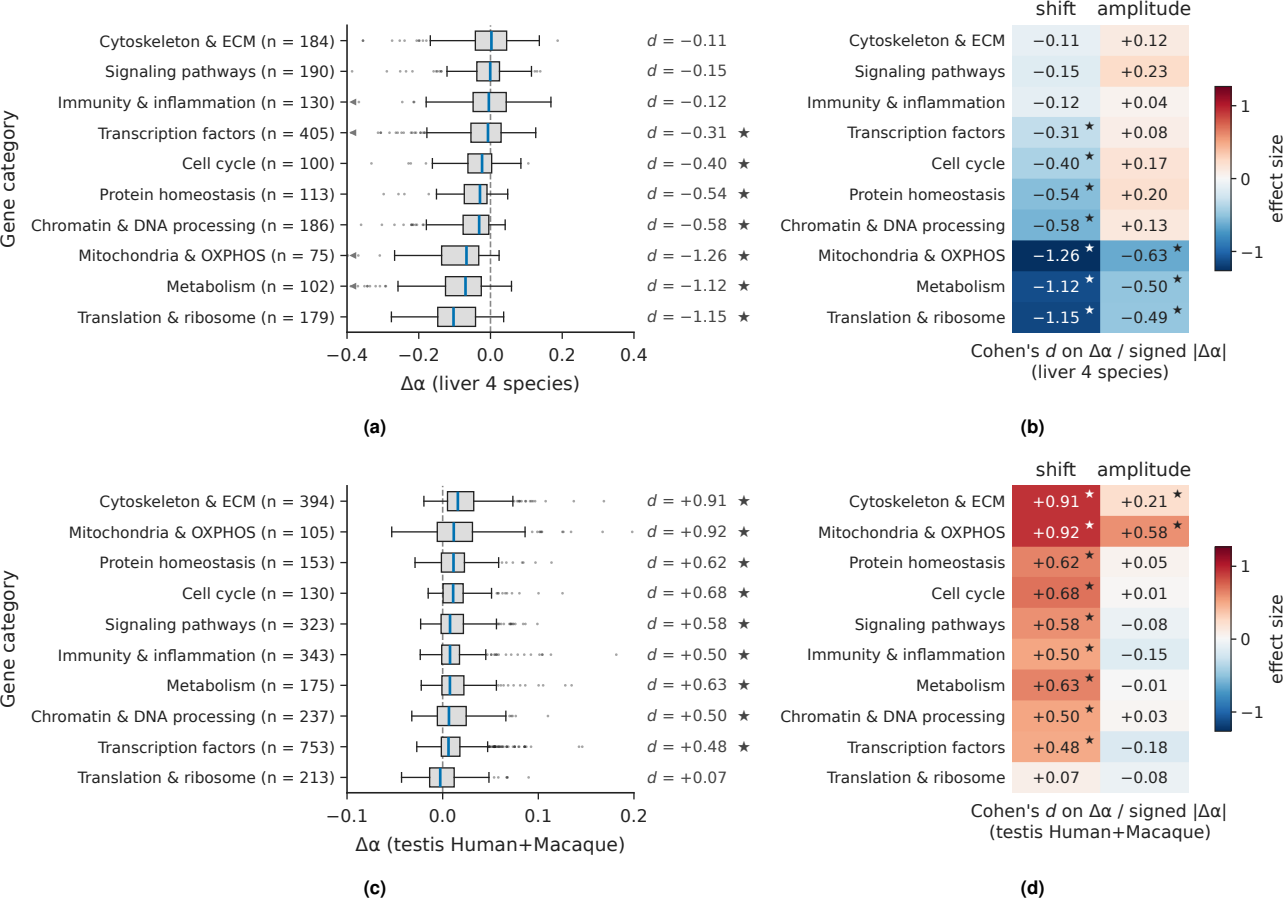

Liver Human+Mouse
Liver 4 species
Testis Human+Mouse
protein identity π
log₁₀ dN/dS
LOEUF
ESM-2 distance
EVEE (−log variance)
profile correlation
specificity similarity
expression level
cell-type specificity τ
coding length
Δα
count evidence S_g
columns in the order of the rows
ρ
gray: no value
gene groups: median and interquartile range
housekeeping
core machinery
mitochondrially encoded
transcription factors
interferon, inflammatory
extracellular
xenobiotic, bile acid
protein identity π
LOEUF
EVEE (−log variance)
profile correlation
ρ = −0.06
ρ = +0.08
ρ = −0.16
ρ = −0.10
ρ = −0.27
ρ = +0.21
ρ = +0.16
ρ = −0.01
ρ = +0.14
genes with count evidence
genes without count evidence (Δα imputed from features)
median in 14 equal-count bins
genes beyond an axis

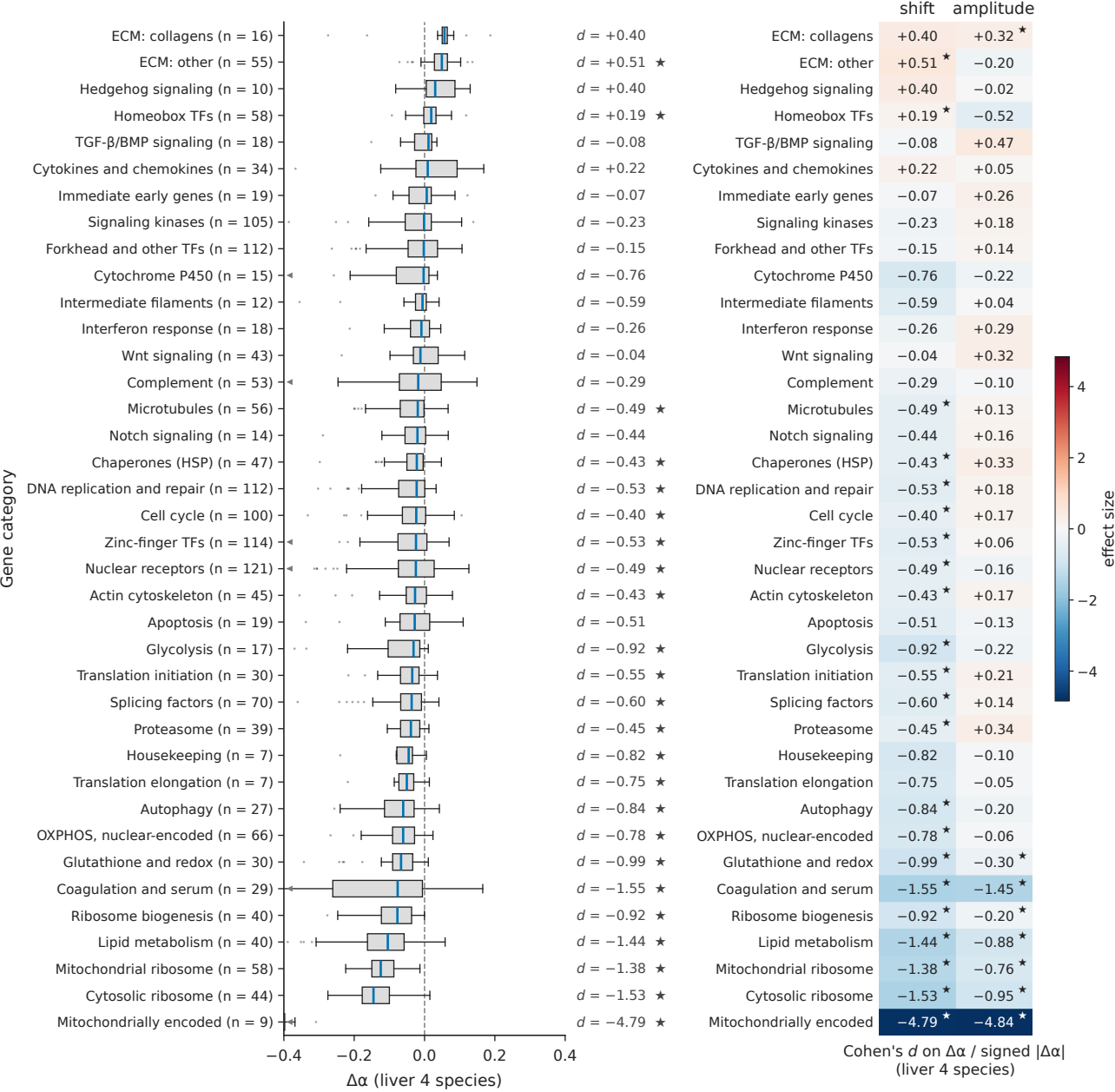


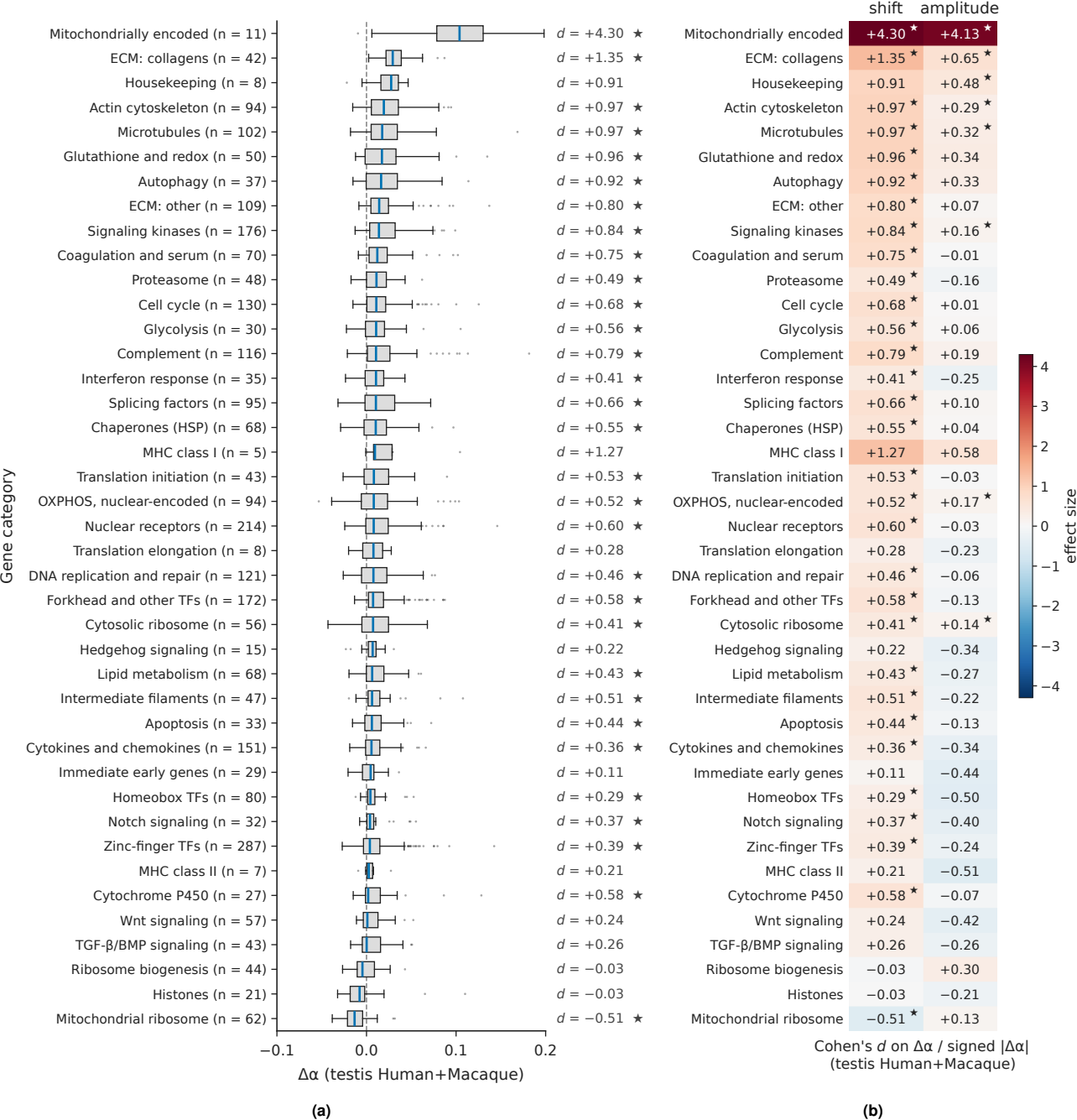
Partial ρ: rank partial correlation given the other covariates (identity, expression, detection, coding length).

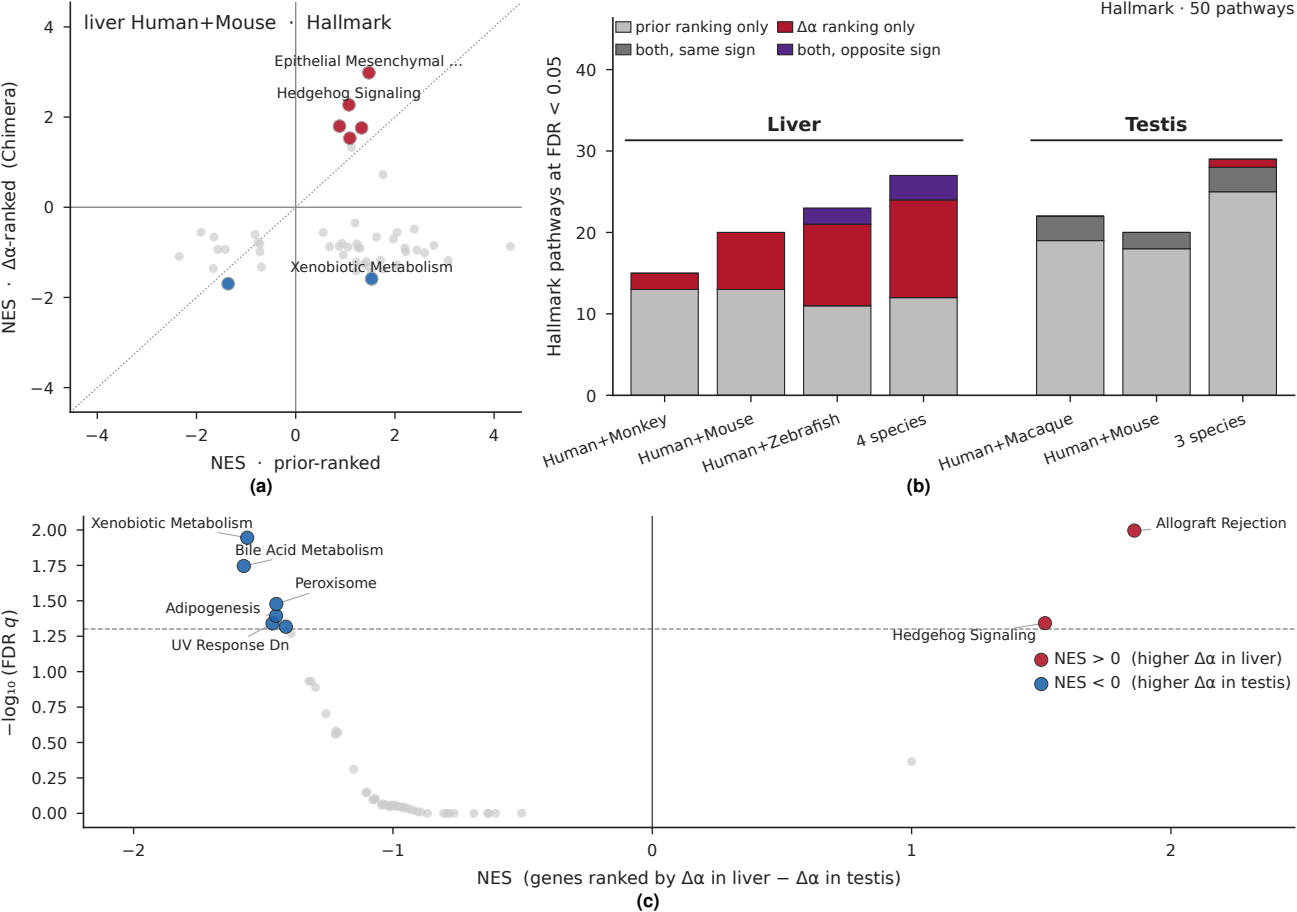
not defined for the cohort. Liver cohorts are anchored on the raw prior: their columns of the middle maps repeat the top maps.

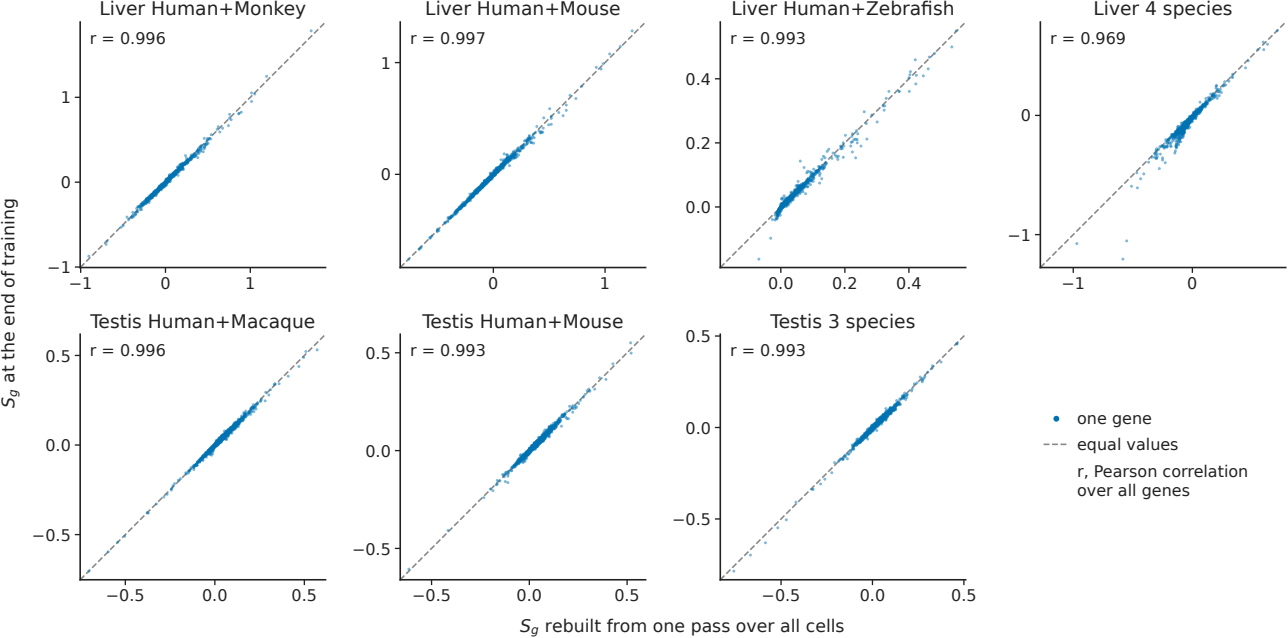

genes per hexagon: 1 most in the plot (log scale)
median of the readout in ten equal-count bins of the property
Liver Human+Mouse
Liver 4 species
Testis Human+Mouse
Δα
S_g
not defined for this cohort
not defined for this cohort
log₁₀ dN/dS
protein identity π
LOEUF
expression level (log CP10K)
cell-type specificity τ
coding length (log₁₀ bp)
profile correlation
specificity similarity

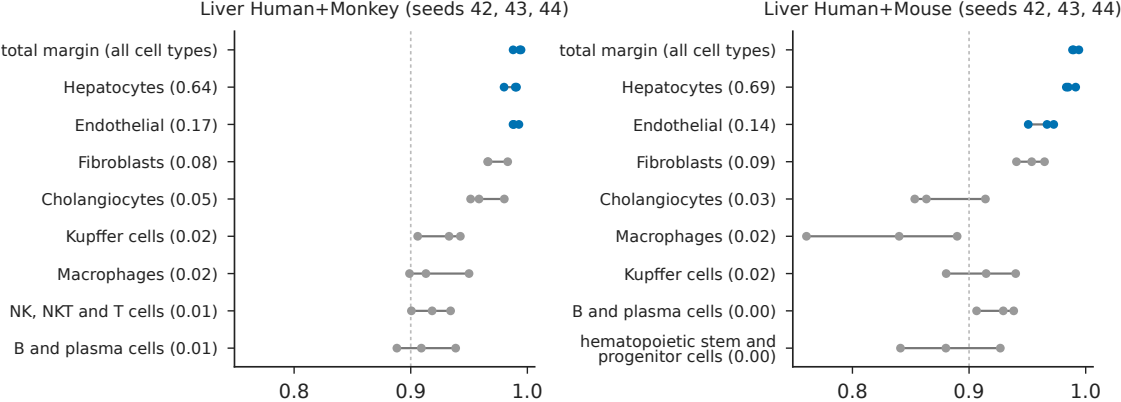

Gene category
Cytoskeleton & ECM (n = 184)
Signaling pathways (n = 190)
Immunity & inflammation (n = 130)
Transcription factors (n = 405)
Cell cycle (n = 100)
Protein homeostasis (n = 113)
Chromatin & DNA processing (n = 186)
Mitochondria & OXPHOS (n = 75)
Metabolism (n = 102)
Translation & ribosome (n = 179)
d = −0.11
d = −0.15
d = −0.12
d = −0.31 ★
d = −0.40 ★
d = −0.54 ★
d = −0.58 ★
d = −1.26 ★
d = −1.12 ★
d = −1.15 ★
−0.4
−0.2
0.0
0.2
0.4
Δα (liver 4 species)
(a)
shift
amplitude
−0.11
+0.12
−0.15
+0.23
−0.12
+0.04
−0.31 ★
+0.08
−0.40 ★
+0.17
−0.54 ★
+0.20
−0.58 ★
+0.13
−1.26 ★
−0.63 ★
−1.12 ★
−0.50 ★
−1.15 ★
−0.49 ★
1
0
−1
effect size
Cohen's d on Δα / signed |Δα|
(liver 4 species)
(b)
Gene category
Cytoskeleton & ECM (n = 394)
Mitochondria & OXPHOS (n = 105)
Protein homeostasis (n = 153)
Cell cycle (n = 130)
Signaling pathways (n = 323)
Immunity & inflammation (n = 343)
Metabolism (n = 175)
Chromatin & DNA processing (n = 237)
Transcription factors (n = 753)
Translation & ribosome (n = 213)
d = +0.91 ★
d = +0.92 ★
d = +0.62 ★
d = +0.68 ★
d = +0.58 ★
d = +0.50 ★
d = +0.63 ★
d = +0.50 ★
d = +0.48 ★
d = +0.07
−0.1
0.0
0.1
0.2
Δα (testis Human+Macaque)
(c)
shift
amplitude
+0.91 ★
+0.21 ★
+0.92 ★
+0.58 ★
+0.62 ★
+0.05
+0.68 ★
+0.01
+0.58 ★
−0.08
+0.50 ★
−0.15
+0.63 ★
−0.01
+0.50 ★
+0.03
+0.48 ★
−0.18
+0.07
−0.08
1
0
−1
effect size
Cohen's d on Δα / signed |Δα|
(testis Human+Macaque)
(d)


Gene category
ECM: collagens (n = 16)
ECM: other (n = 55)
Hedgehog signaling (n = 10)
Homeobox TFs (n = 58)
TGF-β/BMP signaling (n = 18)
Cytokines and chemokines (n = 34)
Immediate early genes (n = 19)
Signaling kinases (n = 105)
Forkhead and other TFs (n = 112)
Cytochrome P450 (n = 15)
Intermediate filaments (n = 12)
Interferon response (n = 18)
Wnt signaling (n = 43)
Complement (n = 53)
Microtubules (n = 56)
Notch signaling (n = 14)
Chaperones (HSP) (n = 47)
DNA replication and repair (n = 112)
Cell cycle (n = 100)
Zinc-finger TFs (n = 114)
Nuclear receptors (n = 121)
Actin cytoskeleton (n = 45)
Apoptosis (n = 19)
Glycolysis (n = 17)
Translation initiation (n = 30)
Splicing factors (n = 70)
Proteasome (n = 39)
Housekeeping (n = 7)
Translation elongation (n = 7)
Autophagy (n = 27)
OXPHOS, nuclear-encoded (n = 66)
Glutathione and redox (n = 30)
Coagulation and serum (n = 29)
Ribosome biogenesis (n = 40)
Lipid metabolism (n = 40)
Mitochondrial ribosome (n = 58)
Cytosolic ribosome (n = 44)
Mitochondrially encoded (n = 9)
d = +0.40
d = +0.51 ★
d = +0.40
d = +0.19 ★
d = −0.08
d = +0.22
d = −0.07
d = −0.23
d = −0.15
d = −0.76
d = −0.59
d = −0.26
d = −0.04
d = −0.29
d = −0.49 ★
d = −0.44
d = −0.43 ★
d = −0.53 ★
d = −0.40 ★
d = −0.53 ★
d = −0.49 ★
d = −0.43 ★
d = −0.51
d = −0.92 ★
d = −0.55 ★
d = −0.60 ★
d = −0.45 ★
d = −0.82 ★
d = −0.75 ★
d = −0.84 ★
d = −0.78 ★
d = −0.99 ★
d = −1.55 ★
d = −0.92 ★
d = −1.44 ★
d = −1.38 ★
d = −1.53 ★
d = −4.79 ★
−0.4
−0.2
0.0
0.2
0.4
Δα (liver 4 species)
(a)
shift
amplitude
+0.40
+0.32 ★
+0.51 ★
−0.20
+0.40
−0.02
+0.19 ★
−0.52
−0.08
+0.47
+0.22
+0.05
−0.07
+0.26
−0.23
+0.18
−0.15
+0.14
−0.76
−0.22
−0.59
+0.04
−0.26
+0.29
−0.04
+0.32
−0.29
−0.10
−0.49 ★
+0.13
−0.44
+0.16
−0.43 ★
+0.33
−0.53 ★
+0.18
−0.40 ★
+0.17
−0.53 ★
+0.06
−0.49 ★
−0.16
−0.43 ★
+0.17
−0.51
−0.13
−0.92 ★
−0.22
−0.55 ★
+0.21
−0.60 ★
+0.14
−0.45 ★
+0.34
−0.82
−0.10
−0.75
−0.05
−0.84 ★
−0.20
−0.78 ★
−0.06
−0.99 ★
−0.30 ★
−1.55 ★
−1.45 ★
−0.92 ★
−0.20 ★
−1.44 ★
−0.88 ★
−1.38 ★
−0.76 ★
−1.53 ★
−0.95 ★
−4.79 ★
−4.84 ★
4
2
0
−2
−4
effect size
Cohen's d on Δα / signed |Δα|
(liver 4 species)
(b)

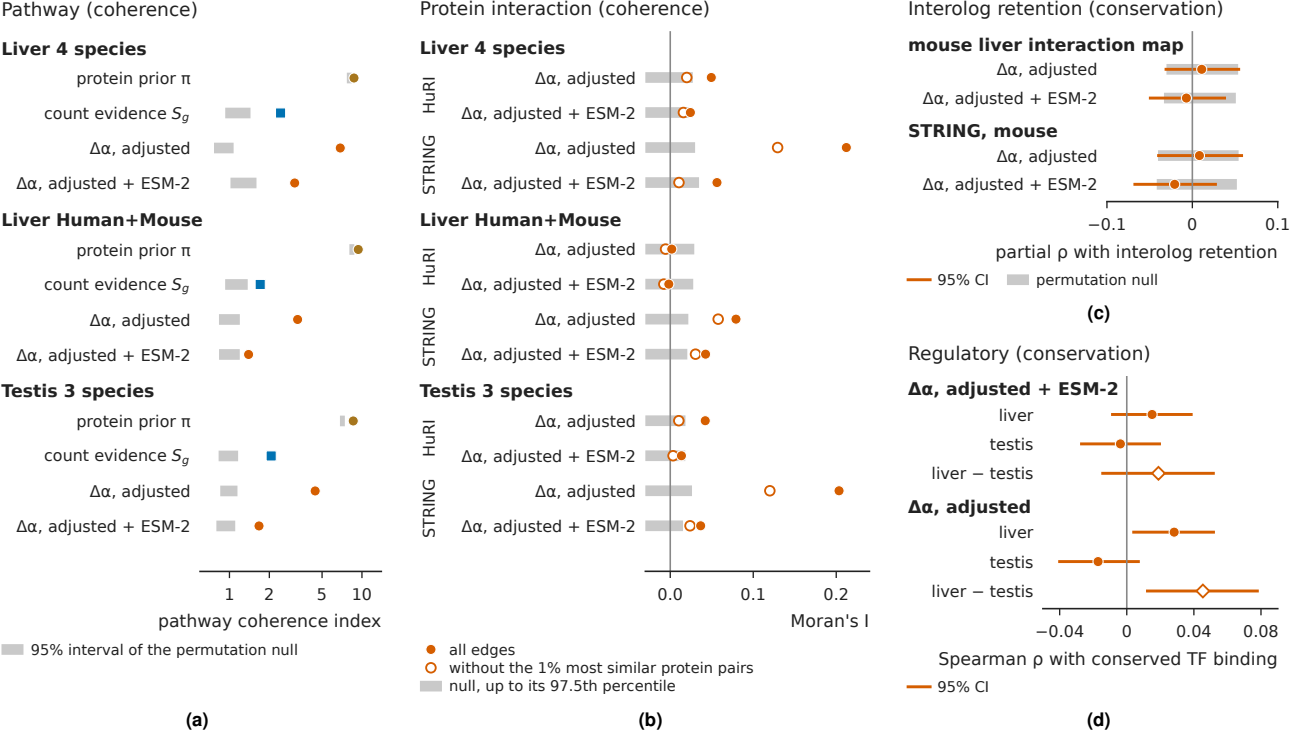

Mitochondrially encoded (n = 11) d = +4.30 ★
ECM: collagens (n = 42) d = +1.35 ★
Housekeeping (n = 8) d = +0.91
Actin cytoskeleton (n = 94) d = +0.97 ★
Microtubules (n = 102) d = +0.97 ★
Glutathione and redox (n = 50) d = +0.96 ★
Autophagy (n = 37) d = +0.92 ★
ECM: other (n = 109) d = +0.80 ★
Signaling kinases (n = 176) d = +0.84 ★
Coagulation and serum (n = 70) d = +0.75 ★
Proteasome (n = 48) d = +0.49 ★
Cell cycle (n = 130) d = +0.68 ★
Glycolysis (n = 30) d = +0.56 ★
Complement (n = 116) d = +0.79 ★
Interferon response (n = 35) d = +0.41 ★
Splicing factors (n = 95) d = +0.66 ★
Chaperones (HSP) (n = 68) d = +0.55 ★
MHC class I (n = 5) d = +1.27
Translation initiation (n = 43) d = +0.53 ★
OXPHOS, nuclear-encoded (n = 94) d = +0.52 ★
Nuclear receptors (n = 214) d = +0.60 ★
Translation elongation (n = 8) d = +0.28
DNA replication and repair (n = 121) d = +0.46 ★
Forkhead and other TFs (n = 172) d = +0.58 ★
Cytosolic ribosome (n = 56) d = +0.41 ★
Hedgehog signaling (n = 15) d = +0.22
Lipid metabolism (n = 68) d = +0.43 ★
Intermediate filaments (n = 47) d = +0.51 ★
Apoptosis (n = 33) d = +0.44 ★
Cytokines and chemokines (n = 151) d = +0.36 ★
Immediate early genes (n = 29) d = +0.11
Homeobox TFs (n = 80) d = +0.29 ★
Notch signaling (n = 32) d = +0.37 ★
Zinc-finger TFs (n = 287) d = +0.39 ★
MHC class II (n = 7) d = +0.21
Cytochrome P450 (n = 27) d = +0.58 ★
Wnt signaling (n = 57) d = +0.24
TGF-β/BMP signaling (n = 43) d = +0.26
Ribosome biogenesis (n = 44) d = −0.03
Histones (n = 21) d = −0.03
Mitochondrial ribosome (n = 62) d = −0.51 ★
Gene category
−0.1
0.0
0.1
0.2
Δα (testis Human+Macaque)
(a)
shift
amplitude
Mitochondrially encoded +4.30 ★ +4.13 ★
ECM: collagens +1.35 ★ +0.65 ★
Housekeeping +0.91 +0.48 ★
Actin cytoskeleton +0.97 ★ +0.29 ★
Microtubules +0.97 ★ +0.32 ★
Glutathione and redox +0.96 ★ +0.34
Autophagy +0.92 ★ +0.33
ECM: other +0.80 ★ +0.07
Signaling kinases +0.84 ★ +0.16 ★
Coagulation and serum +0.75 ★ −0.01
Proteasome +0.49 ★ −0.16
Cell cycle +0.68 ★ +0.01
Glycolysis +0.56 ★ +0.06
Complement +0.79 ★ +0.19
Interferon response +0.41 ★ −0.25
Splicing factors +0.66 ★ +0.10
Chaperones (HSP) +0.55 ★ +0.04
MHC class I +1.27 +0.58
Translation initiation +0.53 ★ −0.03
OXPHOS, nuclear-encoded +0.52 ★ +0.17 ★
Nuclear receptors +0.60 ★ −0.03
Translation elongation +0.28 −0.23
DNA replication and repair +0.46 ★ −0.06
Forkhead and other TFs +0.58 ★ −0.13
Cytosolic ribosome +0.41 ★ +0.14 ★
Hedgehog signaling +0.22 −0.34
Lipid metabolism +0.43 ★ −0.27
Intermediate filaments +0.51 ★ −0.22
Apoptosis +0.44 ★ −0.13
Cytokines and chemokines +0.36 ★ −0.34
Immediate early genes +0.11 −0.44
Homeobox TFs +0.29 ★ −0.50
Notch signaling +0.37 ★ −0.40
Zinc-finger TFs +0.39 ★ −0.24
MHC class II +0.21 −0.51
Cytochrome P450 +0.58 ★ −0.07
Wnt signaling +0.24 −0.42
TGF-β/BMP signaling +0.26 −0.26
Ribosome biogenesis −0.03 +0.30
Histones −0.03 −0.21
Mitochondrial ribosome −0.51 ★ +0.13
4
3
2
1
0
−1
−2
−3
−4
effect size
Cohen's d on Δα / signed |Δα|
(testis Human+Macaque)
(b)

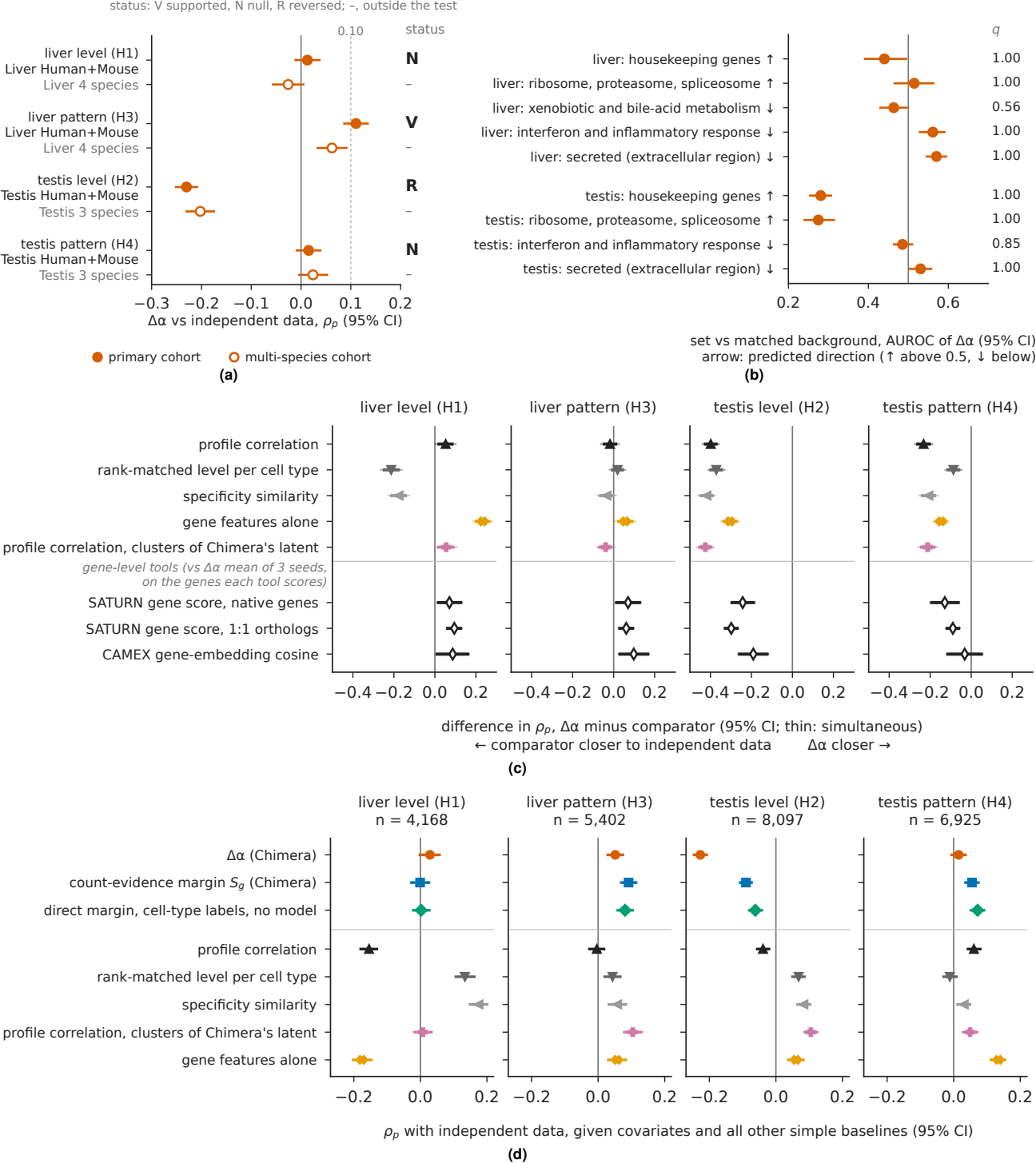

liver Human+Mouse · Hallmark
Epithelial Mesenchymal ...
Hedgehog Signaling
Xenobiotic Metabolism
NES · Δα-ranked (Chimera)
NES · prior-ranked
(a)
Hallmark · 50 pathways
prior ranking only
Δα ranking only
both, same sign
both, opposite sign
Liver
Testis
Hallmark pathways at FDR < 0.05
Human+Monkey
Human+Mouse
Human+Zebrafish
4 species
Human+Macaque
Human+Mouse
3 species
(b)
Xenobiotic Metabolism
Bile Acid Metabolism
Peroxisome
Adipogenesis
UV Response Dn
Allograft Rejection
Hedgehog Signaling
NES > 0 (higher Δα in liver)
NES < 0 (higher Δα in testis)
−log10 (FDR q)
NES (genes ranked by Δα in liver − Δα in testis)
(c)

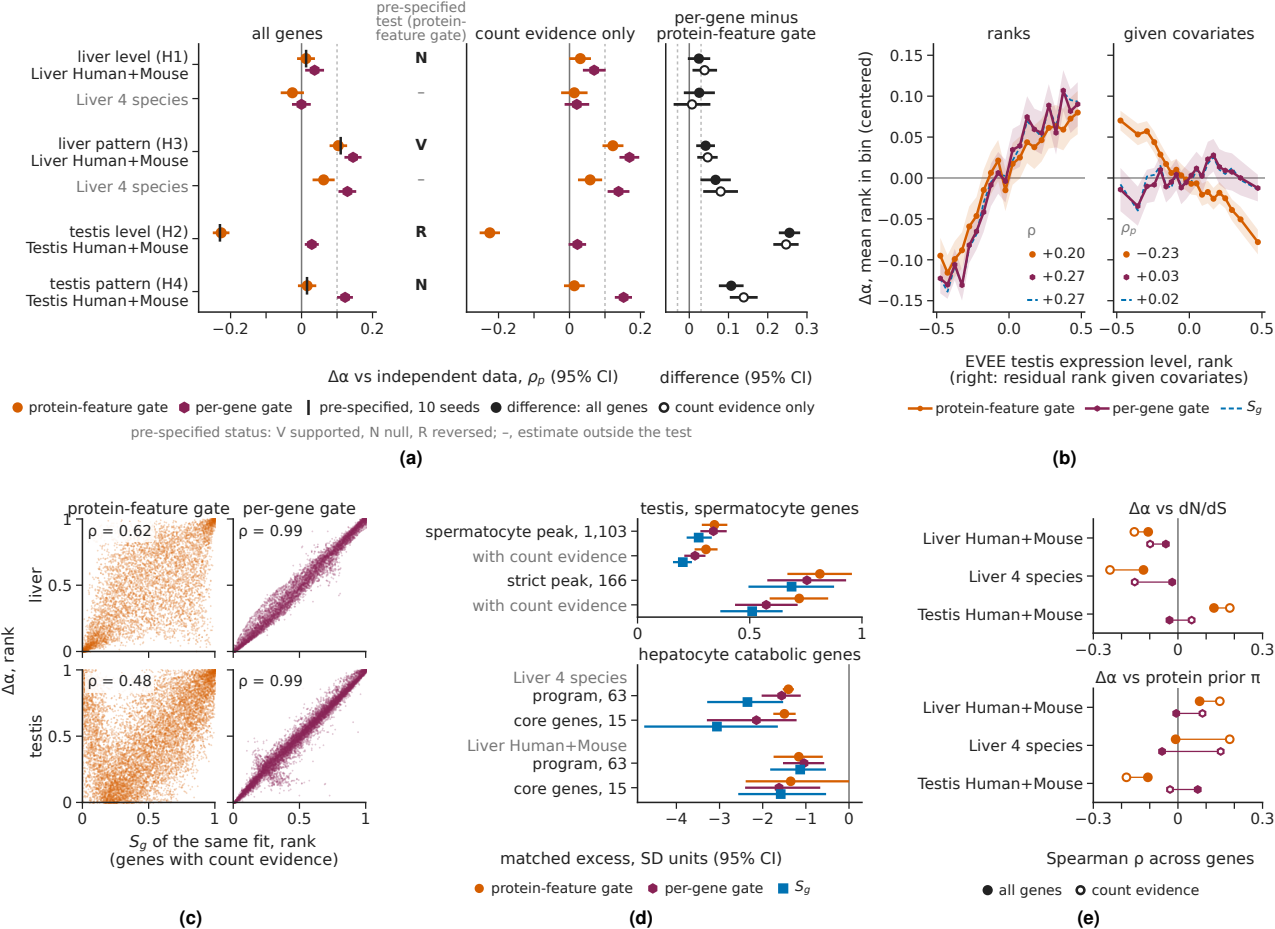

Liver Human+Monkey
r = 0.996
Liver Human+Mouse
r = 0.997
Liver Human+Zebrafish
r = 0.993
Liver 4 species
r = 0.969
Testis Human+Macaque
r = 0.996
Testis Human+Mouse
r = 0.993
Testis 3 species
r = 0.993
S_g at the end of training
S_g rebuilt from one pass over all cells
one gene
equal values
r, Pearson correlation over all genes



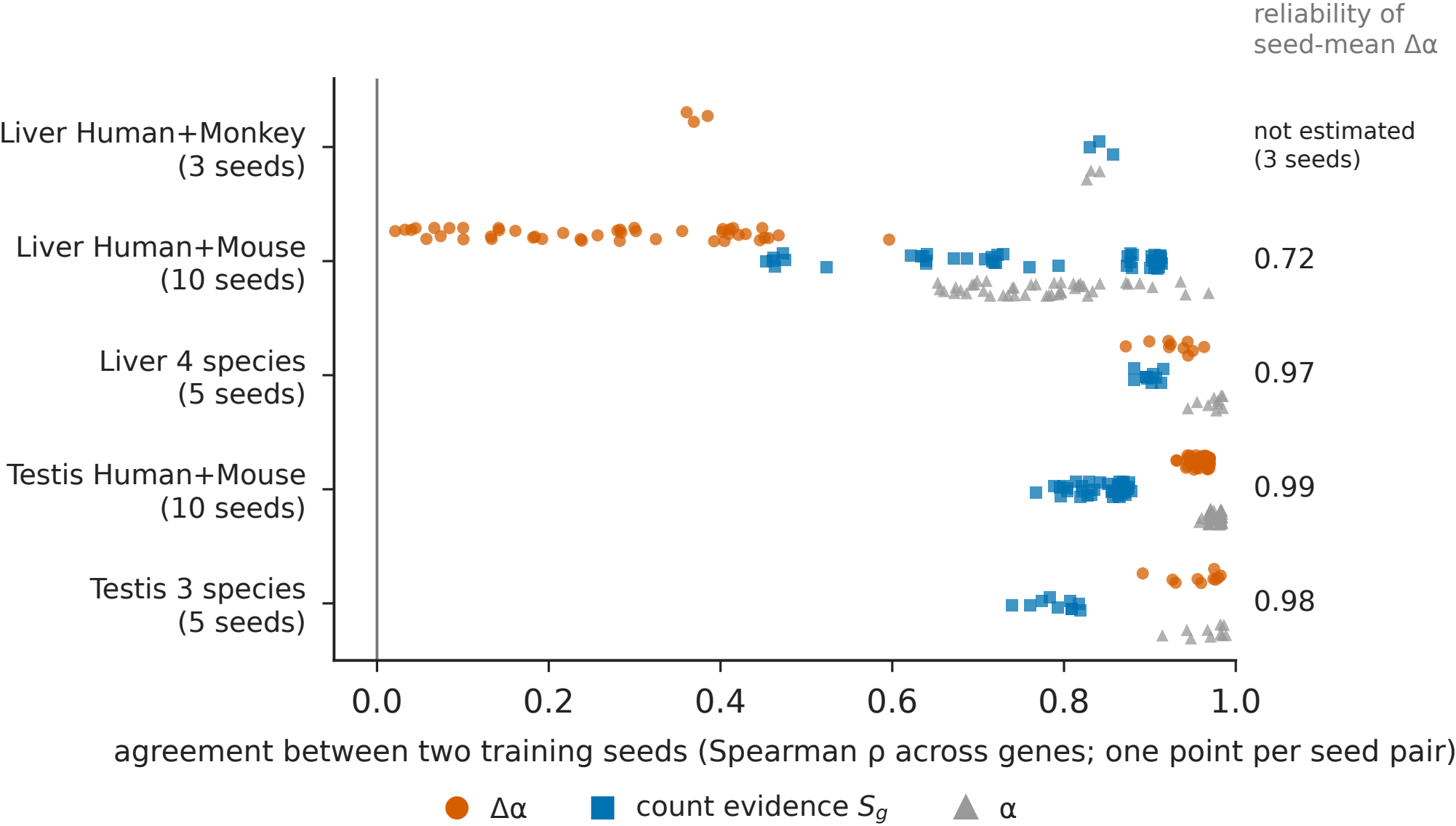
agreement between seeds of a cell type's contribution to S_g (Spearman ρ across genes;
points: seed pairs; in parentheses: cell-type weight; blue: weight at least 0.10)

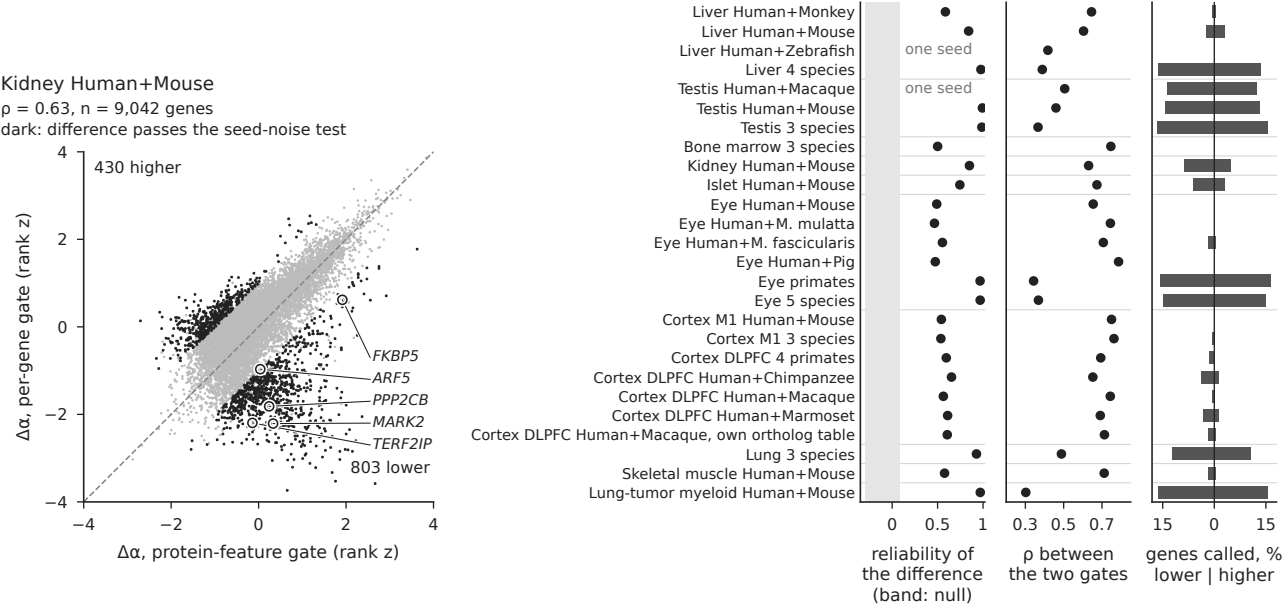

Pathway (coherence)
Liver 4 species
protein prior π
count evidence S_g
Δα, adjusted
Δα, adjusted + ESM-2
Liver Human+Mouse
protein prior π
count evidence S_g
Δα, adjusted
Δα, adjusted + ESM-2
Testis 3 species
protein prior π
count evidence S_g
Δα, adjusted
Δα, adjusted + ESM-2
1
2
5
10
pathway coherence index
95% interval of the permutation null
(a)
Protein interaction (coherence)
Liver 4 species
HuRI
Δα, adjusted
Δα, adjusted + ESM-2
STRING
Δα, adjusted
Δα, adjusted + ESM-2
Liver Human+Mouse
HuRI
Δα, adjusted
Δα, adjusted + ESM-2
STRING
Δα, adjusted
Δα, adjusted + ESM-2
Testis 3 species
HuRI
Δα, adjusted
Δα, adjusted + ESM-2
STRING
Δα, adjusted
Δα, adjusted + ESM-2
0.0
0.1
0.2
Moran's I
all edges
without the 1% most similar protein pairs
null, up to its 97.5th percentile
(b)
Interolog retention (conservation)
mouse liver interaction map
Δα, adjusted
Δα, adjusted + ESM-2
STRING, mouse
Δα, adjusted
Δα, adjusted + ESM-2
−0.1
0
0.1
partial ρ with interolog retention
95% CI
permutation null
(c)
Regulatory (conservation)
Δα, adjusted + ESM-2
liver
testis
liver − testis
Δα, adjusted
liver
testis
liver − testis
−0.04
0
0.04
0.08
Spearman ρ with conserved TF binding
95% CI
(d)


status: V supported, N null, R reversed; –, outside the test
0.10
status
liver level (H1)
Liver Human+Mouse
Liver 4 species
liver pattern (H3)
Liver Human+Mouse
Liver 4 species
testis level (H2)
Testis Human+Mouse
Testis 3 species
testis pattern (H4)
Testis Human+Mouse
Testis 3 species
N
–
V
–
R
–
N
–
−0.3 −0.2 −0.1 0.0 0.1 0.2
Δα vs independent data, ρ_p (95% CI)
primary cohort
multi-species cohort
(a)
q
liver: housekeeping genes ↑
liver: ribosome, proteasome, spliceosome ↑
liver: xenobiotic and bile-acid metabolism ↓
liver: interferon and inflammatory response ↓
liver: secreted (extracellular region) ↓
testis: housekeeping genes ↑
testis: ribosome, proteasome, spliceosome ↑
testis: interferon and inflammatory response ↓
testis: secreted (extracellular region) ↓
1.00
1.00
0.56
1.00
1.00
1.00
1.00
0.85
1.00
0.2 0.4 0.6
set vs matched background, AUROC of Δα (95% CI)
arrow: predicted direction (↑ above 0.5, ↓ below)
(b)
liver level (H1)
liver pattern (H3)
testis level (H2)
testis pattern (H4)
profile correlation
rank-matched level per cell type
specificity similarity
gene features alone
profile correlation, clusters of Chimera's latent
gene-level tools (vs Δα mean of 3 seeds, on the genes each tool scores)
SATURN gene score, native genes
SATURN gene score, 1:1 orthologs
CAMEX gene-embedding cosine
−0.4 −0.2 0.0 0.2
difference in ρ_p, Δα minus comparator (95% CI; thin: simultaneous)
← comparator closer to independent data
Δα closer →
(c)
liver level (H1)
n = 4,168
liver pattern (H3)
n = 5,402
testis level (H2)
n = 8,097
testis pattern (H4)
n = 6,925
Δα (Chimera)
count-evidence margin S_g (Chimera)
direct margin, cell-type labels, no model
profile correlation
rank-matched level per cell type
specificity similarity
profile correlation, clusters of Chimera's latent
gene features alone
−0.2 0.0 0.2
ρ_p with independent data, given covariates and all other simple baselines (95% CI)
(d)

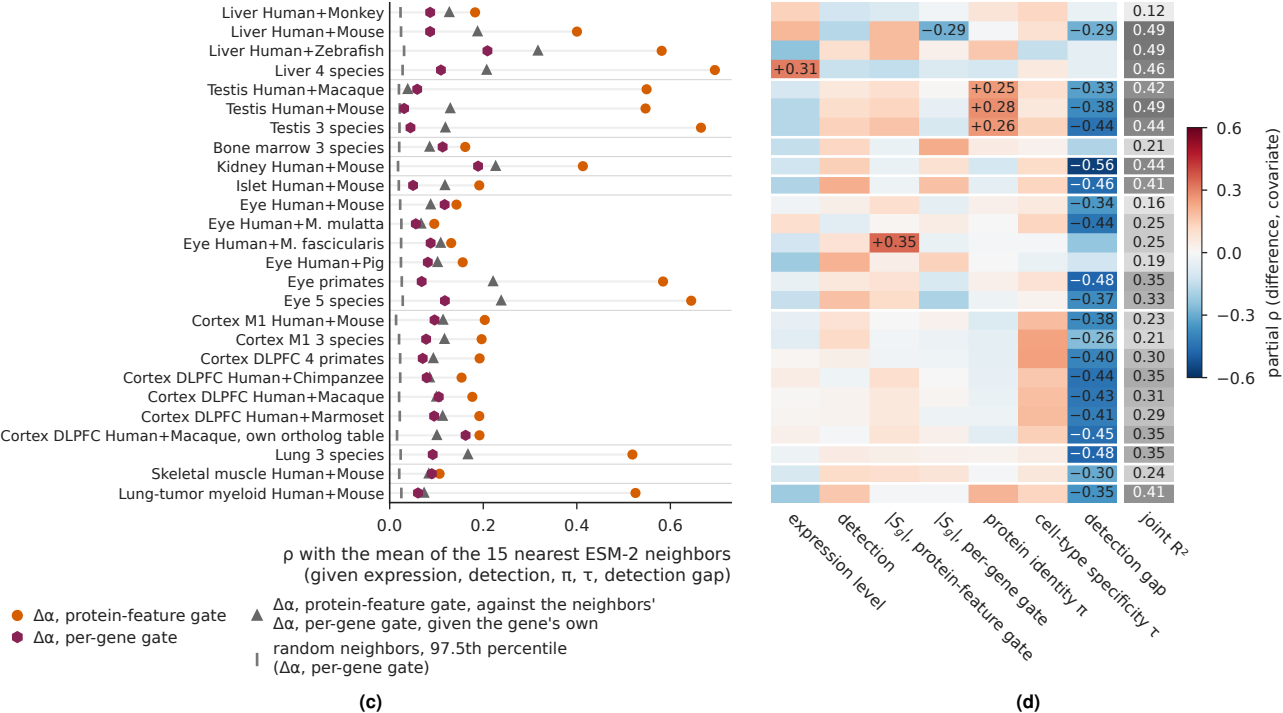

all genes
pre-specified test (protein-feature gate)
count evidence only
per-gene minus protein-feature gate
liver level (H1)
Liver Human+Mouse
Liver 4 species
liver pattern (H3)
Liver Human+Mouse
Liver 4 species
testis level (H2)
Testis Human+Mouse
testis pattern (H4)
Testis Human+Mouse
N
–
V
–
R
N
−0.2
0
0.2
0
0.1
0.2
0.3
Δα vs independent data, ρ_p (95% CI)
difference (95% CI)
protein-feature gate
per-gene gate
pre-specified, 10 seeds
difference: all genes
count evidence only
pre-specified status: V supported, N null, R reversed; –, estimate outside the test
(a)
ranks
given covariates
Δα, mean rank in bin (centered)
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
ρ
+0.20
+0.27
+0.27
ρ_p
−0.23
+0.03
+0.02
−0.5
0.0
0.5
EVEE testis expression level, rank
(right: residual rank given covariates)
protein-feature gate
per-gene gate
S_g
(b)
protein-feature gate
per-gene gate
liver
testis
ρ = 0.62
ρ = 0.99
ρ = 0.48
ρ = 0.99
Δα, rank
S_g of the same fit, rank
(genes with count evidence)
(c)
testis, spermatocyte genes
spermatocyte peak, 1,103
with count evidence
strict peak, 166
with count evidence
hepatocyte catabolic genes
Liver 4 species
program, 63
core genes, 15
Liver Human+Mouse
program, 63
core genes, 15
−4
−3
−2
−1
0
matched excess, SD units (95% CI)
protein-feature gate
per-gene gate
S_g
(d)
Δα vs dN/dS
Liver Human+Mouse
Liver 4 species
Testis Human+Mouse
−0.3
0
0.3
Δα vs protein prior π
Liver Human+Mouse
Liver 4 species
Testis Human+Mouse
Spearman ρ across genes
all genes
count evidence
(e)

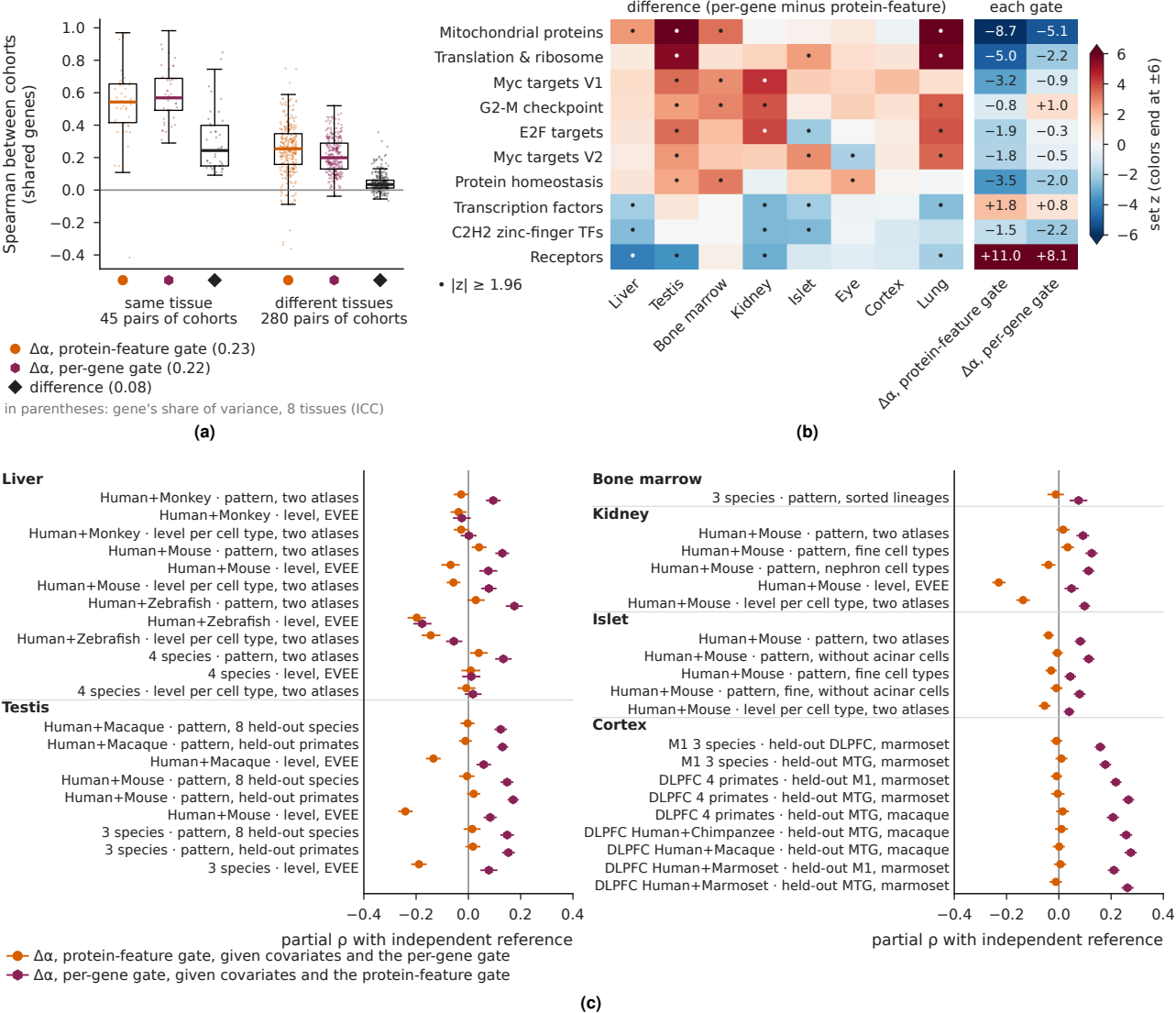
expression, log mean
expression, detection rate
expression, log variance
cell-type specificity τ
protein prior π
count evidence S_g
coding length
coding GC content
count evidence S_g given expression
cross-validated R², 3 expression features
cross-validated R², all 72 gate inputs
−0.04 −0.35 +0.39 −0.63 +0.73 +0.73 +0.67
−0.05 −0.35 +0.35 −0.64 +0.72 +0.71 +0.66
−0.03 −0.34 +0.42 −0.60 +0.74 +0.74 +0.68
+0.08 +0.31 −0.05 +0.55 −0.27 −0.34 −0.35
0.00 +0.19 −0.05 +0.02 −0.16 −0.16 −0.18
+0.41 +0.28 +0.32 +0.48 +0.50 +0.48 +0.35
+0.02 +0.02 +0.28 −0.01 +0.29 +0.21 +0.32
+0.03 +0.15 −0.10 +0.27 −0.31 −0.32 −0.31
+0.39 +0.17 +0.12 +0.11 +0.15 +0.09 +0.14
0.03 0.24 0.61 0.62 0.68 0.69 0.54
0.03 0.48 0.84 0.91 0.92 0.88 0.93
Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
+1.0
+0.5
0.0
−0.5
−1.0
Spearman ρ with Δα

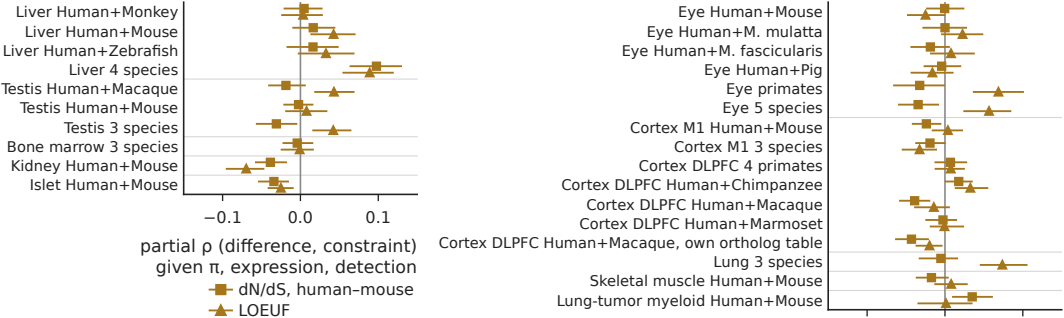

reliability of
seed-mean Δα
Liver Human+Monkey
(3 seeds)
not estimated
(3 seeds)
Liver Human+Mouse
(10 seeds)
0.72
Liver 4 species
(5 seeds)
0.97
Testis Human+Mouse
(10 seeds)
0.99
Testis 3 species
(5 seeds)
0.98
0.0
0.2
0.4
0.6
0.8
1.0
agreement between two training seeds (Spearman ρ across genes; one point per seed pair)
Δα
count evidence S_g
α


Kidney Human+Mouse
ρ = 0.63, n = 9,042 genes
dark: difference passes the seed-noise test
430 higher
803 lower
FKBP5
ARF5
PPP2CB
MARK2
TERF2IP
Δα, per-gene gate (rank z)
Δα, protein-feature gate (rank z)
Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
Bone marrow 3 species
Kidney Human+Mouse
Islet Human+Mouse
Eye Human+Mouse
Eye Human+M. mulatta
Eye Human+M. fascicularis
Eye Human+Pig
Eye primates
Eye 5 species
Cortex M1 Human+Mouse
Cortex M1 3 species
Cortex DLPFC 4 primates
Cortex DLPFC Human+Chimpanzee
Cortex DLPFC Human+Macaque
Cortex DLPFC Human+Marmoset
Cortex DLPFC Human+Macaque, own ortholog table
Lung 3 species
Skeletal muscle Human+Mouse
Lung-tumor myeloid Human+Mouse
one seed
one seed
reliability of the difference (band: null)
ρ between the two gates
genes called, % lower | higher

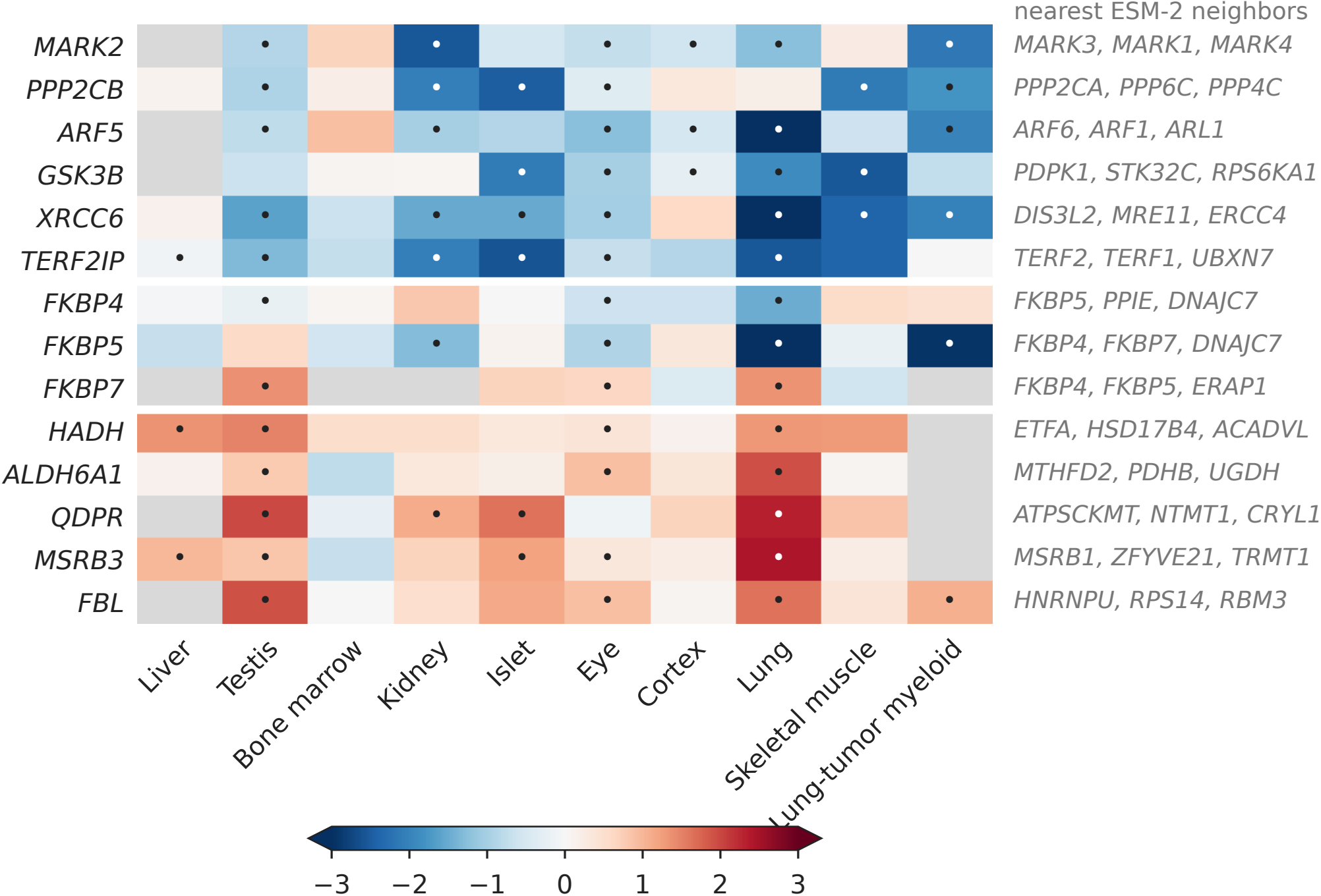
(a) (b)


Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
Bone marrow 3 species
Kidney Human+Mouse
Islet Human+Mouse
Eye Human+Mouse
Eye Human+M. mulatta
Eye Human+M. fascicularis
Eye Human+Pig
Eye primates
Eye 5 species
Cortex M1 Human+Mouse
Cortex M1 3 species
Cortex DLPFC 4 primates
Cortex DLPFC Human+Chimpanzee
Cortex DLPFC Human+Macaque
Cortex DLPFC Human+Marmoset
Cortex DLPFC Human+Macaque, own ortholog table
Lung 3 species
Skeletal muscle Human+Mouse
Lung-tumor myeloid Human+Mouse
0.0
0.2
0.4
0.6
ρ with the mean of the 15 nearest ESM-2 neighbors
(given expression, detection, π, τ, detection gap)
Δα, protein-feature gate
Δα, per-gene gate
Δα, protein-feature gate, against the neighbors' Δα, per-gene gate, given the gene's own
random neighbors, 97.5th percentile (Δα, per-gene gate)
(c)
+0.31
−0.29
−0.29
+0.25
+0.28
+0.26
−0.33
−0.38
−0.44
−0.56
−0.46
−0.34
−0.44
+0.35
−0.48
−0.37
−0.38
−0.26
−0.40
−0.44
−0.43
−0.41
−0.45
−0.48
−0.30
−0.35
0.12
0.49
0.49
0.46
0.42
0.49
0.44
0.21
0.44
0.41
0.16
0.25
0.25
0.19
0.35
0.33
0.23
0.21
0.30
0.35
0.31
0.29
0.35
0.35
0.24
0.41
expression level
detection
|S_g|, protein-feature gate
|S_g|, per-gene gate
protein identity π
cell-type specificity τ
detection gap
joint R²
0.6
0.3
0.0
−0.3
−0.6
partial ρ (difference, covariate)
(d)


Spearman between cohorts (shared genes)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
same tissue
45 pairs of cohorts
different tissues
280 pairs of cohorts
Δα, protein-feature gate (0.23)
Δα, per-gene gate (0.22)
difference (0.08)
in parentheses: gene's share of variance, 8 tissues (ICC)
(a)
difference (per-gene minus protein-feature)
each gate
Mitochondrial proteins
Translation & ribosome
Myc targets V1
G2-M checkpoint
E2F targets
Myc targets V2
Protein homeostasis
Transcription factors
C2H2 zinc-finger TFs
Receptors
−8.7 −5.1
−5.0 −2.2
−3.2 −0.9
−0.8 +1.0
−1.9 −0.3
−1.8 −0.5
−3.5 −2.0
+1.8 +0.8
−1.5 −2.2
+11.0 +8.1
• |z| ≥ 1.96
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
Δα, protein-feature gate
Δα, per-gene gate
set z (colors end at ±6)
6
4
2
0
−2
−4
−6
(b)
Liver
Human+Monkey · pattern, two atlases
Human+Monkey · level, EVEE
Human+Monkey · level per cell type, two atlases
Human+Mouse · pattern, two atlases
Human+Mouse · level, EVEE
Human+Mouse · level per cell type, two atlases
Human+Zebrafish · pattern, two atlases
Human+Zebrafish · level, EVEE
Human+Zebrafish · level per cell type, two atlases
4 species · pattern, two atlases
4 species · level, EVEE
4 species · level per cell type, two atlases
Testis
Human+Macaque · pattern, 8 held-out species
Human+Macaque · pattern, held-out primates
Human+Macaque · level, EVEE
Human+Mouse · pattern, 8 held-out species
Human+Mouse · pattern, held-out primates
Human+Mouse · level, EVEE
3 species · pattern, 8 held-out species
3 species · pattern, held-out primates
3 species · level, EVEE
−0.4 −0.2 0.0 0.2 0.4
partial ρ with independent reference
Bone marrow
3 species · pattern, sorted lineages
Kidney
Human+Mouse · pattern, two atlases
Human+Mouse · pattern, fine cell types
Human+Mouse · pattern, nephron cell types
Human+Mouse · level, EVEE
Human+Mouse · level per cell type, two atlases
Islet
Human+Mouse · pattern, two atlases
Human+Mouse · pattern, without acinar cells
Human+Mouse · pattern, fine cell types
Human+Mouse · pattern, fine, without acinar cells
Human+Mouse · level per cell type, two atlases
Cortex
M1 3 species · held-out DLPFC, marmoset
M1 3 species · held-out MTG, marmoset
DLPFC 4 primates · held-out M1, marmoset
DLPFC 4 primates · held-out MTG, marmoset
DLPFC 4 primates · held-out MTG, macaque
DLPFC Human+Chimpanzee · held-out MTG, macaque
DLPFC Human+Macaque · held-out MTG, macaque
DLPFC Human+Marmoset · held-out M1, marmoset
DLPFC Human+Marmoset · held-out MTG, marmoset
−0.4 −0.2 0.0 0.2 0.4
partial ρ with independent reference
Δα, protein-feature gate, given covariates and the per-gene gate
Δα, per-gene gate, given covariates and the protein-feature gate
(c)

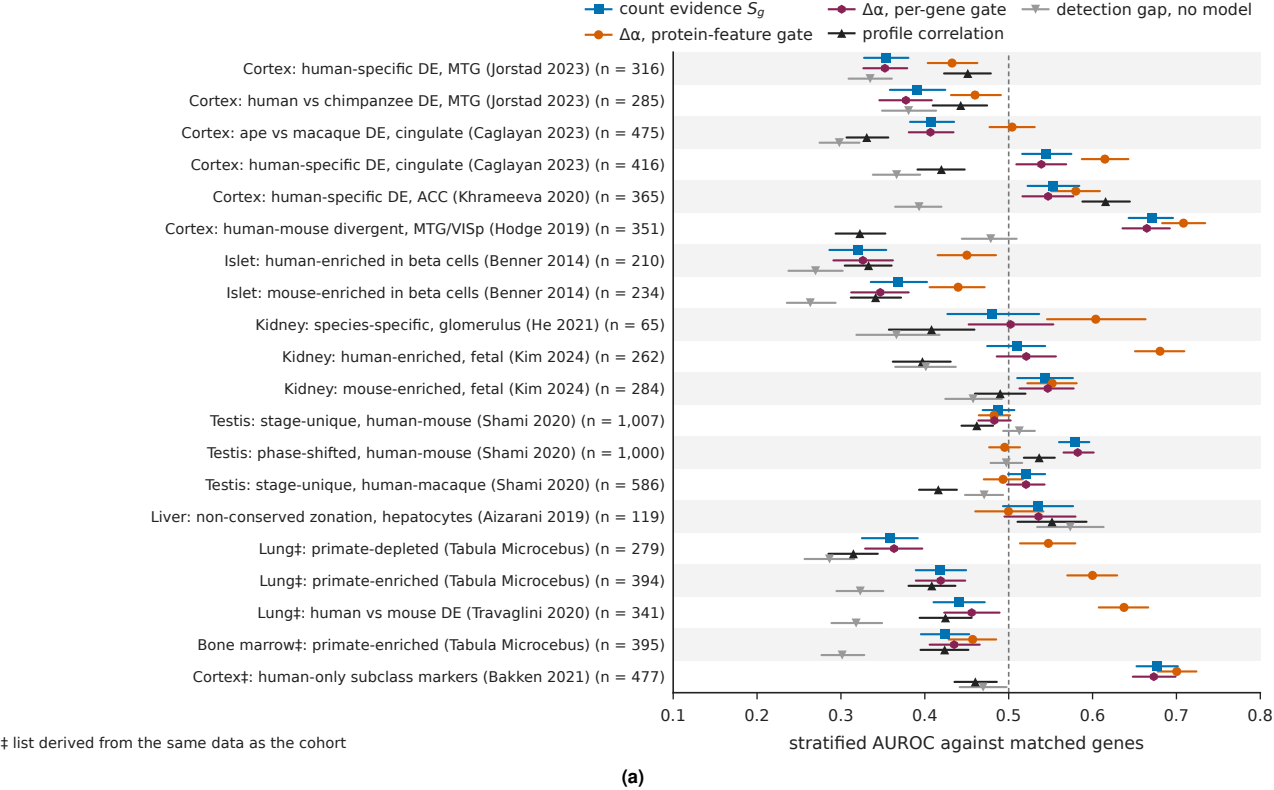

Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
Bone marrow 3 species
Kidney Human+Mouse
Islet Human+Mouse
−0.1
0.0
0.1
partial ρ (difference, constraint)
given π, expression, detection
dN/dS, human–mouse
LOEUF
Eye Human+Mouse
Eye Human+M. mulatta
Eye Human+M. fascicularis
Eye Human+Pig
Eye primates
Eye 5 species
Cortex M1 Human+Mouse
Cortex M1 3 species
Cortex DLPFC 4 primates
Cortex DLPFC Human+Chimpanzee
Cortex DLPFC Human+Macaque
Cortex DLPFC Human+Marmoset
Cortex DLPFC Human+Macaque, own ortholog table
Lung 3 species
Skeletal muscle Human+Mouse
Lung-tumor myeloid Human+Mouse
−0.1
0.0
0.1
partial ρ (difference, constraint)
given π, expression, detection

(d)

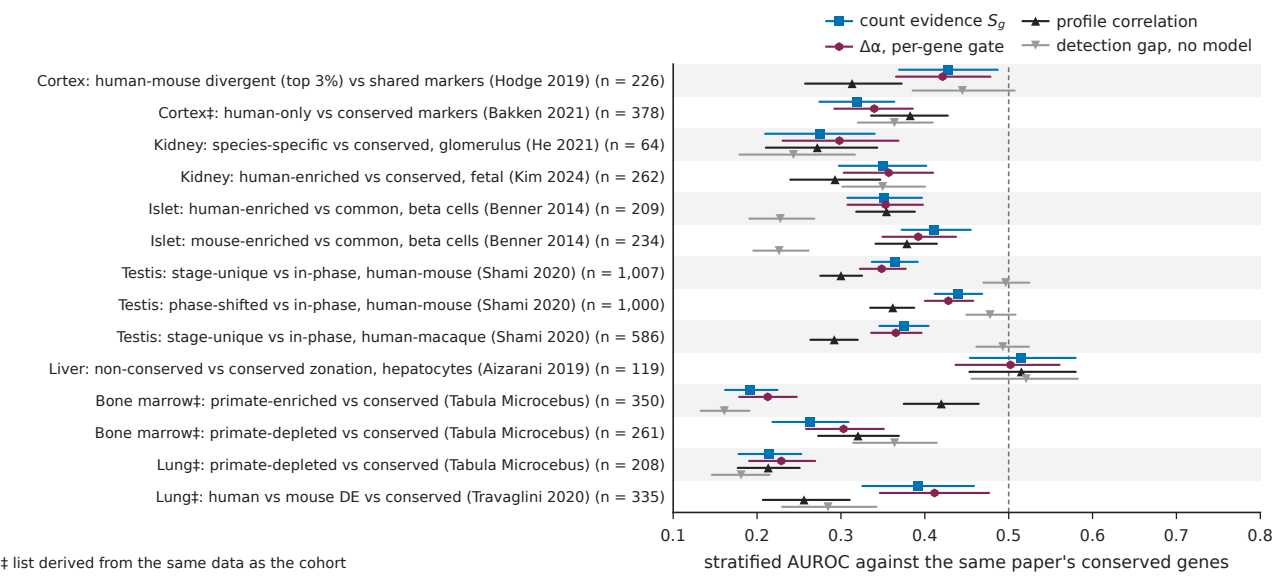


difference of Δα, per-gene gate minus protein-feature gate (rank z); 3 cells beyond ±3

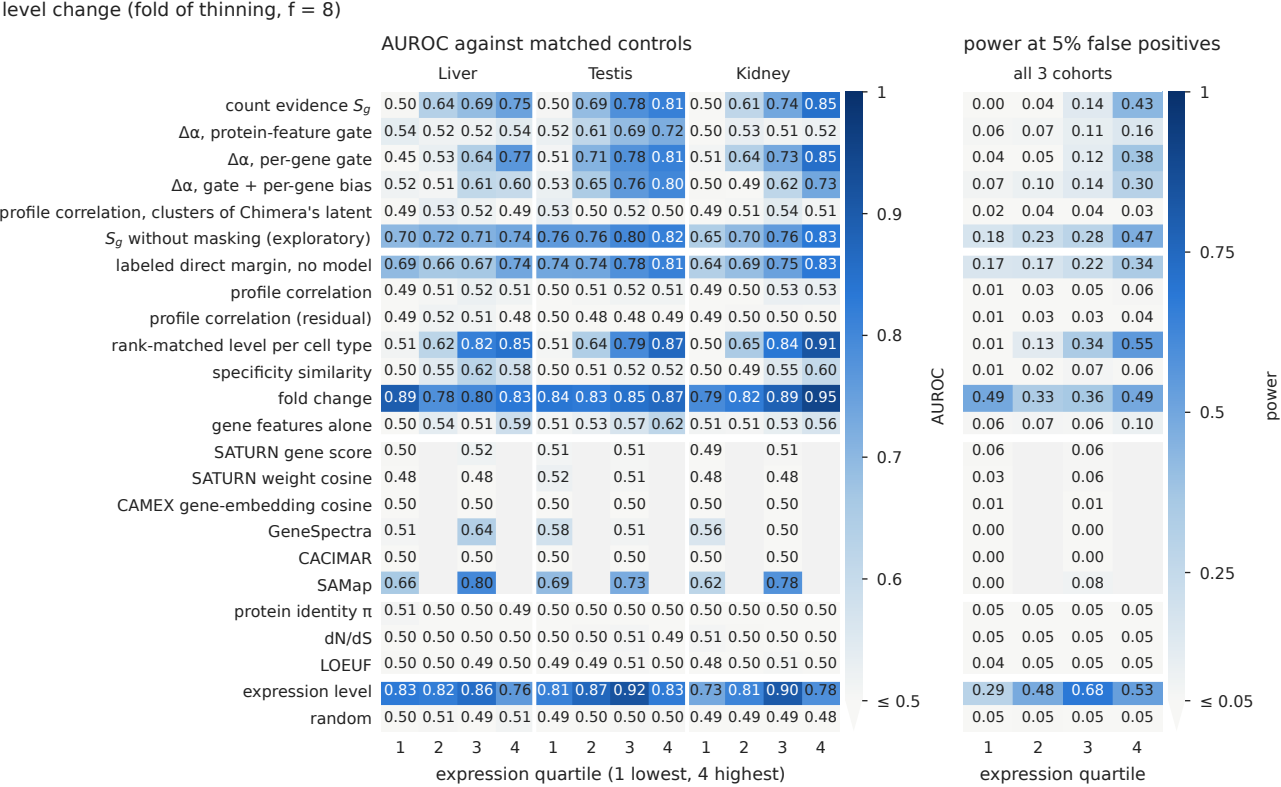
• difference called in a cohort of the tissue gene without count evidence in both fits

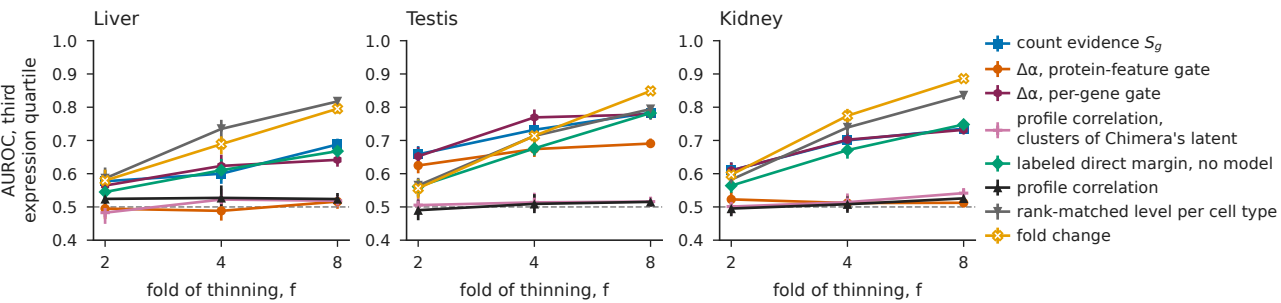

count evidence S_g
Δα, per-gene gate
detection gap, no model
Δα, protein-feature gate
profile correlation
Cortex: human-specific DE, MTG (Jorstad 2023) (n = 316)
Cortex: human vs chimpanzee DE, MTG (Jorstad 2023) (n = 285)
Cortex: ape vs macaque DE, cingulate (Caglayan 2023) (n = 475)
Cortex: human-specific DE, cingulate (Caglayan 2023) (n = 416)
Cortex: human-specific DE, ACC (Khrameeva 2020) (n = 365)
Cortex: human-mouse divergent, MTG/VISp (Hodge 2019) (n = 351)
Islet: human-enriched in beta cells (Benner 2014) (n = 210)
Islet: mouse-enriched in beta cells (Benner 2014) (n = 234)
Kidney: species-specific, glomerulus (He 2021) (n = 65)
Kidney: human-enriched, fetal (Kim 2024) (n = 262)
Kidney: mouse-enriched, fetal (Kim 2024) (n = 284)
Testis: stage-unique, human-mouse (Shami 2020) (n = 1,007)
Testis: phase-shifted, human-mouse (Shami 2020) (n = 1,000)
Testis: stage-unique, human-macaque (Shami 2020) (n = 586)
Liver: non-conserved zonation, hepatocytes (Aizarani 2019) (n = 119)
Lung‡: primate-depleted (Tabula Microcebus) (n = 279)
Lung‡: primate-enriched (Tabula Microcebus) (n = 394)
Lung‡: human vs mouse DE (Travaglini 2020) (n = 341)
Bone marrow‡: primate-enriched (Tabula Microcebus) (n = 395)
Cortex‡: human-only subclass markers (Bakken 2021) (n = 477)
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
‡ list derived from the same data as the cohort
stratified AUROC against matched genes

(a)

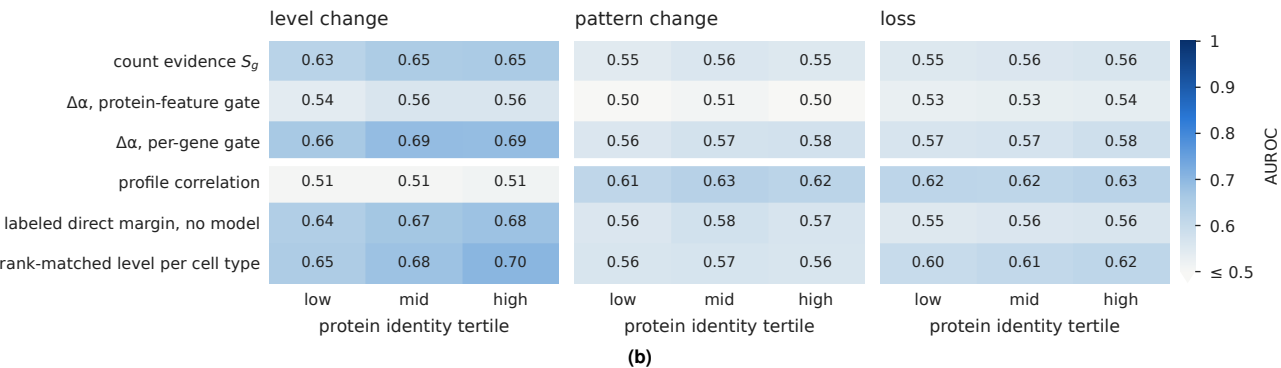

count evidence S_g
profile correlation
Δα, per-gene gate
detection gap, no model
Cortex: human-mouse divergent (top 3%) vs shared markers (Hodge 2019) (n = 226)
Cortex‡: human-only vs conserved markers (Bakken 2021) (n = 378)
Kidney: species-specific vs conserved, glomerulus (He 2021) (n = 64)
Kidney: human-enriched vs conserved, fetal (Kim 2024) (n = 262)
Islet: human-enriched vs common, beta cells (Benner 2014) (n = 209)
Islet: mouse-enriched vs common, beta cells (Benner 2014) (n = 234)
Testis: stage-unique vs in-phase, human-mouse (Shami 2020) (n = 1,007)
Testis: phase-shifted vs in-phase, human-mouse (Shami 2020) (n = 1,000)
Testis: stage-unique vs in-phase, human-macaque (Shami 2020) (n = 586)
Liver: non-conserved vs conserved zonation, hepatocytes (Aizarani 2019) (n = 119)
Bone marrow‡: primate-enriched vs conserved (Tabula Microcebus) (n = 350)
Bone marrow‡: primate-depleted vs conserved (Tabula Microcebus) (n = 261)
Lung‡: primate-depleted vs conserved (Tabula Microcebus) (n = 208)
Lung‡: human vs mouse DE vs conserved (Travaglini 2020) (n = 335)
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
‡ list derived from the same data as the cohort
stratified AUROC against the same paper's conserved genes

(b)

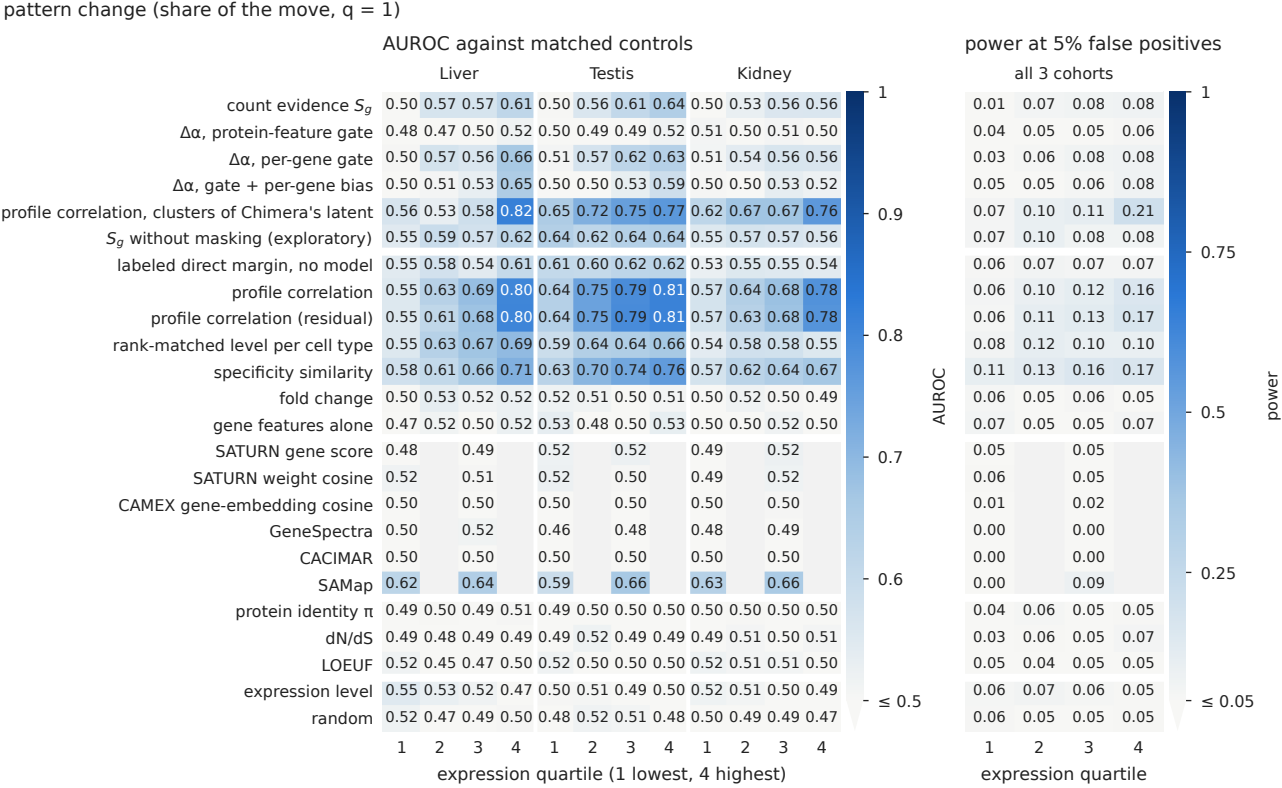

level change (fold of thinning, f = 8)
AUROC against matched controls
power at 5% false positives
Liver
Testis
Kidney
all 3 cohorts
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
Δα, gate + per-gene bias
profile correlation, clusters of Chimera's latent
S_g without masking (exploratory)
labeled direct margin, no model
profile correlation
profile correlation (residual)
rank-matched level per cell type
specificity similarity
fold change
gene features alone
SATURN gene score
SATURN weight cosine
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SAMap
protein identity π
dN/dS
LOEUF
expression level
random
AUROC
power
expression quartile (1 lowest, 4 highest)
expression quartile

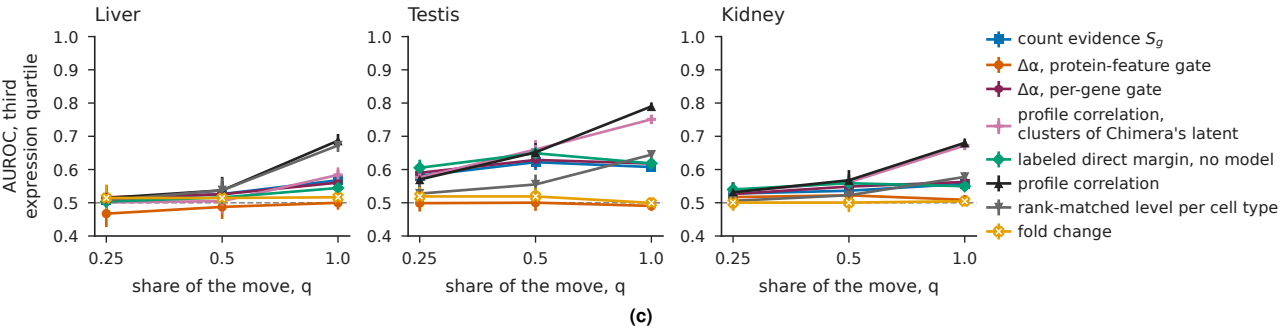

Liver
Testis
Kidney
AUROC, third expression quartile
fold of thinning, f
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation, clusters of Chimera's latent
labeled direct margin, no model
profile correlation
rank-matched level per cell type
fold change

(a)

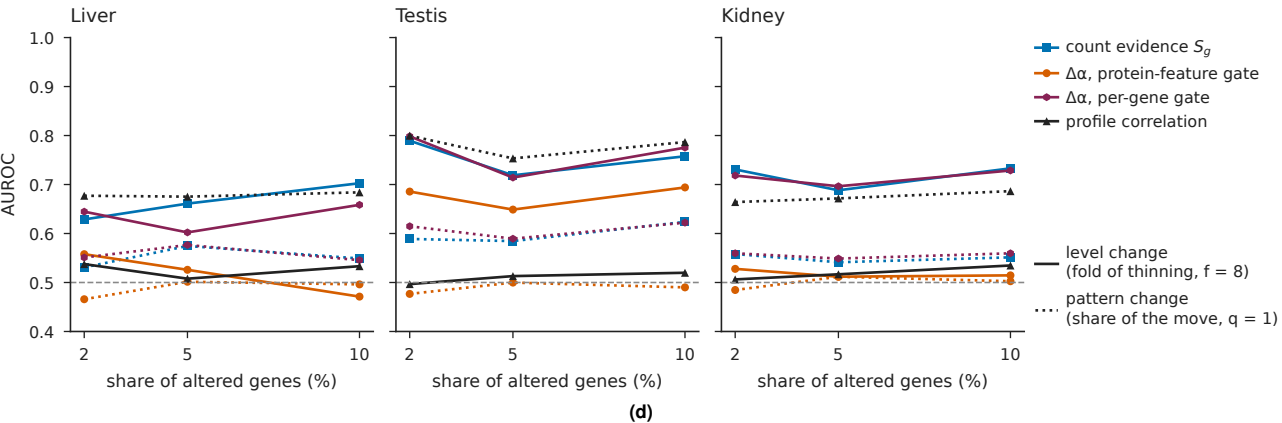

level change
pattern change
loss
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
labeled direct margin, no model
rank-matched level per cell type
low
mid
high
protein identity tertile
AUROC

(b)

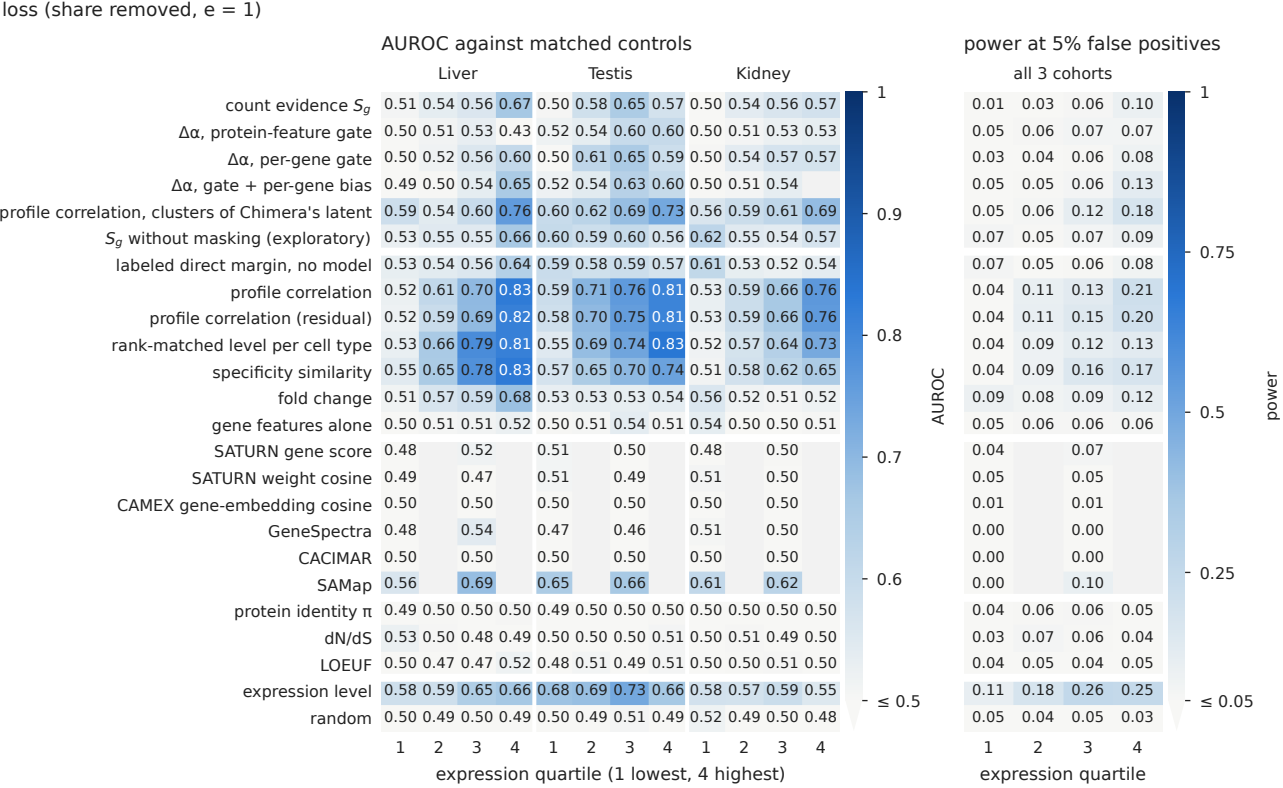

pattern change (share of the move, q = 1)
AUROC against matched controls
power at 5% false positives
Liver
Testis
Kidney
all 3 cohorts
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
Δα, gate + per-gene bias
profile correlation, clusters of Chimera's latent
S_g without masking (exploratory)
labeled direct margin, no model
profile correlation
profile correlation (residual)
rank-matched level per cell type
specificity similarity
fold change
gene features alone
SATURN gene score
SATURN weight cosine
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SAMap
protein identity π
dN/dS
LOEUF
expression level
random
AUROC
power
expression quartile (1 lowest, 4 highest)
expression quartile

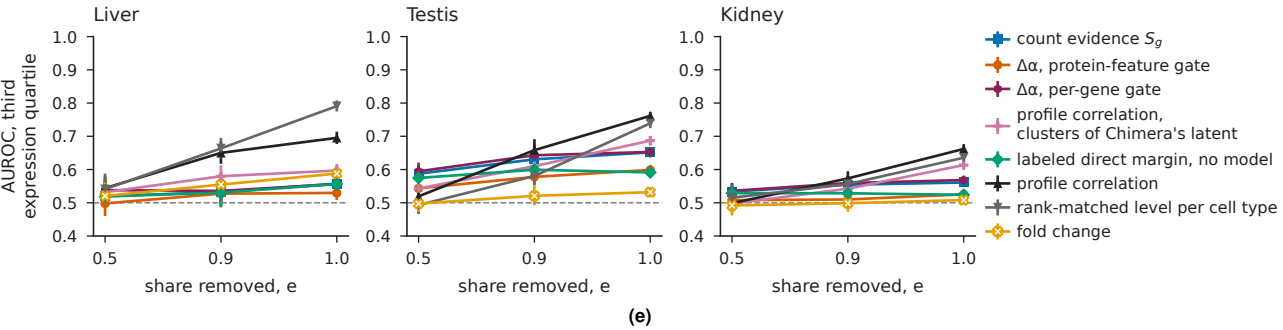

Liver
Testis
Kidney
AUROC, third expression quartile
share of the move, q
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation, clusters of Chimera's latent
labeled direct margin, no model
profile correlation
rank-matched level per cell type
fold change

(c)

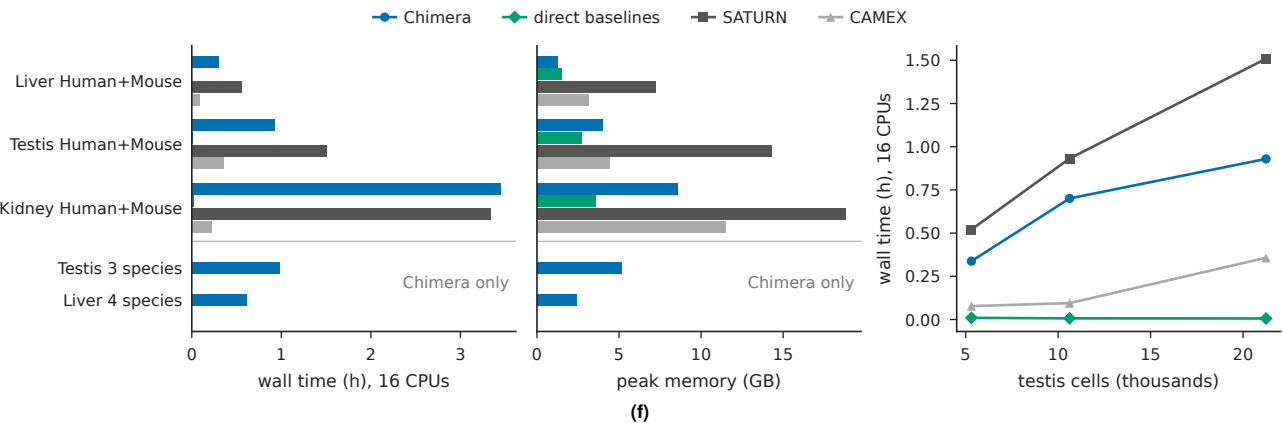

Liver
Testis
Kidney
AUROC
share of altered genes (%)
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
level change (fold of thinning, f = 8)
pattern change (share of the move, q = 1)

(d)

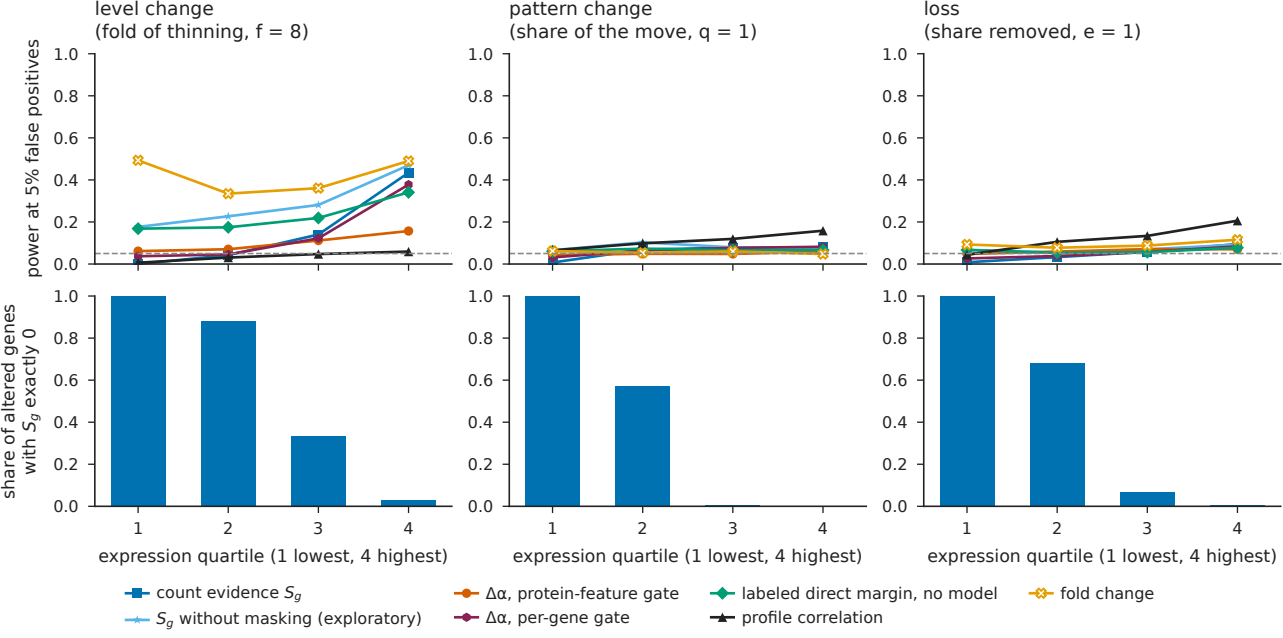

loss (share removed, e = 1)
AUROC against matched controls
Liver
Testis
Kidney
power at 5% false positives
all 3 cohorts
count evidence S_g 0.51 0.54 0.56 0.67 0.50 0.58 0.65 0.57 0.50 0.54 0.56 0.57 0.01 0.03 0.06 0.10
Δα, protein-feature gate 0.50 0.51 0.53 0.43 0.52 0.54 0.60 0.60 0.50 0.51 0.53 0.53 0.05 0.06 0.07 0.07
Δα, per-gene gate 0.50 0.52 0.56 0.60 0.50 0.61 0.65 0.59 0.50 0.54 0.57 0.57 0.03 0.04 0.06 0.08
Δα, gate + per-gene bias 0.49 0.50 0.54 0.65 0.52 0.54 0.63 0.60 0.50 0.51 0.54 0.05 0.05 0.06 0.13
profile correlation, clusters of Chimera's latent 0.59 0.54 0.60 0.76 0.60 0.62 0.69 0.73 0.56 0.59 0.61 0.69 0.05 0.06 0.12 0.18
S_g without masking (exploratory) 0.53 0.55 0.55 0.66 0.60 0.59 0.60 0.56 0.62 0.55 0.54 0.57 0.07 0.05 0.07 0.09
labeled direct margin, no model 0.53 0.54 0.56 0.64 0.59 0.58 0.59 0.57 0.61 0.53 0.52 0.54 0.07 0.05 0.06 0.08
profile correlation 0.52 0.61 0.70 0.83 0.59 0.71 0.76 0.81 0.53 0.59 0.66 0.76 0.04 0.11 0.13 0.21
profile correlation (residual) 0.52 0.59 0.69 0.82 0.58 0.70 0.75 0.81 0.53 0.59 0.66 0.76 0.04 0.11 0.15 0.20
rank-matched level per cell type 0.53 0.66 0.79 0.81 0.55 0.69 0.74 0.83 0.52 0.57 0.64 0.73 0.04 0.09 0.12 0.13
specificity similarity 0.55 0.65 0.78 0.83 0.57 0.65 0.70 0.74 0.51 0.58 0.62 0.65 0.04 0.09 0.16 0.17
fold change 0.51 0.57 0.59 0.68 0.53 0.53 0.53 0.54 0.56 0.52 0.51 0.52 0.09 0.08 0.09 0.12
gene features alone 0.50 0.51 0.51 0.52 0.50 0.51 0.54 0.51 0.54 0.50 0.50 0.51 0.05 0.06 0.06 0.06
SATURN gene score 0.48 0.52 0.51 0.50 0.48 0.50 0.04 0.07
SATURN weight cosine 0.49 0.47 0.51 0.49 0.51 0.50 0.05 0.05
CAMEX gene-embedding cosine 0.50 0.50 0.50 0.50 0.50 0.50 0.01 0.01
GeneSpectra 0.48 0.54 0.47 0.46 0.51 0.50 0.00 0.00
CACIMAR 0.50 0.50 0.50 0.50 0.50 0.50 0.00 0.00
SAMap 0.56 0.69 0.65 0.66 0.61 0.62 0.00 0.10
protein identity π 0.49 0.50 0.50 0.50 0.49 0.50 0.50 0.50 0.50 0.50 0.50 0.50 0.04 0.06 0.06 0.05
dN/dS 0.53 0.50 0.48 0.49 0.50 0.50 0.50 0.51 0.50 0.51 0.49 0.50 0.03 0.07 0.06 0.04
LOEUF 0.50 0.47 0.47 0.52 0.48 0.51 0.49 0.51 0.50 0.50 0.51 0.50 0.04 0.05 0.04 0.05
expression level 0.58 0.59 0.65 0.66 0.68 0.69 0.73 0.66 0.58 0.57 0.59 0.55 0.11 0.18 0.26 0.25
random 0.50 0.49 0.50 0.49 0.50 0.49 0.51 0.49 0.52 0.49 0.50 0.48 0.05 0.04 0.05 0.03
1 2 3 4 1 2 3 4 1 2 3 4
expression quartile (1 lowest, 4 highest)
1 2 3 4
expression quartile
AUROC
1
0.9
0.8
0.7
0.6
≤ 0.5
power
1
0.75
0.5
0.25
≤ 0.05


Liver
Testis
Kidney
AUROC, third expression quartile
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.5
0.9
1.0
share removed, e
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation, clusters of Chimera's latent
labeled direct margin, no model
profile correlation
rank-matched level per cell type
fold change

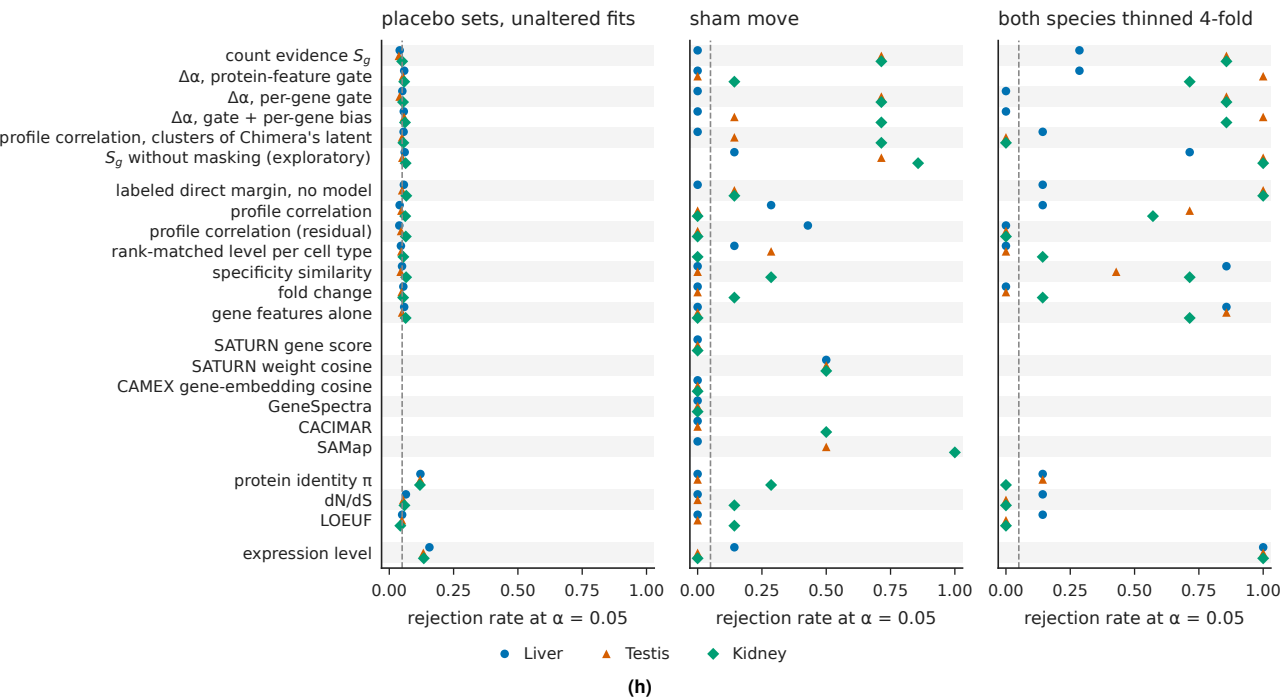
(e)


Chimera
direct baselines
SATURN
CAMEX
Liver Human+Mouse
Testis Human+Mouse
Kidney Human+Mouse
Testis 3 species
Liver 4 species
Chimera only
Chimera only
0
1
2
3
wall time (h), 16 CPUs
0
5
10
15
peak memory (GB)
wall time (h), 16 CPUs
1.50
1.25
1.00
0.75
0.50
0.25
0.00
5
10
15
20
testis cells (thousands)

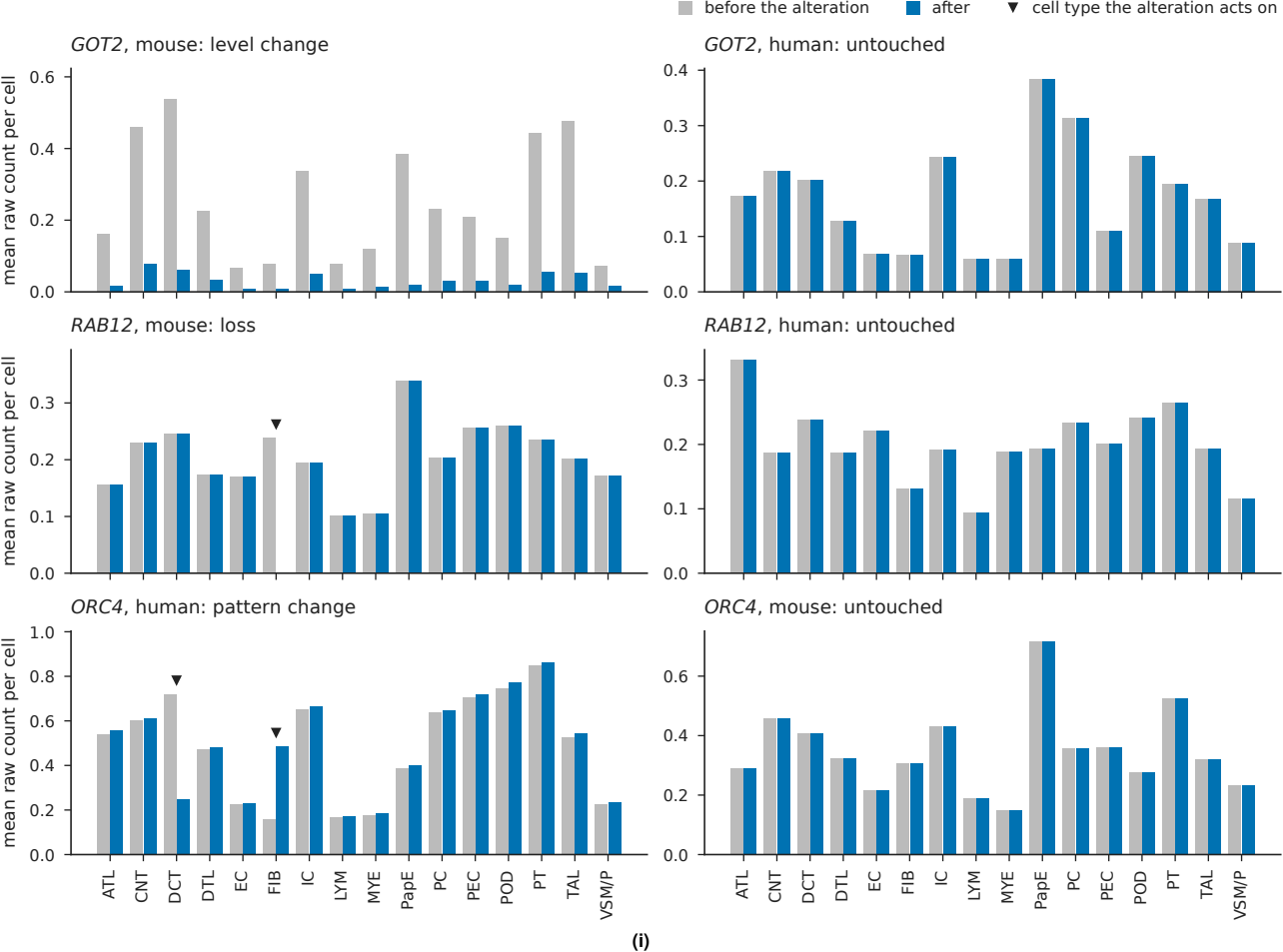
(f)


level change
(fold of thinning, f = 8)
pattern change
(share of the move, q = 1)
loss
(share removed, e = 1)
power at 5% false positives
share of altered genes with S_g exactly 0
expression quartile (1 lowest, 4 highest)
count evidence S_g
S_g without masking (exploratory)
Δα, protein-feature gate
Δα, per-gene gate
labeled direct margin, no model
profile correlation
fold change

(g)

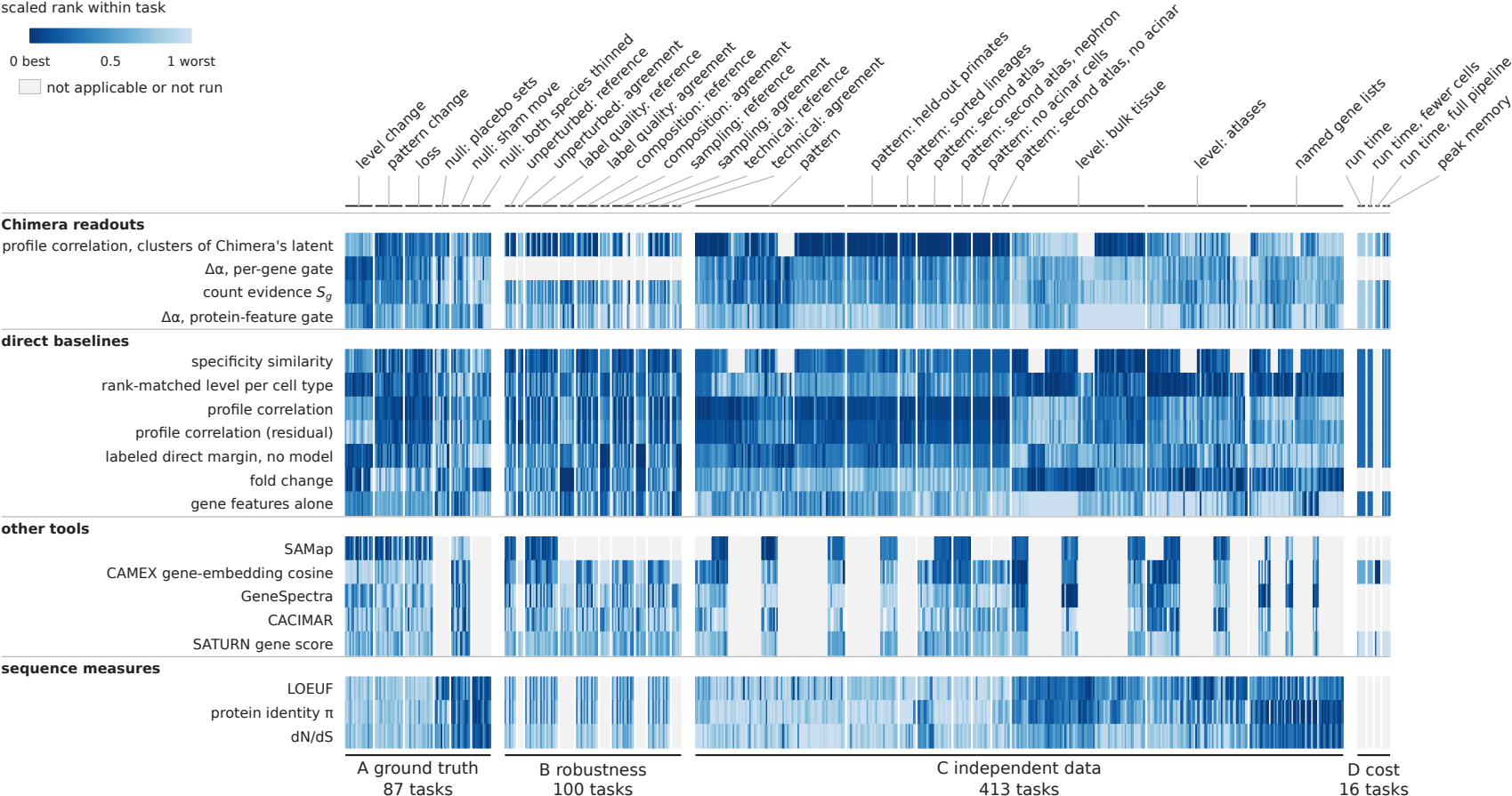

placebo sets, unaltered fits
sham move
both species thinned 4-fold
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
Δα, gate + per-gene bias
profile correlation, clusters of Chimera's latent
S_g without masking (exploratory)
labeled direct margin, no model
profile correlation
profile correlation (residual)
rank-matched level per cell type
specificity similarity
fold change
gene features alone
SATURN gene score
SATURN weight cosine
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SAMap
protein identity π
dN/dS
LOEUF
expression level
rejection rate at α = 0.05
Liver
Testis
Kidney

(h)

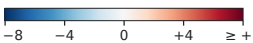

before the alteration
after
cell type the alteration acts on
GOT2, mouse: level change
GOT2, human: untouched
RAB12, mouse: loss
RAB12, human: untouched
ORC4, human: pattern change
ORC4, mouse: untouched
mean raw count per cell
ATL
CNT
DCT
DTL
EC
FIB
IC
LYM
MYE
PapE
PC
PEC
POD
PT
TAL
VSM/P

(i)

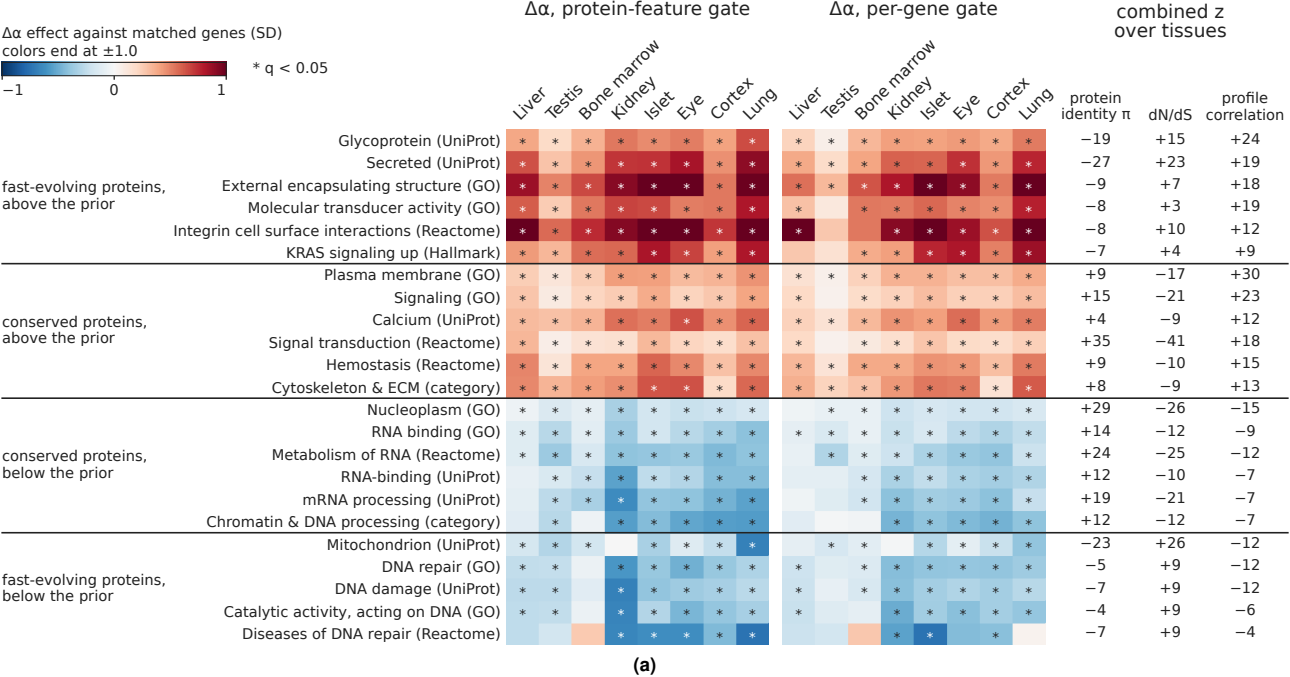
mean scaled rank within task
0 best
0.5
1 worst
not applicable or not run
5/15 runs finished / runs tried
level change
pattern change
loss
null: placebo sets
null: sham move
null: both species thinned
unperturbed: reference
unperturbed: agreement
label quality: reference
label quality: agreement
composition: reference
composition: agreement
sampling: reference
sampling: agreement
technical: reference
technical: agreement
pattern
pattern: held-out primates
pattern: sorted lineages
pattern: second atlas
pattern: second atlas, nephron
pattern: no acinar cells
pattern: second atlas, no acinar
level: bulk tissue
level: atlases
named gene lists
run time
run time, fewer cells
run time, full pipeline
peak memory
number of tasks
18 18 18 9 12 12
A ground truth
7 3 21 9 14 6 14 6 14 6
B robustness
99 33 10 22 11 11 11 88 66 62
C independent data
5 3 3 5
D cost
overall scaled rank
0 best; bar = 1 − rank
Chimera readouts
profile correlation, clusters of Chimera's latent
Δα, per-gene gate
count evidence S_g
Δα, protein-feature gate
5/15 2/6
direct baselines
specificity similarity
rank-matched level per cell type
profile correlation
profile correlation (residual)
labeled direct margin, no model
fold change
gene features alone
other tools
SAMap
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SATURN gene score
sequence measures
LOEUF
protein identity π
dN/dS
0.45
0.47
0.56
0.68
0.29
0.32
0.32
0.36
0.39
0.43
0.50
0.27
0.57
0.61
0.65
0.74
0.60
0.62
0.68


scaled rank within task
0 best
0.5
1 worst
not applicable or not run
level change
pattern change
loss
null: placebo sets
null: sham move
null: both species thinned
unperturbed: reference
unperturbed: agreement
label quality: reference
label quality: agreement
composition: reference
composition: agreement
sampling: reference
sampling: agreement
technical: reference
technical: agreement
pattern
pattern: held-out primates
pattern: sorted lineages
pattern: second atlas
pattern: second atlas, nephron
pattern: no acinar cells
pattern: second atlas, no acinar
level: bulk tissue
level: atlases
named gene lists
run time
run time, fewer cells
run time, full pipeline
peak memory
Chimera readouts
profile correlation, clusters of Chimera's latent
Δα, per-gene gate
count evidence S_g
Δα, protein-feature gate
direct baselines
specificity similarity
rank-matched level per cell type
profile correlation
profile correlation (residual)
labeled direct margin, no model
fold change
gene features alone
other tools
SAMap
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SATURN gene score
sequence measures
LOEUF
protein identity π
dN/dS
A ground truth
87 tasks
B robustness
100 tasks
C independent data
413 tasks
D cost
16 tasks

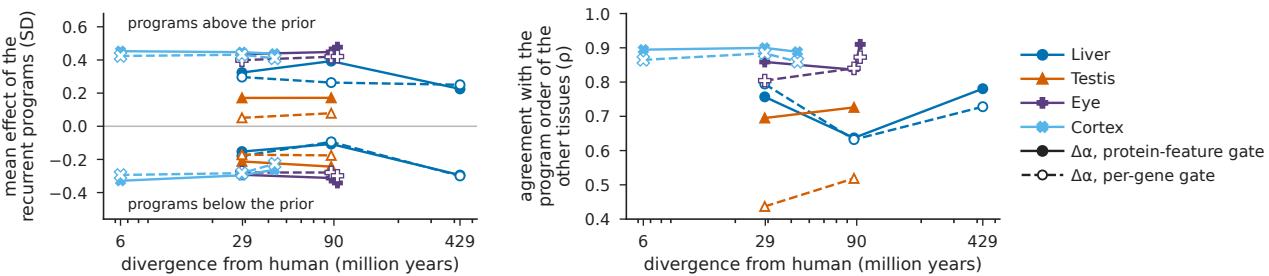
	Liver	Testis	Bone marrow	Kidney	Islet	Eye	Cortex	Lung	q	S_g	profile correlation	π	dN/dS
Recurrent, above the prior (42)													
Receptors (class)	+7.1*	+7.3*	+9.5*	+15.0*	+13.6*	+7.2*	+13.1*	+14.1*	<.001	8↑0↓	7↑0↓	−2.2	+0.9
Epithelial Mesenchymal Transition	+7.8*	+7.6*	+4.0*	+12.1*	+16.7*	+10.0*	+8.1*	+11.6*	<.001	8↑0↓	6↑0↓	+0.4	0.0
Cytoskeleton & ECM (category)	+5.4*	+10.1*	+5.5*	+8.5*	+10.1*	+7.8*	+3.3*	+9.5*	<.001	8↑0↓	6↑1↓	+2.8	−2.9
KRAS Signaling Up	+2.9*	+5.1*	+5.2*	+6.5*	+9.8*	+6.1*	+4.5*	+8.9*	<.001	6↑0↓	5↑0↓	−2.7	+1.5
Inflammatory Response	+3.7*	+3.4*	+6.4*	+6.3*	+9.9*	+4.0*	+6.1*	+9.1*	<.001	7↑0↓	6↑0↓	−4.2	+3.4
Apical Junction	+4.0*	+2.6*	+3.6*	+7.1*	+9.2*	+6.4*	+5.0*	+7.6*	<.001	8↑0↓	6↑0↓	+2.0	−3.2
TNF-alpha Signaling via NF-kB	+1.3	+3.7*	+5.0*	+6.7*	+11.1*	+2.8*	+5.0*	+9.6*	<.001	6↑0↓	6↑0↓	−1.4	+1.5
GPCRs (class)	+3.7*	+4.5*	+1.9	+7.6*	+8.0*	+3.9*	+7.5*	+5.9*	<.001	7↑0↓	6↑0↓	−0.1	−0.5
Coagulation	+4.7*	+4.5*	+2.5*	+6.7*	+8.2*	+4.2*	+4.7*	+6.5*	<.001	7↑0↓	6↑0↓	−1.8	+1.4
IL-2/STAT5 Signaling	+2.7*	+3.2*	+5.1*	+5.5*	+7.4*	+5.6*	+4.1*	+7.3*	<.001	7↑0↓	5↑0↓	−1.5	+1.0
Myogenesis	+4.1*	+4.3*	+2.3*	+6.2*	+7.4*	+5.5*	+4.8*	+6.3*	<.001	8↑0↓	6↑0↓	+0.5	−1.2
UV Response Dn	+3.0*	+4.1*	+1.3	+5.2*	+8.5*	+5.3*	+5.9*	+5.3*	<.001	6↑0↓	7↑0↓	+2.4	−1.9
Complement	+4.5*	+1.4	+4.8*	+5.5*	+7.5*	+3.8*	+4.5*	+6.3*	<.001	7↑0↓	3↑0↓	−0.9	+0.6
Hypoxia	+3.0*	+2.1	+3.8*	+6.1*	+8.0*	+3.9*	+3.5*	+6.3*	<.001	6↑0↓	4↑0↓	+0.7	−1.3
Estrogen Response Early	−0.5	+1.8	+2.3*	+7.7*	+7.5*	+5.1*	+4.2*	+7.1*	<.001	5↑0↓	6↑0↓	0.0	−0.3
Allograft Rejection	+2.6*	+1.3	+6.7*	+4.5*	+6.3*	+2.9*	+2.2*	+7.0*	<.001	5↑0↓	7↑0↓	−3.0	+2.6
Angiogenesis		+3.1*	+3.5*	+5.1*	+6.1*	+2.6*	+4.4*	+6.3*	<.001	7↑0↓	4↑0↓	−0.1	+0.1
Signaling pathways (category)	+4.8*	+4.4*	+1.1	+2.7*	+5.9*	+4.4*	+2.4*	+7.4*	<.001	5↑0↓	6↑0↓	+6.4	−7.7
Ion channels (class)	−0.7	+7.0*	0.0	+5.9*	+4.9*	+2.6*	+9.4*	+3.2*	<.001	4↑0↓	3↑1↓	+2.9	−5.0
Estrogen Response Late	+0.2	+0.8	+3.4*	+7.0*	+6.3*	+4.2*	+3.4*	+6.9*	<.001	6↑0↓	3↑0↓	−0.2	−0.6
Immunity & inflammation (category)	+2.3*	+3.0*	+3.9*	+4.3*	+3.6*	+3.6*	+2.8*	+7.2*	<.001	4↑0↓	6↑0↓	−8.3	+7.0
Apoptosis	+1.5	+2.7*	+4.5*	+4.0*	+6.3*	+2.1	+2.8*	+4.9*	<.001	6↑0↓	4↑0↓	−0.9	+1.2
Protein kinases (class)	+2.9*	+5.0*	+3.2*	+2.1	+5.2*	+0.3	+1.2	+8.3*	<.001	3↑0↓	6↑0↓	+7.0	−7.4
IL-6/JAK/STAT3 Signaling	+2.6*	+1.2	+3.9*	+4.0*	+4.4*	+2.7*	+1.5	+5.0*	<.001	6↑0↓	6↑0↓	−4.8	+3.7
Homeodomain TFs (class)	+1.8	+2.5*	−0.3	+4.5*	+5.4*	+2.0	+5.8*	+3.0*	<.001	5↑0↓	6↑0↓	+3.1	−2.7
Interferon Gamma Response	+1.7	+2.1	+4.3*	+3.4*	+5.3*	+1.8	+0.4	+5.6*	<.001	4↑0↓	4↑0↓	−6.6	+5.3
Androgen Response	+2.2	+1.8	+2.2*	+4.6*	+3.2*	+1.1	+4.6*	+4.0*	<.001	4↑0↓	0↑0↓	+0.4	−0.6
Glycolysis	+2.6*	+1.3	+2.1	+4.3*	+3.3*	+3.4*	+2.1	+2.3*	<.001	5↑0↓	1↑0↓	+1.6	−1.2
Xenobiotic Metabolism	−1.1	−0.2	+3.2*	+6.9*	+4.2*	+2.5*	+2.2*	+2.8*	<.001	5↑0↓	1↑0↓	−1.1	+0.2
Hedgehog Signaling		+2.5*	+1.5	+2.8*	+4.0*	+2.1	+3.0*	+3.1*	<.001	4↑0↓	1↑0↓	+1.7	−2.0
Apical Surface		+1.3	+2.7*	+3.6*	+3.3*		+3.6*	+3.0*	<.001	5↑0↓	2↑0↓	−1.2	+0.5
Mitotic Spindle	+2.9*	+2.5*	+4.4*	−0.7	+4.3*	+1.6	+1.1	+4.0*	<.001	2↑0↓	3↑0↓	+1.1	−2.0
TGF-beta Signaling	+2.4*	+1.4	+0.7	+1.2	+5.7*	+1.5	+2.3*	+4.3*	<.001	3↑0↓	5↑0↓	+2.0	−3.3
Nuclear receptors (class)	+1.3	+4.7*	+0.6	+5.9*	+0.8	+1.9	+0.6	+3.6*	0.002	2↑0↓	1↑0↓	+3.0	−2.8
SLC transporters (class)	−0.3	+5.3*	+2.0	+8.8*	+1.0	+1.3	+1.1	−0.2	0.002	2↑0↓	2↑0↓	+0.5	+0.5
KRAS Signaling Dn	+1.3	+2.2	+1.6	+5.6*	+0.1	+1.4	+3.1*	+2.4*	<.001	4↑0↓	3↑0↓	+0.2	−1.2
p53 Pathway	+1.9	−0.6	+2.9*	+1.9	+4.2*	+0.3	+2.4*	+4.4*	<.001	5↑0↓	1↑0↓	+0.2	+0.1
Cholesterol Homeostasis	−0.5	+0.9	+1.1	+2.5*	+2.7*	+3.2*	+2.7*	+1.9	<.001	4↑0↓	0↑2↓	−1.8	+1.3
Transcription factors (class)	+5.9*	+3.9*	−2.2*	−0.6	+1.0	+0.4	+1.1	+4.4*	0.002	3↑0↓	5↑0↓	+1.4	−1.4
Notch Signaling		−0.1	+1.1	+1.5	+3.0*	+2.7*	+0.9	+2.4*	0.002	2↑0↓	0↑0↓	+2.7	−4.5
mTORC1 Signaling	−1.7	−1.4	+5.6*	+2.3*	+3.8*	−0.1	+0.6	0.0	0.002	2↑0↓	1↑1↓	+3.4	−2.7
UV Response Up	−0.3	−1.4	+2.7*	+2.6*	+1.6	+0.5	+0.3	+2.1*	0.002	2↑0↓	0↑0↓	+1.5	−1.8
Recurrent, below the prior (15)													
G2-M Checkpoint	+1.0	+0.1	+7.2*	−6.1*	+0.5	−1.2	−4.1*	−3.2*	0.002	1↑1↓	1↑1↓	+3.0	−2.4
C2H2 zinc-finger TFs (class)	+3.2*	+0.1	−4.2*	−3.2*	−4.4*	−1.3	−2.9*	+0.6	<.001	0↑4↓	0↑1↓	−1.9	+1.0
Unfolded Protein Response	−0.7	−5.3*	+2.7*	−3.4*	+1.9	−2.2	−3.3*	−3.0*	<.001	2↑2↓	2↑3↓	+3.6	−2.5
Myc Targets V2	−0.9	−4.6*	+2.5*	−2.9*	−0.5	−1.7	−1.8	−3.8*	0.002	1↑1↓	1↑2↓	−1.7	+1.7
E2F Targets	+0.1	−3.5*	+9.0*	−6.4*	−2.7*	−4.0*	−3.4*	−4.1*	<.001	1↑4↓	1↑3↓	+1.9	−1.7
Mitochondria & OXPHOS (category)	−3.2*	−3.1*	−4.7*	+1.7	−3.1*	+1.2	−5.0*	−7.6*	<.001	1↑6↓	0↑3↓	−5.0	+3.1
DNA Repair	0.0	−6.7*	−0.1	−2.8*	−3.6*	−2.2	−4.5*	−4.8*	<.001	0↑5↓	0↑4↓	+2.2	−3.2
Myc Targets V1	−0.6	−7.2*	+4.3*	−4.4*	−2.3*	−4.5*	−5.0*	−5.0*	<.001	1↑3↓	2↑4↓	+7.0	−6.6
Protein homeostasis (category)	−1.9	−4.7*	−1.8	−3.4*	−1.6	−5.5*	−5.8*	−2.7*	<.001	0↑4↓	0↑4↓	+4.1	−3.8
Oxidative Phosphorylation	−2.6*	−9.3*	−3.1*	+1.6	−3.0*	+0.7	−5.3*	−9.4*	<.001	1↑5↓	0↑4↓	−1.7	+2.7
Enzymes (class)	−6.2*	−2.0	+0.2	−4.7*	−7.3*	−1.4	−5.3*	−4.8*	<.001	0↑5↓	0↑6↓	+5.0	−2.8
Translation & ribosome (category)	+0.4	−15.8*	0.0	+2.0	−3.9*	−5.6*	−4.2*	−11.2*	<.001	2↑4↓	1↑3↓	−0.5	+1.8
Chromatin regulators (class)	−3.7*	−1.1	−3.4*	−12.0*	−5.5*	−4.6*	−6.5*	−3.4*	<.001	0↑7↓	1↑6↓	+5.3	−4.9
Chromatin & DNA processing (category)	−1.7	−7.2*	−0.8	−9.0*	−6.5*	−5.7*	−8.7*	−8.8*	<.001	0↑5↓	0↑5↓	+4.3	−4.1
Mitochondrial proteins (class)	−5.0*	−14.2*	−6.0*	−0.4	−10.6*	−2.9*	−8.8*	−21.0*	<.001	0↑7↓	0↑6↓	−8.3	+9.0
Called in one tissue only (3)													
Heme Metabolism	−1.5	−1.4	+8.2*	+0.3	+0.5	+1.1	+0.4	+0.1	0.351	1↑2↓	0↑1↓	+0.7	+0.4
Bile Acid Metabolism	+0.1	+1.0	+1.4	+2.6*	−0.9	+0.1	−1.5	−0.5	0.351	1↑1↓	1↑0↓	−2.5	+2.2
Protein Secretion	+0.4	−2.5*	+0.8	−0.6	+0.7	−0.5	−0.8	−1.2	0.351	0↑0↓	0↑0↓	+2.7	−2.8



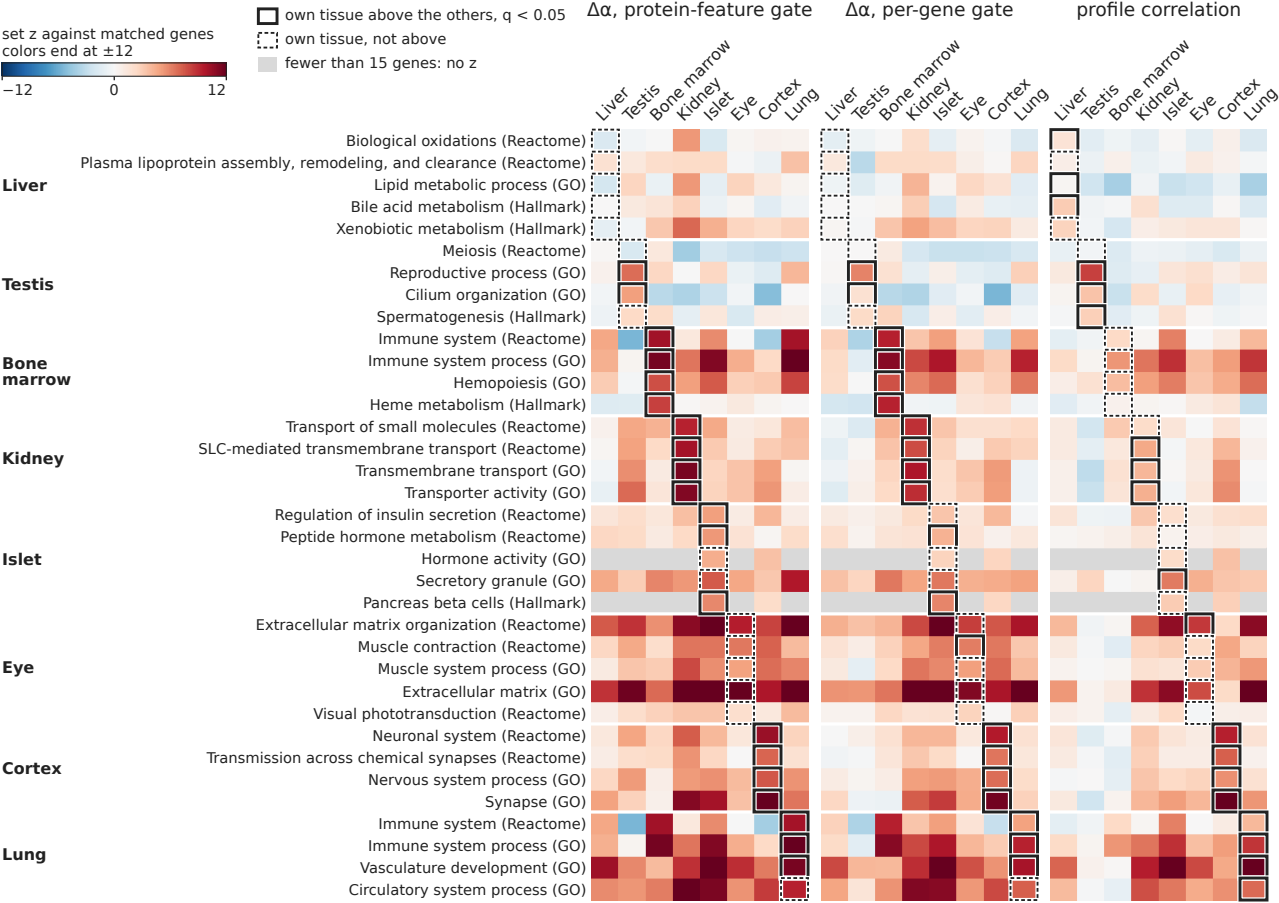
Cells: z of the set against genes permuted within identity and expression bins of the tissue; * $q < 0.05$; gray cell, fewer than 15 genes. Right: q, recurrence across tissues; S_g and profile correlation, tissues with the set above (↑) and below (↓); π and dN/dS, mean z over tissues. Gray type: not recurrent on Δα (q), on that readout (S_g, profile correlation), or |mean z| below 1.96 (π, dN/dS).

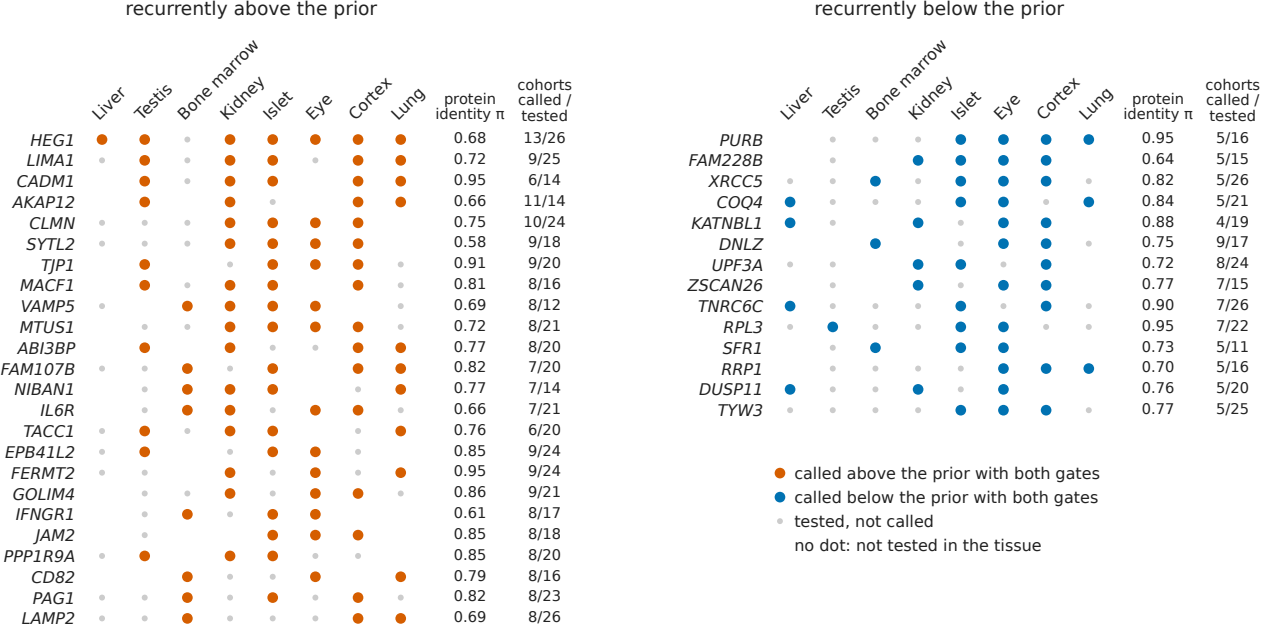

Δα effect against matched genes (SD)
colors end at ±1.0
−1
0
1
* q < 0.05
Δα, protein-feature gate
Δα, per-gene gate
combined z over tissues
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
protein identity π
dN/dS
profile correlation
Glycoprotein (UniProt) −19 +15 +24
Secreted (UniProt) −27 +23 +19
fast-evolving proteins, above the prior
External encapsulating structure (GO) −9 +7 +18
Molecular transducer activity (GO) −8 +3 +19
Integrin cell surface interactions (Reactome) −8 +10 +12
KRAS signaling up (Hallmark) −7 +4 +9
Plasma membrane (GO) +9 −17 +30
Signaling (GO) +15 −21 +23
conserved proteins, above the prior
Calcium (UniProt) +4 −9 +12
Signal transduction (Reactome) +35 −41 +18
Hemostasis (Reactome) +9 −10 +15
Cytoskeleton & ECM (category) +8 −9 +13
Nucleoplasm (GO) +29 −26 −15
RNA binding (GO) +14 −12 −9
conserved proteins, below the prior
Metabolism of RNA (Reactome) +24 −25 −12
RNA-binding (UniProt) +12 −10 −7
mRNA processing (UniProt) +19 −21 −7
Chromatin & DNA processing (category) +12 −12 −7
Mitochondrion (UniProt) −23 +26 −12
DNA repair (GO) −5 +9 −12
fast-evolving proteins, below the prior
DNA damage (UniProt) −7 +9 −12
Catalytic activity, acting on DNA (GO) −4 +9 −6
Diseases of DNA repair (Reactome) −7 +9 −4

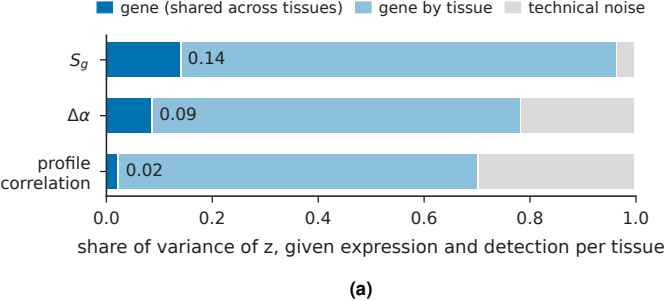
(a)


programs above the prior
programs below the prior
mean effect of the recurrent programs (SD)
0.6
0.4
0.2
0.0
−0.2
−0.4
6
29
90
429
divergence from human (million years)
agreement with the program order of the other tissues (ρ)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Liver
Testis
Eye
Cortex
Δα, protein-feature gate
Δα, per-gene gate

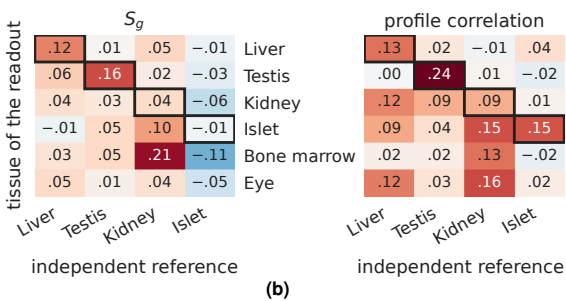
(b)


set z against matched genes
colors end at ±12
−12
0
12
own tissue above the others, q < 0.05
own tissue, not above
fewer than 15 genes: no z
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
Biological oxidations (Reactome)
Plasma lipoprotein assembly, remodeling, and clearance (Reactome)
Lipid metabolic process (GO)
Bile acid metabolism (Hallmark)
Xenobiotic metabolism (Hallmark)
Meiosis (Reactome)
Reproductive process (GO)
Cilium organization (GO)
Spermatogenesis (Hallmark)
Immune system (Reactome)
Immune system process (GO)
Hemopoiesis (GO)
Heme metabolism (Hallmark)
Transport of small molecules (Reactome)
SLC-mediated transmembrane transport (Reactome)
Transmembrane transport (GO)
Transporter activity (GO)
Regulation of insulin secretion (Reactome)
Peptide hormone metabolism (Reactome)
Hormone activity (GO)
Secretory granule (GO)
Pancreas beta cells (Hallmark)
Extracellular matrix organization (Reactome)
Muscle contraction (Reactome)
Muscle system process (GO)
Extracellular matrix (GO)
Visual phototransduction (Reactome)
Neuronal system (Reactome)
Transmission across chemical synapses (Reactome)
Nervous system process (GO)
Synapse (GO)
Immune system (Reactome)
Immune system process (GO)
Vasculature development (GO)
Circulatory system process (GO)
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung

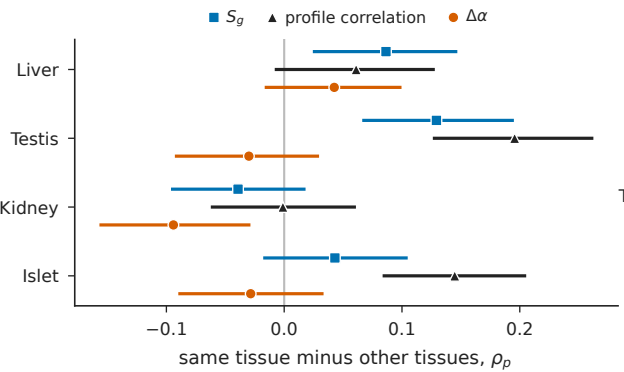

recurrently above the prior
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
protein identity π
cohorts called / tested
HEG1 0.68 13/26
LIMA1 0.72 9/25
CADM1 0.95 6/14
AKAP12 0.66 11/14
CLMN 0.75 10/24
SYTL2 0.58 9/18
TJP1 0.91 9/20
MACF1 0.81 8/16
VAMP5 0.69 8/12
MTUS1 0.72 8/21
ABI3BP 0.77 8/20
FAM107B 0.82 7/20
NIBAN1 0.77 7/14
IL6R 0.66 7/21
TACC1 0.76 6/20
EPB41L2 0.85 9/24
FERMT2 0.95 9/24
GOLIM4 0.86 9/21
IFNGR1 0.61 8/17
JAM2 0.85 8/18
PPP1R9A 0.85 8/20
CD82 0.79 8/16
PAG1 0.82 8/23
LAMP2 0.69 8/26
recurrently below the prior
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
protein identity π
cohorts called / tested
PURB 0.95 5/16
FAM228B 0.64 5/15
XRCC5 0.82 5/26
COQ4 0.84 5/21
KATNBL1 0.88 4/19
DNLZ 0.75 9/17
UPF3A 0.72 8/24
ZSCAN26 0.77 7/15
TNRC6C 0.90 7/26
RPL3 0.95 7/22
SFR1 0.73 5/11
RRP1 0.70 5/16
DUSP11 0.76 5/20
TYW3 0.77 5/25
called above the prior with both gates
called below the prior with both gates
tested, not called
no dot: not tested in the tissue


gene (shared across tissues)
gene by tissue
technical noise
S_g
$\Delta\alpha$
profile correlation
0.14
0.09
0.02
0.0
0.2
0.4
0.6
0.8
1.0
share of variance of z, given expression and detection per tissue

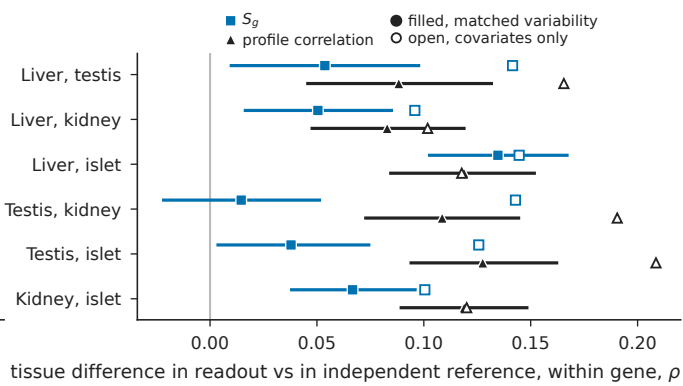
(a)


S_g
profile correlation
tissue of the readout
.12 .01 .05 −.01 Liver
.06 .16 .02 −.03 Testis
.04 .03 .04 −.06 Kidney
−.01 .05 .10 −.01 Islet
.03 .05 .21 −.11 Bone marrow
.05 .01 .04 −.05 Eye
.13 .02 −.01 .04
.00 .24 .01 −.02
.12 .09 .09 .01
.09 .04 .15 .15
.02 .02 .13 −.02
.12 .03 .16 .02
Liver
Testis
Kidney
Islet
independent reference
independent reference

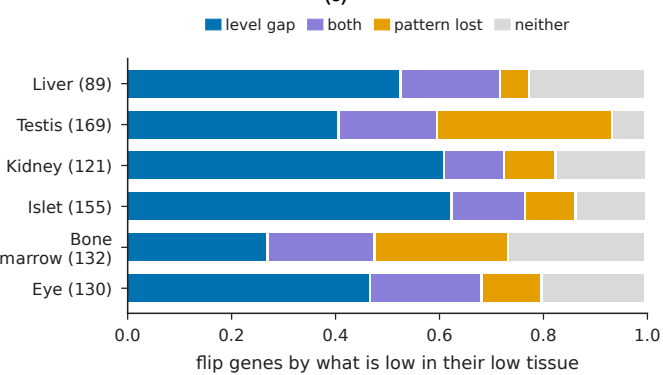
(b)


S_g
profile correlation
$\Delta\alpha$
Liver
Testis
Kidney
Islet
−0.1
0.0
0.1
0.2
same tissue minus other tissues, ρ_p

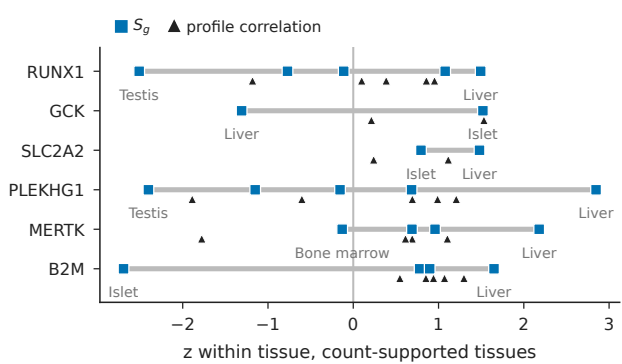
(c)


S_g
profile correlation
filled, matched variability
open, covariates only
Liver, testis
Liver, kidney
Liver, islet
Testis, kidney
Testis, islet
Kidney, islet
0.00
0.05
0.10
0.15
0.20
tissue difference in readout vs in independent reference, within gene, ρ

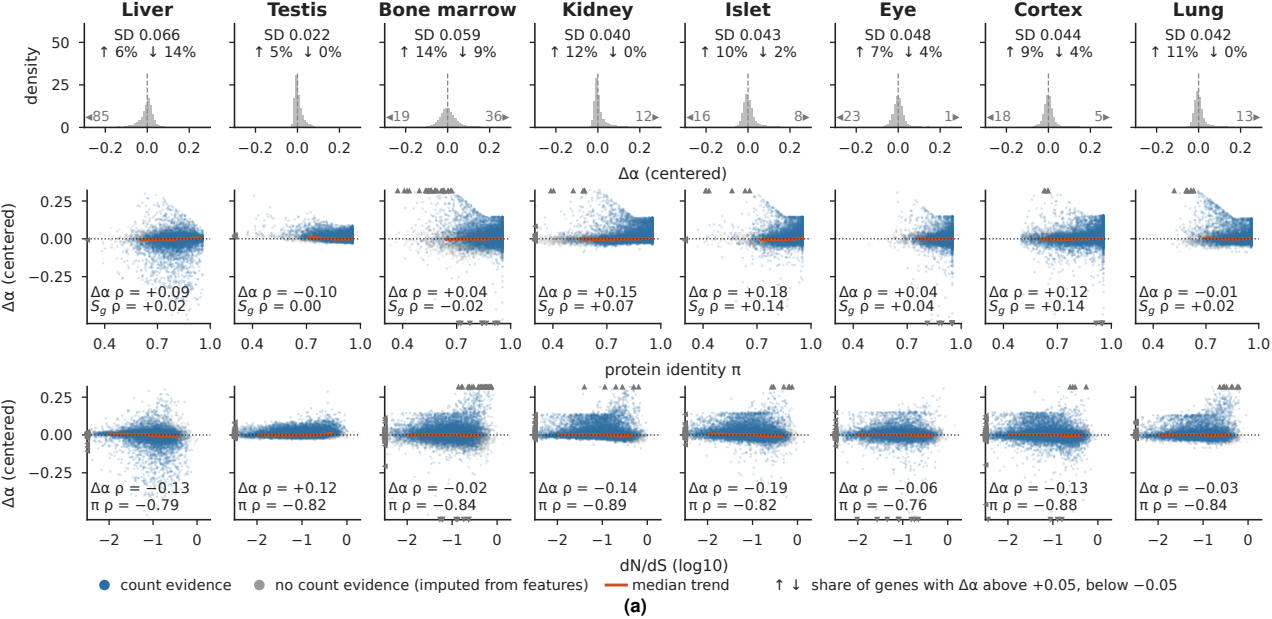
(d)


level gap
both
pattern lost
neither
Liver (89)
Testis (169)
Kidney (121)
Islet (155)
Bone marrow (132)
Eye (130)
0.0
0.2
0.4
0.6
0.8
1.0
flip genes by what is low in their low tissue

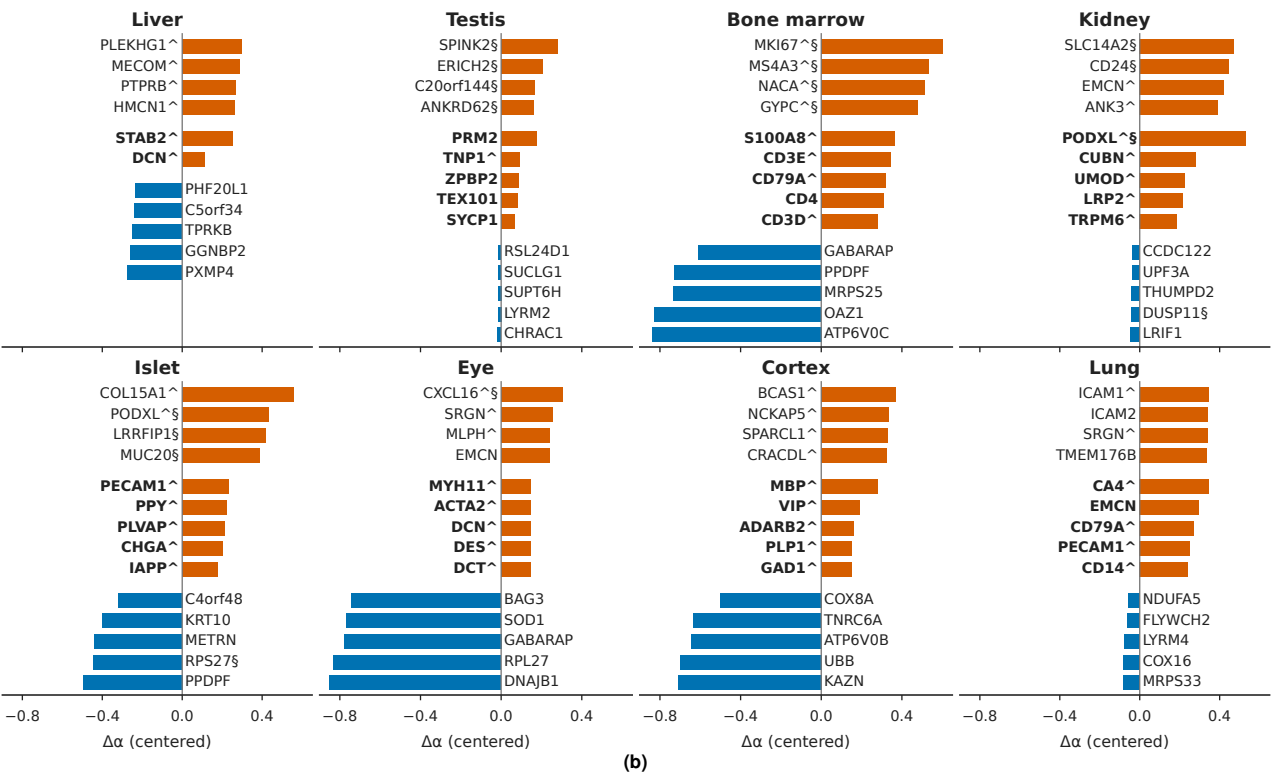
(e)


S_g
profile correlation
RUNX1
Testis
Liver
GCK
Liver
Islet
SLC2A2
Islet
Liver
PLEKHG1
Testis
Liver
MERTK
Bone marrow
Liver
B2M
Islet
Liver
−2
−1
0
1
2
3
z within tissue, count-supported tissues

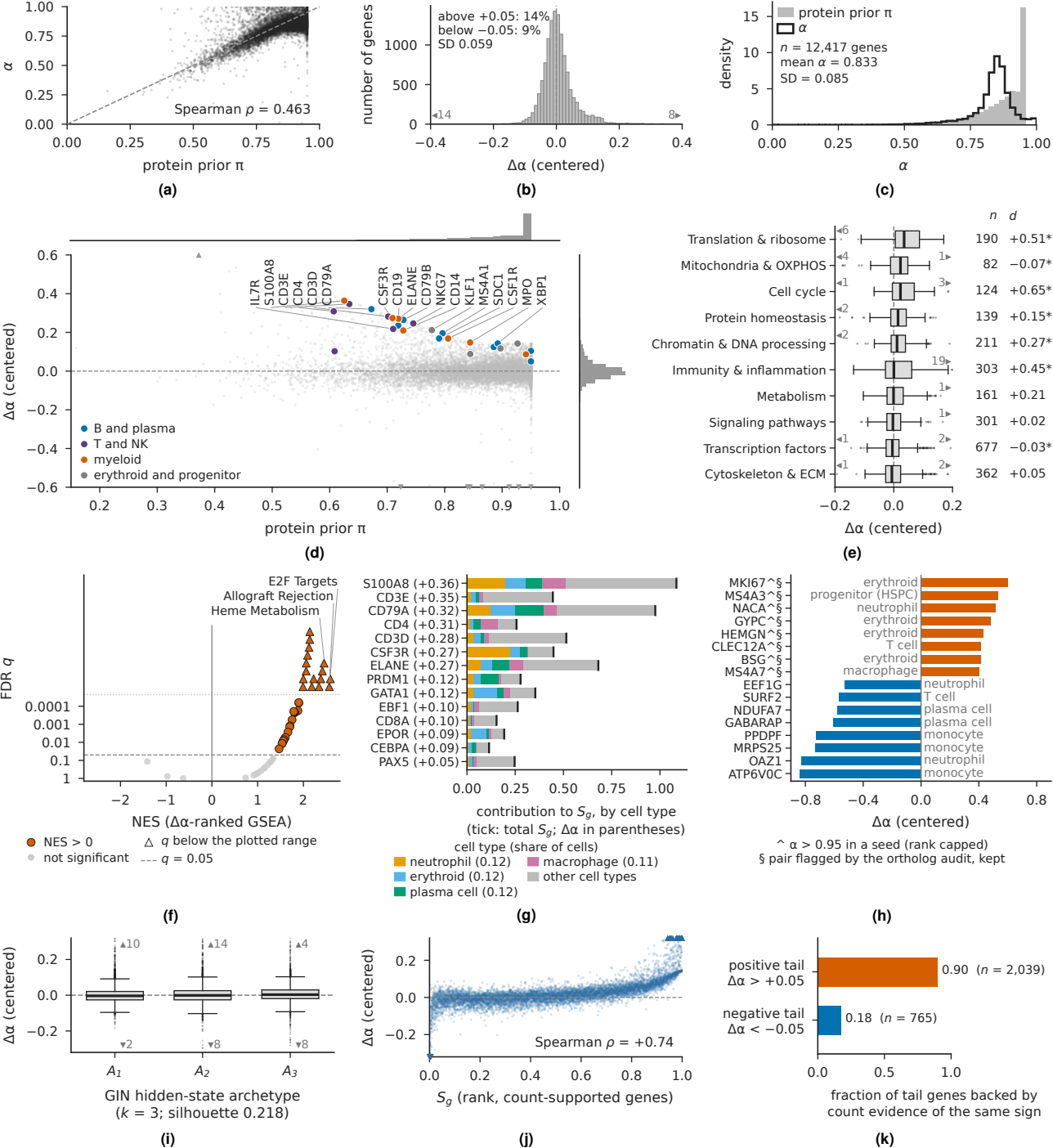
(f)

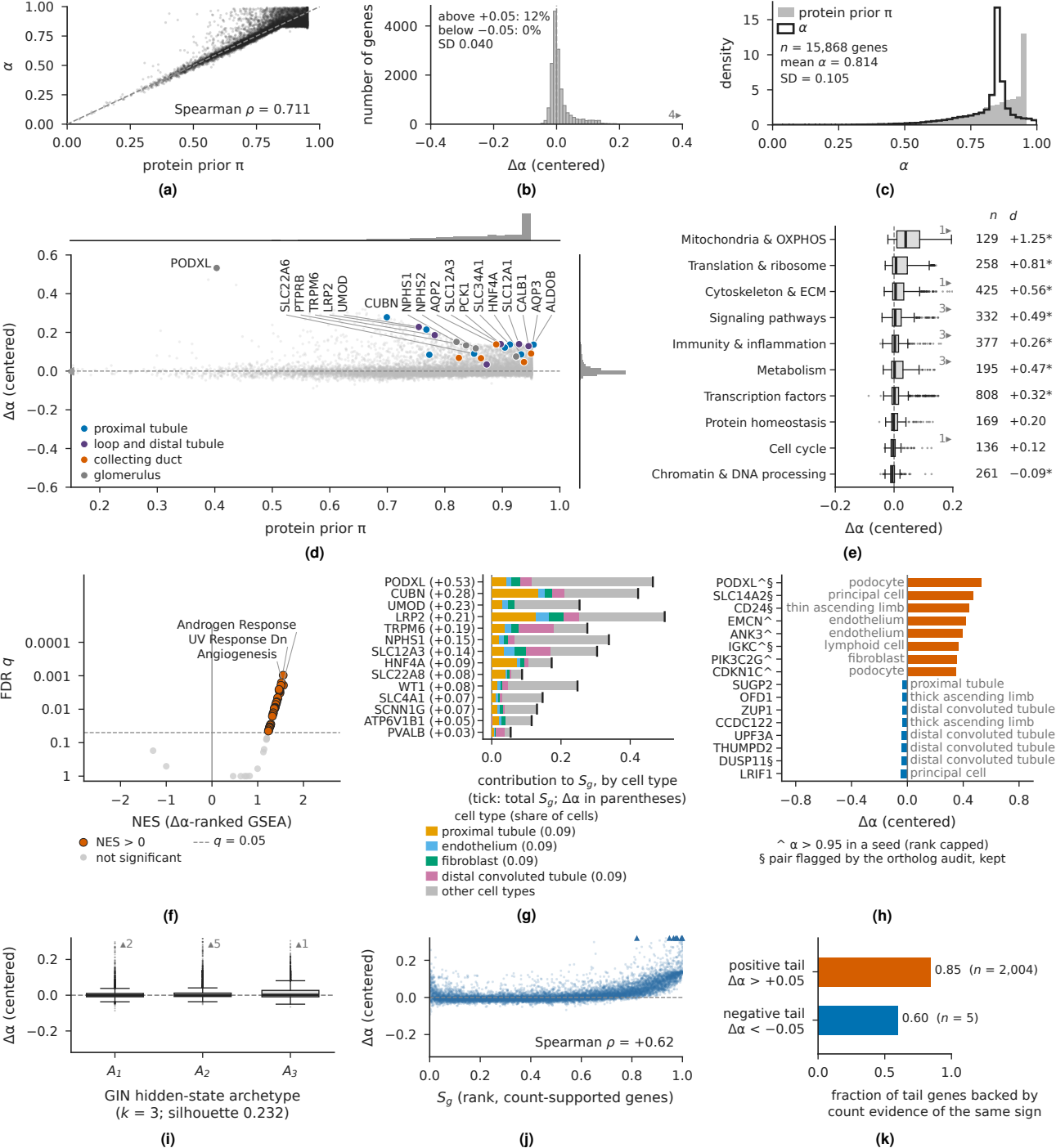

Liver
SD 0.066
↑ 6% ↓ 14%
Testis
SD 0.022
↑ 5% ↓ 0%
Bone marrow
SD 0.059
↑ 14% ↓ 9%
Kidney
SD 0.040
↑ 12% ↓ 0%
Islet
SD 0.043
↑ 10% ↓ 2%
Eye
SD 0.048
↑ 7% ↓ 4%
Cortex
SD 0.044
↑ 9% ↓ 4%
Lung
SD 0.042
↑ 11% ↓ 0%
density
Δα (centered)
Δα ρ = +0.09
S_g ρ = +0.02
Δα ρ = −0.10
S_g ρ = 0.00
Δα ρ = +0.04
S_g ρ = −0.02
Δα ρ = +0.15
S_g ρ = +0.07
Δα ρ = +0.18
S_g ρ = +0.14
Δα ρ = +0.04
S_g ρ = +0.04
Δα ρ = +0.12
S_g ρ = +0.14
Δα ρ = −0.01
S_g ρ = +0.02
protein identity π
Δα ρ = −0.13
π ρ = −0.79
Δα ρ = +0.12
π ρ = −0.82
Δα ρ = −0.02
π ρ = −0.84
Δα ρ = −0.14
π ρ = −0.89
Δα ρ = −0.19
π ρ = −0.82
Δα ρ = −0.06
π ρ = −0.76
Δα ρ = −0.13
π ρ = −0.88
Δα ρ = −0.03
π ρ = −0.84
dN/dS (log10)
count evidence
no count evidence (imputed from features)
median trend
↑ ↓ share of genes with Δα above +0.05, below −0.05

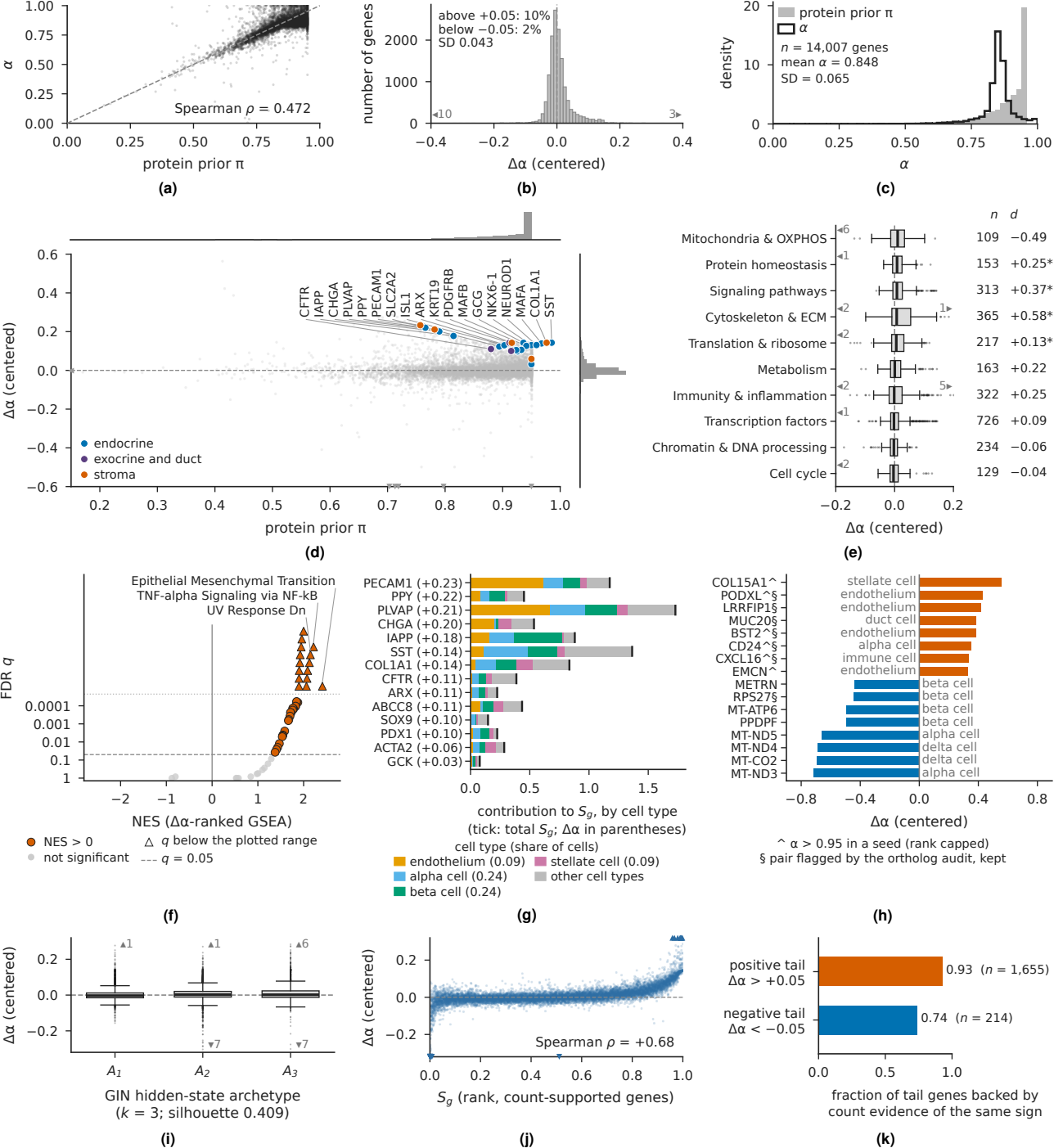
(a)

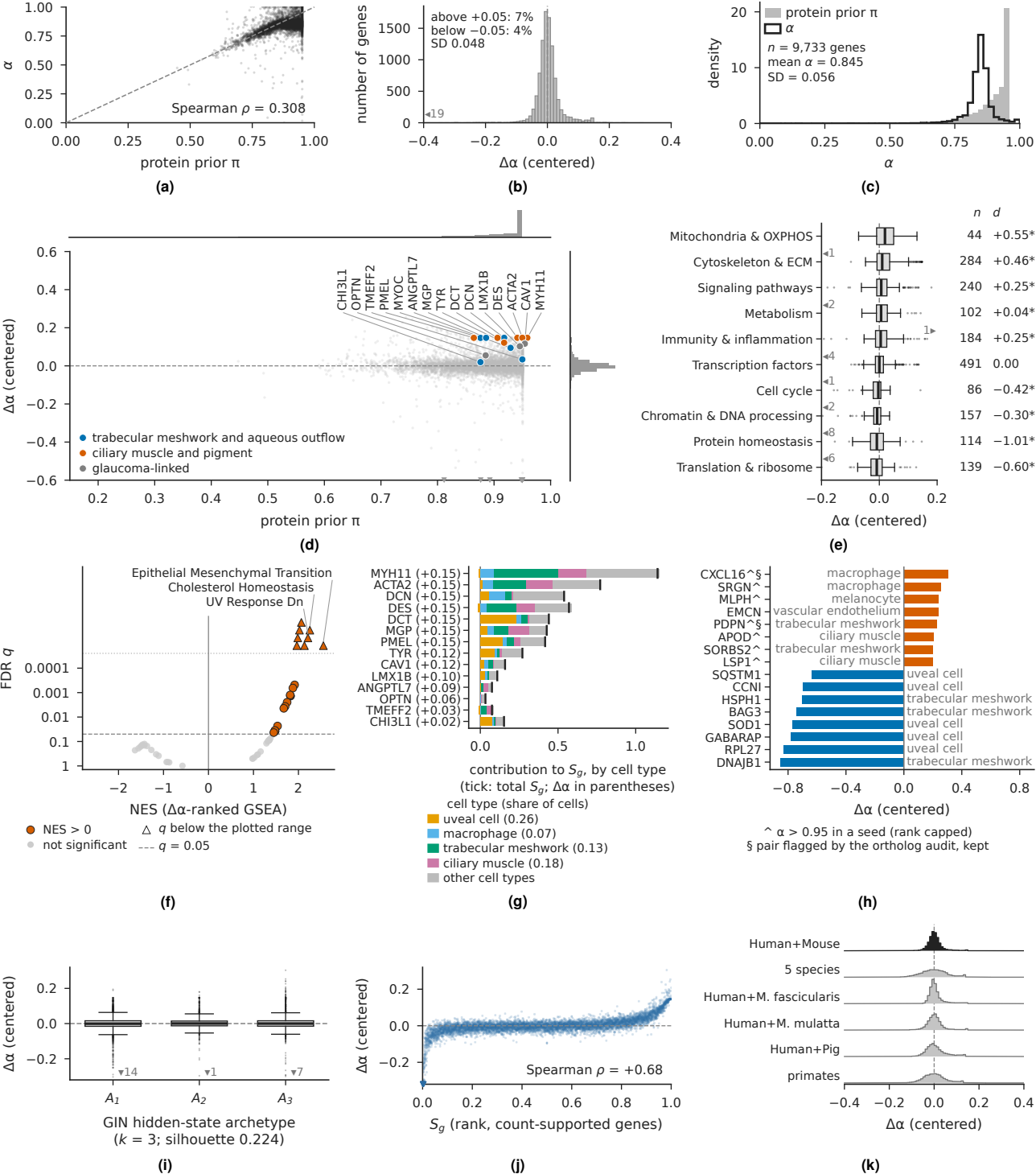

Liver
PLEKHG1^
MECOM^
PTPRB^
HMCN1^
STAB2^
DCN^
PHF20L1
C5orf34
TPRKB
GGNBP2
PXMP4
Testis
SPINK2§
ERICH2§
C20orf144§
ANKRD62§
PRM2
TNP1^
ZPBP2
TEX101
SYCP1
RSL24D1
SUCLG1
SUPT6H
LYRM2
CHRAC1
Bone marrow
MKI67^§
MS4A3^§
NACA^§
GYPC^§
S100A8^
CD3E^
CD79A^
CD4
CD3D^
GABARAP
PPDPF
MRPS25
OAZ1
ATP6V0C
Kidney
SLC14A2§
CD24§
EMCN^
ANK3^
PODXL^§
CUBN^
UMOD^
LRP2^
TRPM6^
CCDC122
UPF3A
THUMPD2
DUSP11§
LRIF1
Islet
COL15A1^
PODXL^§
LRRFIP1§
MUC20§
PECAM1^
PPY^
PLVAP^
CHGA^
IAPP^
C4orf48
KRT10
METRN
RPS27§
PPDPF
Eye
CXCL16^§
SRGN^
MLPH^
EMCN
MYH11^
ACTA2^
DCN^
DES^
DCT^
BAG3
SOD1
GABARAP
RPL27
DNAJB1
Cortex
BCAS1^
NCKAP5^
SPARCL1^
CRACDL^
MBP^
VIP^
ADARB2^
PLP1^
GAD1^
COX8A
TNRC6A
ATP6V0B
UBB
KAZN
Lung
ICAM1^
ICAM2
SRGN^
TMEM176B
CA4^
EMCN
CD79A^
PECAM1^
CD14^
NDUFA5
FLYWCH2
LYRM4
COX16
MRPS33
Δα (centered)

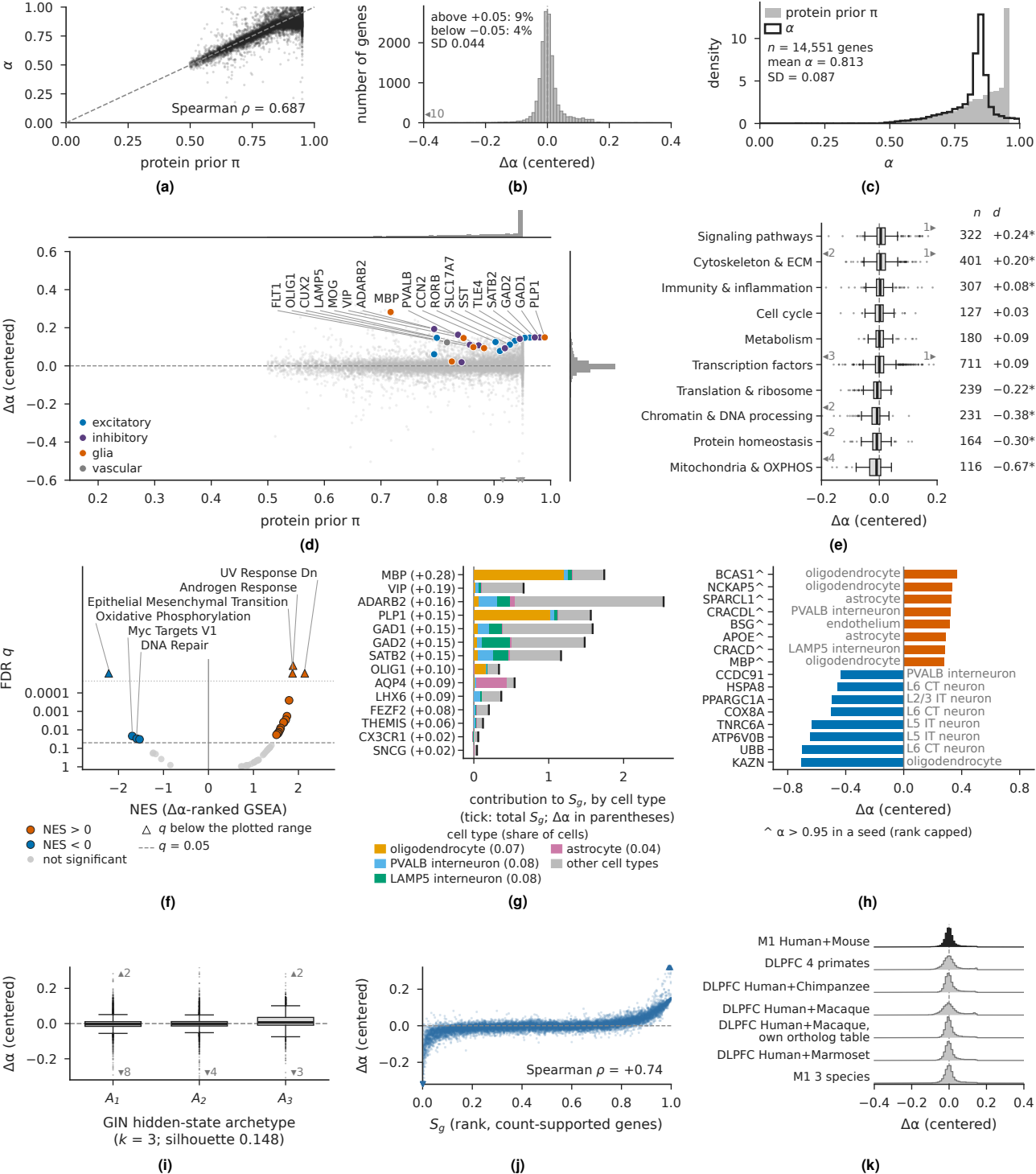
(b)

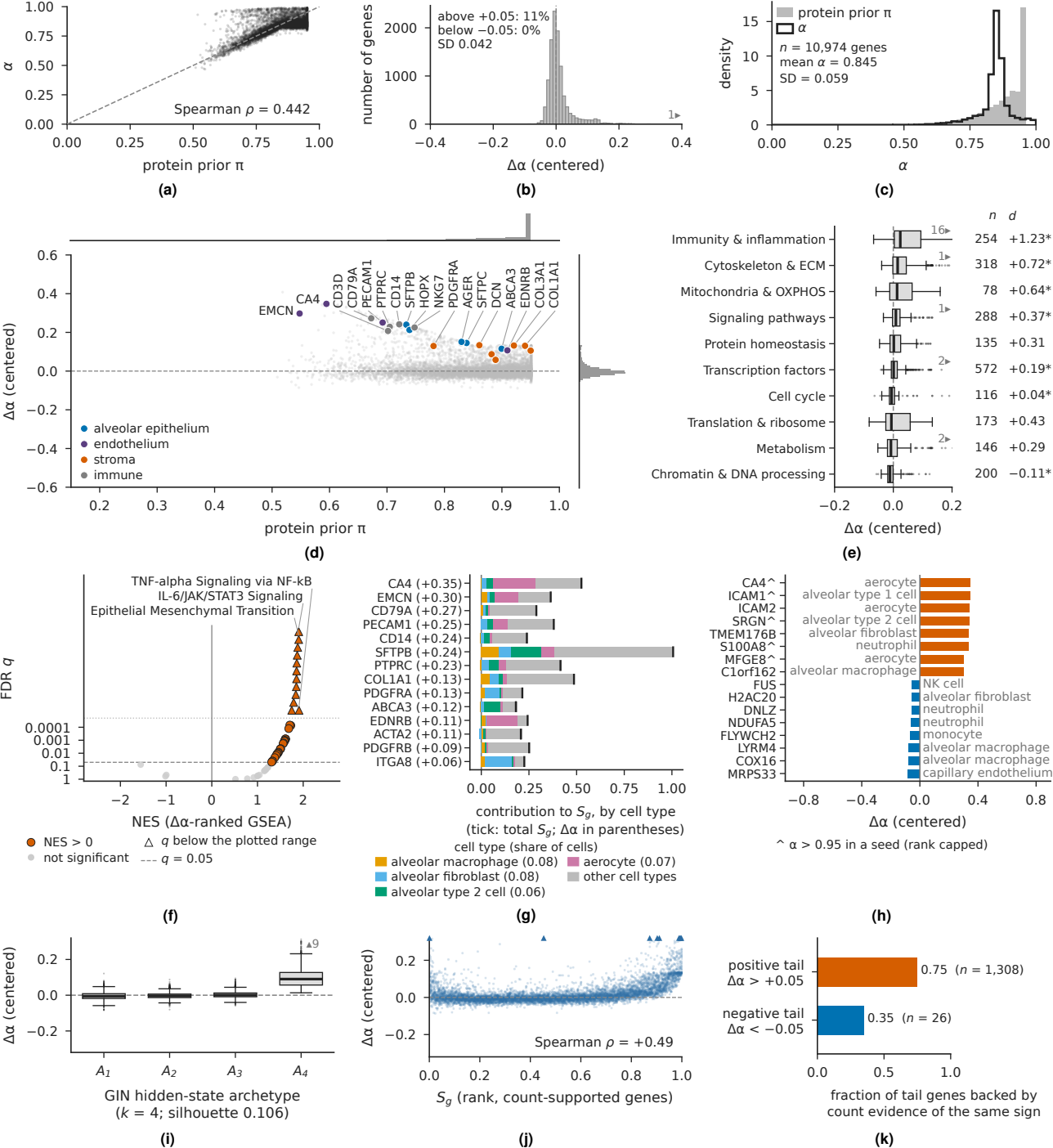

Spearman ρ = 0.463
protein prior π
α
(a)
above +0.05: 14%
below −0.05: 9%
SD 0.059
number of genes
Δα (centered)
(b)
protein prior π
α
n = 12,417 genes
mean α = 0.833
SD = 0.085
density
(c)
IL7R
S100A8
CD3E
CD4
CD3D
CD79A
CSF3R
CD19
ELANE
CD79B
NKG7
CD14
KLF1
MS4A1
SDC1
CSF1R
MPO
XBP1
B and plasma
T and NK
myeloid
erythroid and progenitor
Δα (centered)
protein prior π
(d)
n
d
Translation & ribosome 190 +0.51*
Mitochondria & OXPHOS 82 −0.07*
Cell cycle 124 +0.65*
Protein homeostasis 139 +0.15*
Chromatin & DNA processing 211 +0.27*
Immunity & inflammation 303 +0.45*
Metabolism 161 +0.21
Signaling pathways 301 +0.02
Transcription factors 677 −0.03*
Cytoskeleton & ECM 362 +0.05
Δα (centered)
(e)
E2F Targets
Allograft Rejection
Heme Metabolism
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
△ q below the plotted range
q = 0.05
(f)
S100A8 (+0.36)
CD3E (+0.35)
CD79A (+0.32)
CD4 (+0.31)
CD3D (+0.28)
CSF3R (+0.27)
ELANE (+0.27)
PRDM1 (+0.12)
GATA1 (+0.12)
EBF1 (+0.10)
CD8A (+0.10)
EPOR (+0.09)
CEBPA (+0.09)
PAX5 (+0.05)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
neutrophil (0.12)
macrophage (0.11)
erythroid (0.12)
other cell types
plasma cell (0.12)
(g)
MKI67^§ erythroid
MS4A3^§ progenitor (HSPC)
NACA^§ neutrophil
GYPC^§ erythroid
HEMGN^§ erythroid
CLEC12A^§ T cell
BSG^§ erythroid
MS4A7^§ macrophage
EEF1G neutrophil
SURF2 T cell
NDUFA7 plasma cell
GABARAP plasma cell
PPDPF monocyte
MRPS25 monocyte
OAZ1 neutrophil
ATP6V0C monocyte
Δα (centered)
^ α > 0.95 in a seed (rank capped)
§ pair flagged by the ortholog audit, kept
(h)
Δα (centered)
A_1
A_2
A_3
GIN hidden-state archetype
(k = 3; silhouette 0.218)
(i)
Spearman ρ = +0.74
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.90 (n = 2,039)
negative tail
Δα < −0.05
0.18 (n = 765)
fraction of tail genes backed by
count evidence of the same sign
(k)

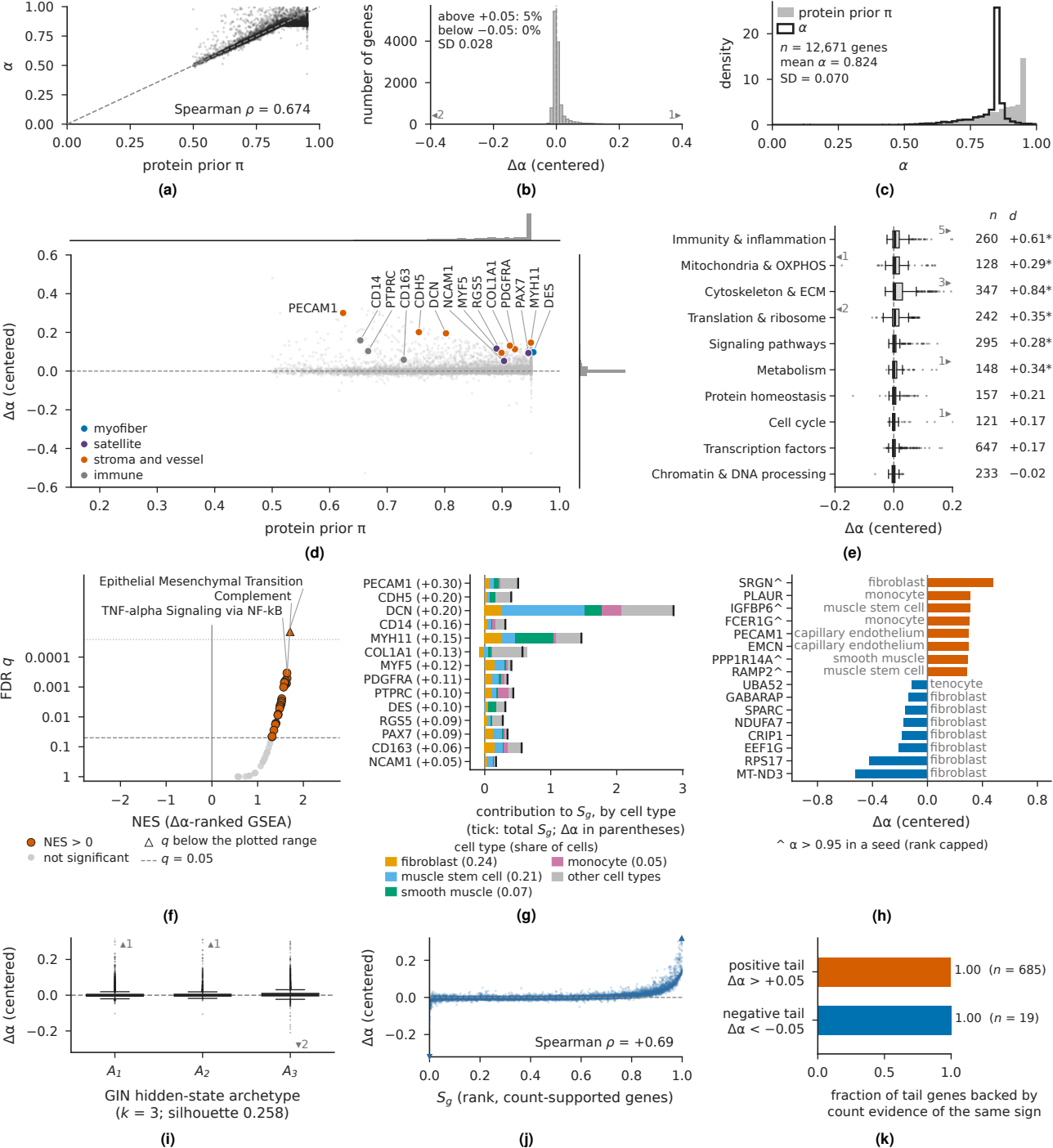

Spearman ρ = 0.711
protein prior π
α
(a)
above +0.05: 12%
below −0.05: 0%
SD 0.040
number of genes
Δα (centered)
(b)
protein prior π
α
n = 15,868 genes
mean α = 0.814
SD = 0.105
density
(c)
PODXL
CUBN
SLC22A6
PTPRB
TRPM6
LRP2
UMOD
NPHS1
NPHS2
AQP2
SLC12A3
PCK1
SLC34A1
HNF4A
SLC12A1
CALB1
AQP3
ALDOB
proximal tubule
loop and distal tubule
collecting duct
glomerulus
Δα (centered)
protein prior π
(d)
n d
Mitochondria & OXPHOS 129 +1.25*
Translation & ribosome 258 +0.81*
Cytoskeleton & ECM 425 +0.56*
Signaling pathways 332 +0.49*
Immunity & inflammation 377 +0.26*
Metabolism 195 +0.47*
Transcription factors 808 +0.32*
Protein homeostasis 169 +0.20
Cell cycle 136 +0.12
Chromatin & DNA processing 261 −0.09*
Δα (centered)
(e)
Androgen Response
UV Response Dn
Angiogenesis
FDR q
NES (Δα-ranked GSEA)
NES > 0
q = 0.05
not significant
(f)
PODXL (+0.53)
CUBN (+0.28)
UMOD (+0.23)
LRP2 (+0.21)
TRPM6 (+0.19)
NPHS1 (+0.15)
SLC12A3 (+0.14)
HNF4A (+0.09)
SLC22A8 (+0.08)
WT1 (+0.08)
SLC4A1 (+0.07)
SCNN1G (+0.07)
ATP6V1B1 (+0.05)
PVALB (+0.03)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
proximal tubule (0.09)
endothelium (0.09)
fibroblast (0.09)
distal convoluted tubule (0.09)
other cell types
(g)
PODXL^§ podocyte
SLC14A2§ principal cell
CD24§ thin ascending limb
EMCN^ endothelium
ANK3^ endothelium
IGKC^§ lymphoid cell
PIK3C2G^ fibroblast
CDKN1C^ podocyte
SUGP2 proximal tubule
OFD1 thick ascending limb
ZUP1 distal convoluted tubule
CCDC122 thick ascending limb
UPF3A distal convoluted tubule
THUMPD2 distal convoluted tubule
DUSP11§ distal convoluted tubule
LRIF1 principal cell
Δα (centered)
^ α > 0.95 in a seed (rank capped)
§ pair flagged by the ortholog audit, kept
(h)
Δα (centered)
GIN hidden-state archetype
(k = 3; silhouette 0.232)
(i)
Δα (centered)
Spearman ρ = +0.62
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.85 (n = 2,004)
negative tail
Δα < −0.05
0.60 (n = 5)
fraction of tail genes backed by
count evidence of the same sign
(k)

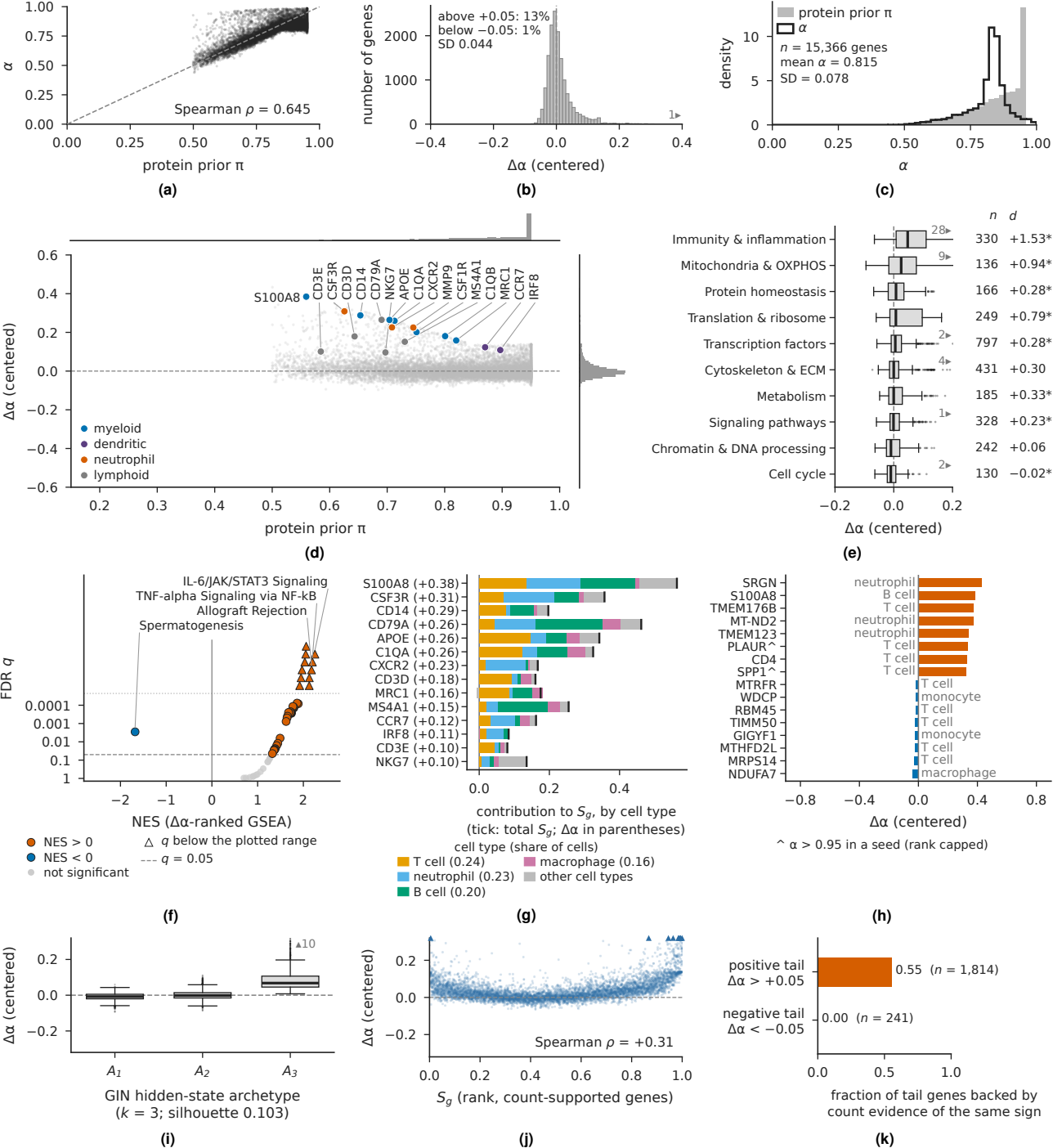

Spearman ρ = 0.472
protein prior π
α
(a)
above +0.05: 10%
below −0.05: 2%
SD 0.043
number of genes
Δα (centered)
(b)
protein prior π
α
n = 14,007 genes
mean α = 0.848
SD = 0.065
density
(c)
CFTR
IAPP
CHGA
PLVAP
PPY
PECAM1
SLC2A2
ISL1
ARX
KRT19
PDGFRB
MAFB
GCG
NKX6-1
NEUROD1
MAFA
COL1A1
SST
endocrine
exocrine and duct
stroma
Δα (centered)
protein prior π
(d)
n d
Mitochondria & OXPHOS 109 −0.49
Protein homeostasis 153 +0.25*
Signaling pathways 313 +0.37*
Cytoskeleton & ECM 365 +0.58*
Translation & ribosome 217 +0.13*
Metabolism 163 +0.22
Immunity & inflammation 322 +0.25
Transcription factors 726 +0.09
Chromatin & DNA processing 234 −0.06
Cell cycle 129 −0.04
Δα (centered)
(e)
Epithelial Mesenchymal Transition
TNF-alpha Signaling via NF-kB
UV Response Dn
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
q below the plotted range
q = 0.05
(f)
PECAM1 (+0.23)
PPY (+0.22)
PLVAP (+0.21)
CHGA (+0.20)
IAPP (+0.18)
SST (+0.14)
COL1A1 (+0.14)
CFTR (+0.11)
ARX (+0.11)
ABCC8 (+0.11)
SOX9 (+0.10)
PDX1 (+0.10)
ACTA2 (+0.06)
GCK (+0.03)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
endothelium (0.09)
stellate cell (0.09)
alpha cell (0.24)
other cell types
beta cell (0.24)
(g)
COL15A1^
PODXL^§
LRRFIP1§
MUC20§
BST2^§
CD24^§
CXCL16^§
EMCN^
METRN
RPS27§
MT-ATP6
PPDPF
MT-ND5
MT-ND4
MT-CO2
MT-ND3
stellate cell
endothelium
endothelium
duct cell
endothelium
alpha cell
immune cell
endothelium
beta cell
beta cell
beta cell
beta cell
alpha cell
delta cell
delta cell
alpha cell
Δα (centered)
^ α > 0.95 in a seed (rank capped)
§ pair flagged by the ortholog audit, kept
(h)
Δα (centered)
A_1
A_2
A_3
GIN hidden-state archetype
(k = 3; silhouette 0.409)
(i)
Spearman ρ = +0.68
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.93 (n = 1,655)
negative tail
Δα < −0.05
0.74 (n = 214)
fraction of tail genes backed by
count evidence of the same sign
(k)

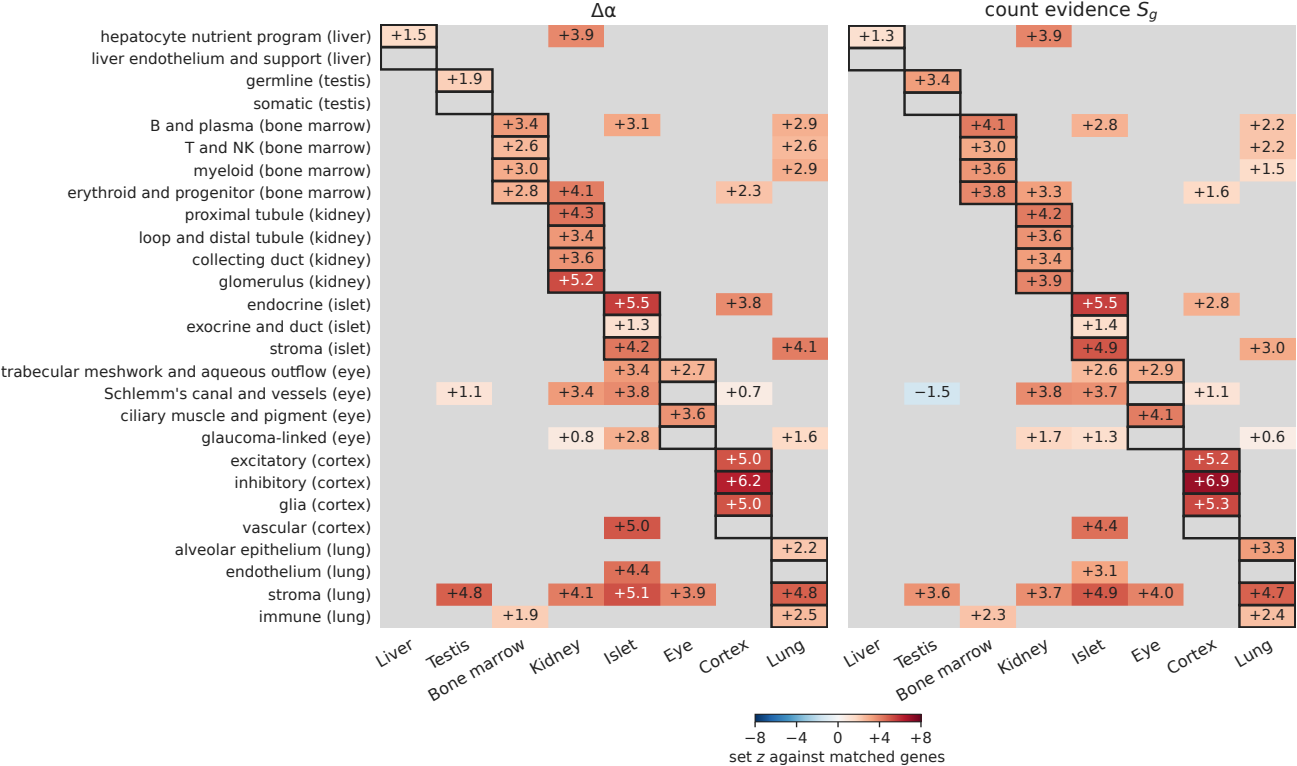

Spearman ρ = 0.308
protein prior π
α
(a)
above +0.05: 7%
below −0.05: 4%
SD 0.048
number of genes
Δα (centered)
(b)
protein prior π
α
n = 9,733 genes
mean α = 0.845
SD = 0.056
density
(c)
CHI3L1
OPTN
TMEFF2
PMEL
MYOC
ANGPTL7
MGP
TYR
DCT
DCN
LMX1B
DES
ACTA2
CAV1
MYH11
trabecular meshwork and aqueous outflow
ciliary muscle and pigment
glaucoma-linked
Δα (centered)
protein prior π
(d)
n
d
Mitochondria & OXPHOS 44 +0.55*
Cytoskeleton & ECM 284 +0.46*
Signaling pathways 240 +0.25*
Metabolism 102 +0.04*
Immunity & inflammation 184 +0.25*
Transcription factors 491 0.00
Cell cycle 86 −0.42*
Chromatin & DNA processing 157 −0.30*
Protein homeostasis 114 −1.01*
Translation & ribosome 139 −0.60*
Δα (centered)
(e)
Epithelial Mesenchymal Transition
Cholesterol Homeostasis
UV Response Dn
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
q below the plotted range
q = 0.05
(f)
MYH11 (+0.15)
ACTA2 (+0.15)
DCN (+0.15)
DES (+0.15)
DCT (+0.15)
MGP (+0.15)
PMEL (+0.15)
TYR (+0.12)
CAV1 (+0.12)
LMX1B (+0.10)
ANGPTL7 (+0.09)
OPTN (+0.06)
TMEFF2 (+0.03)
CHI3L1 (+0.02)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
uveal cell (0.26)
macrophage (0.07)
trabecular meshwork (0.13)
ciliary muscle (0.18)
other cell types
(g)
CXCL16^§ macrophage
SRGN^ macrophage
MLPH^ melanocyte
EMCN vascular endothelium
PDPN^§ trabecular meshwork
APOD^ ciliary muscle
SORBS2^ trabecular meshwork
LSP1^ ciliary muscle
SQSTM1 uveal cell
CCNI uveal cell
HSPH1 trabecular meshwork
BAG3 trabecular meshwork
SOD1 uveal cell
GABARAP uveal cell
RPL27 uveal cell
DNAJB1 trabecular meshwork
Δα (centered)
^ α > 0.95 in a seed (rank capped)
§ pair flagged by the ortholog audit, kept
(h)
Δα (centered)
GIN hidden-state archetype
(k = 3; silhouette 0.224)
(i)
Spearman ρ = +0.68
S_g (rank, count-supported genes)
(j)
Human+Mouse
5 species
Human+M. fascicularis
Human+M. mulatta
Human+Pig
primates
Δα (centered)
(k)

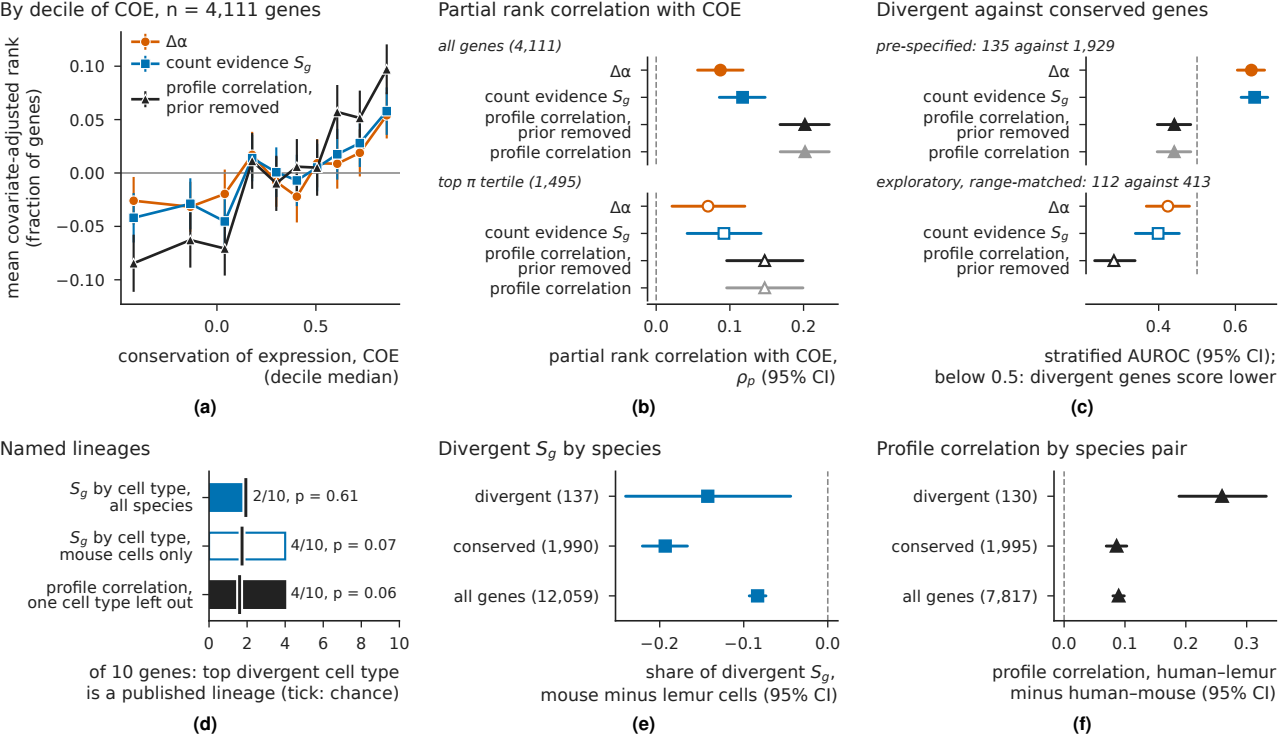

Spearman ρ = 0.687
protein prior π
α
(a)
above +0.05: 9%
below −0.05: 4%
SD 0.044
number of genes
Δα (centered)
(b)
protein prior π
α
n = 14,551 genes
mean α = 0.813
SD = 0.087
density
(c)
FLT1
OLIG1
CUX2
LAMP5
MOG
VIP
ADARB2
MBP
PVALB
CCN2
RORB
SLC17A7
SST
TLE4
SATB2
GAD2
GAD1
PLP1
excitatory
inhibitory
glia
vascular
Δα (centered)
protein prior π
(d)
n
d
Signaling pathways 322 +0.24*
Cytoskeleton & ECM 401 +0.20*
Immunity & inflammation 307 +0.08*
Cell cycle 127 +0.03
Metabolism 180 +0.09
Transcription factors 711 +0.09
Translation & ribosome 239 −0.22*
Chromatin & DNA processing 231 −0.38*
Protein homeostasis 164 −0.30*
Mitochondria & OXPHOS 116 −0.67*
Δα (centered)
(e)
UV Response Dn
Androgen Response
Epithelial Mesenchymal Transition
Oxidative Phosphorylation
Myc Targets V1
DNA Repair
FDR q
NES (Δα-ranked GSEA)
NES > 0
NES < 0
not significant
△ q below the plotted range
--- q = 0.05
(f)
MBP (+0.28)
VIP (+0.19)
ADARB2 (+0.16)
PLP1 (+0.15)
GAD1 (+0.15)
GAD2 (+0.15)
SATB2 (+0.15)
OLIG1 (+0.10)
AQP4 (+0.09)
LHX6 (+0.09)
FEZF2 (+0.08)
THEMIS (+0.06)
CX3CR1 (+0.02)
SNCG (+0.02)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
oligodendrocyte (0.07)
astrocyte (0.04)
PVALB interneuron (0.08)
other cell types
LAMP5 interneuron (0.08)
(g)
BCAS1^ oligodendrocyte
NCKAP5^ oligodendrocyte
SPARCL1^ astrocyte
CRACDL^ PVALB interneuron
BSG^ endothelium
APOE^ astrocyte
CRACD^ LAMP5 interneuron
MBP^ oligodendrocyte
CCDC91 PVALB interneuron
HSPA8 L6 CT neuron
PPARGC1A L2/3 IT neuron
COX8A L6 CT neuron
TNRC6A L5 IT neuron
ATP6V0B L5 IT neuron
UBB L6 CT neuron
KAZN oligodendrocyte
Δα (centered)
^ α > 0.95 in a seed (rank capped)
(h)
Δα (centered)
A1
A2
A3
GIN hidden-state archetype
(k = 3; silhouette 0.148)
(i)
Spearman ρ = +0.74
S_g (rank, count-supported genes)
(j)
M1 Human+Mouse
DLPFC 4 primates
DLPFC Human+Chimpanzee
DLPFC Human+Macaque
DLPFC Human+Macaque, own ortholog table
DLPFC Human+Marmoset
M1 3 species
Δα (centered)
(k)

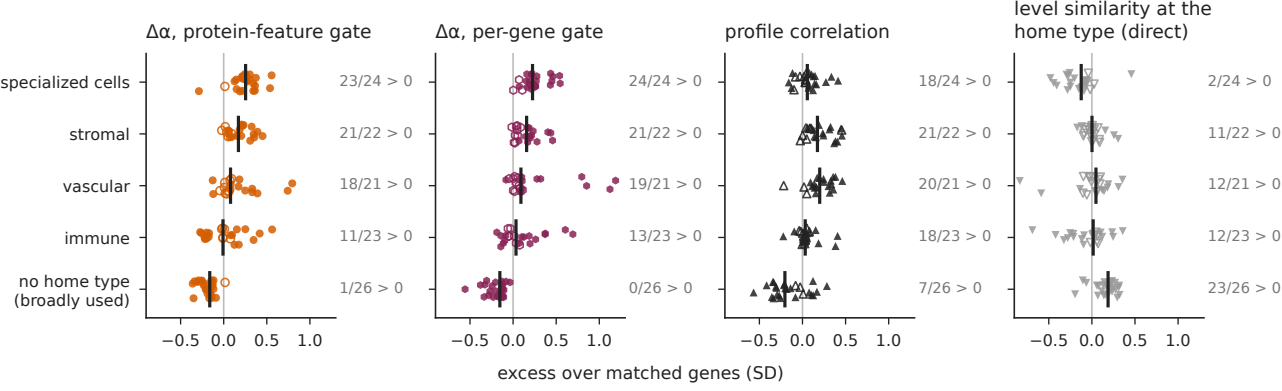

Spearman ρ = 0.442
protein prior π
α
(a)
above +0.05: 11%
below −0.05: 0%
SD 0.042
number of genes
Δα (centered)
(b)
protein prior π
α
n = 10,974 genes
mean α = 0.845
SD = 0.059
density
α
(c)
EMCN
CA4
CD3D
CD79A
PECAM1
PTPRC
CD14
SFTPB
HOPX
NKG7
PDGFRA
AGER
SFTPC
DCN
ABCA3
EDNRB
COL3A1
COL1A1
alveolar epithelium
endothelium
stroma
immune
Δα (centered)
protein prior π
(d)
n
d
Immunity & inflammation 254 +1.23*
Cytoskeleton & ECM 318 +0.72*
Mitochondria & OXPHOS 78 +0.64*
Signaling pathways 288 +0.37*
Protein homeostasis 135 +0.31
Transcription factors 572 +0.19*
Cell cycle 116 +0.04*
Translation & ribosome 173 +0.43
Metabolism 146 +0.29
Chromatin & DNA processing 200 −0.11*
Δα (centered)
(e)
TNF-alpha Signaling via NF-kB
IL-6/JAK/STAT3 Signaling
Epithelial Mesenchymal Transition
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
△ q below the plotted range
q = 0.05
(f)
CA4 (+0.35)
EMCN (+0.30)
CD79A (+0.27)
PECAM1 (+0.25)
CD14 (+0.24)
SFTPB (+0.24)
PTPRC (+0.23)
COL1A1 (+0.13)
PDGFRA (+0.13)
ABCA3 (+0.12)
EDNRB (+0.11)
ACTA2 (+0.11)
PDGFRB (+0.09)
ITGA8 (+0.06)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
alveolar macrophage (0.08)
aerocyte (0.07)
alveolar fibroblast (0.08)
other cell types
alveolar type 2 cell (0.06)
(g)
CA4^ aerocyte
ICAM1^ alveolar type 1 cell
ICAM2 aerocyte
SRGN^ alveolar type 2 cell
TMEM176B alveolar fibroblast
S100A8^ neutrophil
MFGE8^ aerocyte
C1orf162 alveolar macrophage
FUS NK cell
H2AC20 alveolar fibroblast
DNLZ neutrophil
NDUFA5 neutrophil
FLYWCH2 monocyte
LYRM4 alveolar macrophage
COX16 alveolar macrophage
MRPS33 capillary endothelium
Δα (centered)
^ α > 0.95 in a seed (rank capped)
(h)
Δα (centered)
A_1
A_2
A_3
A_4
GIN hidden-state archetype
(k = 4; silhouette 0.106)
(i)
Spearman ρ = +0.49
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.75 (n = 1,308)
negative tail
Δα < −0.05
0.35 (n = 26)
fraction of tail genes backed by
count evidence of the same sign
(k)


Spearman ρ = 0.674
protein prior π
α
(a)
above +0.05: 5%
below −0.05: 0%
SD 0.028
number of genes
Δα (centered)
(b)
protein prior π
α
n = 12,671 genes
mean α = 0.824
SD = 0.070
density
(c)
PECAM1
CD14
PTPRC
CD163
CDH5
DCN
NCAM1
MYF5
RGS5
COL1A1
PDGFRA
PAX7
MYH11
DES
myofiber
satellite
stroma and vessel
immune
Δα (centered)
protein prior π
(d)
n
d
Immunity & inflammation 260 +0.61*
Mitochondria & OXPHOS 128 +0.29*
Cytoskeleton & ECM 347 +0.84*
Translation & ribosome 242 +0.35*
Signaling pathways 295 +0.28*
Metabolism 148 +0.34*
Protein homeostasis 157 +0.21
Cell cycle 121 +0.17
Transcription factors 647 +0.17
Chromatin & DNA processing 233 −0.02
Δα (centered)
(e)
Epithelial Mesenchymal Transition
Complement
TNF-alpha Signaling via NF-kB
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
q below the plotted range
q = 0.05
(f)
PECAM1 (+0.30)
CDH5 (+0.20)
DCN (+0.20)
CD14 (+0.16)
MYH11 (+0.15)
COL1A1 (+0.13)
MYF5 (+0.12)
PDGFRA (+0.11)
PTPRC (+0.10)
DES (+0.10)
RGS5 (+0.09)
PAX7 (+0.09)
CD163 (+0.06)
NCAM1 (+0.05)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
fibroblast (0.24)
monocyte (0.05)
muscle stem cell (0.21)
other cell types
smooth muscle (0.07)
(g)
SRGN^ fibroblast
PLAUR monocyte
IGFBP6^ muscle stem cell
FCER1G^ monocyte
PECAM1 capillary endothelium
EMCN capillary endothelium
PPP1R14A^ smooth muscle
RAMP2^ muscle stem cell
UBA52 tenocyte
GABARAP fibroblast
SPARC fibroblast
NDUFA7 fibroblast
CRIP1 fibroblast
EEF1G fibroblast
RPS17 fibroblast
MT-ND3 fibroblast
Δα (centered)
^ α > 0.95 in a seed (rank capped)
(h)
Δα (centered)
A_1
A_2
A_3
GIN hidden-state archetype
(k = 3; silhouette 0.258)
(i)
Spearman ρ = +0.69
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
1.00 (n = 685)
negative tail
Δα < −0.05
1.00 (n = 19)
fraction of tail genes backed by
count evidence of the same sign
(k)

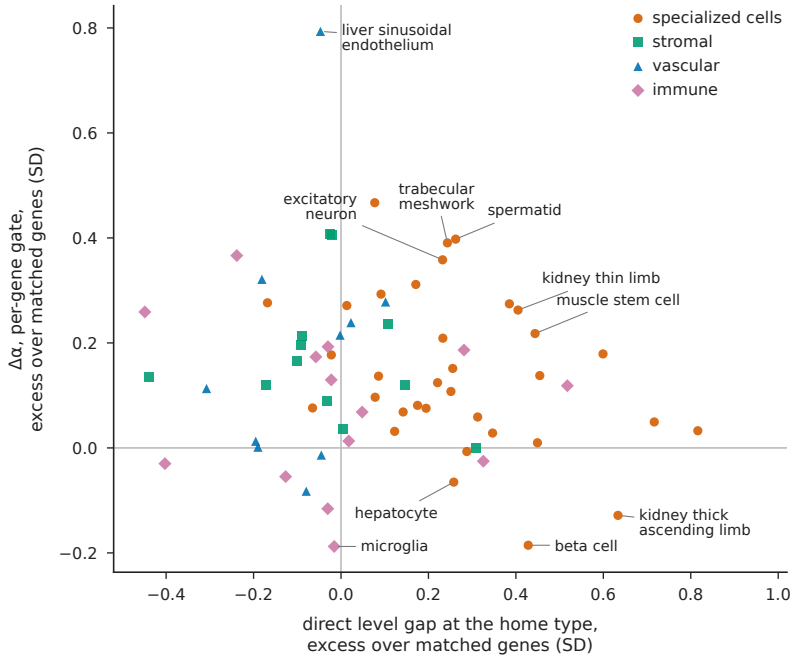

Spearman ρ = 0.645
protein prior π
α
(a)
above +0.05: 13%
below −0.05: 1%
SD 0.044
number of genes
Δα (centered)
(b)
protein prior π
α
n = 15,366 genes
mean α = 0.815
SD = 0.078
density
(c)
S100A8
CD3E
CSF3R
CD3D
CD14
CD79A
NKG7
APOE
C1QA
CXCR2
MMP9
CSF1R
MS4A1
C1QB
MRC1
CCR7
IRF8
myeloid
dendritic
neutrophil
lymphoid
Δα (centered)
protein prior π
(d)
n
d
Immunity & inflammation 330 +1.53*
Mitochondria & OXPHOS 136 +0.94*
Protein homeostasis 166 +0.28*
Translation & ribosome 249 +0.79*
Transcription factors 797 +0.28*
Cytoskeleton & ECM 431 +0.30
Metabolism 185 +0.33*
Signaling pathways 328 +0.23*
Chromatin & DNA processing 242 +0.06
Cell cycle 130 −0.02*
Δα (centered)
(e)
IL-6/JAK/STAT3 Signaling
TNF-alpha Signaling via NF-kB
Allograft Rejection
Spermatogenesis
FDR q
NES (Δα-ranked GSEA)
NES > 0
NES < 0
not significant
△ q below the plotted range
q = 0.05
(f)
S100A8 (+0.38)
CSF3R (+0.31)
CD14 (+0.29)
CD79A (+0.26)
APOE (+0.26)
C1QA (+0.26)
CXCR2 (+0.23)
CD3D (+0.18)
MRC1 (+0.16)
MS4A1 (+0.15)
CCR7 (+0.12)
IRF8 (+0.11)
CD3E (+0.10)
NKG7 (+0.10)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
T cell (0.24)
macrophage (0.16)
neutrophil (0.23)
other cell types
B cell (0.20)
(g)
SRGN neutrophil
S100A8 B cell
TMEM176B T cell
MT-ND2 neutrophil
TMEM123 neutrophil
PLAUR^ T cell
CD4 T cell
SPP1^ T cell
MTRFR T cell
WDCP monocyte
RBM45 T cell
TIMM50 T cell
GIGYF1 monocyte
MTHFD2L T cell
MRPS14 T cell
NDUFA7 macrophage
Δα (centered)
^ α > 0.95 in a seed (rank capped)
(h)
Δα (centered)
GIN hidden-state archetype
(k = 3; silhouette 0.103)
(i)
Spearman ρ = +0.31
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.55 (n = 1,814)
negative tail
Δα < −0.05
0.00 (n = 241)
fraction of tail genes backed by
count evidence of the same sign
(k)


Δα
count evidence S_g
hepatocyte nutrient program (liver)
liver endothelium and support (liver)
germline (testis)
somatic (testis)
B and plasma (bone marrow)
T and NK (bone marrow)
myeloid (bone marrow)
erythroid and progenitor (bone marrow)
proximal tubule (kidney)
loop and distal tubule (kidney)
collecting duct (kidney)
glomerulus (kidney)
endocrine (islet)
exocrine and duct (islet)
stroma (islet)
trabecular meshwork and aqueous outflow (eye)
Schlemm's canal and vessels (eye)
ciliary muscle and pigment (eye)
glaucoma-linked (eye)
excitatory (cortex)
inhibitory (cortex)
glia (cortex)
vascular (cortex)
alveolar epithelium (lung)
endothelium (lung)
stroma (lung)
immune (lung)
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
−8 −4 0 +4 +8
set z against matched genes

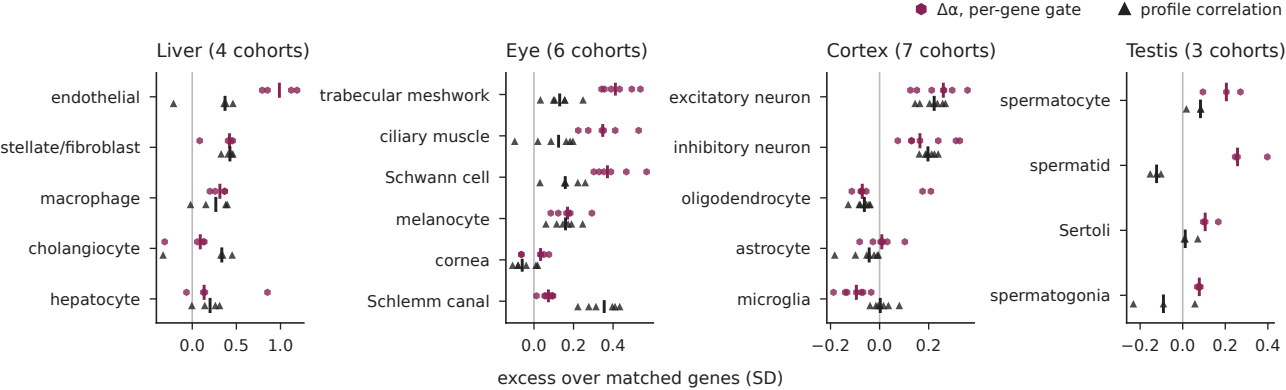

By decile of COE, n = 4,111 genes
Δα
count evidence S_g
profile correlation, prior removed
mean covariate-adjusted rank (fraction of genes)
0.10
0.05
0.00
−0.05
−0.10
0.0
0.5
conservation of expression, COE (decile median)
(a)
Partial rank correlation with COE
all genes (4,111)
Δα
count evidence S_g
profile correlation, prior removed
profile correlation
top π tertile (1,495)
Δα
count evidence S_g
profile correlation, prior removed
profile correlation
0.0
0.1
0.2
partial rank correlation with COE, ρ_p (95% CI)
(b)
Divergent against conserved genes
pre-specified: 135 against 1,929
Δα
count evidence S_g
profile correlation, prior removed
profile correlation
exploratory, range-matched: 112 against 413
Δα
count evidence S_g
profile correlation, prior removed
0.4
0.6
stratified AUROC (95% CI); below 0.5: divergent genes score lower
(c)
Named lineages
S_g by cell type, all species
2/10, p = 0.61
S_g by cell type, mouse cells only
4/10, p = 0.07
profile correlation, one cell type left out
4/10, p = 0.06
0
2
4
6
8
10
of 10 genes: top divergent cell type is a published lineage (tick: chance)
(d)
Divergent S_g by species
divergent (137)
conserved (1,990)
all genes (12,059)
−0.2
−0.1
0.0
share of divergent S_g, mouse minus lemur cells (95% CI)
(e)
Profile correlation by species pair
divergent (130)
conserved (1,995)
all genes (7,817)
0.0
0.1
0.2
0.3
profile correlation, human–lemur minus human–mouse (95% CI)
(f)


Δα, protein-feature gate
Δα, per-gene gate
profile correlation
level similarity at the home type (direct)
specialized cells
stromal
vascular
immune
no home type (broadly used)
23/24 > 0
21/22 > 0
18/21 > 0
11/23 > 0
1/26 > 0
24/24 > 0
21/22 > 0
19/21 > 0
13/23 > 0
0/26 > 0
18/24 > 0
21/22 > 0
20/21 > 0
18/23 > 0
7/26 > 0
2/24 > 0
11/22 > 0
12/21 > 0
12/23 > 0
23/26 > 0
−0.5 0.0 0.5 1.0
excess over matched genes (SD)

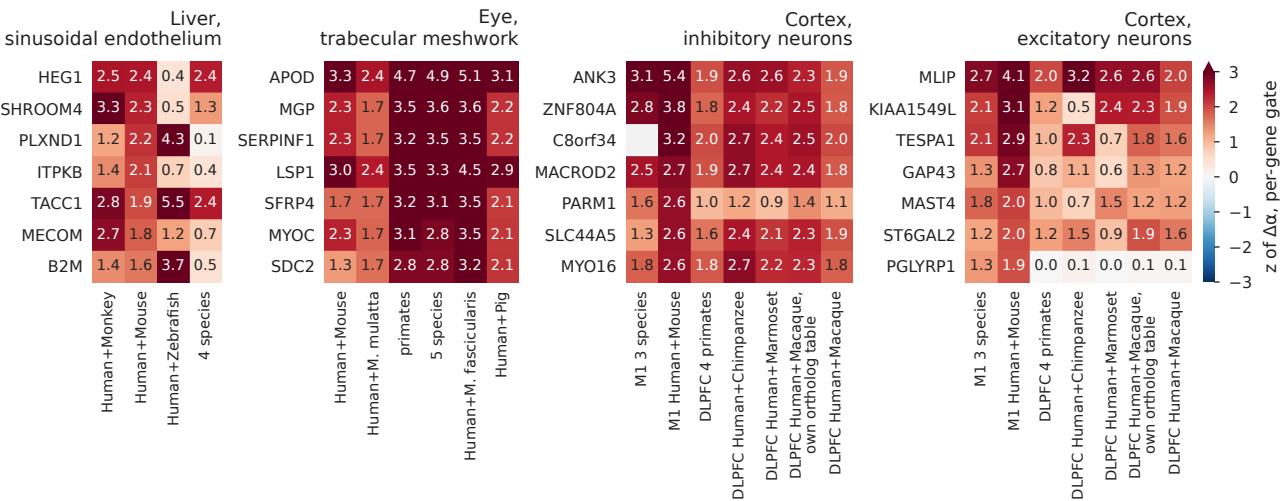
(a)


specialized cells
stromal
vascular
immune
liver sinusoidal endothelium
excitatory neuron
trabecular meshwork
spermatid
kidney thin limb
muscle stem cell
hepatocyte
microglia
kidney thick ascending limb
beta cell
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4 −0.2 0.0 0.2 0.4 0.6 0.8 1.0
Δα, per-gene gate, excess over matched genes (SD)
direct level gap at the home type, excess over matched genes (SD)

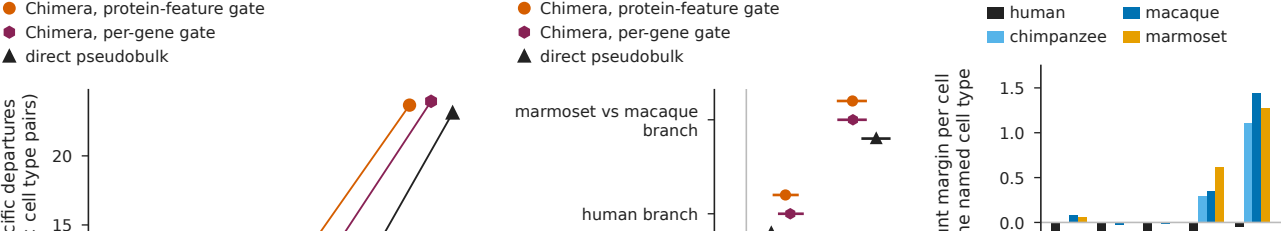
(b)

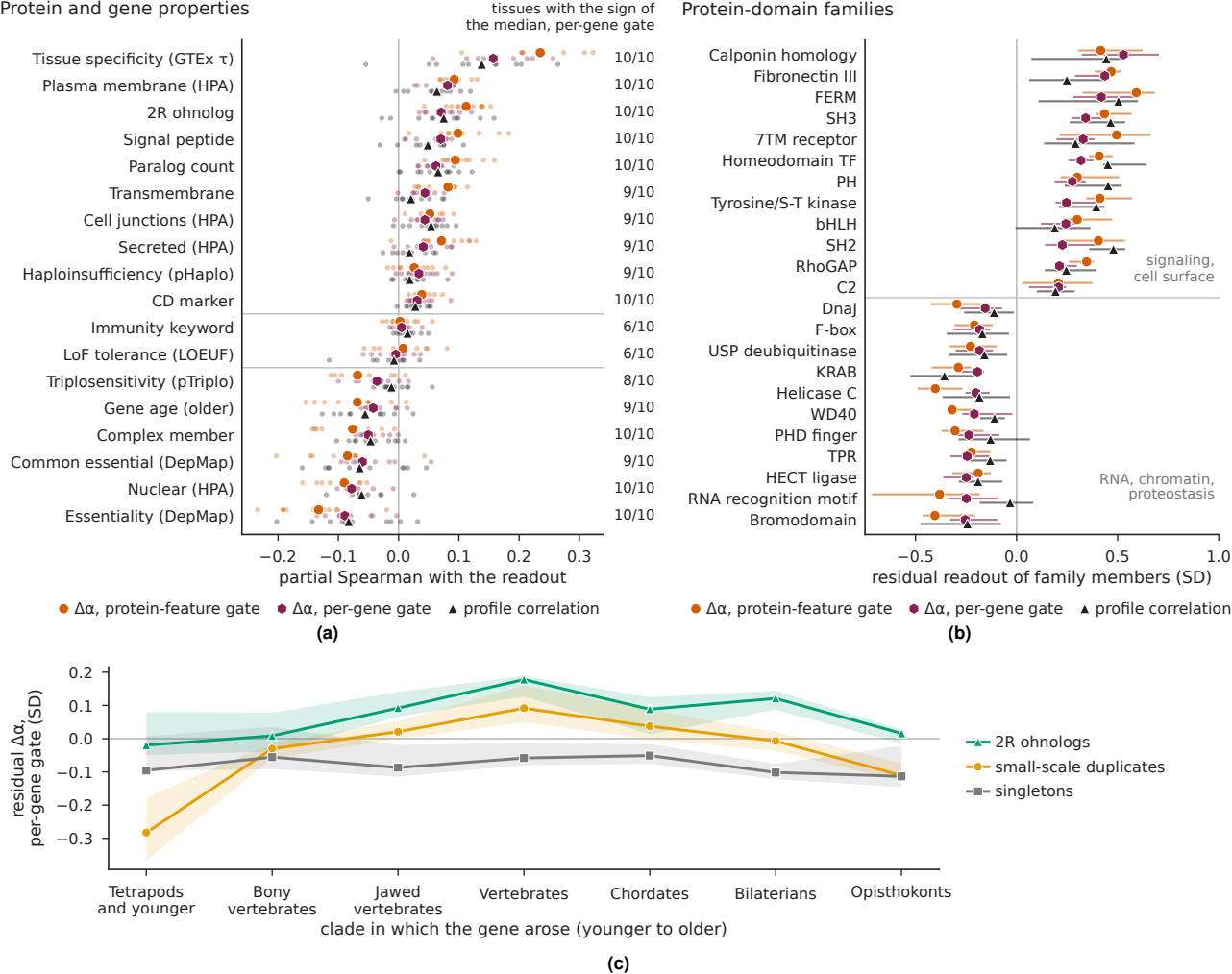

Δα, per-gene gate
profile correlation
Liver (4 cohorts)
endothelial
stellate/fibroblast
macrophage
cholangiocyte
hepatocyte
0.0 0.5 1.0
Eye (6 cohorts)
trabecular meshwork
ciliary muscle
Schwann cell
melanocyte
cornea
Schlemm canal
0.0 0.2 0.4
Cortex (7 cohorts)
excitatory neuron
inhibitory neuron
oligodendrocyte
astrocyte
microglia
−0.2 0.0 0.2
Testis (3 cohorts)
spermatocyte
spermatid
Sertoli
spermatogonia
−0.2 0.0 0.2 0.4
excess over matched genes (SD)

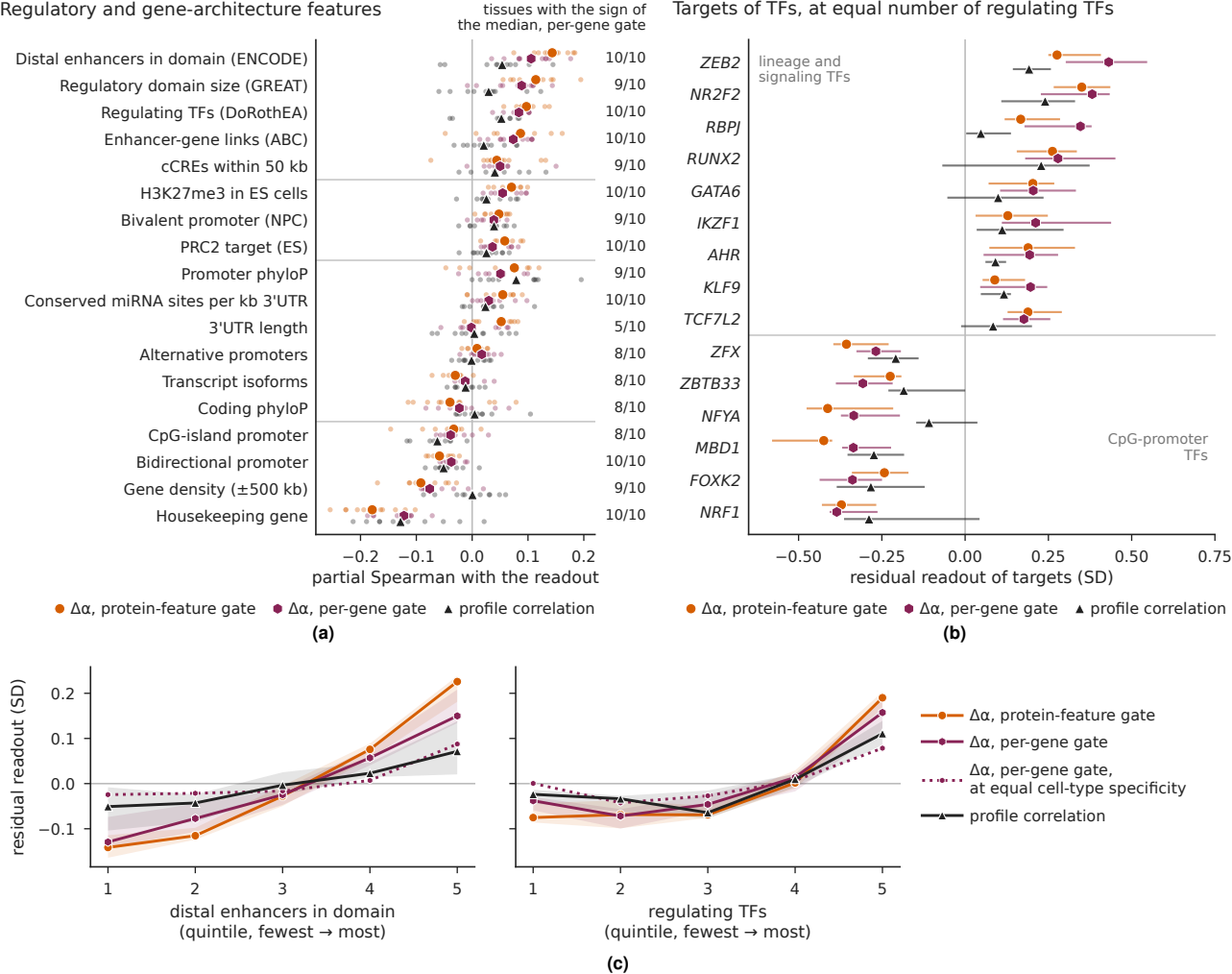
(a)

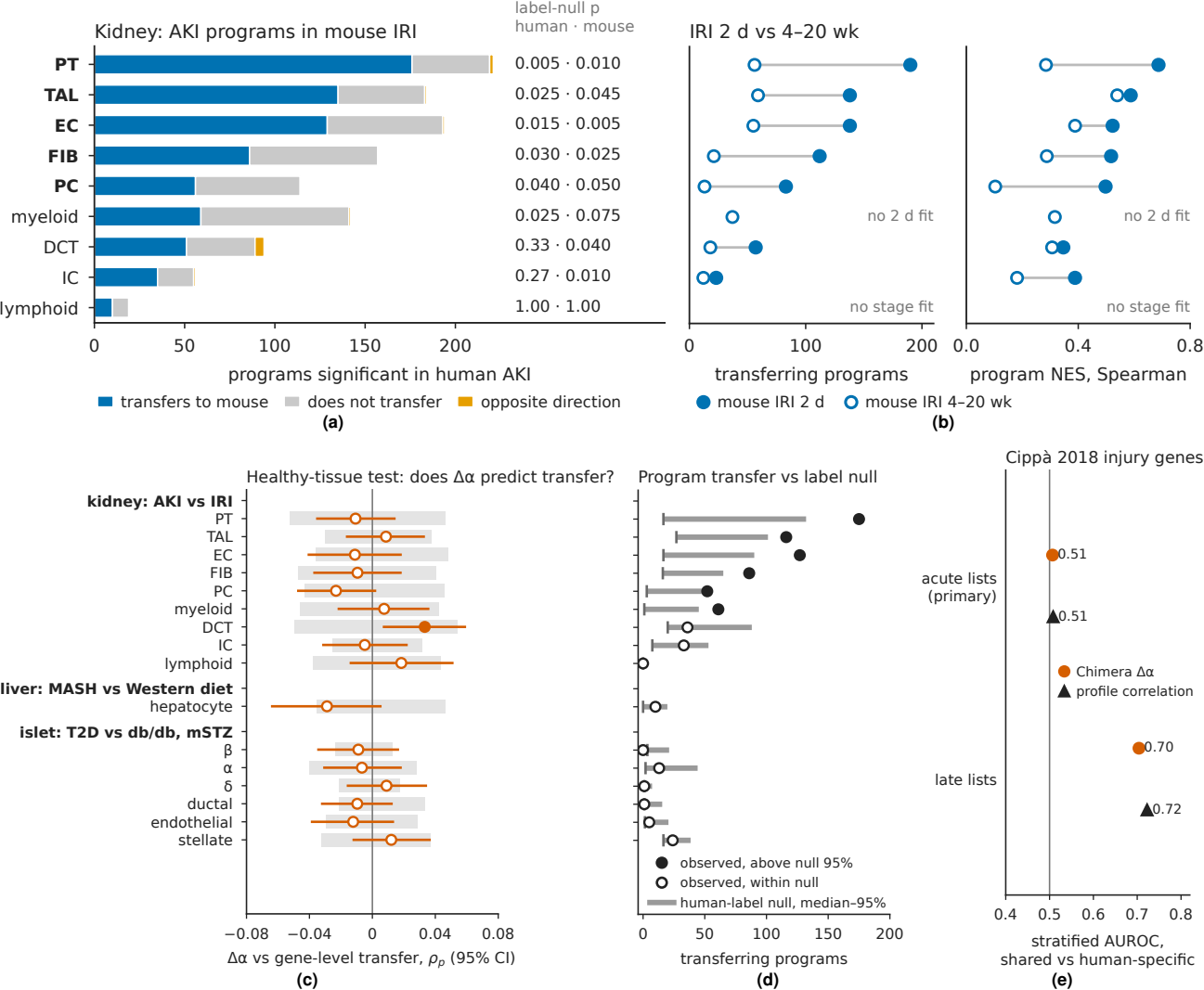

Liver, sinusoidal endothelium
HEG1 2.5 2.4 0.4 2.4
SHROOM4 3.3 2.3 0.5 1.3
PLXND1 1.2 2.2 4.3 0.1
ITPKB 1.4 2.1 0.7 0.4
TACC1 2.8 1.9 5.5 2.4
MECOM 2.7 1.8 1.2 0.7
B2M 1.4 1.6 3.7 0.5
Human+Monkey
Human+Mouse
Human+Zebrafish
4 species
Eye, trabecular meshwork
APOD 3.3 2.4 4.7 4.9 5.1 3.1
MGP 2.3 1.7 3.5 3.6 3.6 2.2
SERPINF1 2.3 1.7 3.2 3.5 3.5 2.2
LSP1 3.0 2.4 3.5 3.3 4.5 2.9
SFRP4 1.7 1.7 3.2 3.1 3.5 2.1
MYOC 2.3 1.7 3.1 2.8 3.5 2.1
SDC2 1.3 1.7 2.8 2.8 3.2 2.1
Human+Mouse
Human+M. mulatta
primates
5 species
Human+M. fascicularis
Human+Pig
Cortex, inhibitory neurons
ANK3 3.1 5.4 1.9 2.6 2.6 2.3 1.9
ZNF804A 2.8 3.8 1.8 2.4 2.2 2.5 1.8
C8orf34 3.2 2.0 2.7 2.4 2.5 2.0
MACROD2 2.5 2.7 1.9 2.7 2.4 2.4 1.8
PARM1 1.6 2.6 1.0 1.2 0.9 1.4 1.1
SLC44A5 1.3 2.6 1.6 2.4 2.1 2.3 1.9
MYO16 1.8 2.6 1.8 2.7 2.2 2.3 1.8
M1 3 species
M1 Human+Mouse
DLPFC 4 primates
DLPFC Human+Chimpanzee
DLPFC Human+Marmoset
DLPFC Human+Macaque, own ortholog table
DLPFC Human+Macaque
Cortex, excitatory neurons
MLIP 2.7 4.1 2.0 3.2 2.6 2.6 2.0
KIAA1549L 2.1 3.1 1.2 0.5 2.4 2.3 1.9
TESPA1 2.1 2.9 1.0 2.3 0.7 1.8 1.6
GAP43 1.3 2.7 0.8 1.1 0.6 1.3 1.2
MAST4 1.8 2.0 1.0 0.7 1.5 1.2 1.2
ST6GAL2 1.2 2.0 1.2 1.5 0.9 1.9 1.6
PGLYRP1 1.3 1.9 0.0 0.1 0.0 0.1 0.1
M1 3 species
M1 Human+Mouse
DLPFC 4 primates
DLPFC Human+Chimpanzee
DLPFC Human+Marmoset
DLPFC Human+Macaque, own ortholog table
DLPFC Human+Macaque
3 2 1 0 −1 −2 −3
z of Δα, per-gene gate

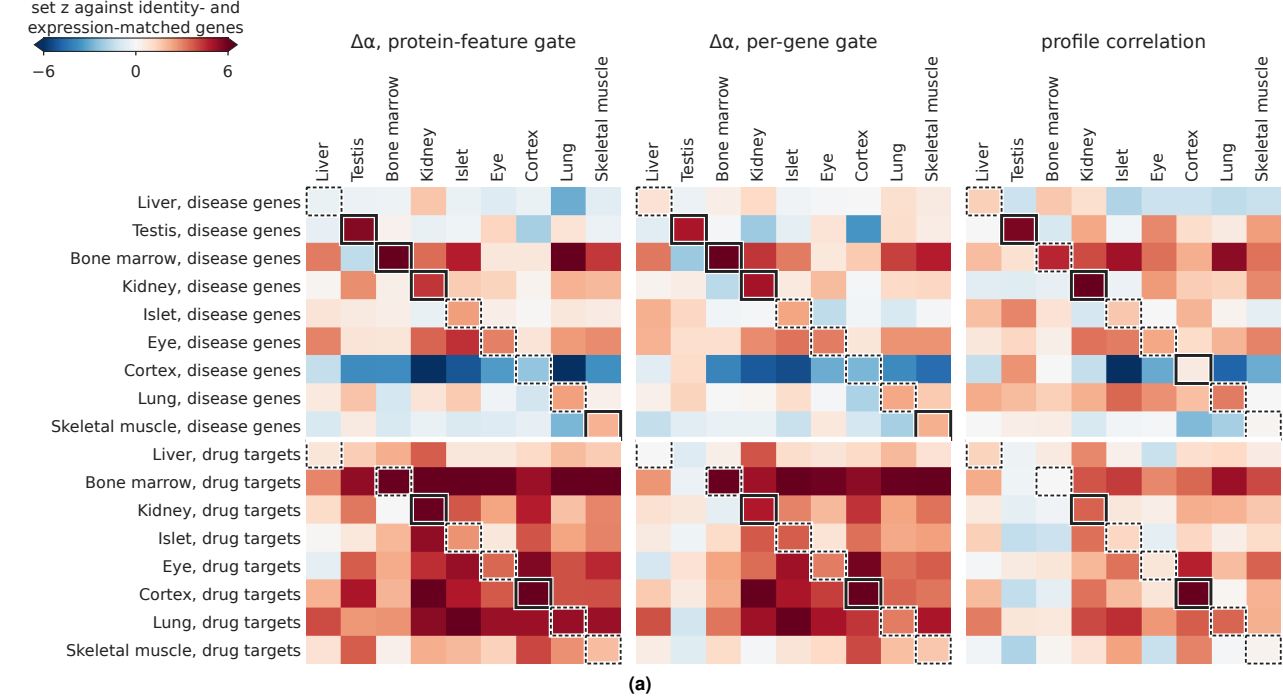
(b)


Chimera, protein-feature gate
Chimera, per-gene gate
direct pseudobulk
species-specific departures
(% of gene × cell type pairs)
20
15
10
human
6.4
chimpanzee
6.4
macaque
28.8
marmoset
43
species, terminal branch length (My)
(a)
Chimera, protein-feature gate
Chimera, per-gene gate
direct pseudobulk
marmoset vs macaque
branch
human branch
chimpanzee branch
(control)
0.0
0.1
0.2
ρ with held-out MTG level gaps
(partial on expression)
(b)
human
macaque
chimpanzee
marmoset
count margin per cell
in the named cell type
1.5
1.0
0.5
0.0
−0.5
FOXP2
microglia
MELTF
PVALB interneurons
PKD2L1
L3-5 IT neurons
CACNA1D
microglia
DSCAM
microglia
(c)

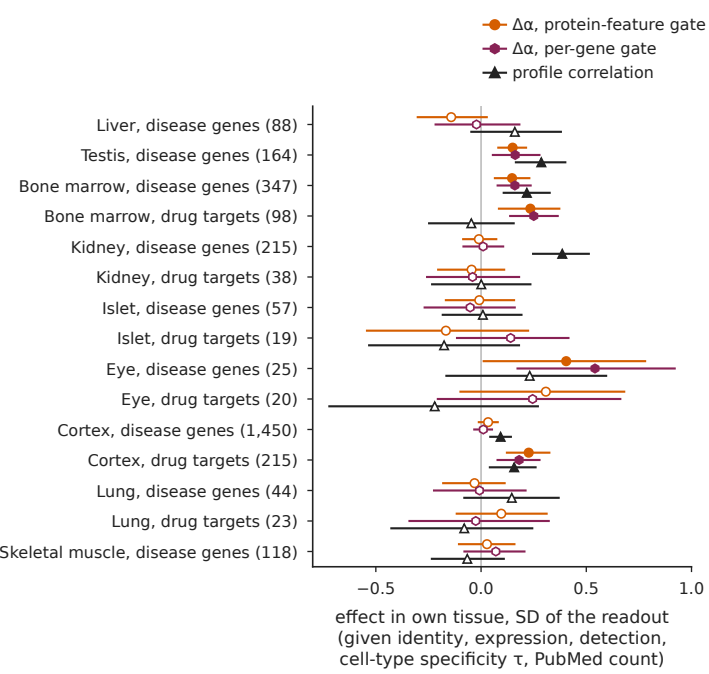

Protein and gene properties
tissues with the sign of the median, per-gene gate
Tissue specificity (GTEx τ)
Plasma membrane (HPA)
2R ohnolog
Signal peptide
Paralog count
Transmembrane
Cell junctions (HPA)
Secreted (HPA)
Haploinsufficiency (pHaplo)
CD marker
Immunity keyword
LoF tolerance (LOEUF)
Triplosensitivity (pTriplo)
Gene age (older)
Complex member
Common essential (DepMap)
Nuclear (HPA)
Essentiality (DepMap)
10/10
10/10
10/10
10/10
10/10
9/10
9/10
9/10
9/10
10/10
6/10
6/10
8/10
9/10
10/10
9/10
10/10
10/10
−0.2
−0.1
0.0
0.1
0.2
0.3
partial Spearman with the readout
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
(a)
Protein-domain families
Calponin homology
Fibronectin III
FERM
SH3
7TM receptor
Homeodomain TF
PH
Tyrosine/S-T kinase
bHLH
SH2
RhoGAP
C2
signaling, cell surface
DnaJ
F-box
USP deubiquitinase
KRAB
Helicase C
WD40
PHD finger
TPR
HECT ligase
RNA recognition motif
Bromodomain
RNA, chromatin, proteostasis
−0.5
0.0
0.5
1.0
residual readout of family members (SD)
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
(b)
residual Δα, per-gene gate (SD)
0.2
0.1
0.0
−0.1
−0.2
−0.3
2R ohnologs
small-scale duplicates
singletons
Tetrapods and younger
Bony vertebrates
Jawed vertebrates
Vertebrates
Chordates
Bilaterians
Opisthokonts
clade in which the gene arose (younger to older)
(c)


Regulatory and gene-architecture features
tissues with the sign of the median, per-gene gate
Distal enhancers in domain (ENCODE)
Regulatory domain size (GREAT)
Regulating TFs (DoRothEA)
Enhancer-gene links (ABC)
cCREs within 50 kb
H3K27me3 in ES cells
Bivalent promoter (NPC)
PRC2 target (ES)
Promoter phyloP
Conserved miRNA sites per kb 3'UTR
3'UTR length
Alternative promoters
Transcript isoforms
Coding phyloP
CpG-island promoter
Bidirectional promoter
Gene density (±500 kb)
Housekeeping gene
10/10
9/10
10/10
10/10
9/10
10/10
9/10
10/10
9/10
10/10
5/10
8/10
8/10
8/10
8/10
10/10
9/10
10/10
−0.2 −0.1 0.0 0.1 0.2
partial Spearman with the readout
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
(a)
Targets of TFs, at equal number of regulating TFs
lineage and signaling TFs
ZEB2
NR2F2
RBPJ
RUNX2
GATA6
IKZF1
AHR
KLF9
TCF7L2
ZFX
ZBTB33
NFYA
MBD1
FOXK2
NRF1
CpG-promoter TFs
−0.50 −0.25 0.00 0.25 0.50 0.75
residual readout of targets (SD)
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
(b)
residual readout (SD)
0.2
0.1
0.0
−0.1
1 2 3 4 5
distal enhancers in domain (quintile, fewest → most)
regulating TFs (quintile, fewest → most)
Δα, protein-feature gate
Δα, per-gene gate
Δα, per-gene gate, at equal cell-type specificity
profile correlation
(c)

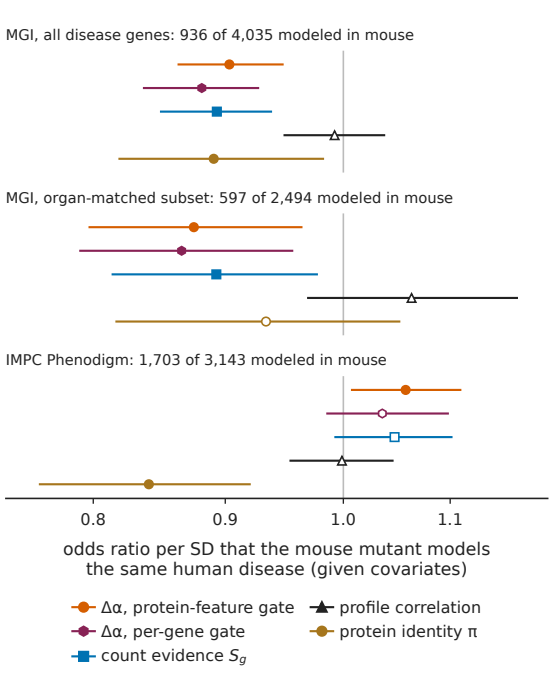

Kidney: AKI programs in mouse IRI
label-null p
human · mouse
PT
TAL
EC
FIB
PC
myeloid
DCT
IC
lymphoid
0.005 · 0.010
0.025 · 0.045
0.015 · 0.005
0.030 · 0.025
0.040 · 0.050
0.025 · 0.075
0.33 · 0.040
0.27 · 0.010
1.00 · 1.00
0
50
100
150
200
programs significant in human AKI
transfers to mouse
does not transfer
opposite direction
(a)
IRI 2 d vs 4–20 wk
no 2 d fit
no stage fit
no 2 d fit
no stage fit
0
100
200
transferring programs
0.0
0.4
0.8
program NES, Spearman
mouse IRI 2 d
mouse IRI 4–20 wk
(b)
Healthy-tissue test: does Δα predict transfer?
kidney: AKI vs IRI
PT
TAL
EC
FIB
PC
myeloid
DCT
IC
lymphoid
liver: MASH vs Western diet
hepatocyte
islet: T2D vs db/db, mSTZ
β
α
δ
ductal
endothelial
stellate
−0.08
−0.04
0
0.04
0.08
Δα vs gene-level transfer, ρ_p (95% CI)
(c)
Program transfer vs label null
observed, above null 95%
observed, within null
human-label null, median–95%
0
50
100
150
200
transferring programs
(d)
Cippà 2018 injury genes
acute lists
(primary)
late lists
0.51
0.51
Chimera Δα
profile correlation
0.70
0.72
0.4
0.5
0.6
0.7
0.8
stratified AUROC,
shared vs human-specific
(e)


set z against identity- and expression-matched genes
−6
0
6
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
Skeletal muscle
Liver, disease genes
Testis, disease genes
Bone marrow, disease genes
Kidney, disease genes
Islet, disease genes
Eye, disease genes
Cortex, disease genes
Lung, disease genes
Skeletal muscle, disease genes
Liver, drug targets
Bone marrow, drug targets
Kidney, drug targets
Islet, drug targets
Eye, drug targets
Cortex, drug targets
Lung, drug targets
Skeletal muscle, drug targets

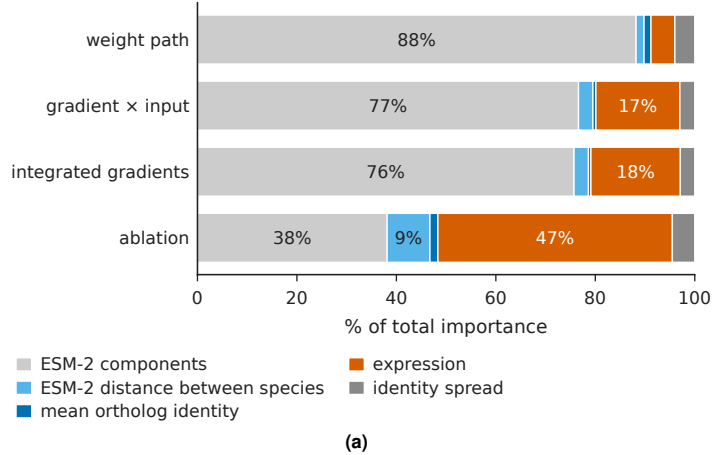
(a)


Δα, protein-feature gate
Δα, per-gene gate
profile correlation
Liver, disease genes (88)
Testis, disease genes (164)
Bone marrow, disease genes (347)
Bone marrow, drug targets (98)
Kidney, disease genes (215)
Kidney, drug targets (38)
Islet, disease genes (57)
Islet, drug targets (19)
Eye, disease genes (25)
Eye, drug targets (20)
Cortex, disease genes (1,450)
Cortex, drug targets (215)
Lung, disease genes (44)
Lung, drug targets (23)
Skeletal muscle, disease genes (118)
−0.5
0.0
0.5
1.0
effect in own tissue, SD of the readout
(given identity, expression, detection,
cell-type specificity τ, PubMed count)

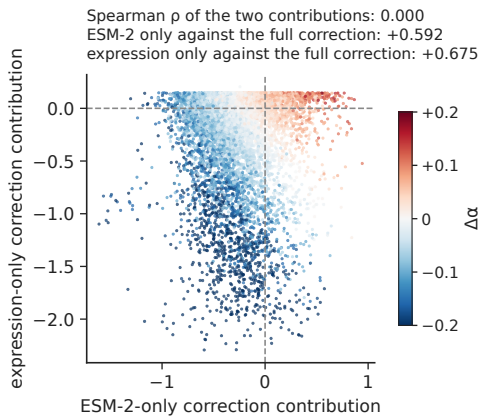
(b)


MGI, all disease genes: 936 of 4,035 modeled in mouse
MGI, organ-matched subset: 597 of 2,494 modeled in mouse
IMPC Phenodigm: 1,703 of 3,143 modeled in mouse
0.8
0.9
1.0
1.1
odds ratio per SD that the mouse mutant models
the same human disease (given covariates)
Δα, protein-feature gate
Δα, per-gene gate
count evidence S_g
profile correlation
protein identity π

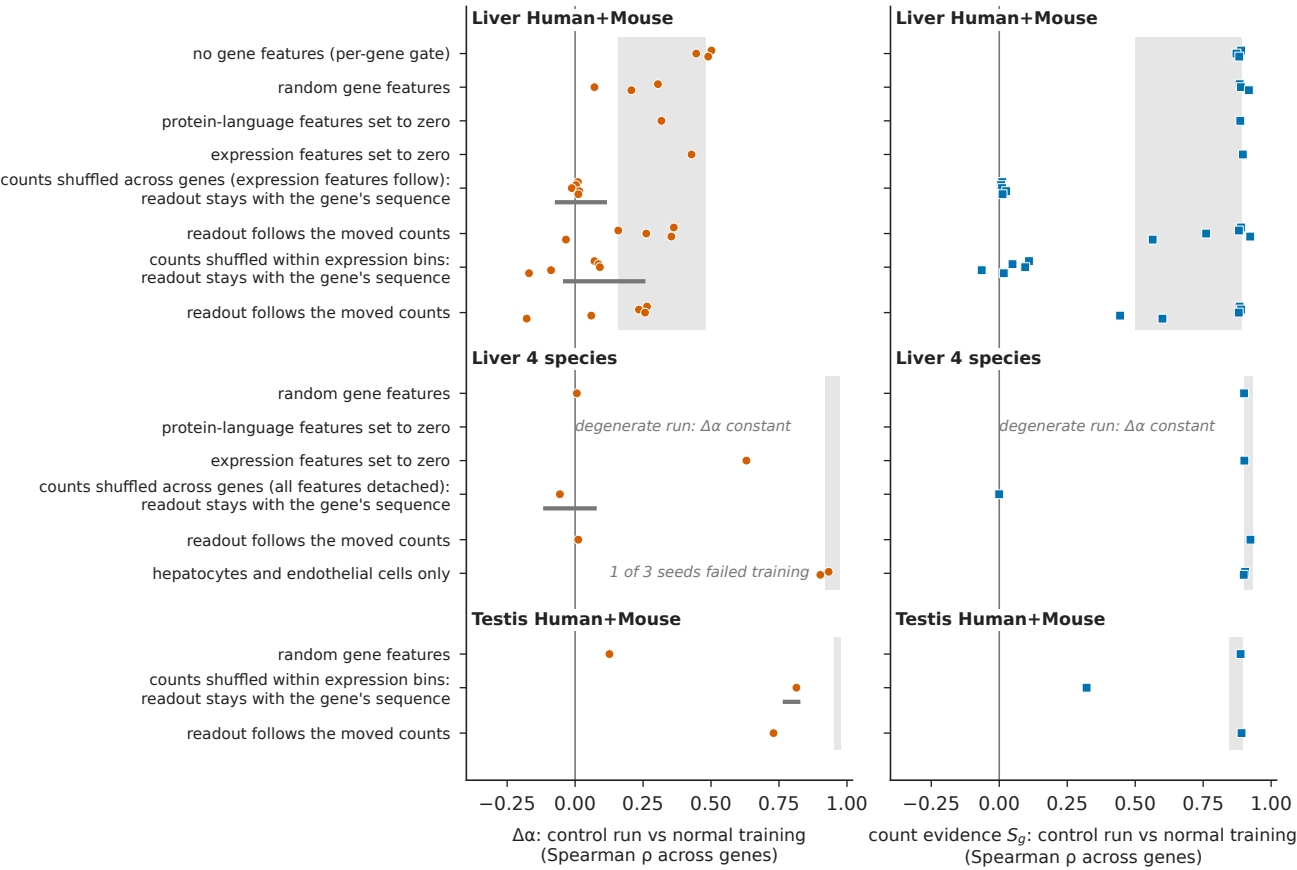
(c)


weight path
gradient × input
integrated gradients
ablation
88%
77%
17%
76%
18%
38%
9%
47%
0
20
40
60
80
100
% of total importance
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread

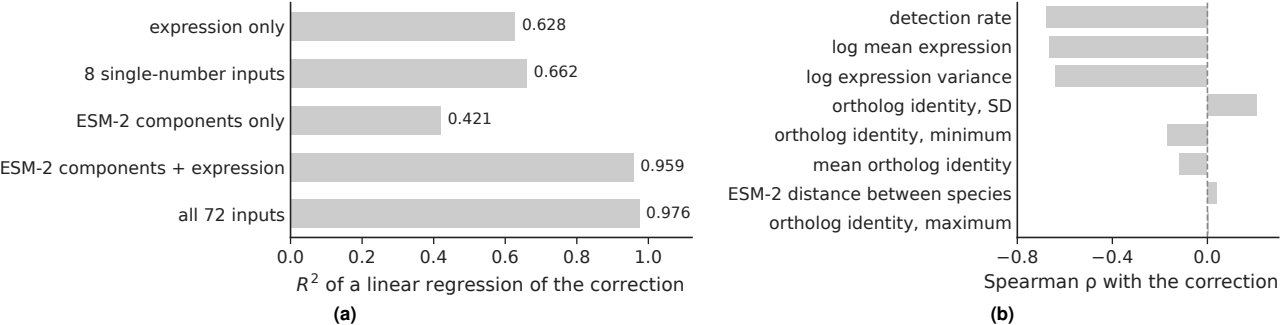
(a)

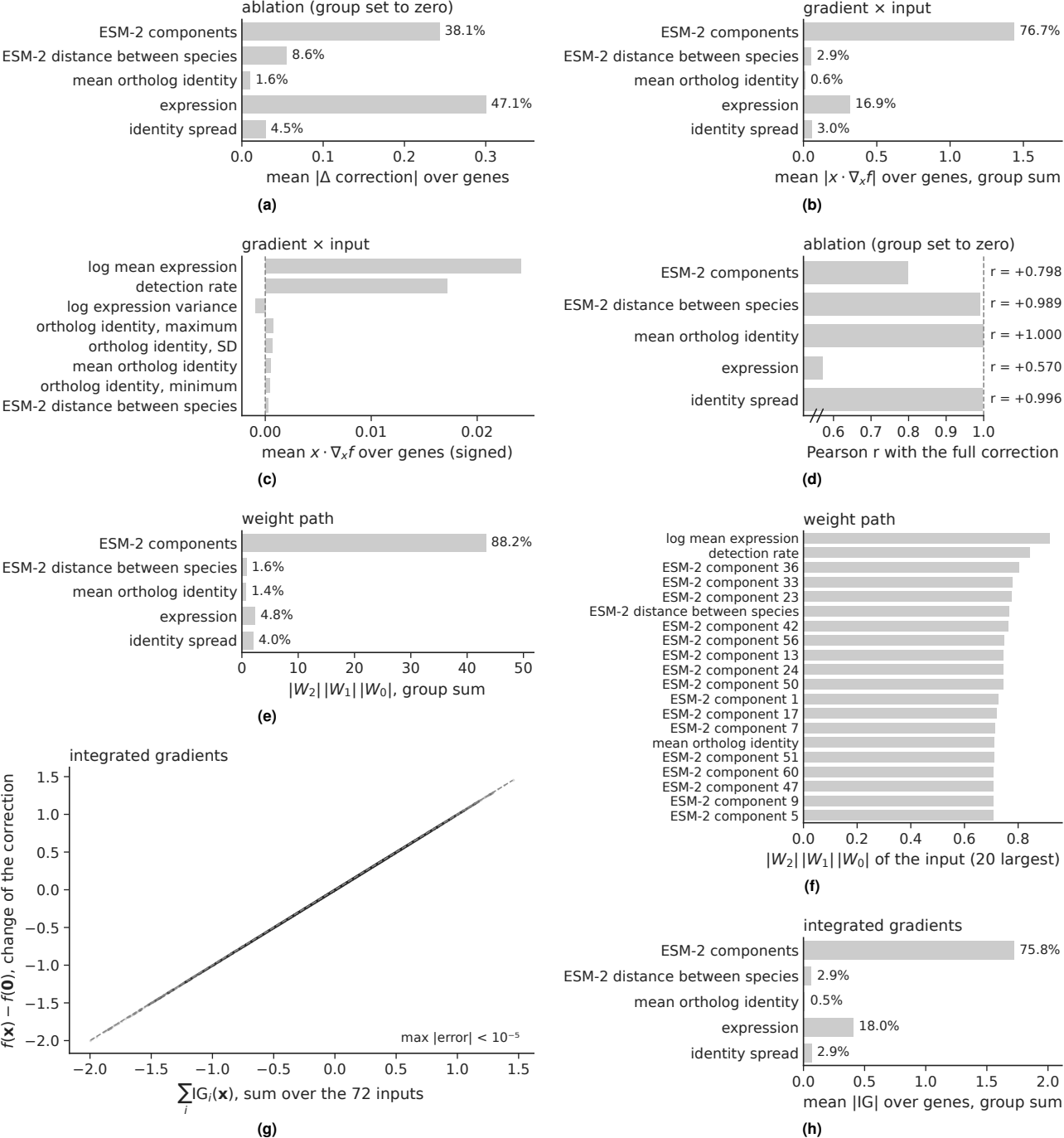

Spearman ρ of the two contributions: 0.000
ESM-2 only against the full correction: +0.592
expression only against the full correction: +0.675
expression-only correction contribution
ESM-2-only correction contribution
Δα
+0.2
+0.1
0
−0.1
−0.2

(b)


Liver Human+Mouse
no gene features (per-gene gate)
random gene features
protein-language features set to zero
expression features set to zero
counts shuffled across genes (expression features follow): readout stays with the gene's sequence
readout follows the moved counts
counts shuffled within expression bins: readout stays with the gene's sequence
readout follows the moved counts
Liver 4 species
random gene features
protein-language features set to zero
expression features set to zero
counts shuffled across genes (all features detached): readout stays with the gene's sequence
readout follows the moved counts
hepatocytes and endothelial cells only
degenerate run: Δα constant
1 of 3 seeds failed training
Testis Human+Mouse
random gene features
counts shuffled within expression bins: readout stays with the gene's sequence
readout follows the moved counts
−0.25
0.00
0.25
0.50
0.75
1.00
Δα: control run vs normal training (Spearman ρ across genes)
count evidence S_g: control run vs normal training (Spearman ρ across genes)
seed floor: one normal seed vs the others (range)
projection null for 'stays with sequence' (2.5-97.5%)
Δα, one control run
S_g, one control run

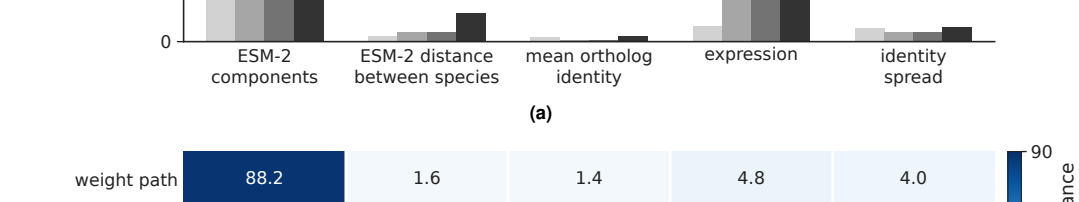
(c)


expression only
8 single-number inputs
ESM-2 components only
ESM-2 components + expression
all 72 inputs
0.628
0.662
0.421
0.959
0.976
0.0
0.2
0.4
0.6
0.8
1.0
R^2 of a linear regression of the correction
(a)
detection rate
log mean expression
log expression variance
ortholog identity, SD
ortholog identity, minimum
mean ortholog identity
ESM-2 distance between species
ortholog identity, maximum
−0.8
−0.4
0.0
Spearman ρ with the correction
(b)

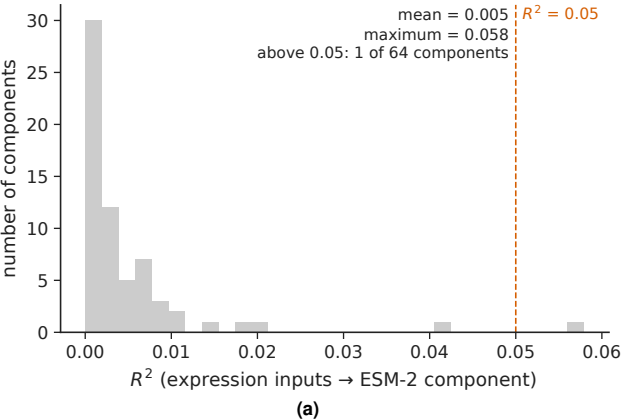

ablation (group set to zero)
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
38.1%
8.6%
1.6%
47.1%
4.5%
0.0
0.1
0.2
0.3
mean |Δ correction| over genes
(a)
gradient × input
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
76.7%
2.9%
0.6%
16.9%
3.0%
0.0
0.5
1.0
1.5
mean $|x \cdot \nabla_x f|$ over genes, group sum
(b)
gradient × input
log mean expression
detection rate
log expression variance
ortholog identity, maximum
ortholog identity, SD
mean ortholog identity
ortholog identity, minimum
ESM-2 distance between species
0.00
0.01
0.02
mean $x \cdot \nabla_x f$ over genes (signed)
(c)
ablation (group set to zero)
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
r = +0.798
r = +0.989
r = +1.000
r = +0.570
r = +0.996
0.6
0.7
0.8
0.9
1.0
Pearson r with the full correction
(d)
weight path
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
88.2%
1.6%
1.4%
4.8%
4.0%
0
10
20
30
40
50
$|W_2||W_1||W_0|$, group sum
(e)
weight path
log mean expression
detection rate
ESM-2 component 36
ESM-2 component 33
ESM-2 component 23
ESM-2 distance between species
ESM-2 component 42
ESM-2 component 56
ESM-2 component 13
ESM-2 component 24
ESM-2 component 50
ESM-2 component 1
ESM-2 component 17
ESM-2 component 7
mean ortholog identity
ESM-2 component 51
ESM-2 component 60
ESM-2 component 47
ESM-2 component 9
ESM-2 component 5
0.0
0.2
0.4
0.6
0.8
$|W_2||W_1||W_0|$ of the input (20 largest)
(f)
integrated gradients
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
$f(\mathbf{x}) - f(\mathbf{0})$, change of the correction
max |error| < 10^{-5}
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
$\sum_i \mathrm{IG}_i(\mathbf{x})$, sum over the 72 inputs
(g)
integrated gradients
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
75.8%
2.9%
0.5%
18.0%
2.9%
0.0
0.5
1.0
1.5
2.0
mean |IG| over genes, group sum
(h)

weight path
gradient × input
integrated gradients
ablation
% of total importance
0
20
40
60
80
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread

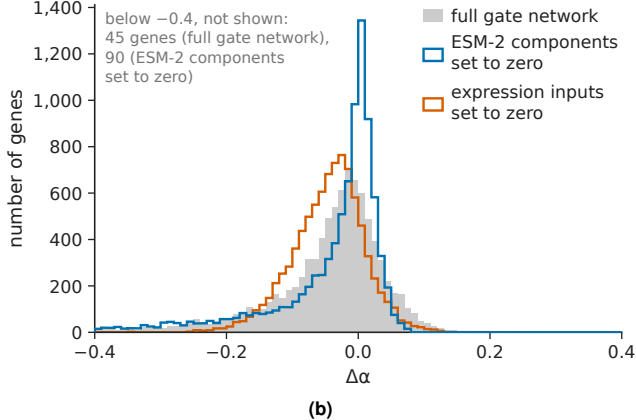
(a)


weight path
gradient × input
integrated gradients
ablation
88.2
1.6
1.4
4.8
4.0
76.7
2.9
0.6
16.9
3.0
75.8
2.9
0.5
18.0
2.9
38.1
8.6
1.6
47.1
4.5
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
% of total importance
0
30
60
90

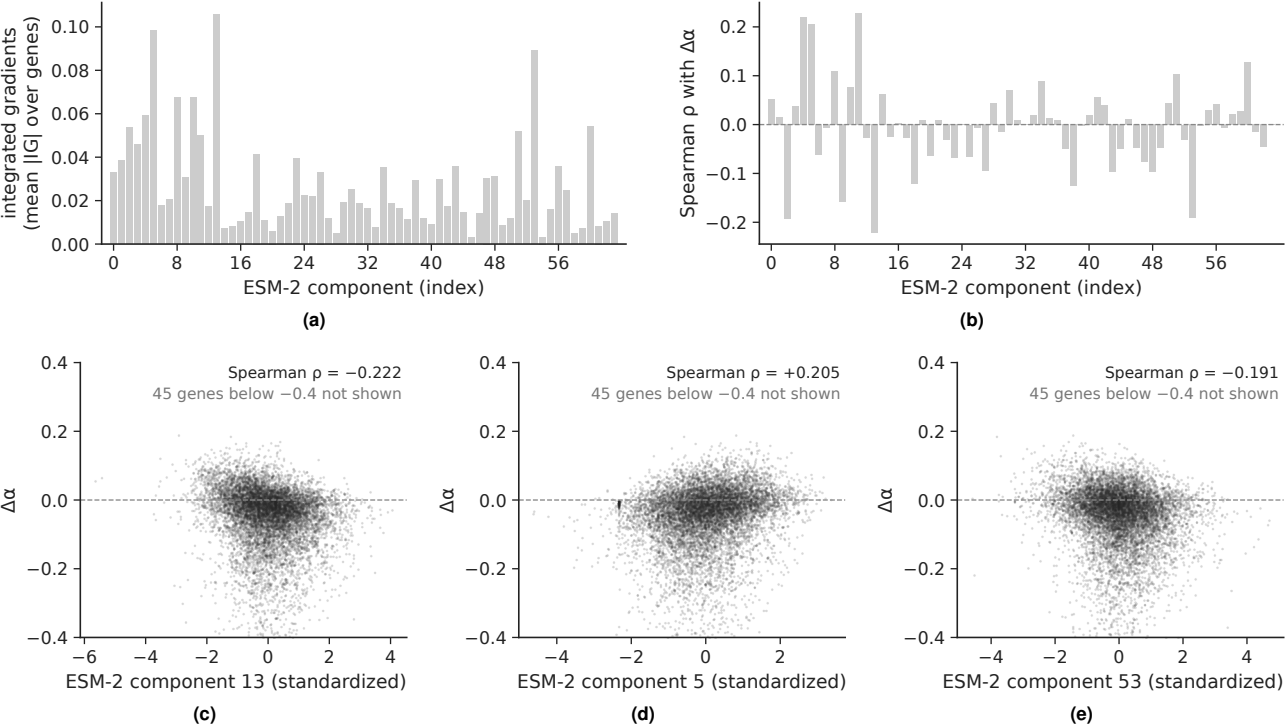
(b)

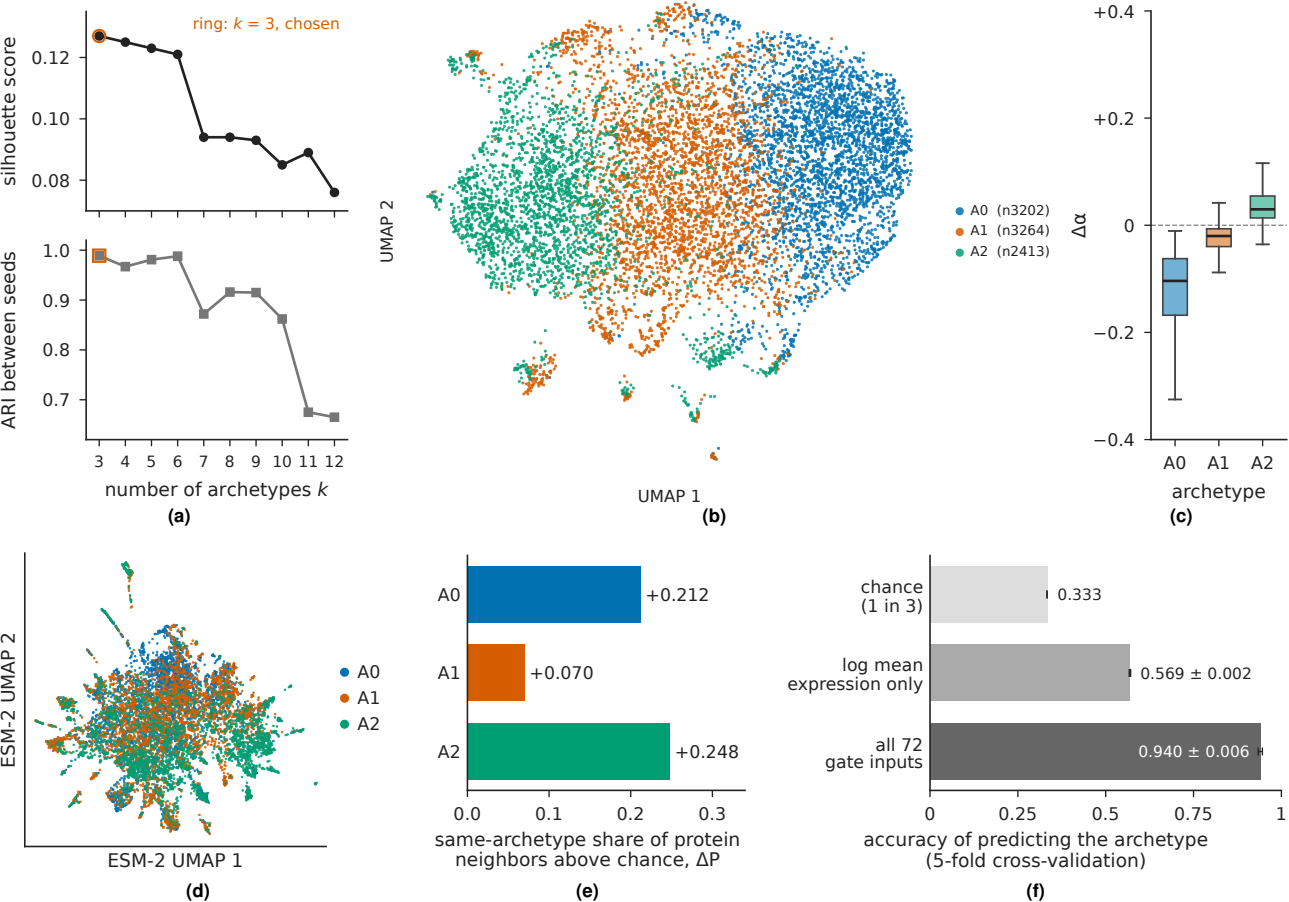

mean = 0.005
maximum = 0.058
above 0.05: 1 of 64 components
R^2 = 0.05
number of components
0
5
10
15
20
25
30
0.00
0.01
0.02
0.03
0.04
0.05
0.06
R^2 (expression inputs → ESM-2 component)

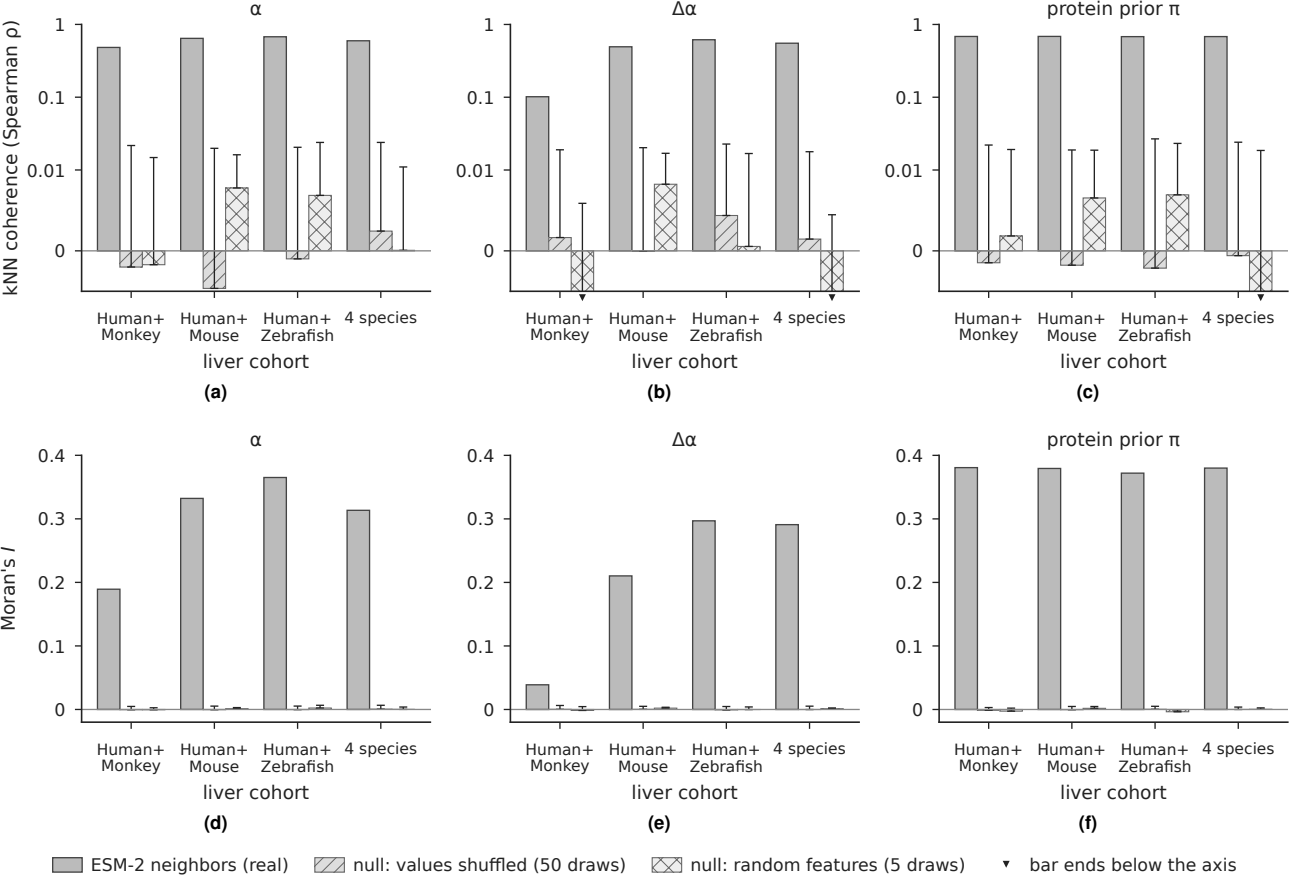
(a)

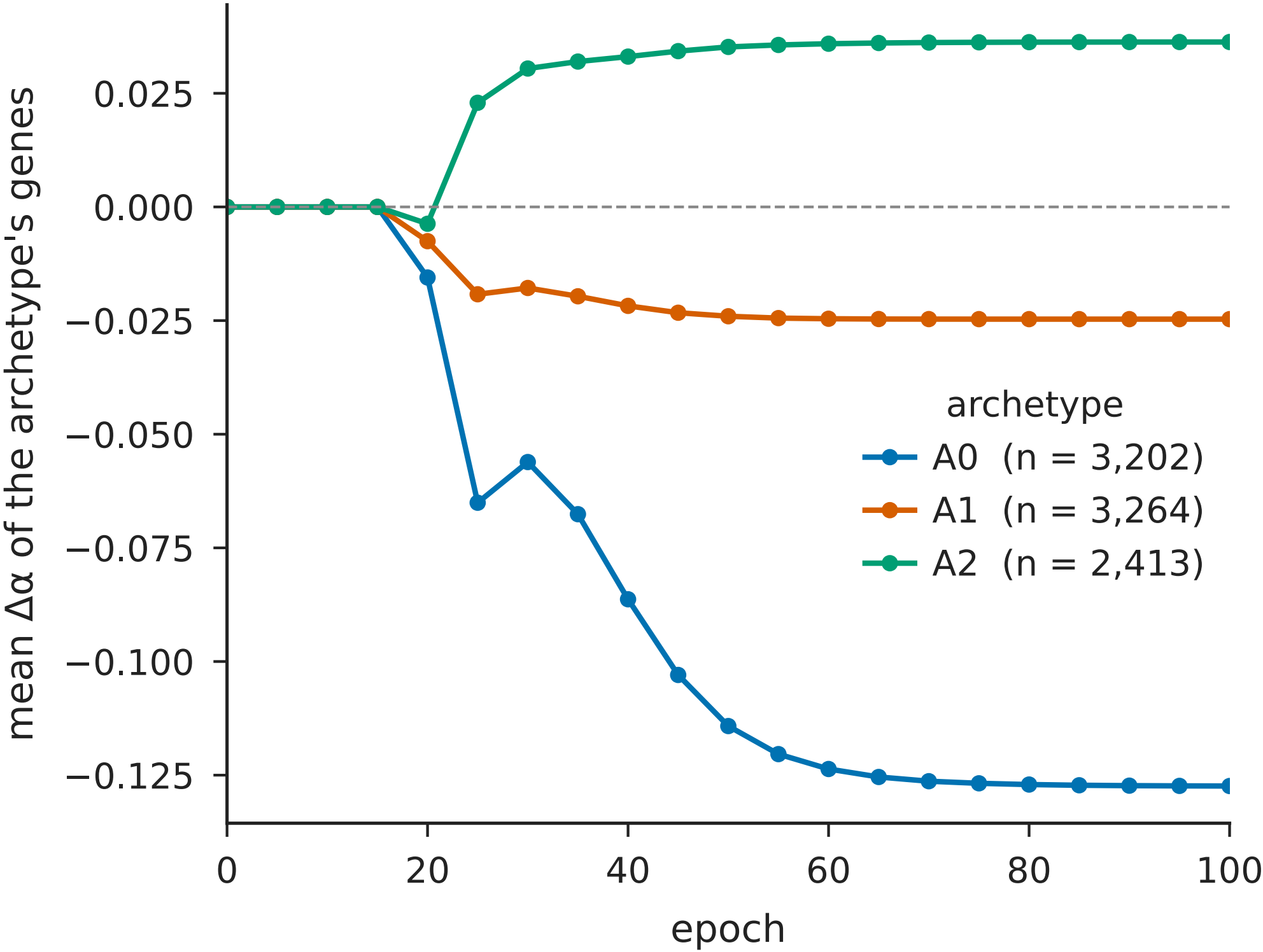

below −0.4, not shown:
45 genes (full gate network),
90 (ESM-2 components
set to zero)
full gate network
ESM-2 components
set to zero
expression inputs
set to zero
number of genes
0
200
400
600
800
1,000
1,200
1,400
−0.4
−0.2
0.0
0.2
0.4
Δα

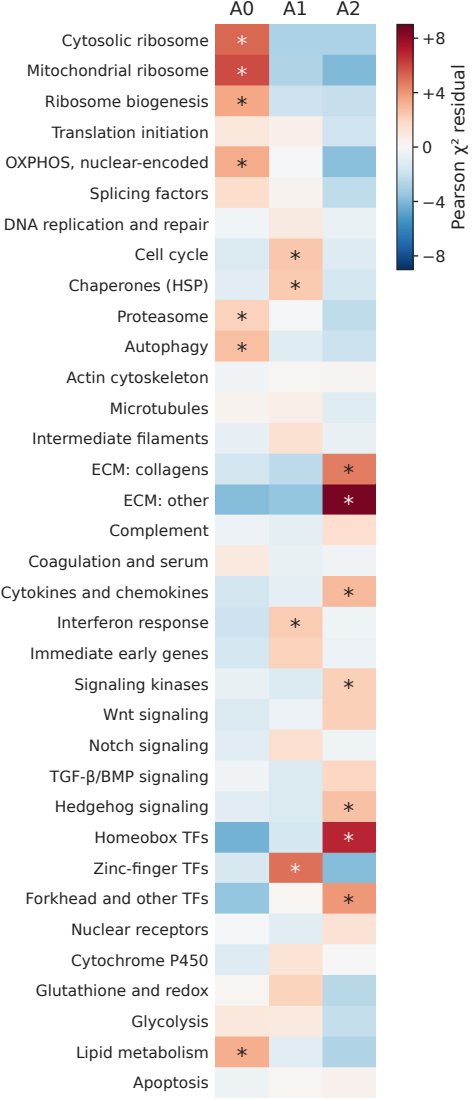
(b)


integrated gradients
(mean |IG| over genes)
0.10
0.08
0.06
0.04
0.02
0.00
0
8
16
24
32
40
48
56
ESM-2 component (index)
(a)
Spearman ρ with Δα
0.2
0.1
0.0
−0.1
−0.2
0
8
16
24
32
40
48
56
ESM-2 component (index)
(b)
Spearman ρ = −0.222
45 genes below −0.4 not shown
Δα
0.4
0.2
0.0
−0.2
−0.4
−6
−4
−2
0
2
4
ESM-2 component 13 (standardized)
(c)
Spearman ρ = +0.205
45 genes below −0.4 not shown
Δα
0.4
0.2
0.0
−0.2
−0.4
−4
−2
0
2
ESM-2 component 5 (standardized)
(d)
Spearman ρ = −0.191
45 genes below −0.4 not shown
Δα
0.4
0.2
0.0
−0.2
−0.4
−4
−2
0
2
4
ESM-2 component 53 (standardized)
(e)

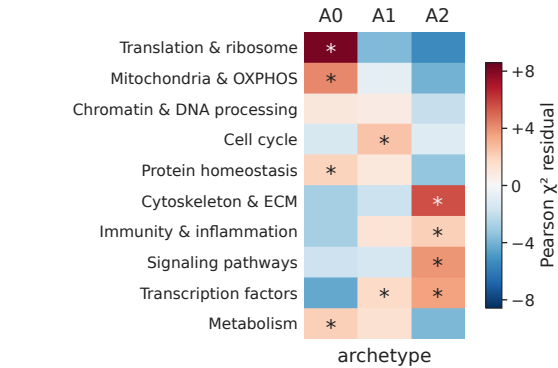

ring: k = 3, chosen
silhouette score
0.12
0.10
0.08
ARI between seeds
1.0
0.9
0.8
0.7
3 4 5 6 7 8 9 10 11 12
number of archetypes k
(a)
UMAP 2
UMAP 1
A0 (n3202)
A1 (n3264)
A2 (n2413)
(b)
+0.4
+0.2
0
−0.2
−0.4
Δα
A0 A1 A2
archetype
(c)
ESM-2 UMAP 2
ESM-2 UMAP 1
A0
A1
A2
(d)
A0 +0.212
A1 +0.070
A2 +0.248
0.0 0.1 0.2 0.3
same-archetype share of protein neighbors above chance, ΔP
(e)
chance (1 in 3) 0.333
log mean expression only 0.569 ± 0.002
all 72 gate inputs 0.940 ± 0.006
0 0.25 0.5 0.75 1
accuracy of predicting the archetype (5-fold cross-validation)
(f)


α
Δα
protein prior π
kNN coherence (Spearman ρ)
1
0.1
0.01
0
Human+ Monkey
Human+ Mouse
Human+ Zebrafish
4 species
liver cohort
(a)
(b)
(c)
Moran's I
0.4
0.3
0.2
0.1
(d)
(e)
(f)
ESM-2 neighbors (real)
null: values shuffled (50 draws)
null: random features (5 draws)
bar ends below the axis

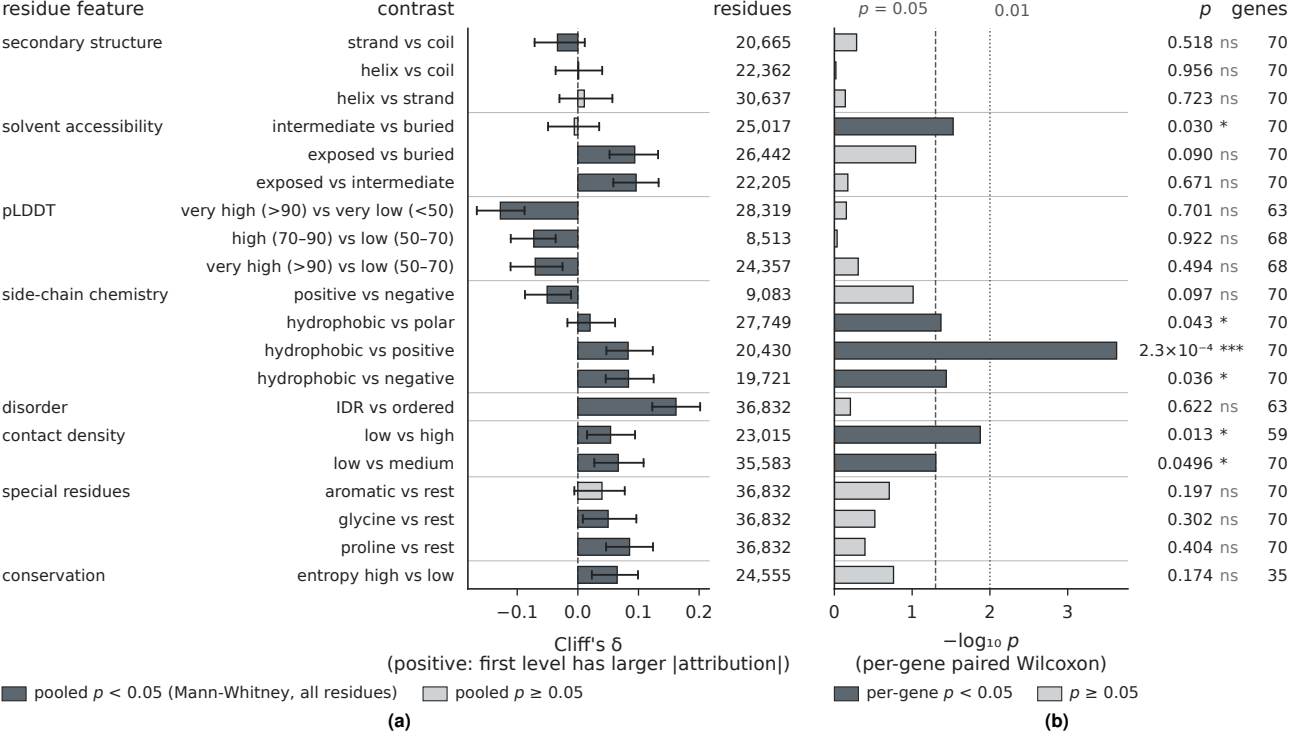

mean Δα of the archetype's genes
0.025
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
0
20
40
60
80
100
epoch
archetype
A0 (n = 3,202)
A1 (n = 3,264)
A2 (n = 2,413)

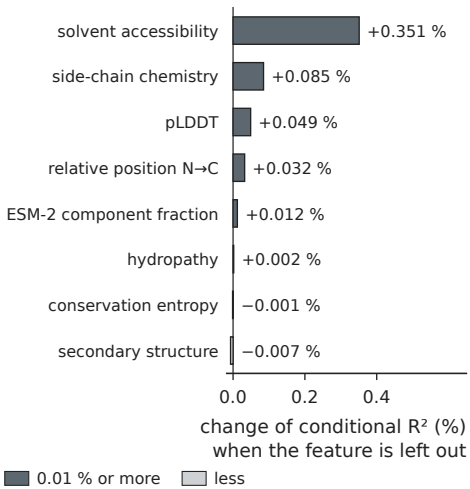

A0
A1
A2
Cytosolic ribosome
Mitochondrial ribosome
Ribosome biogenesis
Translation initiation
OXPHOS, nuclear-encoded
Splicing factors
DNA replication and repair
Cell cycle
Chaperones (HSP)
Proteasome
Autophagy
Actin cytoskeleton
Microtubules
Intermediate filaments
ECM: collagens
ECM: other
Complement
Coagulation and serum
Cytokines and chemokines
Interferon response
Immediate early genes
Signaling kinases
Wnt signaling
Notch signaling
TGF-β/BMP signaling
Hedgehog signaling
Homeobox TFs
Zinc-finger TFs
Forkhead and other TFs
Nuclear receptors
Cytochrome P450
Glutathione and redox
Glycolysis
Lipid metabolism
Apoptosis
archetype
+8
+4
0
−4
−8
Pearson χ² residual

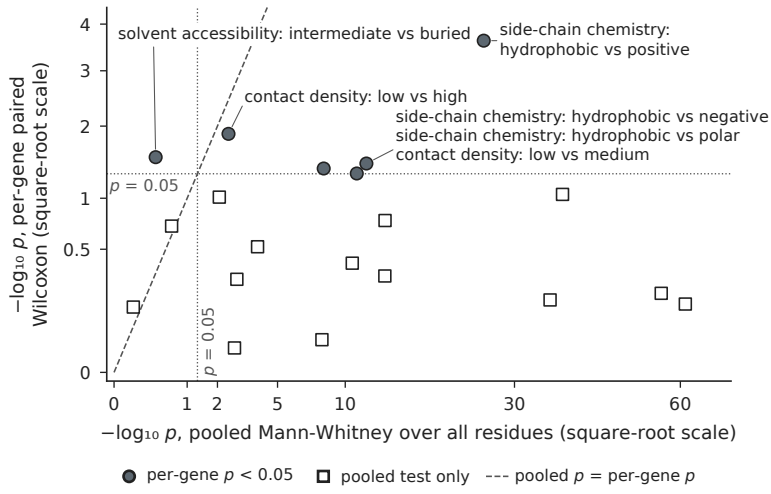
(a)

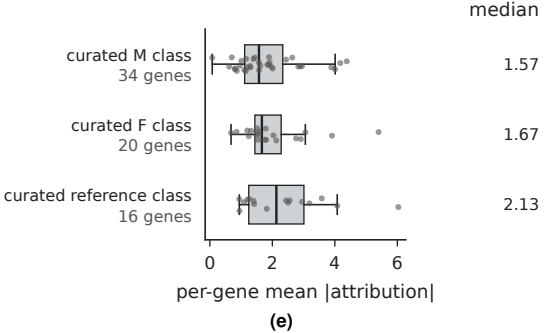

A0
A1
A2
Translation & ribosome
Mitochondria & OXPHOS
Chromatin & DNA processing
Cell cycle
Protein homeostasis
Cytoskeleton & ECM
Immunity & inflammation
Signaling pathways
Transcription factors
Metabolism
archetype
+8
+4
0
−4
−8
Pearson χ² residual

(b)

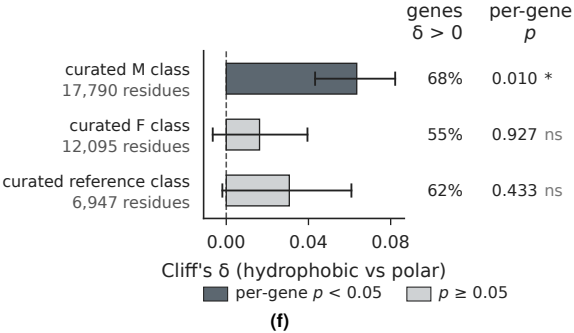

residue feature
contrast
residues
p
genes
p = 0.05
0.01
secondary structure
strand vs coil
20,665
0.518 ns 70
helix vs coil
22,362
0.956 ns 70
helix vs strand
30,637
0.723 ns 70
solvent accessibility
intermediate vs buried
25,017
0.030 * 70
exposed vs buried
26,442
0.090 ns 70
exposed vs intermediate
22,205
0.671 ns 70
pLDDT
very high (>90) vs very low (<50)
28,319
0.701 ns 63
high (70–90) vs low (50–70)
8,513
0.922 ns 68
very high (>90) vs low (50–70)
24,357
0.494 ns 68
side-chain chemistry
positive vs negative
9,083
0.097 ns 70
hydrophobic vs polar
27,749
0.043 * 70
hydrophobic vs positive
20,430
2.3×10⁻⁴ *** 70
hydrophobic vs negative
19,721
0.036 * 70
disorder
IDR vs ordered
36,832
0.622 ns 63
contact density
low vs high
23,015
0.013 * 59
low vs medium
35,583
0.0496 * 70
special residues
aromatic vs rest
36,832
0.197 ns 70
glycine vs rest
36,832
0.302 ns 70
proline vs rest
36,832
0.404 ns 70
conservation
entropy high vs low
24,555
0.174 ns 35
−0.1
0.0
0.1
0.2
Cliff's δ
(positive: first level has larger |attribution|)
0
1
2
3
−log₁₀ p
(per-gene paired Wilcoxon)
pooled p < 0.05 (Mann-Whitney, all residues)
pooled p ≥ 0.05
per-gene p < 0.05
p ≥ 0.05
(a)
(b)


solvent accessibility
+0.351 %
side-chain chemistry
+0.085 %
pLDDT
+0.049 %
relative position N→C
+0.032 %
ESM-2 component fraction
+0.012 %
hydropathy
+0.002 %
conservation entropy
−0.001 %
secondary structure
−0.007 %
0.0
0.2
0.4
change of conditional R² (%)
when the feature is left out
0.01 % or more
less
(c)

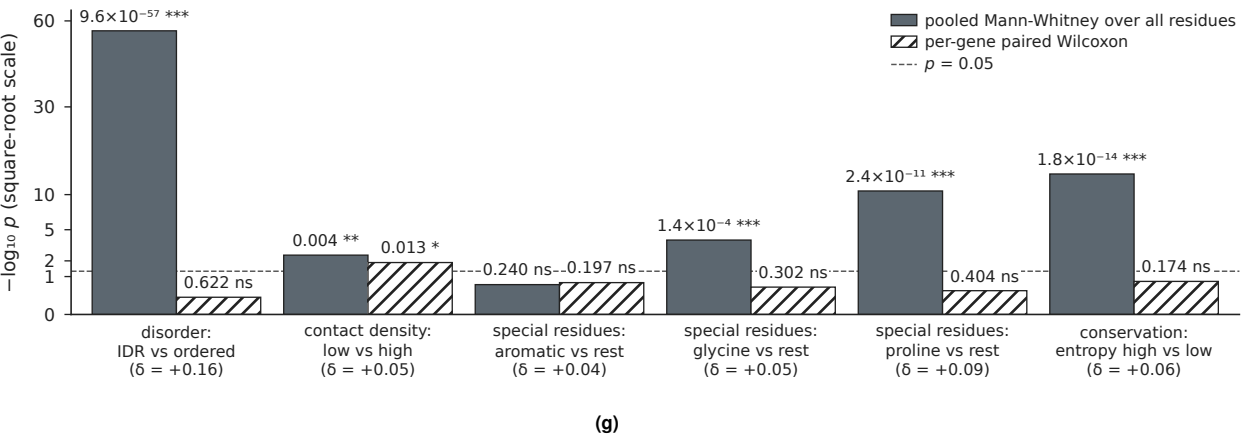

solvent accessibility: intermediate vs buried
side-chain chemistry:
hydrophobic vs positive
contact density: low vs high
side-chain chemistry: hydrophobic vs negative
side-chain chemistry: hydrophobic vs polar
contact density: low vs medium
p = 0.05
p = 0.05
4
3
2
1
0.5
0
0
1
2
5
10
30
60
−log₁₀ p, per-gene paired
Wilcoxon (square-root scale)
−log₁₀ p, pooled Mann-Whitney over all residues (square-root scale)
per-gene p < 0.05
pooled test only
pooled p = per-gene p
(d)


median
curated M class
34 genes
1.57
curated F class
20 genes
1.67
curated reference class
16 genes
2.13
0
2
4
6
per-gene mean |attribution|

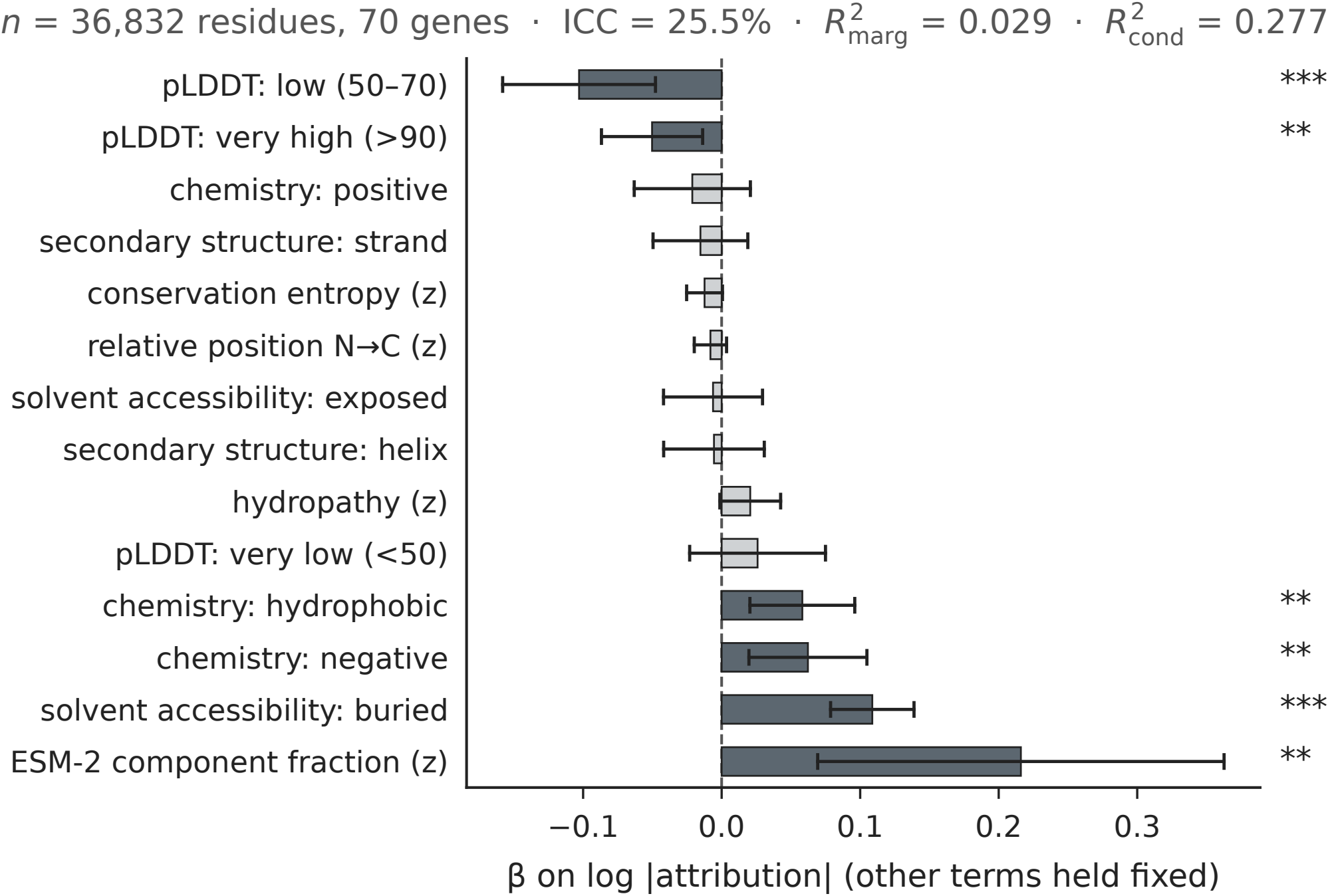
(e)

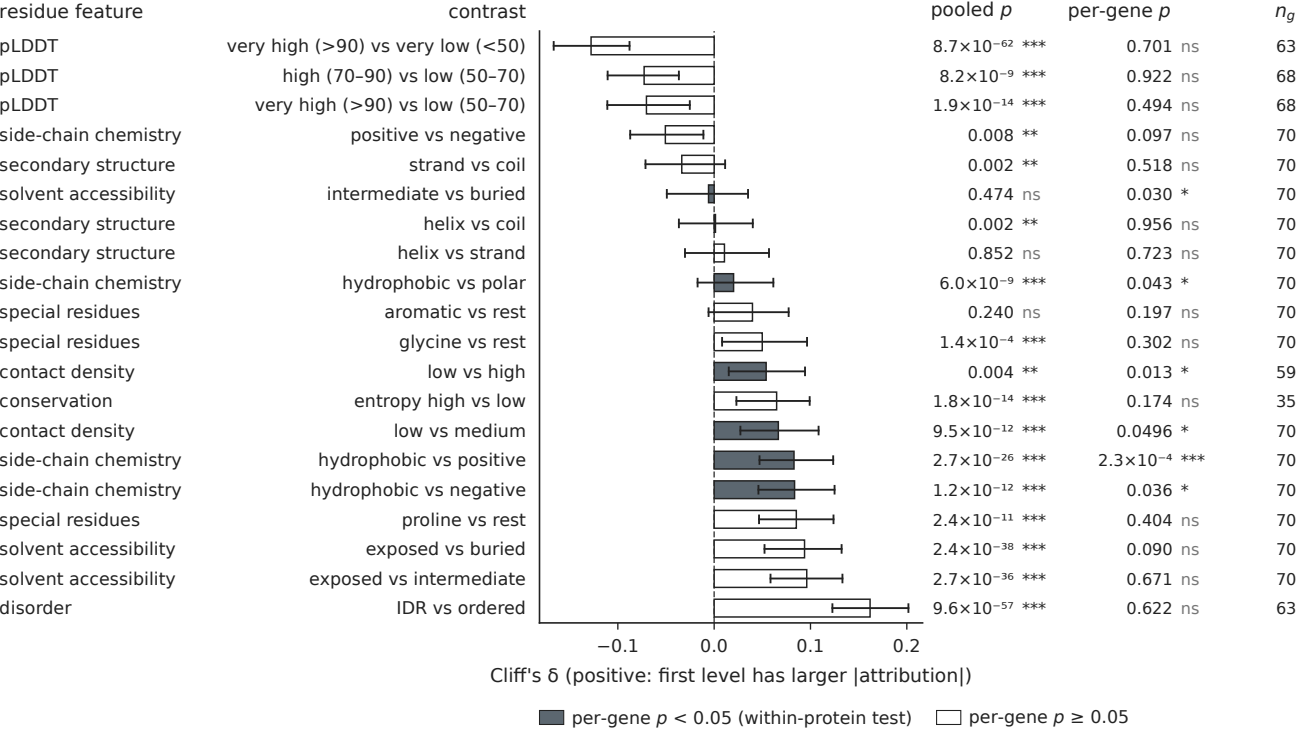

genes
δ > 0
per-gene
p
curated M class
17,790 residues
68%
0.010 *
curated F class
12,095 residues
55%
0.927 ns
curated reference class
6,947 residues
62%
0.433 ns
0.00
0.04
0.08
Cliff's δ (hydrophobic vs polar)
per-gene p < 0.05
p ≥ 0.05

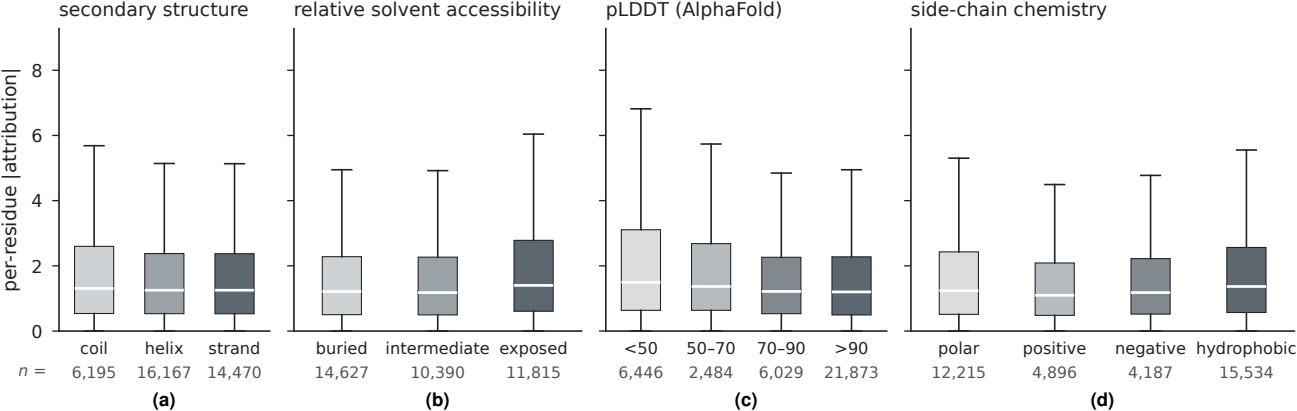
(f)

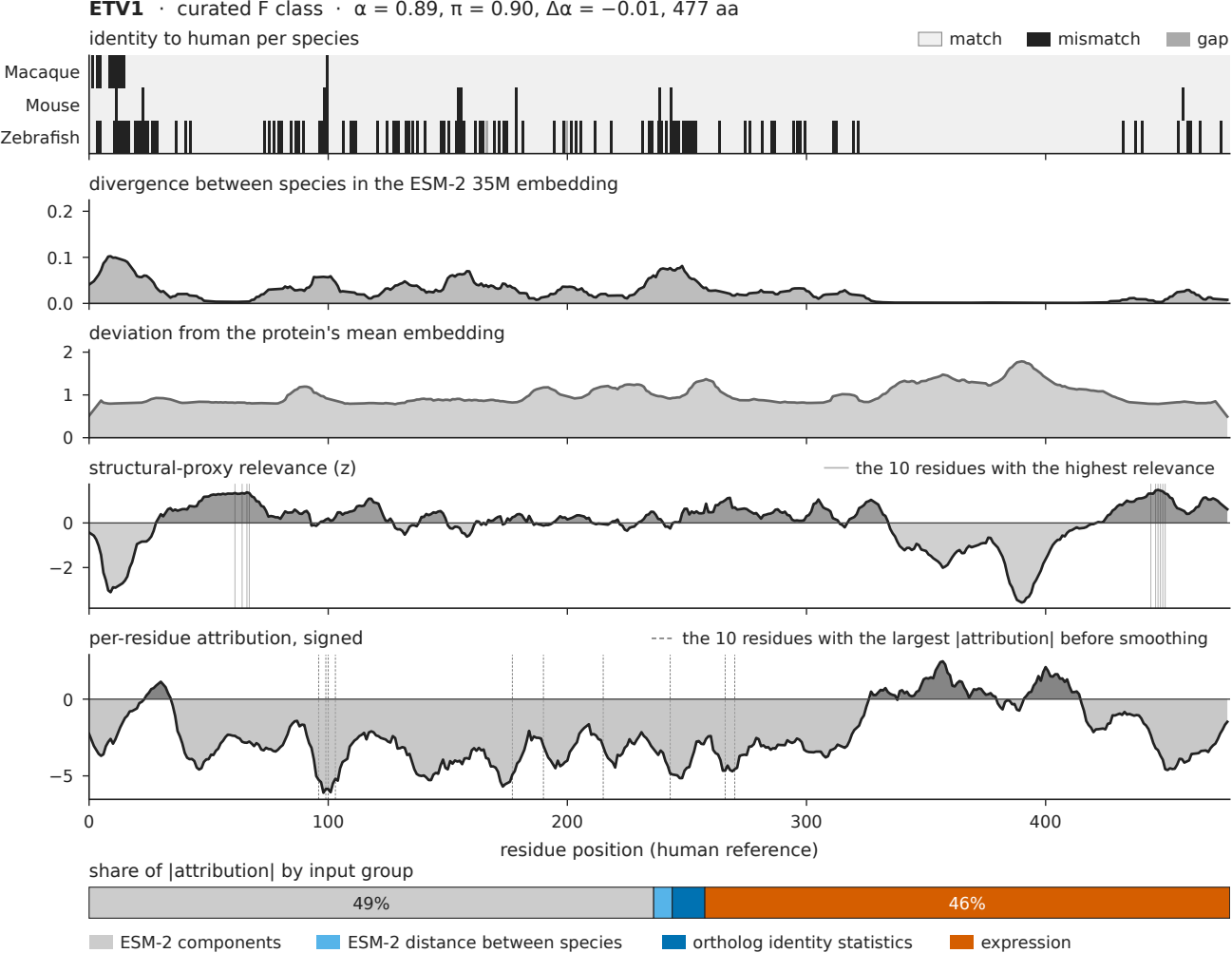

pooled Mann-Whitney over all residues
per-gene paired Wilcoxon
p = 0.05
−log10 p (square-root scale)
60
30
10
5
2
1
0
9.6×10−57 ***
0.622 ns
0.004 **
0.013 *
0.240 ns
0.197 ns
1.4×10−4 ***
0.302 ns
2.4×10−11 ***
0.404 ns
1.8×10−14 ***
0.174 ns
disorder:
IDR vs ordered
(δ = +0.16)
contact density:
low vs high
(δ = +0.05)
special residues:
aromatic vs rest
(δ = +0.04)
special residues:
glycine vs rest
(δ = +0.05)
special residues:
proline vs rest
(δ = +0.09)
conservation:
entropy high vs low
(δ = +0.06)

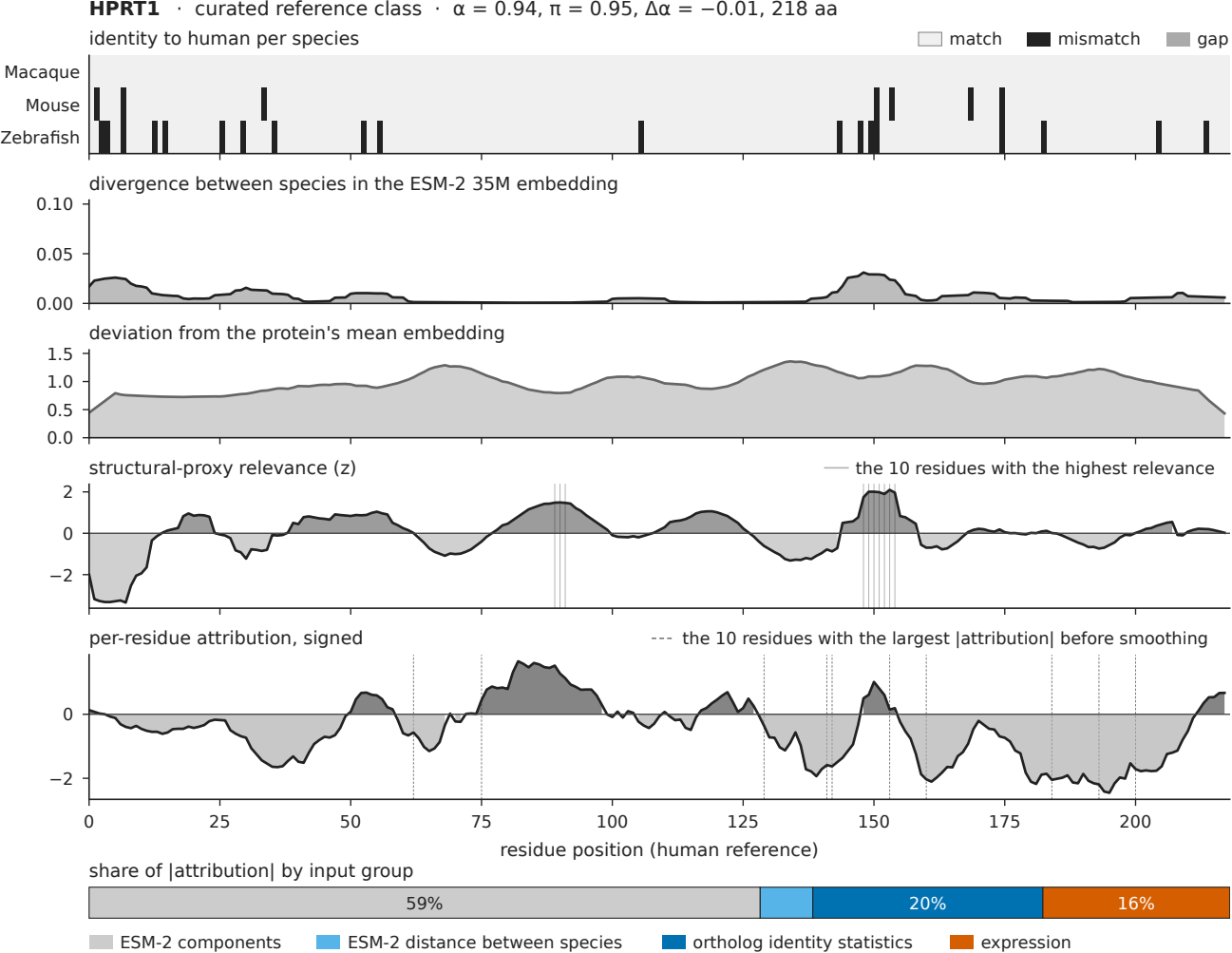
(g)

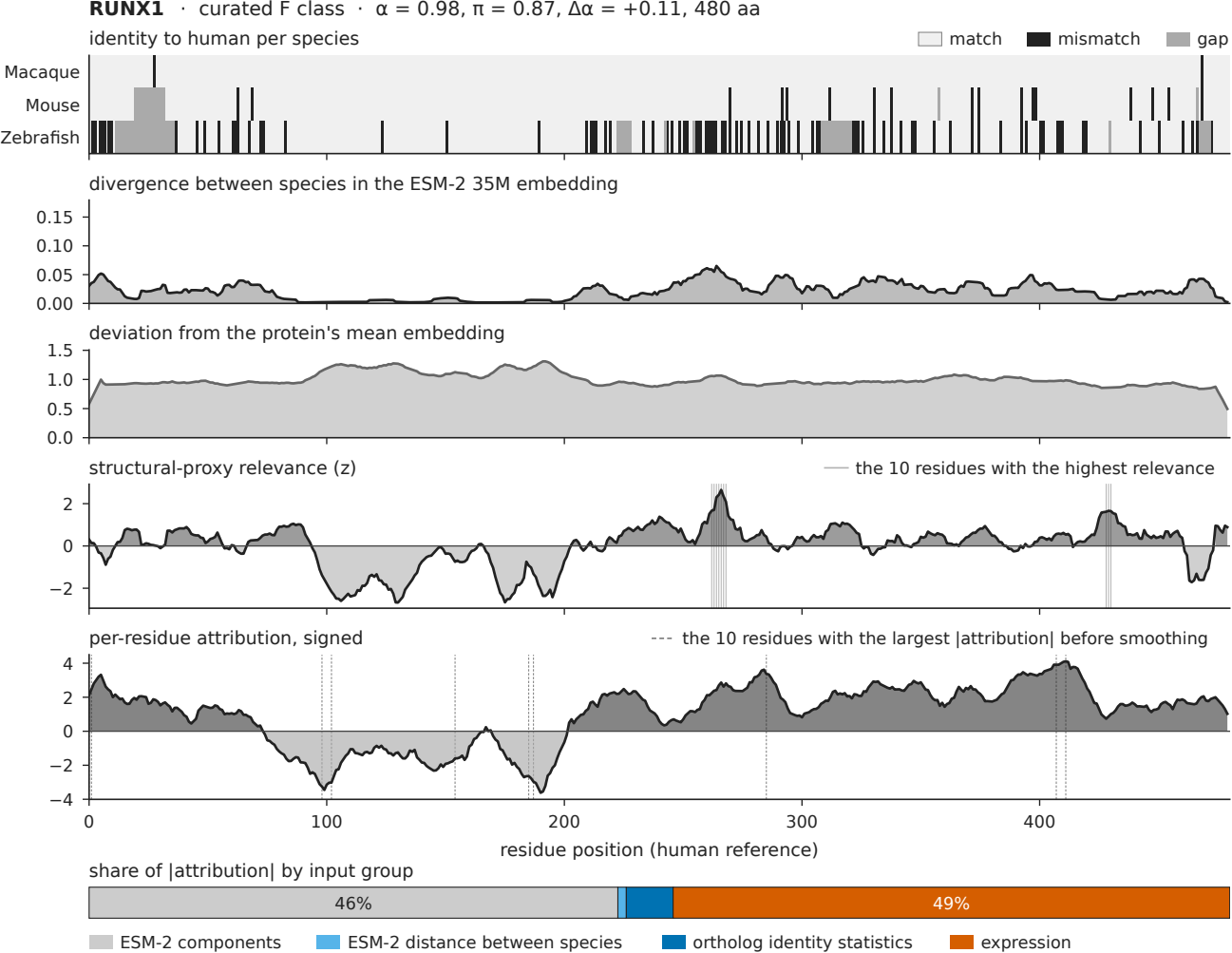

n = 36,832 residues, 70 genes · ICC = 25.5% · R^2_{marg} = 0.029 · R^2_{cond} = 0.277
pLDDT: low (50–70)
pLDDT: very high (>90)
chemistry: positive
secondary structure: strand
conservation entropy (z)
relative position N→C (z)
solvent accessibility: exposed
secondary structure: helix
hydropathy (z)
pLDDT: very low (<50)
chemistry: hydrophobic
chemistry: negative
solvent accessibility: buried
ESM-2 component fraction (z)
***
**
**
**
***
**
−0.1
0.0
0.1
0.2
0.3
β on log |attribution| (other terms held fixed)


residue feature
contrast
pooled p
per-gene p
n_g
pLDDT
very high (>90) vs very low (<50)
8.7×10⁻⁶² ***
0.701 ns
63
pLDDT
high (70–90) vs low (50–70)
8.2×10⁻⁹ ***
0.922 ns
68
pLDDT
very high (>90) vs low (50–70)
1.9×10⁻¹⁴ ***
0.494 ns
68
side-chain chemistry
positive vs negative
0.008 **
0.097 ns
70
secondary structure
strand vs coil
0.002 **
0.518 ns
70
solvent accessibility
intermediate vs buried
0.474 ns
0.030 *
70
secondary structure
helix vs coil
0.002 **
0.956 ns
70
secondary structure
helix vs strand
0.852 ns
0.723 ns
70
side-chain chemistry
hydrophobic vs polar
6.0×10⁻⁹ ***
0.043 *
70
special residues
aromatic vs rest
0.240 ns
0.197 ns
70
special residues
glycine vs rest
1.4×10⁻⁴ ***
0.302 ns
70
contact density
low vs high
0.004 **
0.013 *
59
conservation
entropy high vs low
1.8×10⁻¹⁴ ***
0.174 ns
35
contact density
low vs medium
9.5×10⁻¹² ***
0.0496 *
70
side-chain chemistry
hydrophobic vs positive
2.7×10⁻²⁶ ***
2.3×10⁻⁴ ***
70
side-chain chemistry
hydrophobic vs negative
1.2×10⁻¹² ***
0.036 *
70
special residues
proline vs rest
2.4×10⁻¹¹ ***
0.404 ns
70
solvent accessibility
exposed vs buried
2.4×10⁻³⁸ ***
0.090 ns
70
solvent accessibility
exposed vs intermediate
2.7×10⁻³⁶ ***
0.671 ns
70
disorder
IDR vs ordered
9.6×10⁻⁵⁷ ***
0.622 ns
63
−0.1
0.0
0.1
0.2
Cliff's δ (positive: first level has larger |attribution|)
per-gene p < 0.05 (within-protein test)
per-gene p ≥ 0.05


secondary structure
relative solvent accessibility
pLDDT (AlphaFold)
side-chain chemistry
per-residue |attribution|
0
2
4
6
8
coil
helix
strand
buried
intermediate
exposed
<50
50–70
70–90
>90
polar
positive
negative
hydrophobic
n =
6,195
16,167
14,470
14,627
10,390
11,815
6,446
2,484
6,029
21,873
12,215
4,896
4,187
15,534
(a)
(b)
(c)
(d)


ETV1 · curated F class · α = 0.89, π = 0.90, Δα = −0.01, 477 aa
identity to human per species
match
mismatch
gap
Macaque
Mouse
Zebrafish
divergence between species in the ESM-2 35M embedding
0.2
0.1
0.0
deviation from the protein's mean embedding
2
1
0
structural-proxy relevance (z)
the 10 residues with the highest relevance
0
−2
per-residue attribution, signed
the 10 residues with the largest |attribution| before smoothing
0
−5
0
100
200
300
400
residue position (human reference)
share of |attribution| by input group
49%
46%
ESM-2 components
ESM-2 distance between species
ortholog identity statistics
expression


HPRT1 · curated reference class · α = 0.94, π = 0.95, Δα = −0.01, 218 aa
identity to human per species
match
mismatch
gap
Macaque
Mouse
Zebrafish
divergence between species in the ESM-2 35M embedding
0.10
0.05
0.00
deviation from the protein's mean embedding
1.5
1.0
0.5
0.0
structural-proxy relevance (z)
the 10 residues with the highest relevance
2
0
−2
per-residue attribution, signed
the 10 residues with the largest |attribution| before smoothing
0
−2
0
25
50
75
100
125
150
175
200
residue position (human reference)
share of |attribution| by input group
59%
20%
16%
ESM-2 components
ESM-2 distance between species
ortholog identity statistics
expression


RUNX1 · curated F class · α = 0.98, π = 0.87, Δα = +0.11, 480 aa
identity to human per species
match
mismatch
gap
Macaque
Mouse
Zebrafish
divergence between species in the ESM-2 35M embedding
0.15
0.10
0.05
0.00
deviation from the protein's mean embedding
1.5
1.0
0.5
0.0
structural-proxy relevance (z)
the 10 residues with the highest relevance
2
0
−2
per-residue attribution, signed
the 10 residues with the largest |attribution| before smoothing
4
2
0
−2
−4
0
100
200
300
400
residue position (human reference)
share of |attribution| by input group
46%
49%
ESM-2 components
ESM-2 distance between species
ortholog identity statistics
expression